# NEKHOROSHEV STABILITY FOR QUASI-CONVEX HAMILTONIANS: BLUEPRINT (TRACK N)

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Abstract. A proof plan, step by step, for the three frozen statements of track N of the KAM formalization: the exponential stability estimate for strictly quasi-convex nearly integrable Hamiltonians with a complex Hessian bound (N-HEAD-C, after Pöschel 1993, Theorem 1), the exact integrable dynamics for zero perturbation (N-ZERO) and the nonsingular isoenergetic derivative (N-QC). Every result is labelled and cites its pinned source. This version contains Part 0 only; the plan of the remaining parts is in `PLAN.md`.

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## 1. Conventions, definitions and the frozen statements

This part fixes notation and restates, in mathematical form, the three frozen theorems of track N (owner approval D20, 27 October 2026) and the eleven frozen definitions of `KAM/Frozen/Nekhoroshev/Defs/` together with the five track-K definitions they use. The Lean declarations are authoritative; the statements below are transcriptions of them, with the Lean names recorded under each. N-HEAD with real bounds only is *not* frozen (source gap F-N1, D17/D18/D20); the frozen headline is N-HEAD-C. The additions of D22 (N-PARTIAL-CONVEX, N-AVERAGING) were frozen by D25 and D28 and are transcribed below. Results of track K's blueprint are cited as [K:`bp:p1:...`].

### 1.1. Conventions.

- $n \in \mathbb{N}$ is the number of degrees of freedom; N-HEAD-C assumes $n \geq 3$ (D17), N-ZERO and N-QC are stated for every $n$. A phase point is $z = (\theta, I)$ with angles $\theta$ and actions $I$, in this order.
- Ambient spaces are those of the Lean text: $\mathbb{R}^n = (\texttt{Fin}\ n \to \mathbb{R})$, $\mathbb{C}^n = (\texttt{Fin}\ n \to \mathbb{C})$, with the max norms $|v| = \max_i |v_i|$ on both. The embedding $\mathbb{R}^n \subset \mathbb{C}^n$ is componentwise. The norm $\|\cdot\|$ in every frozen statement is the max norm; on the phase space $\mathbb{R}^n \times \mathbb{R}^n$ it is $\|(\theta, I)\| = \max(|\theta|, |I|)$ (Lean product norm). The only Euclidean structure is the dot

product $\sum_i \omega_i v_i$ used for orthogonality in quasi-convexity; Euclidean norms used later for comparison with the sources come with explicit inequalities ($|v| \le \|v\|_2 \le \sqrt{n}|v|$). For $n = 0$ all these spaces are $\{0\}$ and the norm is 0.
- Operator norms of linear and multilinear maps are the induced norms of these max norms. In particular $\|D^2 h(z)\|$ for complex $h$ is the norm of $D^2 h(z) \in \mathcal{L}(\mathbb{C}^n, \mathcal{L}(\mathbb{C}^n, \mathbb{C}))$ (Lean `fderiv` applied twice), i.e. $\sup_{|u|,|v|\le 1} |D^2 h(z)[u, v]|$.
- Angles are *not* reduced modulo $\mathbb{Z}^n$ in the statements: $\theta \in \mathbb{R}^n$ or $\theta \in T_s \subset \mathbb{C}^n$, and 1-periodicity is a hypothesis on $f$ (Definition 1.12). So the torus is $\mathbb{R}^n/\mathbb{Z}^n$ and the Fourier characters are $e^{2\pi i k\cdot\theta}$ as in track K ([K:`bp:p1:decay`], [K:`bp:p1:summation`]). Pöschel's torus is $\mathbb{R}^n/2\pi\mathbb{Z}^n$ and his norms are Euclidean; the conversion is a separate step of the proof (Part 1 of `PLAN.md`).
- The complex strip is $T_s = \{\theta \in \mathbb{C}^n : |\operatorname{Im}\theta_i| < s \ \forall i\}$ (Lean `KAM.complexStrip`), as in track K.

## 1.2. **Track-K definitions used.**

**Definition 1.1** (Complex action neighbourhood)**.** For $B \subset \mathbb{R}^n$ and $r \in \mathbb{R}$, $B_r = \{I \in \mathbb{C}^n : \exists J \in B,\ |I - J| < r\}$ (max norm, $J$ viewed in $\mathbb{C}^n$). Thus $B_r = \emptyset$ for $r \le 0$, and for $B$ open $B_r$ is open.
Frozen Lean declaration: `KAM.actionNbhd`
Source: BM11 p. 4 ($D_s$); track K, [K:`bp:p0:domains`].

**Definition 1.2** (Frequency map)**.** For $h : \mathbb{C}^n \to \mathbb{C}$ and $I \in \mathbb{R}^n$, $\omega_h(I) = \nabla h(I) \in \mathbb{R}^n$ is the vector with $\omega_h(I)_i = D\big(J \mapsto \operatorname{Re} h(J)\big)(I)[e_i]$, the real partial derivatives of $\operatorname{Re} h$ on $\mathbb{R}^n$ at $I$. It is 0 if $\operatorname{Re} h|_{\mathbb{R}^n}$ is not differentiable at $I$ (Lean totality, Remark 1.18).
Frozen Lean declaration: `KAM.frequencyMap`
Source: BM11 p. 1; Pos93 p. 190 ($\omega(I) = \partial_I h(I)$).

**Definition 1.3** (Hamiltonian vector field)**.** For $H : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$ and $z = (\theta, I)$, $X_H(z) = \big((\partial_{I_i} H(z))_i,\ (-\partial_{\theta_i} H(z))_i\big)$, where $\partial_{I_i} H(z) = DH(z)[(0, e_i)]$ and $\partial_{\theta_i} H(z) = DH(z)[(e_i, 0)]$ are the Fréchet derivatives in the Lean sense (0 if $H$ is not differentiable at $z$).
Frozen Lean declaration: `KAM.hamiltonianVectorField`
Source: BM11 p. 1; track K, [K:`bp:p0:hamiltonian`].

**Definition 1.4** (Real Hamiltonian)**.** For $h : \mathbb{C}^n \to \mathbb{C}$, $f : \mathbb{C}^n \times \mathbb{C}^n \to \mathbb{C}$ and real $z = (\theta, I)$, $H_{h,f}(z) = \operatorname{Re}\big(h(I) + f(\theta, I)\big)$, with $\theta, I$ viewed in $\mathbb{C}^n$.
Frozen Lean declaration: `KAM.realHamiltonian`
Source: BM11 p. 5 ($H = h + f$).

*Remark* 1.5 (Strip and neighbourhood notation). `KAM.complexStrip` $n\, s = T_s$ as above; for $n > 0$ it is empty when $s \le 0$, and for $n = 0$ it is the whole (one-point) space $\mathbb{C}^0$ for every $s$. No hypothesis $n > 0$ is added to N-ZERO or N-QC.
Frozen Lean declaration: `KAM.complexStrip`

## 1.3. **Frozen track-N definitions.**

**Definition 1.6** (Action ball)**.** $B(0, R) = \{I \in \mathbb{R}^n : |I| < R\}$, the open ball of the max norm; written `actionBall` $n\, R$.
Frozen Lean declaration: `KAM.Nekhoroshev.actionBall`
Source: BM11 p. 4.

**Definition 1.7** (Complex domain)**.** $D_s(R) = T_s \times B(0, R)_s \subset \mathbb{C}^n \times \mathbb{C}^n$; `complexDomain` $n\, R\, s$. It is the period-one lift of BM11's $D_s$, with a single width $s$ for angles and actions. It is open.
Frozen Lean declaration: `KAM.Nekhoroshev.complexDomain`
Source: BM11 p. 4 ($D_s$); ours for the lift.

**Definition 1.8** (Real action domain). $\mathcal{B}_s(R) = \{I \in \mathbb{R}^n : I \in B(0,R)_s\}$; `realActionDomain` $n\,R\,s$.
Frozen Lean declaration: `KAM.Nekhoroshev.realActionDomain`
Source: BM11 p. 4 ($B_s$).

*Remark* 1.9 (Computing $\mathcal{B}_s(R)$). Since $|I-J|$ for real $I, J$ is the real max norm, $\mathcal{B}_s(R) = \{I \in \mathbb{R}^n : \exists J,\ |J| < R,\ |I-J| < s\} = B(0, R+s)$ for $R, s > 0$. This identity is ours and is proved from the definitions (it is not stated in the source); it shows that $B(0,R) \subset \mathcal{B}_s(R)$ and that $\mathcal{B}_s(R)$ is open.

**Definition 1.10** (Real part of the integrable part). For $h : \mathbb{C}^n \to \mathbb{C}$ and $I \in \mathbb{R}^n$, $h_{\mathbb{R}}(I) = \operatorname{Re} h(I)$; `realPart` $h$.
Frozen Lean declaration: `KAM.Nekhoroshev.realPart`
Source: BM11 p. 4 ($h : B_s \to \mathbb{R}$).

**Definition 1.11** (Analytic integrable part, $h \in A_s(D)$). `IsAnalyticIntegrable` $R\,s\,h$ holds if

(i) $h$ is complex differentiable on $B(0,R)_s$;
(ii) $h$ is continuous on $\overline{B(0,R)_s}$ (closure in $\mathbb{C}^n$);
(iii) there is $A \geq 0$ with $|h(I)| \leq A$ for all $I \in B(0,R)_s$;
(iv) $\operatorname{Im} h(I) = 0$ for all $I \in \mathcal{B}_s(R)$;
(v) $h_{\mathbb{R}}$ is of class $C^3$ on $\mathcal{B}_s(R)$ (real Fréchet sense; the set is open).

No numerical bound on $h$ is fixed. Clause (v) is redundant by holomorphy ([K:`bp:p1:holomorphic-regularity`], [K:`bp:p1:real-restriction`]) and records the regularity that guards every real derivative.
Frozen Lean declaration: `KAM.Nekhoroshev.IsAnalyticIntegrable`
Source: BM11 p. 4 ($A_s(D)$, integrable case, with continuity on the closure).

**Definition 1.12** (Analytic perturbation of size $\varepsilon$). `IsAnalyticPerturbation` $R\,s\,\varepsilon\,f$ holds if $\varepsilon \geq 0$ and

(i) $f$ is complex differentiable on $D_s(R)$ and continuous on $\overline{D_s(R)}$;
(ii) $f(\theta + k, I) = f(\theta, I)$ for all $(\theta, I) \in D_s(R)$ and $k \in \mathbb{Z}^n$;
(iii) $\operatorname{Im} f(\theta, I) = 0$ for all $\theta \in \mathbb{R}^n$ and $I \in \mathcal{B}_s(R)$;
(iv) $|f(z)| \leq \varepsilon$ for all $z \in D_s(R)$ (complex absolute value).

The size is measured by $\leq \varepsilon$, not the source's strict bound; the source is applied with $2\varepsilon$. For $\varepsilon = 0$ this forces $f = 0$ on $D_s(R)$.
Frozen Lean declaration: `KAM.Nekhoroshev.IsAnalyticPerturbation`
Source: BM11 pp. 4–5 ($f \in A_s(D)$, $|f|_s < \varepsilon$); deviation: $\leq \varepsilon$ (ruling Q2, D18).

**Definition 1.13** (Derivative bounds $B(M)$). For $U \subset \mathbb{R}^n$, $h : \mathbb{C}^n \to \mathbb{C}$ and $M \in \mathbb{R}$, `HasDerivativeBounds` $U\,h\,M$ holds if $M > 0$, $U$ is open, $h_{\mathbb{R}} \in C^3(U)$, and for all $I \in U$, all $k \in \{1,2,3\}$ and all $(i_1, \ldots, i_k) \in \{1, \ldots, n\}^k$,

$$\left|D^k h_{\mathbb{R}}(I)[e_{i_1}, \ldots, e_{i_k}]\right| \leq M.$$

These are all ordered coordinate partials of orders one to three; for $C^3$ functions they coincide with the multi-index partials $\partial^k h$ of the source (symmetry, [K:`bp:p1:ext-schwarz`]). The bound is on coordinate partials, not on operator norms.
Frozen Lean declaration: `KAM.Nekhoroshev.HasDerivativeBounds`
Source: BM11 p. 4, $B(M)$.

**Definition 1.14** (Strict quasi-convexity $QC(m)$). For $U \subset \mathbb{R}^n$, $h : \mathbb{C}^n \to \mathbb{C}$, $m \in \mathbb{R}$, `IsStrictlyQuasiConvex` $U\,h\,m$ holds if $m > 0$, $U$ is open, $h_{\mathbb{R}} \in C^2(U)$, and for every $I \in U$: $\omega_h(I) \neq 0$ and

$$\sum_i \omega_h(I)_i v_i = 0 \implies m|v|^2 \leq D^2 h_{\mathbb{R}}(I)[v,v] \qquad (v \in \mathbb{R}^n),$$

with $|v|$ the max norm on the left and the Euclidean dot product in the hypothesis.

Frozen Lean declaration: `KAM.Nekhoroshev.IsStrictlyQuasiConvex`
Source: BM11 p. 4, $QC(m)$ (the source's norm on the right is the max norm, orthogonality is Euclidean).

**Definition 1.15** (Complex Hessian bound)**.** `HasComplexHessianBound` $R\,s\,h\,M_{\mathbb{C}}$ holds if $M_{\mathbb{C}} > 0$, $h$ is $C^2$ in the complex sense on $B(0,R)_s$ and $\|D^2h(z)\| \le M_{\mathbb{C}}$ for all $z \in B(0,R)_s$, the norm being that of $D(Dh)(z) \in \mathcal{L}(\mathbb{C}^n, \mathcal{L}(\mathbb{C}^n, \mathbb{C}))$ for the max norm. Every entry $|\partial_i\partial_j h|$ is then $\le M_{\mathbb{C}}$, and the Euclidean operator norm of the Hessian matrix is $\le nM_{\mathbb{C}}$. This is not the spectral radius of a complex symmetric matrix.
Frozen Lean declaration: `KAM.Nekhoroshev.HasComplexHessianBound`
Source: Pos93 p. 190, "The Hessian of the integrable hamiltonian $h$" (conversion by $n$ recorded in `docs/tracks/N/sources.md`, anchor POS190-HESS).

**Definition 1.16** (Solutions on a time set)**.** For $H : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$, $U \subset \mathbb{R}^n$, $J \subset \mathbb{R}$ and $z : \mathbb{R} \to \mathbb{R}^n \times \mathbb{R}^n$, `IsSolutionOn` $H\,U\,J\,z$ holds if for every $t \in J$: $z(t)_2 \in U$, $H$ is differentiable at $z(t)$, and $z$ is differentiable at $t$ (two-sided, also at endpoints of $J$, since $z$ is defined on $\mathbb{R}$) with $z'(t) = X_H(z(t))$.
Frozen Lean declaration: `KAM.Nekhoroshev.IsSolutionOn`
Source: BM11 Theorem 2.1 p. 5 and the maximal-solution argument p. 13.

**Definition 1.17** (The integrable orbit)**.** For $h : \mathbb{C}^n \to \mathbb{C}$, $\theta_0, I_0 \in \mathbb{R}^n$ and $t \in \mathbb{R}$: $\gamma_{\theta_0,I_0}(t) = (\theta_0 + t\,\omega_h(I_0),\, I_0)$.
Frozen Lean declaration: `KAM.Nekhoroshev.integrableOrbit`
Source: BM11 p. 1 ($H(\theta, I) = h(I)$).

*Remark* 1.18 (Lean totality). The frozen definitions and theorems use `fderiv` (which is 0 at a point of non-differentiability), `iteratedFDeriv`, the operator norm, `rpow` ($\varepsilon^{\pm 1/(2n)}$), `Real.exp` and the max norm; there is no matrix inverse, no Bochner integral and no division by a quantity that may vanish. The guards are:

(a) In N-HEAD-C and N-ZERO, $\omega_h$ and $D^2h_{\mathbb{R}}$ are used only on the open set $\mathcal{B}_s(R)$ where $h_{\mathbb{R}}$ is $C^3$ (Definitions 1.11(v), 1.13, 1.14), so the derivatives are the genuine ones. N-QC uses them on its arbitrary open $U$, where its hypothesis gives $C^2$ (item (e)). The definitions `frequencyMap`, `integrableOrbit` themselves are total and keep Lean's default values outside the differentiability locus.
(b) $D^2h$ for complex $h$ is used only on $B(0,R)_s$, open, where $h$ is $C^2$ (Definition 1.15).
(c) $X_H$ is used only inside `IsSolutionOn`, which requires differentiability of $H$ at every visited point; in N-ZERO and N-HEAD-C this is a proof obligation: $H_{h,f}$ is real-differentiable on $\mathbb{R}^n \times \mathcal{B}_s(R) \supseteq \mathbb{R}^n \times B(0,R)$ (angles first, actions restricted) because $h, f$ are holomorphic on open sets containing the real points ([K:`bp:p1:hamiltonian-real`]).
(d) $\varepsilon^{-1/(2n)}$ and $\varepsilon^{1/(2n)}$ occur only with $\varepsilon > 0$ in N-HEAD-C, and $2n \neq 0$.
(e) In N-QC the map $\Psi(I', \lambda) = (h_{\mathbb{R}}(I'), \lambda\omega_h(I'))$ is differentiable at $(I, \lambda)$ because $h_{\mathbb{R}} \in C^2$ on the open set $U \ni I$; `fderiv` is then the true derivative.

## 1.4. **The frozen theorems.**

**Theorem 1.19** (N-HEAD-C: exponential stability, complex Hessian bound)**.** *Let $n \ge 3$ and $R, s, M, m, M_{\mathbb{C}}, \omega_{\min} > 0$. Then there are $C, T, c, \varepsilon_0 > 0$ with*

$$C\,\varepsilon_0^{1/(2n)} \le R/4$$

*such that the following holds for every $h : \mathbb{C}^n \to \mathbb{C}$, $f : \mathbb{C}^n \times \mathbb{C}^n \to \mathbb{C}$ and $\varepsilon \in \mathbb{R}$ with*

*(H1)* `IsAnalyticIntegrable` $R\,s\,h$ *(Definition 1.11);*
*(H2)* `HasComplexHessianBound` $R\,s\,h\,M_{\mathbb{C}}$ *(Definition 1.15);*
*(H3)* $\omega_{\min} \le |\omega_h(I)|$ *for every $I \in B(0,R)$ (max norm);*
*(H4)* `IsStrictlyQuasiConvex` $\mathcal{B}_s(R)\,h\,m$ *(Definition 1.14);*

*(H5)* `HasDerivativeBounds` $\mathcal{B}_s(R)\, h\, M$ *(Definition 1.13);*
*(H6)* `IsAnalyticPerturbation` $R\, s\, \varepsilon\, f$ *(Definition 1.12);*
*(H7)* $0 < \varepsilon \le \varepsilon_0$.

*For every* $\theta_0 \in \mathbb{R}^n$ *and* $I_0 \in B(0, R/2)$, *with* $\tau := T\exp\bigl(c\,\varepsilon^{-1/(2n)}\bigr)$ *and* $H = H_{h,f}$, *there is* $z : \mathbb{R} \to \mathbb{R}^n \times \mathbb{R}^n$ *with*

*(a)* $z(0) = (\theta_0, I_0)$;
*(b)* `IsSolutionOn` $H\, B(0,R)\, [-\tau, \tau]\, z$;
*(c)* $|z(t)_2 - I_0| \le C\varepsilon^{1/(2n)}$ *for all* $t$ *with* $|t| \le \tau$;
*(d) (uniqueness) for every* $L \in [0, \tau]$ *and every* $y : \mathbb{R} \to \mathbb{R}^n \times \mathbb{R}^n$ *with* $y(0) = (\theta_0, I_0)$ *and* `IsSolutionOn` $H\, B(0,R)\, [-L, L]\, y$: $y(t) = z(t)$ *for all* $t \in [-L, L]$.

*The order of choice is:* $n$; $R, s, M, m, M_{\mathbb{C}}, \omega_{\min}$; *then* $C, T, c, \varepsilon_0$, *depending only on these seven data; then* $h, f, \varepsilon$, *then* $\theta_0, I_0$; *the solution* $z$ *may depend on all of them. There is no dependence on* $A$ *in Definition 1.11(iii). In (H4)–(H5) the set is* $\mathcal{B}_s(R) = B(0, R+s)$ *(Remark 1.9), in (H3) the smaller ball* $B(0, R)$.

*Frozen Lean declaration:* `KAM.Nekhoroshev.quasiConvex_nekhoroshev_complexBound`
*Source: Pos93 Theorem 1, pp. 190–191 (with the conversions and anchors of* `docs/tracks/N/sources.md`*); exponent* $1/(2n)$ *as BM11 p. 5 with* $\delta = (2n(n-1))^{-1}$; *ruling D18 Q2, Q4; deviation D20: the extra inputs* $M_{\mathbb{C}}$ *and* $\omega_{\min}$.
*Source gap: The printed theorem is for a* $2\pi$*-periodic torus, Euclidean norms and exceptional initial frequencies* $\|\omega(I_0)\| \le mr_0/8$; *the conversion to the frozen statement (rescaling* $H \mapsto 2\pi H(\varphi/2\pi, I)$, *choice of* $r_0, s_0$ *from the seven data, exclusion of the exceptional branch by* $\omega_{\min}$, *existence and uniqueness of the solution) is given only in* `sources.md` *and is proved in Part 10 of* `PLAN.md`.

**Theorem 1.20** (N-ZERO: exact integrable dynamics). *Let* $n \in \mathbb{N}$, $R, s > 0$, $h$ *with* `IsAnalyticIntegrable` $R\, s\, h$ *and* $f$ *with* `IsAnalyticPerturbation` $R\, s\, 0\, f$. *For every* $\theta_0 \in \mathbb{R}^n$ *and* $I_0 \in B(0, R/2)$, *with* $\gamma = \gamma_{\theta_0, I_0}$ *(Definition 1.17):*

$$\gamma(0) = (\theta_0, I_0), \qquad \texttt{IsSolutionOn}\, H_{h,f}\, B(0,R)\, \mathbb{R}\, \gamma, \qquad \gamma(t)_2 = I_0 \quad (t \in \mathbb{R}).$$

*Frozen Lean declaration:* `KAM.Nekhoroshev.zero_perturbation`
*Source: Ours (supplement to BM11 Theorem 2.1, p. 5, whose strict bound excludes* $\varepsilon = 0$*); from* $H = h(I)$, *BM11 p. 1.*

**Theorem 1.21** (N-QC: nonsingular isoenergetic derivative). *Let* $n \in \mathbb{N}$, $U \subset \mathbb{R}^n$, $h : \mathbb{C}^n \to \mathbb{C}$, $m \in \mathbb{R}$ *with* `IsStrictlyQuasiConvex` $U\, h\, m$. *For* $I \in U$ *and* $\lambda > 0$, *the derivative at* $(I, \lambda)$ *of*

$$\Psi : \mathbb{R}^n \times \mathbb{R} \to \mathbb{R} \times \mathbb{R}^n, \qquad \Psi(I', \lambda') = \bigl(h_{\mathbb{R}}(I'),\ \lambda'\omega_h(I')\bigr)$$

*is a bijective linear map* $\mathbb{R}^n \times \mathbb{R} \to \mathbb{R} \times \mathbb{R}^n$. *Explicitly it is* $D\Psi(I,\lambda)[v,u] = \bigl(\omega_h(I)\cdot v,\ u\,\omega_h(I) + \lambda D^2 h_{\mathbb{R}}(I)v\bigr)$. *No uniform radius of invertibility is asserted.*
*Frozen Lean declaration:* `KAM.Nekhoroshev.quasiConvex_isoenergetic_derivative`
*Source: BM11 Lemma 3.3 and its proof, pp. 9–10 (only the derivative assertion).*

1.5. **D25 additions: the partial (slow–fast) convex theorem.** Owner approval D25 (27 October 2026) froze one further theorem, N-PARTIAL-CONVEX, and seven definitions. The Lean declarations are authoritative (`KAM/Frozen/Nekhoroshev/Theorems/PartialConvexNekhoroshev.lean`, `KAM/Frozen/Nekhoroshev/Defs/{ComplexifyPhase,InnerCore,PartialComplexDomain,HasDefiniteFastPart,IsPartialPerturbation,PartialTimeExponent,PartialDriftExponent}.lean`). Notation: $\mathrm{Ph}_d = \mathbb{R}^d \times \mathbb{R}^d$ (Lean `KAM.Planetary.Phase` $d$, positions first, then momenta) with the product of max norms, $\|(x,y)\| = \max(|x|,|y|)$. For $v \in \mathbb{R}^{n+k}$ (or $\mathbb{R}^{n+k}$-valued vectors), $\mathrm{fast}(v) = (v_1, \dots, v_n)$ is the first block and $\mathrm{slow}(v) = (v_{n+1}, \dots, v_{n+k})$ the second block (`KAM.Planetary.fast`, `KAM.Planetary.slow`). A phase point of the partial system is $z = (\varphi, \mathsf{I}) \in \mathrm{Ph}_{n+k}$: $\varphi = (\varphi_{\mathrm{f}}, q)$ with $n$ fast angles $\varphi_{\mathrm{f}} = \mathrm{fast}(\varphi)$ and $k$ slow positions $q = \mathrm{slow}(\varphi)$; $\mathsf{I} = (I, p)$ with $n$ fast actions $I = \mathrm{fast}(\mathsf{I})$

and $k$ slow momenta $p = \mathrm{slow}(\mathsf{I})$. The slow pair is $(q,p) = \big(\mathrm{slow}(\varphi), \mathrm{slow}(\mathsf{I})\big) \in \mathrm{Ph}_k$. The fast angles have period $2\pi$ (`KAM.Planetary.fastShift` $\ell = (2\pi\ell, 0)$ in the fast block).

**Definition 1.22** (`complexifyPhase`)**.** For $z = (x,y) \in \mathrm{Ph}_d$: $\mathrm{cx}(z) = \big((x_i)_i, (y_i)_i\big) \in \mathbb{C}^d \times \mathbb{C}^d$, componentwise inclusion $\mathbb{R} \subset \mathbb{C}$; no choices.

Frozen Lean declaration: `KAM.Nekhoroshev.complexifyPhase`

Source: Ours (notation for Nek77 §4.3, p. 29: "for real values of the arguments $H$ is real-valued").

**Definition 1.23** (`innerCore`: closed max-norm erosion)**.** For a normed group $E$, $U \subset E$ and $d \in \mathbb{R}$: $\mathrm{core}(U,d) = \{x \in U : \bar{B}(x,d) \subset U\}$ (closed ball of the norm of $E$). Used only with $d > 0$ (for $d < 0$ the ball is empty and $\mathrm{core}(U,d) = U$). It is used on $E = \mathbb{R}^n$ (max norm) and $E = \mathrm{Ph}_k$ (product of max norms).

Frozen Lean declaration: `KAM.Nekhoroshev.innerCore`

Source: Nek77 §4.2, p. 29 ($X - \varepsilon$: points of $X$ whose $\varepsilon$-neighbourhoods lie in $X$); here closed balls of the max norm, a stronger margin than the Euclidean erosion of the source.

**Definition 1.24** (`partialComplexDomain`)**.** For $G \subset \mathbb{R}^n$, $D \subset \mathrm{Ph}_k$ and $\rho \in \mathbb{R}$,

$$\mathcal{F}(G,D,\rho) = \Big\{(\varphi,\mathsf{I}) \in \mathbb{C}^{n+k} \times \mathbb{C}^{n+k} : \ \mathrm{fast}(\mathrm{Re}\,\mathsf{I}) \in G,\ \big(\mathrm{slow}(\mathrm{Re}\,\varphi), \mathrm{slow}(\mathrm{Re}\,\mathsf{I})\big) \in D,\ \varphi, \mathsf{I} \in T^{n+k}_\rho\Big\},$$

$T^N_\rho = \{u \in \mathbb{C}^N : |\mathrm{Im}\,u_i| < \rho\ \forall i\}$ (Lean `KAM.complexStrip`). The fast real angles and the slow real positions carry no condition other than $(\mathrm{slow}\,\mathrm{Re}\,\varphi, \mathrm{slow}\,\mathrm{Re}\,\mathsf{I}) \in D$.

Frozen Lean declaration: `KAM.Nekhoroshev.partialComplexDomain`

Source: Nek77 §4.3.A, p. 29 (complex tube over arbitrary fast/slow real domains $G$, $D$; there the imaginary bounds are closed, $\le \rho$, and the slow variables are $(p,q) \in D \subset \mathbb{R}^{2(n-s)}$).

**Definition 1.25** (`HasDefiniteFastPart`)**.** For $G \subset \mathbb{R}^n$, $\rho, g, m \in \mathbb{R}$ and $h : \mathbb{C}^n \to \mathbb{C}$, with $U = \{I \in \mathbb{R}^n : \ I \in G_\rho\}$ ($G_\rho =$ `actionNbhd` $G\,\rho$, Definition 1.1; its real points): `HasDefiniteFastPart` $G\,\rho\,g\,m\,h$ holds iff $g > 0$, $m > 0$, the function $h_{\mathbb{R}} = \mathrm{Re}\,h|_{\mathbb{R}^n}$ is $C^3$ on $U$ (Lean `ContDiffOn` $\mathbb{R}\,3$, with $U$ the set above) and there is $\sigma \in \{1,-1\}$ with, for every $I \in U$:

$$|\omega_h(I)| \ge g \qquad \text{and} \qquad \forall v \in \mathbb{R}^n : \quad m \sum_i v_i^2 \le \sigma\, D^2 h_{\mathbb{R}}(I)[v,v],$$

where $\omega_h = \nabla h_{\mathbb{R}}$ (Definition 1.2; the norm of $\omega_h(I) \in \mathbb{R}^n$ is the max norm), and $D^2 h_{\mathbb{R}}(I)[v,v]$ is the second `fderiv` of $h_{\mathbb{R}}$ at $I$ applied to $v$ and $v$ (Lean `fderiv` $\mathbb{R}$ twice).

Frozen Lean declaration: `KAM.Nekhoroshev.HasDefiniteFastPart`

Source: Specialisation of Nek77 §1.18 and §4.1 (index-one steep class) to a definite fast Hessian of one global sign $\sigma$; deviation D-N-ADD-1 (`docs/tracks/N/REVIEW-additions.md`); not a source definition.

**Definition 1.26** (`IsPartialPerturbation`)**.** For $G \subset \mathbb{R}^n$, $D \subset \mathrm{Ph}_k$, $\rho \in \mathbb{R}$ and $P : \mathbb{C}^{n+k} \times \mathbb{C}^{n+k} \to \mathbb{C}$, `IsPartialPerturbation` $G\,D\,\rho\,P$ holds iff

- (P1) $P$ is complex differentiable on $\mathcal{F}(G,D,\rho)$ (Lean `DifferentiableOn` $\mathbb{C}$);
- (P2) $|P(z)| \le 1$ for $z \in \mathcal{F}(G,D,\rho)$;
- (P3) for real $z \in \mathrm{Ph}_{n+k}$ with $\mathrm{cx}(z) \in \mathcal{F}(G,D,\rho)$: $\mathrm{Im}\,P(\mathrm{cx}(z)) = 0$;
- (P4) for every $z \in \mathcal{F}(G,D,\rho)$ and $\ell \in \mathbb{Z}^n$: $P\big(z + \mathrm{cx}(\mathrm{fastShift}\,\ell)\big) = P(z)$, i.e. $P$ is $2\pi$-periodic in each fast angle; there is no periodicity in the slow variables.

Frozen Lean declaration: `KAM.Nekhoroshev.IsPartialPerturbation`

Source: Nek77 §4.3 and Remark 4.5.A, p. 30 (the perturbation $H_1$ is analytic on $F$, $2\pi$-periodic in $\varphi$, real for real arguments; its size $M = \sup|H_1|$ is $\le \mu$ for $H_1 = \mu P$).

**Definition 1.27** (`partialTimeExponent`, `partialDriftExponent`)**.** For $n \in \mathbb{N}$: $a_n = \dfrac{2}{6n^2 - 3n + 14}$ and $b_n = \dfrac{3}{6n^2 - 3n + 14} = \frac{3}{2} a_n$ (real arithmetic; for $n \ge 2$ the denominator is $\ge 6 \cdot 4 - 6 + 14 = 32 > 0$).

Frozen Lean declaration: `KAM.Nekhoroshev.partialTimeExponent`, `KAM.Nekhoroshev.partialDriftExponent`
Source: Nek77 (4.1), (4.4), (4.5), p. 29–30, with all steepness indices $\alpha_i = 1$ and $s = n$ (the formulas (4.4), (4.5) are images that the text layer of the scan loses; the frozen definitions and `docs/tracks/N/statement.md` are used); for $n = 2$: $a = \frac{1}{16}$, $b = \frac{3}{32}$ (`docs/tracks/N/statement.md`).

**Theorem 1.28** (N-PARTIAL-CONVEX: exponential stability of the fast actions, definite fast part, slow–fast)**.** *Let $n \geq 2$, $k \geq 1$, $G, G' \subset \mathbb{R}^n$ open and convex, $D \subset \mathrm{Ph}_k$ open, $\rho, g, m, M_{\mathbb{C}} \in \mathbb{R}$ with $\rho > 0$, $M_{\mathbb{C}} > 0$ and*

$$\bar{B}(I, \rho) \subset G' \quad \text{for every } I \in G \qquad (\text{closed max-norm ball}),$$

*and let $h : \mathbb{C}^n \to \mathbb{C}$ with: $h$ complex differentiable and complex $C^2$ on $G'_\rho$; $\mathrm{Im}\, h(I) = 0$ for every real $I$ with $I \in G'_\rho$; $\|D^2 h(z)\| \leq M_{\mathbb{C}}$ for every $z \in G'_\rho$ (operator norm of the complex Hessian, max norms); and `HasDefiniteFastPart` $G'\, \rho\, g\, m\, h$. Then there is $\mu_0 > 0$ such that for all $\mu \in (0, \mu_0)$, all $P$ with `IsPartialPerturbation` $G'\, D\, \rho\, P$, all $C > 0$ and all $z : \mathbb{R} \to \mathrm{Ph}_{n+k}$ with*

- *(Z1)* `IsSolutionOn` $H_\mu\ \{\mathsf{I} : \mathrm{fast}(\mathsf{I}) \in G\}\ [0, C]\ z$, *where* $H_\mu(\varphi, \mathsf{I}) = \mathrm{Re}\, h\big(\mathrm{fast}(\mathsf{I})\big) + \mu\, \mathrm{Re}\, P\big(\mathrm{cx}(\varphi, \mathsf{I})\big)$ *(Lean* `KAM.Planetary.degenerateHamiltonian (realPart h) (λ_ w. Re P (cx w)) μ`*, Definition 1.10);*
- *(Z2)* $\mathrm{fast}(z(0)_2) \in \mathrm{core}(G, \mu^{b_n})$,

*and every $t \in \big[0, \min\big(C,\ \mu^{-1} \exp(\mu^{-a_n})\big)\big]$ with*

$$\Big(\mathrm{slow}(z(u)_1), \mathrm{slow}(z(u)_2)\Big) \in \mathrm{core}\big(D, \mu^{b_n}\big) \quad \text{for all } u \in [0, t],$$

*one has*

$$\big|\mathrm{fast}(z(t)_2) - \mathrm{fast}(z(0)_2)\big| \leq \tfrac{1}{2}\mu^{b_n}.$$

*The order of choice is: $n, k$; $G, G', D, \rho, g, m, M_{\mathbb{C}}, h$ (together with their hypotheses); then $\mu_0$; then $\mu, P, C, z$; then $t$. The slow confinement is tested only on $[0, t]$ (never assumed up to the exponential time).*

*Frozen Lean declaration:* `KAM.Nekhoroshev.partialConvex_nekhoroshev`
*Source: Nek77 Theorem 4.4, p. 30, with Remarks 4.5.A–B; proof specialised to a definite fast Hessian as in* `docs/tracks/N/notes.md`*, F-NADD-1 (Lemma 10.6, p. 47, replaced by its definite-Hessian endpoint argument); deviation D-N-ADD-1 (collar $G \Subset G'$, $2\pi$ period, stopped quantifier, maximum-norm erosion).*
*Source gap: The printed theorem assumes steepness of $H_0$ on the whole source domain with explicit coefficients, a Euclidean erosion, closed imaginary bounds, and the confinement of $(p, q)$ on the whole $[0, C]$; the frozen statement (collar, closed max-norm erosion, confinement tested on $[0, t]$ only, imaginary width $\rho$ open) is proved here via the route of Part 12. The initial erosion $\mathrm{core}(G, \mu^{b_n})$ is used only through $\mathrm{fast}(z(0)_2) \in G$.*

*Remark* 1.29 (Lean totality for the D25 objects). (a) $\mu^{b_n}$, $\mu^{-a_n}$, $\mu^{-1}$ occur only for $\mu > 0$ (rpow of a positive base), and $6n^2 - 3n + 14 > 0$. (b) $\omega_h$, $D^2 h_{\mathbb{R}}$ in `HasDefiniteFastPart` are guarded: $h_{\mathbb{R}}$ is $C^3$ on the real slice $U$ of the open set $G'_\rho$, which is open in $\mathbb{R}^n$ ($G'$ open), so the `fderiv` are true derivatives at every $I \in U$ (Lemma 2.3 for the relation with the complex derivatives). (c) `IsSolutionOn` asserts differentiability of $H_\mu$ at $z(u)$, $u \in [0, C]$; where $\mathrm{slow}(z(u)) \notin D$ the value $P(\mathrm{cx}\, w)$ is the arbitrary value of the function $P$ outside $\mathcal{F}$, and the theorem uses the solution only for $u \in [0, t]$ where the slow variables lie in $\mathrm{core}(D, \mu^{b_n}) \subset D$, an open set on which $H_\mu$ is real analytic. (d) If $\mathrm{core}(D, \mu^{b_n})$ is empty the confinement hypothesis fails for $t \geq 0$ and the statement is vacuous. (e) `fderiv` $\mathbb{C}$ (`fderiv` $\mathbb{C}\, h$) is the true second derivative on the open set $G'_\rho$ where $h$ is complex $C^2$.

1.6. **D28 additions: multi-frequency averaging.** Owner approval D28 (27 October 2026) froze two further theorems, N-AVERAGING-MEAN and N-AVERAGING-MOST, and two definitions (`KAM/Frozen/Nekhoroshev/Theorems/{AveragingMean,AveragingMost}.lean`, `KAM/Frozen/Nekhoroshev/Defs/{HasAveragingFrequencyRank,AveragingError}.lean`). The Lean text is

authoritative. Notation of the Lean statements: a *phase point* is $x = (\theta, I) \in \mathbb{R}^n \times \mathbb{R}^n$, the action is `x.2`$= x_2 = I$, norms are max norms, `vol` is Lebesgue measure on $\mathbb{R}^n \times \mathbb{R}^n$ ("unnormalised phase volume": no division by $\operatorname{vol} K$ and no probability), and for $\varepsilon > 0$ the Hamiltonian is $H_\varepsilon = H_{h,\varepsilon f} = \operatorname{Re}\big(h(I) + \varepsilon f(\theta, I)\big)$ (Definition 1.4 with the perturbation $w \mapsto (\varepsilon : \mathbb{C})\, f(w)$). With the averaged action orbit $J(t) \equiv I_0$ (Remark 13.2) the error of the averaging method is $\sup_{0 \le t \le 1/\varepsilon} |I(t) - I_0|$.

**Definition 1.30** (`HasAveragingFrequencyRank`: the two rank alternatives)**.** For $U \subset \mathbb{R}^n$ and $h : \mathbb{C}^n \to \mathbb{C}$ write $\omega = \omega_h$ (Definition 1.2). `HasAveragingFrequencyRank` $U\, h$ holds iff

(A) $\forall I \in U :\ D\omega(I) : \mathbb{R}^n \to \mathbb{R}^n$ is surjective  or  (B) $\forall I \in U :\ \omega(I) \neq 0$ and $(v, c) \mapsto D\omega(I)v + c\,\omega(I),\ \mathbb{R}^n \times \mathbb{R}$

with $D\omega(I) =$ `fderiv` $\mathbb{R}\, \omega\, I$. The alternative is global: the same branch (A) or (B) holds at every point of $U$ (the quantifier $\forall I \in U$ is inside each branch). (B) says that the projective class $(\omega_1 : \cdots : \omega_n)$ has differential of rank $n - 1$ at $I$ (its image is $(\operatorname{range} D\omega(I) + \mathbb{R}\omega(I))/\mathbb{R}\omega(I)$); no chart $\omega_1 \neq 0$ is chosen. Branch (A) may hold at a point where $\omega = 0$.
Frozen Lean declaration: `KAM.Nekhoroshev.HasAveragingFrequencyRank`
Source: AKN06 Theorem 6.11, p. 243 (PDF p. 254): "either the rank of the map $I \mapsto \omega(I)$ is equal to the number of frequencies, or the rank of the map $I \mapsto (\omega_1(I) : \cdots : \omega_m(I))$ is by one less than the number of frequencies"; Nei76 p. 1296 hypothesis 3° is the chart form.

**Definition 1.31** (`averagingError`)**.** For $n \in \mathbb{N}$, $\varepsilon \in \mathbb{R}$, $\Phi : (\mathbb{R}^n \times \mathbb{R}^n) \to \mathbb{R} \to (\mathbb{R}^n \times \mathbb{R}^n)$ and $x = (\theta_0, I_0)$:

$$e_\varepsilon^\Phi(x) = \texttt{averagingError}\, \varepsilon\, \Phi\, x = \sup\Big\{ |(\Phi\, x\, t)_2 - I_0| :\ t \in [0, 1/\varepsilon] \Big\} \in \mathbb{R},$$

the Lean real supremum `sSup` (which is 0 for an empty or unbounded-above set).
Frozen Lean declaration: `KAM.Nekhoroshev.averagingError`
Source: AKN06 (6.27), p. 243: $\max_{0 \le t \le 1/\varepsilon} |I(t) - J(t)|$ with $J(t) = I_0$; Nei76 p. 1295 (4).

*Remark* 1.32 (Lean totality for `averagingError`). `sSup` on $\mathbb{R}$ is the true supremum only for a nonempty set bounded above; otherwise it is the junk value 0. The theorems use the definition only under the following guards, each verified in Part 13:

(a) $\varepsilon > 0$ (the quantifier $0 < \varepsilon \le \varepsilon_0$), so $1/\varepsilon > 0$ and $0 \in [0, 1/\varepsilon]$: the interval is a nonempty compact interval (for $\varepsilon < 0$ it is empty, for $\varepsilon = 0$ it is $\{0\}$ since $1/0 = 0$ in Lean);
(b) the family is a genuine solution on $[0, 1/\varepsilon]$, so $t \mapsto (\Phi x\, t)_2$ is continuous on the compact interval (a solution is differentiable at every $t \in J$), whence the image of $t \mapsto |(\Phi x\, t)_2 - x_2|$ is compact, nonempty and bounded; moreover $(\Phi x\, t)_2 \in B(0, R)$ and $x_2 \in B(0, R/2)$ bound the set by $3R/2$ (the bound is also available without continuity);
(c) then the supremum is attained, is $\ge 0$ (it is $\ge$ the value 0 at $t = 0$ when $\Phi x\, 0 = x$), and $|(\Phi x\, t)_2 - x_2| \le e_\varepsilon^\Phi(x)$ for every $t \in [0, 1/\varepsilon]$.

No division by a quantity that may vanish occurs: $1/\varepsilon$ with $\varepsilon > 0$, and $\sqrt{\varepsilon}$ with $\varepsilon > 0$, $A\sqrt{\varepsilon}/\kappa$ with $\kappa > 0$, $A\sqrt{\varepsilon}/\eta$ with $\eta > 0$. The derivatives behind $\omega_h$ and $D\omega_h$ in Definition 1.30 are genuine on $B(0, R) \subset \mathcal{B}_s(R)$ because $h_{\mathbb{R}} \in C^3$ there (Definition 1.11(v)), so $\omega_h$ is $C^2$ and `fderiv` is the true derivative. As a stand-alone predicate on an arbitrary $h$ or on $U = \emptyset$ the definition has its usual vacuity/off-locus behaviour; no theorem uses it there.

**Theorem 1.33** (N-AVERAGING-MEAN: mean error and Markov form)**.** *Let $n \ge 2$, $R, s > 0$, $h : \mathbb{C}^n \to \mathbb{C}$, $f : \mathbb{C}^n \times \mathbb{C}^n \to \mathbb{C}$ with* `IsAnalyticIntegrable` $R\, s\, h$, `IsAnalyticPerturbation` $R\, s\, 1\, f$, `HasAveragingFrequencyRank` $B(0, R)\, h$, *and $K \subset \mathbb{R}^n \times \mathbb{R}^n$ compact with $x_2 \in B(0, R/2)$ for every $x \in K$. Then there are $A, \varepsilon_0 > 0$ such that for every $\varepsilon \in (0, \varepsilon_0]$ and every $\Phi : (\mathbb{R}^n \times \mathbb{R}^n) \to \mathbb{R} \to (\mathbb{R}^n \times \mathbb{R}^n)$ with*

$$\forall x \in K :\quad \Phi\, x\, 0 = x \quad \textit{and} \quad \texttt{IsSolutionOn}\ H_{h,\varepsilon f}\ B(0, R)\ [0, 1/\varepsilon]\ (\Phi\, x),$$

*the function $e_\varepsilon^\Phi$ is integrable on $K$,*

$$\int_K^- \mathrm{ofReal}\big(e_\varepsilon^\Phi(x)\big)\,dx \le \mathrm{ofReal}(A\sqrt{\varepsilon}) \qquad (\textit{Lean } \texttt{lintegral} \textit{ over } K),$$

*and for every $\kappa > 0$ there is a measurable $E \subset K$ with $\mathrm{vol}\,E \le \kappa$ and $|(\Phi\, x\, t)_2 - x_2| \le A\sqrt{\varepsilon}/\kappa$ for all $x \in K \setminus E$ and $t \in [0, 1/\varepsilon]$. The order of choice: $n, R, s, h, f, K$ and their hypotheses; then $A, \varepsilon_0$; then $\varepsilon$, $\Phi$; then $\kappa$ and $E$. The flow premise is only existence and typing on $[0, 1/\varepsilon]$; it contains no error bound and no exceptional set.*

*Frozen Lean declaration:* `KAM.Nekhoroshev.averaging_mean`

*Source: AKN06 Theorem 6.11 and Corollary 6.1 with (6.27), (6.28), p. 243 (printed pages; PDF = printed +11); Nei76 pp. 1295–1298, Theorem 1; analytic specialisation, period-one angles, max norms and unnormalised volume as in* `docs/tracks/N/statement.md`*, §N-AVERAGING v0.5.*

*Source gap: Nei76 p. 1298 (17) writes $\sup_t z(t) = \int_{U'} h$, interchanging a supremum in time with an integral over initial data (ledger E-NAVG-NEI-17). The proof in Part 13 bounds the integral of the maximum directly (Remark 13.19).*

**Theorem 1.34** (N-AVERAGING-MOST: existence and small drift for most initial data)**.** *With the same $n, R, s, h, f, K$ and hypotheses as in Theorem 1.33, there are $A, \varepsilon_0, \eta_0 > 0$ with $\eta_0 \le R/4$ such that for every $\varepsilon \in (0, \varepsilon_0]$ and every $\eta \in (0, \eta_0]$ there is a measurable $E \subset K$ with*

$$\mathrm{vol}\,E \le \mathrm{ofReal}\big(A\sqrt{\varepsilon}/\eta\big),$$

*and for every $x \in K \setminus E$ there is $z : \mathbb{R} \to \mathbb{R}^n \times \mathbb{R}^n$ with $z(0) = x$, $\texttt{IsSolutionOn}\ H_{h,\varepsilon f}\ B(0,R)\ [0,1/\varepsilon]\ z$ and $|z(t)_2 - x_2| \le \eta$ for all $t \in [0, 1/\varepsilon]$. The order of choice: the fixed data; then $A, \varepsilon_0, \eta_0$; then $\varepsilon$, then $\eta$, then $E$, then $x$ and $z$. Existence of the solution on the whole window and confinement are* conclusions *on the good data, not premises.*

*Frozen Lean declaration:* `KAM.Nekhoroshev.averaging_most`

*Source: AKN06 p. 243 (6.28) and Nei76 Theorems 1–2, pp. 1295–1298 (the set $\Phi(\rho, \varepsilon)$ of p. 1296 includes existence on $[0, \varepsilon^{-1}]$ and its complement includes data that reach $\partial G$); AKN06 p. 3 footnote 1 (smoothness convention) and Remark 6.4, pp. 217–218 (averaged orbit away from the boundary).*

*Source gap: As for Theorem 1.33; in addition the printed Theorem 2 of Nei76 imposes $\rho < \rho_1/2$ with an auxiliary collar $\rho_1$; the route of Part 13 uses the exit radius $\rho = R/8$ and $\eta_0 = R/8$ (a proof choice inside the existential constants; no strict inequality $\eta < \rho/2$ is needed there).*

## 2. Interfaces and conversion of hypotheses

This part connects the frozen hypotheses of track N (Definitions of Part 0) to the form in which the proofs use them. It proves: the real action domain and the inclusions of tubes (§2.2); that holomorphy gives a smooth real Hamiltonian with $X_H = X_{h+f}$ (§2.3); the translation of the derivative bounds $B(M)$, the complex Hessian bound and $QC(m)$ into the Euclidean, uniform form of Pöschel's Theorem 1 (§2.4); the change of normalisation $2\pi$-periodic$\leftrightarrow$ 1-periodic and the sup-to-weighted Fourier norm (§2.5); and the dictionary of constants (§2.6). *Convention ruling.* All of Parts 1–10 work in the frozen conventions (period 1, max norms, characters $e^{2\pi i k\cdot\theta}$). Pöschel's normalisation (period $2\pi$, Euclidean norms on actions, weights $e^{|k|_1 s}$) enters only through the labelled conversions Lemmas 2.8, 2.9, 2.7, 2.10; quantities in his normalisation carry a tilde or the subscript $P$. This overrules the suggestion in `PLAN.md` to work in Pöschel's normalisation. The Euclidean norm is written $|\cdot|_2$, the max norm $|\cdot|$ (or $|\cdot|_\infty$), and $|k|_1 = \sum_i |k_i|$; the phase-space norm is $\max(|\theta|_\infty, |I|_\infty)$, Euclidean structure is used only for the dot product and where stated.

### 2.1. External facts.

(E1) *Finite-dimensional linear algebra.* An injective linear map $L : V \to W$ between finite-dimensional real vector spaces with $\dim V = \dim W$ is bijective (rank–nullity; Mathlib `LinearMap.injective_iff_surjective_of_finrank_eq_finrank`,

in `LinearAlgebra/FiniteDimensional/Lemmas.lean`; the endomorphism form `LinearMap.injective_iff_surjective` does not apply directly to $V \neq W$). The fin-ranks used below are $\dim(\mathbb{R}^n \times \mathbb{R}) = \dim(\mathbb{R} \times \mathbb{R}^n) = n+1$ (`Module.finrank_prod`, `Module.finrank_fin_fun`). Exists.

(E2) *Norm comparison and Cauchy–Schwarz.* For $x \in \mathbb{R}^n$ or $\mathbb{C}^n$: $|x|_\infty \le |x|_2 \le \sqrt{n}|x|_\infty$ and $|x|_1 \le \sqrt{n}|x|_2$, $|x|_2 \le |x|_1$; $|x \cdot y| \le |x|_2|y|_2$ (for the bilinear, not Hermitian, product too). Exists.

(E3) *Restriction of scalars and chain rule.* A $\mathbb{C}$-differentiable (resp. $C^k$) map is $\mathbb{R}$-differentiable (resp. $C^k$) with the same derivative; the chain rule for the $\mathbb{R}$-linear inclusion $\iota : \mathbb{R}^N \to \mathbb{C}^N$ and for Re. (Mathlib `DifferentiableAt.restrictScalars`, `ContDiffOn.restrict_scalars`, `HasFDerivAt.comp`.) Exists.

(E4) *Product rule* for $(x, \lambda) \mapsto \lambda\, v(x)$ with $v$ differentiable, and differentiability of a vector function coordinatewise. Exists.

Track-K results used: [K:`bp:p1:holomorphic-regularity`], [K:`bp:p1:real-restriction`], [K:`bp:p1:hamiltonian-real`], [K:`bp:p1:ext-schwarz`], [K:`bp:p1:decay`], [K:`bp:p1:summation`], [K:`bp:p1:exp-sums`].

### 2.2. **Domains and norms.**

**Lemma 2.1** (Norm and lattice comparisons)**.** *(a) For $x \in \mathbb{R}^n$ (or $\mathbb{C}^n$), every $n \ge 0$ (all norms are 0 when $n = 0$): $|x|_\infty \le |x|_2 \le \sqrt{n}\,|x|_\infty$, i.e. $|x|_2^2 \le n|x|_\infty^2$; for $n > 0$ this gives $|x|_\infty^2 \ge |x|_2^2/n$. (b) For $k \in \mathbb{Z}^n$: $|k|_2 \le |k|_1 \le \sqrt{n}|k|_2$. Consequently $\{|k|_1 \le K\} \subset \{|k|_2 \le K\}$ and $\{|k|_2 \le K\} \subset \{|k|_1 \le \sqrt{n}K\}$; so nonresonance known on the Euclidean ball $\{|k|_2 \le K_g\}$ (geometric cutoff $K_g$) holds on the $\ell^1$ ball $\{|k|_1 \le K_a\}$ (analytic cutoff $K_a$) as soon as $K_a \le K_g$ (the first inclusion); the factor $\sqrt{n}$ enters only where the max-norm Lipschitz constant of $\omega$ is compared with the Euclidean covering radius, giving the compatibility $K_g \ge \sqrt{n}\,K_a$ (Part 9, Remark 9.1, Corollary 9.16). (c) The bilinear bound: if $|Q_{ij}| \le M_0$ for a matrix $Q$ then $|x^\top Q y| \le nM_0|x|_2|y|_2$ for $x, y \in \mathbb{C}^n$.*
*Source: Standard ((E2)); the lattice inclusions correct the direction stated in `PLAN.md` (review N06).*

*Proof.* **Step S2.1: Vectors.** (a),(b) are (E2): $|x|_2^2 = \sum|x_i|^2 \le n\max|x_i|^2$, $|x|_1 \le \sqrt{n}|x|_2$ is Cauchy–Schwarz with $(1, \dots, 1)$.

**Step S2.2: Matrices.** $|x^\top Q y| \le M_0|x|_1|y|_1 \le M_0\, n|x|_2|y|_2$ by (b). □

**Lemma 2.2** (Domains)**.** *Let $R, s > 0$, $n \ge 0$.*

- *(a) $\mathcal{B}_s(R) = B(0, R+s)$; in particular it is open and $B(0,R) \subset \mathcal{B}_s(R)$ (Remark 1.9); $W_s := \mathbb{R}^n \times \mathcal{B}_s(R)$ is open in $\mathbb{R}^n \times \mathbb{R}^n$ and $D_s(R) \cap (\mathbb{R}^n \times \mathbb{R}^n) = W_s$ (real points of $D_s(R)$: $\theta \in \mathbb{R}^n \subset T_s$ since $s > 0$).*
- *(b) $B(0,R)_s \cap \mathbb{R}^n = \mathcal{B}_s(R)$; $D_s(R)$ and $B(0,R)_s$ are open.*
- *(c) (Max-norm tubes, as in Definition 5.1) Let $P = \{J \in \mathbb{R}^n : |J| \le R/2\}$ and $0 < r_0 \le \min(s/2, R/8)$. If $I \in \mathbb{C}^n$ and $J \in P$ with $|I - J|_\infty < r_0$, then $I \in B(0,R)_s$ and $|I-J|_\infty < s$. If moreover $I \in \mathbb{R}^n$ then $|I|_\infty < 5R/8 \le 3R/4$. Since $|x|_\infty \le |x|_2$, the same holds (as a consequence) for the smaller Euclidean tube $|I - J|_2 < r_0$.*
- *(d) $T_\rho \subset T_s$ for $\rho \le s$; $T_s = \mathbb{C}^0$ is the universal singleton for $n = 0$ (no $n > 0$ hypothesis is needed anywhere in N-ZERO, N-QC).*

*Source: Ours (Remark 1.9); BM11 p. 4 for the notation.*

*Proof.* **Step S2.3: Real tube.** ($\subset$) $|I| \le |J| + |I - J| < R + s$. ($\supset$) If $|I| < R + s$ put $J = \frac{R}{R+s} I$; then $|J| < R$ and $|I - J| = \frac{s}{R+s}|I| < s$ (for $I = 0$ take $J = 0$).

**Step S2.4: Openness and real points.** $B(0,R)_s = \bigcup_{J \in B(0,R)} \{|I - J| < s\}$ is a union of open balls; $T_s$ is open as a preimage of an open set; products of open sets are open; the rest is the definition of $\iota$ and $\mathcal{B}_s(R) = B(0,R)_s \cap \mathbb{R}^n$ by Step S2.3.

**Step S2.5: Max-norm tubes.** $|I-J|_\infty < r_0 \le s/2 < s$, and $|J|_\infty \le R/2 < R$ since $J \in P$; so $I \in B(0,R)_s$ (direct triangle inequality, no norm comparison needed). For real $I$: $|I|_\infty \le |J|_\infty + |I-J|_\infty < R/2 + R/8$. The Euclidean tube is contained in the max-norm tube by (E2) ($|x|_\infty \le |x|_2$). □

2.3. **Holomorphy implies real smoothness.** Throughout this subsection $R, s > 0$, $h$ satisfies `IsAnalyticIntegrable` $R\,s\,h$, $f$ satisfies `IsAnalyticPerturbation` $R\,s\,\varepsilon\,f$ ($\varepsilon \ge 0$), $G(\theta,I) = h(I) + f(\theta,I)$ on $D_s(R)$, $H = H_{h,f}$ and $\iota : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{C}^n \times \mathbb{C}^n$ the componentwise inclusion. Complex partials are $\partial_{\theta_i} G(w) = DG(w)[(e_i,0)]$, $\partial_{I_i} G(w) = DG(w)[(0,e_i)]$.

**Lemma 2.3** (Real smoothness and real Hamiltonian field)**.** *(a) $G$ is holomorphic and $C^\infty$ on $D_s(R)$; $h$ is holomorphic and $C^\infty$ on $B(0,R)_s$.*

*(b) $H$ is $C^\infty$ on the open set $W_s = \mathbb{R}^n \times B(0,R+s)$ (Lemma 2.2); on $W_s$ it is real and $H = G \circ \iota$, and for all orders $D^k H(z)[\zeta_1,\dots,\zeta_k] = D^k G(\iota z)[\iota\zeta_1,\dots,\iota\zeta_k] \in \mathbb{R}$. Likewise $h_\mathbb{R} = h \circ \iota$ is $C^\infty$ on $\mathcal{B}_s(R)$, with $D^k h_\mathbb{R}(I)[a_1,\dots,a_k] = D^k h(I)[\iota a_1,\dots]$.*

*(c) (Hamiltonian field) For $z = (\theta, I) \in W_s$: $H$ is differentiable at $z$ and $X_H(z) = \big((\partial_{I_i} G(\iota z))_i, (-\partial_{\theta_i} G(\iota z))_i\big) = X_{h+f}(\iota z)$, the complex Hamiltonian field of $G$ at the real point $\iota z$. The frequency map is $\omega_h(I)_i = \partial_i h(I) \in \mathbb{R}$ and $D^2 h_\mathbb{R}(I)[e_i,e_j] = \partial_i\partial_j h(I) = \partial_j\partial_i h(I) \in \mathbb{R}$ for $I \in \mathcal{B}_s(R)$.*

*(d) (Periodicity) $H(\theta + k, I) = H(\theta, I)$ for $(\theta, I) \in W_s$, $k \in \mathbb{Z}^n$; consequently $DH$, $D^k H$ and $X_H$ are 1-periodic in $\theta$ on $W_s$.*

*(e) If $\varepsilon = 0$ then $f = 0$ on $D_s(R)$ and $H(\theta, I) = h_\mathbb{R}(I)$ on $W_s$.*

*Source: BM11 p. 4 (regularity of $h$ on $B_s$);* [K:`bp:p1:holomorphic-regularity`], [K:`bp:p1:hamiltonian-real`]*; ours for the packaging. Part 3 uses this as* [N:`bpN:p3:holo-input`].

*Proof.* **Step S2.6: Holomorphy.** $h$ is complex differentiable on the open set $B(0,R)_s$ and $f$ on $D_s(R)$ (Definitions 1.11(i), 1.12(i); openness Lemma 2.2(b)); the map $(\theta, I) \mapsto h(I)$ is holomorphic on $D_s(R)$ (composition with the linear projection; $B(0,R)_s$ is the second factor). So $G$ is holomorphic on $D_s(R)$, hence $C^\infty$ with holomorphic derivatives of all orders ([K:`bp:p1:holomorphic-regularity`]), which is (a); by (E3) $G$ is $C^\infty$ in the real sense.

**Step S2.7: Reality.** $\iota(W_s) \subset D_s(R)$ (Lemma 2.2(a)) and $W_s = \iota^{-1}(D_s(R))$ is open. For $z = (\theta, I) \in W_s$, $\operatorname{Im} h(I) = 0$ (Definition 1.11(iv), $I \in \mathcal{B}_s(R)$) and $\operatorname{Im} f(\theta, I) = 0$ (Definition 1.12(iii)); so $G(\iota z) \in \mathbb{R}$ and $H(z) = \operatorname{Re} G(\iota z) = G(\iota z)$. Likewise $h_\mathbb{R} = h \circ \iota$ on $\mathcal{B}_s(R)$.

**Step S2.8: Derivatives of all orders.** On the open set $W_s$, $H = G \circ \iota$ with $\iota$ $\mathbb{R}$-linear and $G$ real-$C^\infty$ (Step S2.6): chain rule (E3) gives $H \in C^\infty(W_s)$ and the displayed formula for $D^k H$, with values real because $H$ is real-valued (the derivative of a real-valued function along real directions is a limit of real difference quotients). The same for $h_\mathbb{R}$.

**Step S2.9: The field.** By Definition 1.3, $X_H(z) = ((DH(z)[(0,e_i)])_i, (-DH(z)[(e_i,0)])_i)$; by Step S2.8 these are $\partial_{I_i} G(\iota z)$ and $-\partial_{\theta_i} G(\iota z)$. This is [K:`bp:p1:hamiltonian-real`] (with $V = D_s(R)$, $H$ agreeing with $\operatorname{Re} G$ on the real neighbourhood $W_s$). Taking $k = 1, 2$ for $h_\mathbb{R}$ gives $\omega_h(I)_i = Dh_\mathbb{R}(I)[e_i] = \partial_i h(I)$ and $D^2 h_\mathbb{R}(I)[e_i,e_j] = \partial_i\partial_j h(I)$, symmetric by [K:`bp:p1:holomorphic-regularity`] ([K:`bp:p1:ext-schwarz`]).

**Step S2.10: Periodicity.** For $(\theta, I) \in W_s$, $k \in \mathbb{Z}^n$ and real $\theta$: $(\theta, I), (\theta + k, I) \in D_s(R)$ and $f(\theta + k, I) = f(\theta, I)$ (Definition 1.12(ii)), $h(I)$ does not depend on $\theta$. So $H \circ \tau_k = H$ on $W_s$, $\tau_k(\theta, I) = (\theta + k, I)$, which maps $W_s$ onto itself with $D\tau_k = \mathrm{id}$; differentiating (chain rule) gives $D^j H(\tau_k z) = D^j H(z)$ and hence $X_H(\tau_k z) = X_H(z)$.

**Step S2.11: Zero perturbation.** $|f(w)| \le 0$ on $D_s(R)$ gives $f = 0$ there, so $H = \operatorname{Re} h(I) = h_\mathbb{R}(I)$ on $W_s$. □

**Lemma 2.4** (Ordered partials and first-order bounds)**.** *Let $U \subset \mathbb{R}^n$ be open with* `HasDerivativeBounds` $U\, h\, M$ *(Definition 1.13). Then for $I \in U$, $a, b \in \mathbb{R}^n$: $|\omega_h(I)_i| \le M$, $|\omega_h(I)|_\infty \le M$, $|\omega_h(I)|_2 \le \sqrt{n}\, M$, and $|D^2 h_{\mathbb{R}}(I)[e_i, e_j]| \le M$. The ordered partial $D^k h_{\mathbb{R}}(I)[e_{i_1}, \dots, e_{i_k}]$ ($k \le 3$) is invariant under permutations of $(i_1, \dots, i_k)$, so it equals the multi-index partial of the source.*
*Source: BM11 p. 4 (B(M));* [K:bp:p1:ext-schwarz].

*Proof.* **Step S2.12: First order.** $k = 1$ in the definition gives $|Dh_{\mathbb{R}}(I)[e_i]| = |\omega_h(I)_i| \le M$ (Definition 1.2); then $|\omega|_2 \le \sqrt{n}|\omega|_\infty$ (Lemma 2.1).

**Step S2.13: Symmetry.** $h_{\mathbb{R}} \in C^3$ on the open set $U$: $D^2 h_{\mathbb{R}}(I)$ and $D^3 h_{\mathbb{R}}(I)$ are symmetric ([K:bp:p1:ext-schwarz] applied to $h_{\mathbb{R}}$, $Dh_{\mathbb{R}}$, $D^2 h_{\mathbb{R}}$), which is the permutation invariance; the entry bound is $k = 2$. □

**Lemma 2.5** (Complex Hessian bound in Euclidean form)**.** *Let* `HasComplexHessianBound` $R\, s\, h\, M_{\mathbb{C}}$. *For $I \in B(0,R)_s$ let $Q(I) = (\partial_i \partial_j h(I))_{ij}$ (complex symmetric). Then $|Q(I)_{ij}| \le M_{\mathbb{C}}$ and $|x^\top Q(I) y| \le n M_{\mathbb{C}} |x|_2 |y|_2$ for $x, y \in \mathbb{C}^n$; hence the operator norm of $Q(I) : (\mathbb{C}^n, |\cdot|_2) \to (\mathbb{C}^n, |\cdot|_2)$ is $\le n M_{\mathbb{C}}$. For real $I \in \mathcal{B}_s(R)$ this matrix is the real Hessian $D^2 h_{\mathbb{R}}(I)[e_i, e_j]$ (Lemma 2.3(c)), and for real $x, y$: $|D^2 h_{\mathbb{R}}(I)[x, y]| \le n M_{\mathbb{C}} |x|_2 |y|_2$.*
*Source: Pos93 p. 190 (Hessian bound, anchor POS190-HESS); the conversion factor n is ours (`docs/tracks/N/sources.md`).*
*Source gap: Pos93 p. 190 says that for the symmetric Q the Euclidean operator norm "amounts to the spectral radius norm". For complex symmetric matrices this is false ($Q = \left(\begin{smallmatrix} 1 & i \\ i & -1 \end{smallmatrix}\right)$ has $Q^2 = 0$). The frozen definition uses the operator norm of $D^2 h$ for the max norm, and so does this blueprint; no spectral-radius claim is used.*

*Proof.* **Step S2.14: Entries.** $D^2 h(I) = D(Dh)(I) \in \mathcal{L}(\mathbb{C}^n, \mathcal{L}(\mathbb{C}^n, \mathbb{C}))$ has norm $\le M_{\mathbb{C}}$ by hypothesis and $|e_i| = 1$ for the max norm, so $|\partial_i \partial_j h(I)| = |D^2 h(I)[e_i, e_j]| \le M_{\mathbb{C}}$ (the order of arguments is immaterial by symmetry, [K:bp:p1:holomorphic-regularity]).

**Step S2.15: Euclidean bound.** Lemma 2.1(c). The operator-norm statement: $|Qx|_2 = \sup_{|y|_2=1} |\sum_i \bar{y}_i (Qx)_i| \le n M_{\mathbb{C}} |x|_2$ by the same estimate applied to $\bar{y}$. □

## 2.4. **Quasi-convexity: the uniform Euclidean form.**

**Lemma 2.6** (Transverse coercivity gives conic coercivity (linear algebra core))**.** *Let $n \ge 1$, $B, q > 0$, $Q : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$ bilinear (not necessarily symmetric) with $|Q[x, y]| \le B|x|_2|y|_2$, $\omega \in \mathbb{R}^n \setminus \{0\}$, and*

$$\omega \cdot v = 0 \implies Q[v, v] \ge q|v|_2^2 \qquad (v \in \mathbb{R}^n).$$

*If $\ell' \ge 0$ and $\ell' \le |\omega|_2\, q/(8(B+q))$, then for every $\xi \in \mathbb{R}^n$:*

$$|\omega \cdot \xi| \le \ell' |\xi|_2 \implies Q[\xi, \xi] \ge \tfrac{5}{8} q |\xi|_2^2 \quad (\ge \tfrac{q}{2}|\xi|_2^2).$$

*Source: Ours; the computation is in `docs/tracks/N/sources.md` ("Uniform conversion"); all inequalities re-checked here.*

*Proof.* **Step S2.16: Reduction to unit vectors.** Both hypothesis and conclusion are homogeneous (degrees 1 and 2) in $\xi$, and the case $\xi = 0$ is trivial; so let $|\xi|_2 = 1$.

**Step S2.17: Decomposition.** Put $e = \omega/|\omega|_2$, $t = \xi \cdot e$, $u = \xi - te$. Then $u \cdot \omega = 0$, $|u|_2^2 = 1 - t^2$, $|t| \le 1$, and $|t| = |\omega \cdot \xi|/|\omega|_2 \le \ell'/|\omega|_2 \le \kappa := q/(8(B+q)) < \frac{1}{8}$.

**Step S2.18: Expansion.** $Q[\xi, \xi] = Q[u, u] + tQ[u, e] + tQ[e, u] + t^2 Q[e, e]$. By hypothesis $Q[u, u] \ge q(1 - t^2)$; by the bound on $Q$ and $|u|_2, |e|_2 \le 1$: $|Q[u, e]|, |Q[e, u]|, |Q[e, e]| \le B$. So $Q[\xi, \xi] \ge q - (q + B)t^2 - 2B|t|$.

**Step S2.19: Numerics.** $(q+B)t^2 \le (q+B)\kappa\cdot\kappa = \frac{q}{8}\kappa \le \frac{q}{8}$ and $2B|t| \le 2B\kappa = \frac{2Bq}{8(B+q)} \le \frac{q}{4}$ (as $B \le B+q$). Hence $Q[\xi,\xi] \ge q - \frac{q}{8} - \frac{q}{4} = \frac{5}{8}q$. □

**Lemma 2.7** (Quasi-convexity in Pöschel's quantified form)**.** *Let $n \ge 1$, $R, s, m, M_{\mathbb{C}}, \omega_{\min} > 0$ and put*

$$q = \frac{m}{n}, \qquad B_c = nM_{\mathbb{C}} + q, \qquad \mu = \frac{q}{2}, \qquad \ell = \frac{\omega_{\min}\, q}{8(B_c+q)}.$$

*Let $h$ satisfy* `IsAnalyticIntegrable` $R\, s\, h$ *and* `HasComplexHessianBound` $R\, s\, h\, M_{\mathbb{C}}$, `IsStrictlyQuasiConvex` $\mathcal{B}_s(R)\, h\, m$, *and $\omega_{\min} \le |\omega_h(I)|_\infty$ for all $I \in B(0,R)$. Then for every $I \in B(0,R)$, writing $\omega = \omega_h(I)$, $Q = D^2 h_{\mathbb{R}}(I)$:*

*(a) $|Q[x,y]| \le B_c|x|_2|y|_2$; $\omega\cdot v = 0 \Rightarrow Q[v,v] \ge q|v|_2^2$;*

*(b) (Pöschel's alternative,* conditional*) for every $\xi \in \mathbb{R}^n$: $|\omega\cdot\xi| \le \ell|\xi|_2 \implies Q[\xi,\xi] \ge \mu|\xi|_2^2$; equivalently, either $|\omega\cdot\xi| > \ell|\xi|_2$ or $Q[\xi,\xi] \ge \mu|\xi|_2^2$ (Pos93 p. 190, "$\ell, m$-quasi-convex" with $m := \mu$);*

*(c) $|\omega|_2 \ge |\omega|_\infty \ge \omega_{\min}$ and $|\omega|_2 \le \sqrt{n}|\omega|_\infty$.*

*The constants $q, B_c, \mu, \ell$ depend only on $(n, m, M_{\mathbb{C}}, \omega_{\min})$. In Pöschel's normalisation ($\tilde h = 2\pi h$, Lemma 2.8) the same holds with $(\tilde\ell, \tilde\mu, \tilde B_c, \tilde\omega, \tilde Q) = 2\pi(\ell, \mu, B_c, \omega, Q)$.* Warning. *Full positive definiteness $Q[\xi,\xi] \ge \mu|\xi|_2^2$ for all $\xi$ is false in general and is not claimed: for $h(I) = I_1 + \frac12(I_2^2+I_3^2)$ one has $\omega(I) = (1, I_2, I_3) \ne 0$ (equal to $e_1$ on $I_2 = I_3 = 0$), $QC(m)$ holds on every bounded set for some $m > 0$ depending on the set (for $\omega\cdot v = 0$: $Q[v,v] = v_2^2 + v_3^2$ and $|v_1| \le (|I_2| + |I_3|)\max(|v_2|, |v_3|)$, so $m = (1+2K)^{-2}$ works on $|I|_\infty \le K$), but $Q[e_1, e_1] = 0$ at every point. The estimate (b) is only needed (and only true) in the cone $|\omega\cdot\xi| \le \ell|\xi|_2$.*

*Source: Pos93 p. 190 (anchors POS190-QC, POS190-HESS, POS191-LOW; I read the scan, p. 190, $\ell, m$-quasi-convexity, hypothesis $r_0 \le 4\ell/m$); conversion:* `docs/tracks/N/sources.md`*, "Uniform conversion"; BM11 p. 4 for $QC(m)$. Review N05.*

*Proof.* **Step S2.20: Regularity and Hessian.** Since $B(0,R) \subset \mathcal{B}_s(R)$, $I \in \mathcal{B}_s(R)$; $Q$ is the real Hessian of Lemma 2.5, so $|Q[x,y]| \le nM_{\mathbb{C}}|x|_2|y|_2 \le B_c|x|_2|y|_2$, the first half of (a).

**Step S2.21: Transverse coercivity in the Euclidean norm.** Definition 1.14 at $I \in \mathcal{B}_s(R)$ gives $\omega\cdot v = 0 \Rightarrow Q[v,v] \ge m|v|_\infty^2 \ge \frac{m}{n}|v|_2^2 = q|v|_2^2$ (Lemma 2.1(a)). Also $\omega \ne 0$.

**Step S2.22: Frequency size.** (c): $|\omega|_\infty \ge \omega_{\min}$ on $B(0,R)$ by hypothesis, $|\omega|_2 \ge |\omega|_\infty$; $|\omega|_2 \le \sqrt{n}|\omega|_\infty$ by Lemma 2.1.

**Step S2.23: Conic coercivity.** Apply Lemma 2.6 with $B = B_c$, $q$ and $\ell' = \ell$: $\ell = \omega_{\min} q/(8(B_c + q)) \le |\omega|_2 q/(8(B_c+q))$ by (c). The conclusion is $Q[\xi,\xi] \ge \frac58 q|\xi|_2^2 \ge \mu|\xi|_2^2$.

**Step S2.24: Scaling.** Multiplying $h$ by $2\pi$ multiplies $\omega, Q, B_c$ and the three of $\mu, q, \ell$ by $2\pi$, and the implications are homogeneous, so they persist. □

## 2.5. **Normalisations.**

**Lemma 2.8** (Period conversion)**.** *Let $\Lambda(\theta, I) = (2\pi\theta, I)$ and, for $H : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$ and arbitrary $h : \mathbb{C}^n \to \mathbb{C}$, $f : \mathbb{C}^n \times \mathbb{C}^n \to \mathbb{C}$, define*

$$\tilde H(\varphi, I) = 2\pi H(\varphi/2\pi, I), \qquad \tilde h = 2\pi h, \qquad \tilde f(\varphi, I) = 2\pi f(\varphi/2\pi, I),$$

*so $\tilde H = H_{\tilde h, \tilde f}$ when $H = H_{h,f}$. Clauses (a), (b) are unconditional (no analyticity or periodicity is assumed). Clauses (c) (domains), (d) (periodicity) and (e) (Fourier coefficients) are about $f$ satisfying* `IsAnalyticPerturbation` $R\, s\, \varepsilon\, f$ *(Definition 1.12), $R, s > 0$. Then:*

*(a) (Dynamics) For $U \subset \mathbb{R}^n$, $J \subset \mathbb{R}$: $z$ satisfies* `IsSolutionOn` $H\, U\, J\, z$ *iff $\Lambda \circ z$ satisfies* `IsSolutionOn` $\tilde H\, U\, J$ *(same time, same actions). In particular $|(\Lambda z)_2 - I_0| = |z_2 - I_0|$.*

(b) *(Quantities)* $\omega_{\tilde h} = 2\pi\omega_h$, $D^2\tilde h_{\mathbb R} = 2\pi D^2 h_{\mathbb R}$, *and so:* `IsStrictlyQuasiConvex`$U\,h\,m \Rightarrow$ `IsStrictlyQuasiConvex`$U\,\tilde h\,2\pi m$; $B(M) \Rightarrow$ *coordinate partials of* $\tilde h_{\mathbb R}$ *of order* $1,2,3$ *bounded by* $2\pi M$; *the complex Hessian bound* $M_{\mathbb C} \to 2\pi M_{\mathbb C}$; $\omega_{\min} \to 2\pi\omega_{\min}$; *Euclidean sizes of* $\omega$: $|\tilde\omega|_2 = 2\pi|\omega|_2$. *Time is unchanged.*

(c) *(Domains; needs holomorphy of* $f$ *on* $D_s(R)$ *and the bound* $|f| \le \varepsilon$*)* $\tilde f$ *is holomorphic on* $\{|\operatorname{Im}\varphi| < 2\pi s\} \times B(0,R)_s$ *and* $|\tilde f| \le 2\pi\varepsilon$ *there; the strip* $T_\rho$ *(*$\rho$ *in* $\theta$*) corresponds to the strip of width* $2\pi\rho$ *in* $\varphi$.

(d) *(Periodicity; needs* $f(\theta + k, I) = f(\theta, I)$ *on* $D_s(R)$*)* $\tilde f(\varphi + 2\pi k, I) = \tilde f(\varphi, I)$ *for* $k \in \mathbb Z^n$ *and* $\varphi$ *in that strip. (Without this premise the claim fails:* $f = \theta_1$ *gives* $\tilde f = \varphi_1$*.)*

(e) *(Fourier; needs (c), (d))* $\tilde f(\varphi, I) = \sum_k \hat{\tilde f}_k(I) e^{ik\cdot\varphi}$ *with* $\hat{\tilde f}_k(I) = 2\pi\hat f_k(I)$, $\hat f_k(I) = \int_{[0,1]^n} f(\theta, I) e^{-2\pi ik\cdot\theta}\, d\theta$, *whenever the Fourier series of* $f(\cdot, I)$ *converges to it (*[K:`bp:p1:summation`]*); the coefficient radius* $s'$ *of Pöschel's weight* $e^{|k|_1 s'}$ *equals* $2\pi\rho$.

*Source:* `docs/tracks/N/sources.md` *("For the period conversion"); Pos93 p. 189 (set-up:* $\theta \in \mathbb R^n/2\pi\mathbb Z^n$, *strip* $W_{s_0}$ *of width* $s_0$ *in the max norm of* $\operatorname{Im}\theta$*).*

*Proof.* **Step S2.25: Chain rule.** $\tilde H = 2\pi\, H \circ \Lambda^{-1}$; $\Lambda$ is a linear isomorphism, so $\tilde H$ is differentiable at $\Lambda z$ iff $H$ is at $z$, with $\partial_{I_i}\tilde H(\Lambda z) = 2\pi\partial_{I_i} H(z)$ and $\partial_{\varphi_i}\tilde H(\Lambda z) = \partial_{\theta_i} H(z)$.

**Step S2.26: Vector field.** Hence $X_{\tilde H}(\Lambda z) = (2\pi\partial_I H(z), -\partial_\theta H(z)) = (2\pi\dot\theta, \dot I) = \frac{d}{dt}\Lambda z$ when $\dot z = X_H(z)$, and conversely; the condition $z(t)_2 \in U$ is unchanged. This is (a). (Rescaling the angle without the factor $2\pi$ in $H$ would change time.)

**Step S2.27: Scalars.** (b) is linearity of $h \mapsto \omega_h, D^2 h_{\mathbb R}, D^k h_{\mathbb R}$ and of the operator norm. (c)–(f) are, under the stated premises, the substitution $\varphi = 2\pi\theta$, $|\operatorname{Im}\varphi| = 2\pi|\operatorname{Im}\theta|$, and $e^{ik\cdot\varphi} = e^{2\pi ik\cdot\theta}$. □

**Lemma 2.9** (From sup norm on a strip to the weighted Fourier norm)**.** *Let* $u : T_\rho \to \mathbb C$ *be holomorphic, bounded and* $1$*-periodic,* $\|u\|_\rho = \sup_{T_\rho}|u|$*, and* $0 < \delta < \rho$*. Then*

$$\sum_{k\in\mathbb Z^n} |\hat u_k|\, e^{2\pi|k|_1(\rho-\delta)} \;\le\; \coth(\pi\delta)^n\, \|u\|_\rho,$$

*the left side is Pöschel's weighted norm with weight* $e^{|k|_1 s'}$, $s' = 2\pi(\rho - \delta)$*, and the factor is* $\coth^n\sigma$ *with* $\sigma = \pi\delta$ *(Pos93 p. 190:* $|u|_{D,r,s} \le \coth^n\sigma\, |u|_{D,r,s+2\sigma}$*, where the right side is the sup norm on the strip of width* $s + 2\sigma$ *in* $\varphi$*, i.e.* $\rho = (s' + 2\sigma)/2\pi$*). Conversely* $\|u\|_\rho \le \sum_k |\hat u_k| e^{2\pi|k|_1\rho}$*. For* $I$*-dependent* $u(\theta, I)$ *the statement holds for each fixed* $I$ *with the same constant, so the sup over* $I$ *in the weighted norm is* $\le \coth(\pi\delta)^n \sup_{\theta,I}|u|$.

*Source: Pos93 p. 190 (the displayed relation between the weighted and sup norms); frozen-convention statement:* [K:`bp:p1:summation`]*,* [K:`bp:p1:decay`]*,* [K:`bp:p1:exp-sums`]*(b).*

*Proof.* **Step S2.28: Decay.** For $\rho' \in (\rho - \delta, \rho)$, [K:`bp:p1:decay`] gives $|\hat u_k| \le e^{-2\pi|k|_1\rho'}\|u\|_\rho$; hence $|\hat u_k| e^{2\pi|k|_1(\rho-\delta)} \le e^{-2\pi|k|_1(\rho'-\rho+\delta)}\|u\|_\rho$, and letting $\rho' \uparrow \rho$ (termwise, then summing, or directly [K:`bp:p1:summation`], converse part) the bound is $e^{-2\pi\delta|k|_1}\|u\|_\rho$ per $k$.

**Step S2.29: Sum.** $\sum_{k\in\mathbb Z^n} e^{-2\pi\delta|k|_1} = \coth(\pi\delta)^n$ ([K:`bp:p1:exp-sums`](b) with $x = 2\pi\delta$). The case with parameters is pointwise in $I$. □

## 2.6. **The dictionary of constants.**

**Lemma 2.10** (Constants dictionary for N-HEAD-C)**.** *Let $n \geq 3$ (only $n \geq 2$ is used by Pöschel) and $R, s, M, m, M_{\mathbb{C}}, \omega_{\min} > 0$, $a = \frac{1}{2n}$. Define successively*

$$q = \tfrac{m}{n},\ B_c = nM_{\mathbb{C}} + q,\ \mu = \tfrac{q}{2},\ \ell = \tfrac{\omega_{\min} q}{8(B_c+q)}, \qquad r_0 = \min\Big(\tfrac{s}{2}, \tfrac{R}{8}, \tfrac{\ell}{2B_c}, \tfrac{4\omega_{\min}}{\mu}\Big), \qquad A = \tfrac{11B_c}{\mu},$$

$$s_0 = \pi s, \quad \sigma = \tfrac{\pi s}{4}, \quad F = 2\pi \coth^n\big(\tfrac{\pi s}{4}\big), \quad \varepsilon_* = \frac{2\pi\mu\, r_0^2}{2^{10} A^{2n}},$$

$$C = \tfrac{r_0}{A}\Big(\tfrac{F}{\varepsilon_*}\Big)^a, \quad c = \tfrac{s_0}{6\sqrt{n}}\Big(\tfrac{\varepsilon_*}{F}\Big)^a, \quad T = \frac{A^2 s_0}{4\pi n M},$$

$$\varepsilon_0 = \min\Big(\tfrac{\varepsilon_*}{F},\ \big(\tfrac{R}{4C}\big)^{2n}\Big).$$

*Pöschel's data (his normalisation, for $\tilde{h} = 2\pi h$): $m_P = 2\pi\mu$, $\ell_P = 2\pi\ell$, $M_P = 2\pi B_c$, $A_P = 11M_P/m_P = A$, $\varepsilon_{0,P} = m_P r_0^2/(2^{10}A_P^{2n}) = \varepsilon_*$, $R_0 = r_0/A$, and the set $P = \{J \in \mathbb{R}^n : |J| \leq R/2\}$. Then:*

- *(a) All constants are positive and finite and depend only on $(n, R, s, M, m, M_{\mathbb{C}}, \omega_{\min})$ — not on $h, f, \varepsilon, \theta_0, I_0$, nor on the bound $A$ of Definition 1.11(iii). Order of choice: $q, B_c, \mu, \ell, r_0, A$; then $s_0, \sigma, F$; $\varepsilon_*$; $C, c$; $T$; $\varepsilon_0$. And $C\varepsilon_0^a \leq R/4$, $0 < \varepsilon_0 \leq \varepsilon_*/F$.*
- *(b) $r_0 \leq \ell_P/(2M_P)$ $(= \ell/(2B_c)$, hypothesis (S3) of Definition 10.1); consequently $r_0 \leq 4\ell_P/m_P$ $(= 4\ell/\mu)$; $A \geq 22$; $R_0 = r_0/A \leq r_0/22 \leq R/176$.*
- *(c) (Tubes) With the max-norm tube $V_{r_0}P = \{I \in \mathbb{C}^n : \exists J \in P,\ |I-J|_\infty < r_0\}$ of Definition 5.1 and $U_{r_0}P = V_{r_0}P \cap \mathbb{R}^n$: $V_{r_0}P \subset B(0,R)_s$ and $U_{r_0}P \subset B(0, 5R/8) \subset B(0,R)$ (the Euclidean tubes are contained in these).*
- *(d) (Hypotheses of Theorem 1) If $h$ satisfies (H1)–(H5) of Theorem 1.19, then for $I \in U_{r_0}P$: $\tilde{h}$ is $\ell_P, \mu_P$-quasi-convex (Lemma 2.7) with $\mu_P = 2\pi\mu$; the complex Hessian operator norm (Euclidean) of $\tilde{h}$ on $V_{r_0}P$ is $\leq M_P$ (Lemma 2.5); Pöschel's $m, M$ are $m_P, M_P$ and his hypothesis $r_0 \leq 4\ell_P/m_P$ holds, together with the added hypothesis $r_0 \leq \ell_P/(2M_P)$ of Part 10 (needed on the block $\mathbb{Z}^n$, Remark 10.9(1)).*
- *(e) (Perturbation) If $f$ satisfies (H6) with $\varepsilon$, then Pöschel's norm of $\tilde{f}$ on $V_{r_0}P \times$(strip of width $s_0$) is $\leq F\varepsilon$; and $F\varepsilon \leq \varepsilon_{0,P}$ if $\varepsilon \leq \varepsilon_0$.*
- *(f) (Exceptional branch is empty) For $I_0 \in B(0, R/2)$: $|\tilde{\omega}(I_0)|_2 \geq 2\pi\omega_{\min} > \pi\omega_{\min} \geq m_P r_0/8$.*
- *(g) (Time prefactor) For $I_0 \in B(0, R/2)$, $0 < 2\pi\omega_{\min} \leq \Omega_0 = \sup_{|I-I_0|_2 \leq R_0} |\tilde{\omega}(I)|_2 \leq 2\pi\sqrt{n}M \leq 2\pi n M$, so Pöschel's $T_0 = A_P^2 s_0/\Omega_0 \geq 2T$.*
- *(h) (Conclusion shape, for $\varepsilon > 0$; $\varepsilon = 0$ is N-ZERO) Pöschel's $R_0(\varepsilon_P/\varepsilon_{0,P})^a$ (the radius of Theorem 10.7(a) is half of it, in the max norm, hence $\leq C\varepsilon^a$) and the time scale $T_0 \exp\big(\frac{s_0}{6\sqrt{n}}(\varepsilon_{0,P}/\varepsilon_P)^a\big)$ with $\varepsilon_P = F\varepsilon$ are $C\varepsilon^a$ and $T_0 \exp(c\varepsilon^{-a})$ (the $\sqrt{n}$ in the exponent is that of Theorem 10.7(a), Remark 10.10(ii)), and $T\exp(c\varepsilon^{-a}) \leq \frac{1}{2}T_0 \exp(c\varepsilon^{-a})$.*

*Source:* `docs/tracks/N/sources.md` *"Uniform conversion" (all constants), checked against Pos93 p. 190 (hypotheses, $A = 11M/m$, $\varepsilon_0 = mr_0^2/(2^{10}A^{2n})$, $r_0 \leq 4\ell/m$), p. 191 (parameters $a, R_0, T_0$, exceptional branch) and p. 204 (assembly: $K = (\varepsilon_0/\varepsilon)^a$). Pos93 requires $\|f_\varepsilon\| \leq \varepsilon$ (p. 190) with $\varepsilon \leq \varepsilon_{0,P}$.*
*Source gap: Pos93 states no theorem about the $\omega_{\min}$-conversion and the frozen-to-source dictionary: it exists only in* `sources.md` *and here (Part 0* `bpN:p0:head`*). The choice of $T$ with the factor $1/2$ is slack, not needed by the source.*

*Proof.* **Step S2.30: Positivity and dependence.** Each constant is a positive finite expression in earlier ones (denominators: $n, \mu, 8(B_c+q), A, F, \varepsilon_*, 4C, 4\pi nM$, all $> 0$); $\varepsilon_0 > 0$ as a minimum of positive numbers, and $C\varepsilon_0^a \leq C \cdot \frac{R}{4C} = R/4$ since $((R/4C)^{2n})^{1/(2n)} = R/(4C)$. The list of inputs in each formula contains only $n, R, s, M, m, M_{\mathbb{C}}, \omega_{\min}$. This is (a).

**Step S2.31: Size relations.** $B_c = nM_{\mathbb{C}} + q \geq q = 2\mu$ so $A = 11B_c/\mu \geq 22$. $r_0 \leq \ell/(2B_c)$ by definition, i.e. $r_0 \leq \ell_P/(2M_P)$ (the factors $2\pi$ cancel); since $B_c \geq 2\mu$, $r_0 \leq \ell/(4\mu) \leq 4\ell/\mu = 4\ell_P/m_P$; and $R_0 = r_0/A \leq r_0/22 \leq R/176$. This is (b).

**Step S2.32: Tubes.** $r_0 \leq \min(s/2, R/8)$: Lemma 2.2(c) gives (c).

**Step S2.33: Quasi-convexity and Hessian.** For (d): $U_{r_0}P \subset B(0,R)$ by (c), so Lemma 2.7(b) applies at every point of $U_{r_0}P$ and, after Lemma 2.8(b), gives the $(\ell_P, \mu_P)$-alternative for $\tilde{h}$. For $I \in V_{r_0}P \subset B(0,R)_s$ the Euclidean operator norm of $\tilde{Q}(I)$ is $\leq 2\pi n M_{\mathbb{C}} \leq 2\pi B_c = M_P$ (Lemma 2.5). Pöschel's $A = 11M/m$ with $M = M_P$ and $m = \mu_P = m_P$ is $11B_c/\mu = A$. The hypothesis (S3) of Part 10 is $r_0 \leq \ell_P/(2M_P) = \ell/(2B_c)$, the third entry of the minimum defining $r_0$.

**Step S2.34: Perturbation.** By (H6) $|f| \leq \varepsilon$ on $D_s(R) \supset T_{3s/4} \times V_{r_0}P$ ($V_{r_0}P \subset B(0,R)_s$, (c)). For each fixed $I \in V_{r_0}P$ apply Lemma 2.9 to $u = f(\cdot, I)$ with $\rho = 3s/4$, $\delta = s/4$: $\sum_k |\hat{f}_k(I)| e^{2\pi|k|_1 s/2} \leq \coth(\pi s/4)^n \varepsilon$. Multiplying by $2\pi$ (Lemma 2.8(e); $f$ satisfies (H6), so (c), (d) apply) and noting $2\pi \cdot s/2 = \pi s = s_0$ gives $\leq F\varepsilon$. (In Pöschel's terms $s_0 + 2\sigma = 3\pi s/2 = 2\pi \cdot \frac{3s}{4}$.) $\varepsilon \leq \varepsilon_*/F$ gives $F\varepsilon \leq \varepsilon_* = \varepsilon_{0,P}$. This is (e).

**Step S2.35: Exceptional branch.** $I_0 \in B(0, R/2) \subset B(0,R)$ so $|\omega(I_0)|_2 \geq \omega_{\min}$ (Lemma 2.7(c)) and $|\tilde{\omega}|_2 = 2\pi|\omega|_2$. And $m_P r_0/8 = 2\pi\mu r_0/8 \leq 2\pi\mu \cdot \frac{4\omega_{\min}}{8\mu} = \pi\omega_{\min}$ by $r_0 \leq 4\omega_{\min}/\mu$. This is (f).

**Step S2.36: Time.** The Euclidean ball $\{|I - I_0|_2 \leq R_0\}$ with $R_0 \leq r_0$ lies in $\{|I|_\infty < R/2 + R_0\} \subset B(0, 5R/8)$ ($|I - I_0|_\infty \leq |I - I_0|_2$), inside $\mathcal{B}_s(R)$ where Lemma 2.4 gives $|\omega(I)|_2 \leq \sqrt{n}M$. So $\Omega_0 \leq 2\pi\sqrt{n}M \leq 2\pi n M$ (finite); also $I_0$ belongs to the closed ball, $I_0 \in B(0,R)$, so $\Omega_0 \geq |\tilde{\omega}(I_0)|_2 \geq 2\pi\omega_{\min} > 0$ (Lemma 2.7(c)). Inverting the positive quantities, $T_0 = A^2 s_0/\Omega_0 \geq A^2 s_0/(2\pi n M) = 2T$. (g).

**Step S2.37: Exponent and constants.** $R_0(F\varepsilon/\varepsilon_*)^a = \frac{r_0}{A}(F/\varepsilon_*)^a \varepsilon^a = C\varepsilon^a$ and $\frac{s_0}{6\sqrt{n}}(\varepsilon_*/(F\varepsilon))^a = c\varepsilon^{-a}$ (the exponent of Theorem 10.7(a) is $\frac{\pi s'}{3\sqrt{n}}(\varepsilon_0/\varepsilon')^a$ with geometric width $s' = s/2$, i.e. $\frac{s_0}{6\sqrt{n}}$, $\varepsilon' = \coth^n(\pi s/4)\varepsilon$, $\varepsilon_0^{\mathrm{fr}} = \varepsilon_*/2\pi$, so $(\varepsilon_0^{\mathrm{fr}}/\varepsilon')^a = (\varepsilon_*/F\varepsilon)^a$; Remark 10.10). The radius of Theorem 10.7(a) is $\frac{1}{2}R_0(F\varepsilon/\varepsilon_*)^a = \frac{1}{2}C\varepsilon^a \leq C\varepsilon^a$ in the max norm. (h) follows from $T \leq T_0/2$. □

*Remark* 2.11 (Forward use and an unused constant). Part 10 chooses $C, T, c, \varepsilon_0$ as in Lemma 2.10, verifies the hypotheses of Theorem 10.7 (Pos93's, plus $r_0 \leq \ell/(2B_c)$) by (b),(d),(e) and excludes the exceptional branch by (f). (Corrections relative to the first draft, from Part 10, Remark 10.9(1),(2): the entry $2\ell/\mu$ of $r_0$ became $\ell/(2B_c)$; $c$ became $\frac{s_0}{6\sqrt{n}}(\varepsilon_*/F)^a$. Then $\varepsilon_*$, $C$, $\varepsilon_0$ change through $r_0$ only, and $T$, $F$, $A$ are unchanged.) The time prefactor $T$ enters only through (g); the remaining slack $\varepsilon_0 \leq (R/4C)^{2n}$ gives $C\varepsilon^a \leq R/4$ so that the a-priori confinement keeps $I(t) \in B(0, R/2 + R/4) = B(0, 3R/4)$ (continuation, [N:`bpN:p3:continuation`]). The value $M_{\mathbb{C}}$ enters only through $B_c$; $M$ only through $T$.

## 3. N-QC and N-ZERO

This part proves two frozen theorems completely: N-QC (Theorem 1.21, the nonsingular isoenergetic derivative) and N-ZERO (Theorem 1.20, the exact integrable dynamics). Neither uses the normal-form machinery. The linear-algebra core of N-QC is stated in a reusable form (Lemma 3.1, cited by track I as the isoenergetic gate, [N:`bpN:p2:qc-kernel`]). Frozen conventions throughout (max norms, period 1); Euclidean structure appears only as the dot product $\omega \cdot v = \sum \omega_i v_i$.

### 3.1. N-QC.

**Lemma 3.1** (Kernel core: bordered Hessian is nonsingular)**.** *Let $n \geq 1$, $\omega \in \mathbb{R}^n$, $A$ a real $n \times n$ matrix (not necessarily symmetric, no norm bound), $\lambda \neq 0$, and suppose*

$$\omega \cdot v = 0,\ v \neq 0 \implies v^\top A v \neq 0 \qquad \textit{and} \qquad \omega \neq 0.$$

*Let $L : \mathbb{R}^n \times \mathbb{R} \to \mathbb{R} \times \mathbb{R}^n$, $L(v,u) = (\omega \cdot v,\ u\,\omega + \lambda A v)$. Then $\ker L = 0$, hence $L$ is bijective. In particular this holds if $\lambda > 0$, $\omega \neq 0$ and $v^\top A v \geq m|v|^2$ for some norm and $m > 0$ whenever $\omega \cdot v = 0$. (For $n = 0$ no $\omega \neq 0$ exists; N-QC is then vacuous, Remark 3.2.)*
*Source: BM11 Lemma 3.3 and its proof, pp. 9–10 (kernel computation); linear algebra (E1).*

*Proof.* **Step S3.1: Kernel equations.** Let $L(v,u) = 0$: $\omega \cdot v = 0$ and $u\omega + \lambda A v = 0$.

**Step S3.2: Pairing with $v$.** Dot the second equation with $v$: $u(\omega \cdot v) + \lambda v^\top A v = 0$, so $\lambda v^\top A v = 0$, i.e. $v^\top A v = 0$ as $\lambda \neq 0$. Since also $\omega \cdot v = 0$, the hypothesis forces $v = 0$.

**Step S3.3: The scalar.** With $v = 0$: $u\omega = 0$; as $\omega \neq 0$, $u = 0$.

**Step S3.4: Bijectivity.** $L$ is a linear map between spaces of dimension $n+1$ with trivial kernel, hence bijective by (E1) (`LinearMap.injective_iff_surjective_of_finrank_eq_finrank`, with $\dim(\mathbb{R}^n \times \mathbb{R}) = \dim(\mathbb{R} \times \mathbb{R}^n) = n+1$). □

*Remark* 3.2 ($n = 0$). For $n = 0$ the space $\mathbb{R}^0$ is a point, so $\omega = 0$ and Definition 1.14 requires $\omega_h(I) \neq 0$: no $I$ satisfies it. Hence for $n = 0$ the hypotheses of N-QC are never met with $I \in U$ and the theorem is vacuous; no restriction $n \geq 1$ is added to it (review N03).

**Lemma 3.3** (The derivative of $\Psi$)**.** *Let $U \subset \mathbb{R}^n$ be open, $h$ with $h_\mathbb{R} \in C^2(U)$, $I \in U$, $\lambda \in \mathbb{R}$, $\Psi(I', \lambda') = (h_\mathbb{R}(I'), \lambda' \omega_h(I'))$. Then $\Psi$ is differentiable at $(I, \lambda)$ (so `fderiv` is the genuine derivative; Remark 1.18(e)) and*

$$D\Psi(I,\lambda)[v,u] = \big(\omega \cdot v,\ u\,\omega + \lambda Q v\big), \qquad \omega = \omega_h(I),\ Q_{ij} = D^2 h_\mathbb{R}(I)[e_i, e_j] = Q_{ji}.$$

*Source: BM11 p. 10; (E4),* [K:`bp:p1:ext-schwarz`].

*Proof.* **Step S3.5: First component.** $(I', \lambda') \mapsto h_\mathbb{R}(I')$ is differentiable at $(I, \lambda)$ ($h_\mathbb{R}$ is $C^2$ on the open set $U \ni I$) with derivative $[v,u] \mapsto Dh_\mathbb{R}(I)[v] = \sum_i v_i Dh_\mathbb{R}(I)[e_i] = \omega \cdot v$.

**Step S3.6: The frequency map is $C^1$.** $\omega_h(I')_i = Dh_\mathbb{R}(I')[e_i]$ is the composition of $I' \mapsto Dh_\mathbb{R}(I') \in \mathcal{L}(\mathbb{R}^n, \mathbb{R})$ (which is $C^1$ on $U$ since $h_\mathbb{R} \in C^2(U)$ and $U$ is open) with the continuous linear evaluation $\ell \mapsto \ell(e_i)$. So $\omega_h$ is differentiable on $U$ with $D\omega_h(I)[v]_i = D(Dh_\mathbb{R})(I)[v][e_i] = D^2 h_\mathbb{R}(I)[v, e_i]$.

**Step S3.7: Symmetry.** $D^2 h_\mathbb{R}(I)$ is symmetric ([K:`bp:p1:ext-schwarz`]), so $D\omega_h(I)[v] = Qv$ with $Q$ the symmetric Hessian matrix.

**Step S3.8: Product.** By the product rule (E4), $(I', \lambda') \mapsto \lambda' \omega_h(I')$ is differentiable at $(I, \lambda)$ with derivative $[v,u] \mapsto u\,\omega + \lambda Q v$. Both components are differentiable, so $\Psi$ is, with the stated derivative. □

**Theorem 3.4** (N-QC)**.** *Theorem 1.21 holds: for `IsStrictlyQuasiConvex` $U\,h\,m$, $I \in U$, $\lambda > 0$, the map $D\Psi(I,\lambda)$ is bijective.*
*Frozen Lean declaration: `KAM.Nekhoroshev.quasiConvex_isoenergetic_derivative`*
*Source: BM11 Lemma 3.3 and proof, pp. 9–10.*

*Proof.* **Step S3.9: Unpack the hypothesis.** Definition 1.14: $m > 0$, $U$ open, $h_\mathbb{R} \in C^2(U)$, $\omega := \omega_h(I) \neq 0$, and $\omega \cdot v = 0 \Rightarrow m|v|^2 \leq D^2 h_\mathbb{R}(I)[v,v]$ (max norm on $v$). Note $n \geq 1$ because $\omega \neq 0$ (Remark 3.2).

**Step S3.10: The derivative.** By Lemma 3.3, $D\Psi(I,\lambda) = L$ of Lemma 3.1 with $A = Q$ the Hessian matrix, $v^\top Q v = D^2 h_\mathbb{R}(I)[v,v]$ (bilinearity).

**Step S3.11: Positivity on $\omega^\perp$.** For $\omega \cdot v = 0$, $v \neq 0$: $v^\top Q v \geq m|v|_\infty^2 > 0$, since $m > 0$ and $|v|_\infty > 0$. So the hypothesis of Lemma 3.1 holds (with $\lambda > 0$).

**Step S3.12: Conclusion.** Lemma 3.1 gives bijectivity of $L = D\Psi(I,\lambda) = \mathtt{fderiv}\,\mathbb{R}\,\Psi\,(I,\lambda)$. □

*Remark* 3.5 (Reusable form for track I). Lemma 3.1 needs no norm, no symmetry of $A$ and no uniform constant: any quasi-convexity-type hypothesis ("$v^\top Av > 0$ on $\omega^\perp \setminus 0$") together with $\omega \neq 0$ and $\lambda \neq 0$ gives nonsingularity of the bordered matrix $\begin{pmatrix} \omega^\top & 0 \\ \lambda A & \omega \end{pmatrix}$. A quantitative inverse bound is not asserted by N-QC and is not provided here; it would follow, if needed, from the compactness of $\{v : |v|_2 = 1,\ \omega \cdot v = 0\}$ for a fixed $(I, \lambda)$ but is not uniform in $(I, \lambda)$.

### 3.2. **N-ZERO.**

**Lemma 3.6** (The field at the integrable orbit)**.** *Let $R, s > 0$, `IsAnalyticIntegrable R s h`, `IsAnalyticPerturbation R s 0 f`, $H = H_{h,f}$, and $W_s = \mathbb{R}^n \times B(0, R+s)$. Then $H(\theta, I) = h_{\mathbb{R}}(I)$ on $W_s$; $H$ is differentiable at every $z = (\theta, I) \in W_s$ with $DH(z)[(a,b)] = Dh_{\mathbb{R}}(I)[b]$ and*

$$X_H(\theta, I) = (\omega_h(I), 0).$$

*Source: BM11 p. 1 ($H = h(I)$ when $f = 0$); ours.*

*Proof.* **Step S3.13: $f$ vanishes.** Definition 1.12(iv) with $\varepsilon = 0$: $|f| \leq 0$ on $D_s(R)$, so $f = 0$ there. For $z = (\theta, I) \in W_s$, $(\theta, I) \in D_s(R)$ (Lemma 2.2(a)), so $H(z) = \operatorname{Re} h(I) = h_{\mathbb{R}}(I)$ (Lemma 2.3(e)).

**Step S3.14: Differentiability.** $W_s$ is open and on it $H = h_{\mathbb{R}} \circ \mathrm{pr}_2$ with $h_{\mathbb{R}} \in C^3(\mathcal{B}_s(R))$ (Definition 1.11(v)), $\mathcal{B}_s(R) = B(0, R+s)$ open. So $H$ is differentiable at each point of $W_s$ with $DH(z)[(a,b)] = Dh_{\mathbb{R}}(I)[b]$ (chain rule with the continuous linear projection). The point-wise hypotheses of the totality remark 1.18 are met.

**Step S3.15: Field.** By Definition 1.3, $\partial_{I_i} H = Dh_{\mathbb{R}}(I)[e_i] = \omega_h(I)_i$ (Definition 1.2) and $\partial_{\theta_i} H = DH(z)[(e_i, 0)] = 0$. □

**Theorem 3.7** (N-ZERO)**.** *Theorem 1.20 holds.*
*Frozen Lean declaration: `KAM.Nekhoroshev.zero_perturbation`*
*Source: Ours (supplement to BM11 Theorem 2.1, p. 5).*

*Proof.* **Step S3.16: Initial value and action.** $\gamma(t) = (\theta_0 + t\omega_h(I_0), I_0)$ (Definition 1.17) gives $\gamma(0) = (\theta_0, I_0)$ and $\gamma(t)_2 = I_0$ for all $t$: the first and third conclusions.

**Step S3.17: The action stays in $B(0,R)$.** $I_0 \in B(0, R/2) \subset B(0, R)$ since $R > 0$ (max norm: $|I_0| < R/2 < R$).

**Step S3.18: Differentiability of $H$ along the orbit.** $\gamma(t) \in \mathbb{R}^n \times B(0,R) \subset W_s$ (Lemma 2.2(a): $B(0,R) \subset B(0, R+s)$). By Lemma 3.6 $H$ is differentiable at $\gamma(t)$ for each $t$.

**Step S3.19: Derivative of the orbit.** $\gamma$ is affine in $t$, so differentiable at every $t \in \mathbb{R}$ (two-sided) with $\gamma'(t) = (\omega_h(I_0), 0)$.

**Step S3.20: The equation.** Lemma 3.6 gives $X_H(\gamma(t)) = (\omega_h(I_0), 0) = \gamma'(t)$ for every $t \in \mathbb{R}$. With Steps S3.17–S3.19, every clause of Definition 1.16 holds with $U = B(0,R)$ and $J = \mathbb{R}$. □

*Remark* 3.8 (Scope). N-ZERO uses neither the holomorphy of Part 1 (beyond the stated clauses of the definitions) nor the ODE layer of Part 3; Part 3 supplies uniqueness if ever required for $\varepsilon = 0$ ($\gamma$ is the unique solution by [N:`bpN:p3:gen-uniqueness`]), which is not asserted by the frozen statement. The case $\varepsilon = 0$ is excluded from N-HEAD-C by (H7).

## 4. Hamiltonian ODE infrastructure (shared by tracks I, N, P)

This part is the single ODE layer of the three blueprints I, N, P. It has two levels. The *general layer* (§4.2–§4.6) is about an autonomous vector field $F$ of class $C^1$ on an open subset $W$ of a finite-dimensional real normed space $E$: local existence, uniqueness on *every* interval, maximal solutions, continuation (compact sets, and closed sets with uniform bounds), first exit from an open set.

The *Hamiltonian layer* (§4.7) adds $F = X_H$: smoothness of $X_H$, energy conservation, translation in the angles. §4.8 shows that the frozen solution predicates of the three tracks (`IsSolutionOn`, `IsHamiltonianSolutionOn`, `IsRestrictedSolutionOn`, `IsNewtonSolution`, `IsFixedSolution`) are instances of the general predicate, and §4.9 gives the six track-N results of `PLAN.md`. Tracks I and P cite the general results as [N:`bpN:p3:gen-...`].

### 4.1. **Conventions and external facts.**

- $E$ is a finite-dimensional real normed space (hence complete and proper: closed bounded sets are compact), $F : E \to E$ a *total* function, $W \subset E$ open. In the applications $E = \mathbb{R}^n \times \mathbb{R}^n$ with the max norm $|(\theta, I)| = \max(|\theta|, |I|)$ (the product norm of the Lean text, equal to the max norm on the $2n$ coordinates), or $E = (\mathbb{R}^2)^N \times (\mathbb{R}^2)^N$ for the Newtonian problems.
- $F$ is $C^1$ *on* $W$ means `ContDiffOn` $\mathbb{R}$ 1 $F$ $W$; as $W$ is open this gives $F \in C^1$ at each point of $W$, and no information off $W$. Off $W$ the values of $F$ never enter (Lean totality, below).
- An *interval* is an order-connected subset of $\mathbb{R}$ (Lean `OrdConnected`); open, closed, half-open, bounded or not, possibly a single point. $\overline{B}(x, \delta)$ is the closed ball.
- All derivatives $\dot{z}(t)$ in solution predicates are two-sided (Lean `HasDerivAt`) at every $t \in J$, including endpoints of $J$, exactly as in the frozen predicates ("$z$ is defined on $\mathbb{R}$").

External facts. Mathlib-level statements, with the declarations that exist (checked in `Mathlib/Analysis/ODE/`); what is *not* in Mathlib is listed after them.

(E1) *Picard–Lindelöf, quantitative form.* Let $F : E \to E$, $x \in E$, $a, M, L \geq 0$ with $\|F(y)\| \leq M$ and $F$ $L$-Lipschitz on $\overline{B}(x, a)$, $T \geq 0$ with $MT \leq a$. Then there is $\alpha : \mathbb{R} \to E$ with $\alpha(t_0) = x$ and $\alpha$ has derivative $F(\alpha(t))$ within $[t_0 - T, t_0 + T]$ at each $t$ of that interval. (Mathlib: the root-namespace structure `IsPicardLindelof` $f\, t_0\, x_0\, a\, r\, L\, K$ of `Analysis/ODE/PicardLindelof.lean`, with $r = 0$ and a time-independent field, and `IsPicardLindelof.exists_eq_forall_mem_Icc_hasDerivWithinAt`$_0$ (Unicode subscript zero) of `Analysis/ODE/ExistUnique.lean`; Mathlib's norm bound is the parameter $L$ (our $M$) and its Lipschitz parameter is $K$ (our $L$); the time constraint is the field `mul_max_le`: $M \max(t_{\max} - t_0, t_0 - t_{\min}) \leq a - r$. Parameters are `NNReal`. The contraction on iterates means no smallness of $LT$ is needed.) Exists.

(E2) *Uniqueness with a Lipschitz constant on a set.* If $S \subset E$ and $F$ is $L$-Lipschitz on $S$, $f, g : [a, b] \to E$ are continuous, have right derivative $F(f(t))$, $F(g(t))$ on $[a, b)$, take values in $S$ and $f(a) = g(a)$, then $f = g$ on $[a, b]$; and the mirror statement with $f(b) = g(b)$ and left derivatives. (Mathlib: `ODE_solution_unique_of_mem_Icc_right`, `ODE_solution_unique_of_mem_Icc_left`.) Exists.

(E3) $C^1$ *implies locally Lipschitz.* If $F$ is $C^1$ at $p$ there are $L$ and a neighbourhood on which $F$ is $L$-Lipschitz (Mathlib: `ContDiffAt.exists_lipschitzOnWith`); and a function locally Lipschitz on a compact set is Lipschitz there (Mathlib: `LocallyLipschitzOn.exists_lipschitzOnWith_of_compact`, `Topology/Algebra/MetricSpace/Lipschitz.lean`), which packages Lemma 4.4 once local Lipschitzness on $K$ is extracted from $C^1$ on $W$. Exists.

(E4) *Mean value inequality*: [K:`bp:p1:ext-mvt`]; for a differentiable map on a convex set with $\|DF\| \leq L$, $F$ is $L$-Lipschitz there; for a function of one variable with derivative bounded by $M$ on an interval, $\|\alpha(u) - \alpha(s)\| \leq M|u - s|$. Exists.

(E5) *Topology.* Heine–Borel in $E$ (`ProperSpace`); continuous images of compact sets are compact; sequential compactness; for a *nonempty* closed set $A$, $x \mapsto \operatorname{dist}(x, A)$ is continuous and positive exactly on $E \setminus A$ (`Metric.infDist`, `IsClosed.notMem_iff_infDist_pos`, which requires `A.Nonempty`; for $A = \emptyset$ Mathlib has $\operatorname{infDist}(x, \emptyset) = 0$, `Metric.infDist_empty`, so no statement below uses infDist of an empty set: thickenings are defined by $K_\delta = \{y : \exists x \in K,\ \|y - x\| \leq \delta\}$, which is empty for $K = \emptyset$). Exists.

(E6) *Calculus.* Chain rule `HasFDerivAt.comp_hasDerivAt`; a function continuous on $[s,t]$ with right derivative 0 on $[s,t)$ is constant (`constant_of_has_deriv_right_zero`); two one-sided derivatives agree ⇒ two-sided one (`HasDerivWithinAt.union` with $(-\infty,b]\cup[b,\infty)=\mathbb{R}$ and `hasDerivWithinAt_univ`); `ContDiffOn.fderiv_of_isOpen`: if $H$ is $C^{k+1}$ on an open set then $DH$ is $C^k$ there. Exists.

(E7) *Uniform existence time near a point.* If $F$ is $C^1$ at $x_0$ and $t_0\in\mathbb{R}$ there are $r,\varepsilon>0$ such that for every $x\in\overline{B}(x_0,r)$ there is $\alpha$ with $\alpha(t_0)=x$ and derivative $F(\alpha(t))$ for all $t\in(t_0-\varepsilon,t_0+\varepsilon)$ (Mathlib: `ContDiffAt.exists_forall_mem_closedBall_exists_eq_forall_mem_Ioo_hasDerivAt`, `ExistUnique.lean:145`; note the solution is for the total $F$ and is not asserted to stay in the neighbourhood of $x_0$). Exists. It covers Theorem 4.10 for a single point after the staying-in-$W$ step; the explicit $\delta/\max(M,1)$ time of Proposition 4.9, uniform over arbitrary sets $\Phi$ with uniform data, is derived from (E1).

**Not in Mathlib** (checked in `Analysis/ODE` and `Geometry/Manifold/IntegralCurve`: no maximal-solution, continuation or first-exit statement, and no uniqueness on arbitrary intervals with only local Lipschitz constants): the following must be proved from the blueprint below: uniform data on arbitrary sets and its compact and periodic instances (Lemmas 4.6, 4.7), Lemma 4.8, Proposition 4.9 (derived from (E1); the existing statement (E7) may be reused for the single-point case), Theorems 4.11, 4.13, 4.14, Lemma 4.19 and the Hamiltonian layer.

*Remark* 4.1 (Lean totality). `KAM.hamiltonianVectorField` $H$ uses `fderiv`, which is 0 at a point of non-differentiability, and the Newtonian fields use $1/\ell^3$ with $\ell=$ `planeLength` or `radius`. In every use below these are guarded: $X_H$ occurs only on an open set where $H$ is $C^2$ (or inside the frozen predicates, which assert differentiability at every visited point), and $\ell>0$ at every visited point because the frozen predicates require it. The general layer never uses $F$ off $W$.

## 4.2. **The general solution predicate; Lipschitz data.**

**Definition 4.2** (Solutions in $W$)**.** For $F:E\to E$, $W\subset E$, $J\subset\mathbb{R}$ and $z:\mathbb{R}\to E$: `IsSolIn` $F\,W\,J\,z$ holds if for every $t\in J$, $z(t)\in W$ and $z$ has derivative $F(z(t))$ at $t$ (two-sided). Suggested Lean name `KAM.ODE.IsSolutionIn`; it is *not* a frozen declaration.
Source: Standard; shape dictated by the frozen predicates of Definition 1.16 and §4.8.

Elementary consequences (each a one-line Lean lemma, (E6)): (i) restriction: `IsSolIn` $F\,W\,J\,z$ and $J'\subset J$ give `IsSolIn` $F\,W\,J'\,z$; (ii) $z$ is continuous on $J$ (differentiable at each point of $J$); (iii) *local nature*: if $J'$ is a neighbourhood of each of its points and $z=y$ on it, then `IsSolIn` $F\,W\,J'\,y\Leftrightarrow$ `IsSolIn` $F\,W\,J'\,z$ for open $J'$.

**Lemma 4.3** (Symmetries)**.** *Let* `IsSolIn` $F\,W\,J\,z$.

*(a) (time shift) For $c\in\mathbb{R}$, $t\mapsto z(t+c)$ is a solution on $J-c$.*
*(b) (time reversal) $t\mapsto z(-t)$ is a solution of $-F$ on $-J$, and $-F$ is $C^1$ on $W$ if $F$ is.*
*(c) (space translation) If $v\in E$ satisfies $W+v=W$ and $F(x+v)=F(x)$ for $x\in W$, then $t\mapsto z(t)+v$ is a solution of $F$ in $W$ on $J$.*

*Source: Standard.*

*Proof.* **Step S4.1: Chain rule.** (a) and (b) are the chain rule for $t\mapsto t+c$, $t\mapsto -t$ ((E6)); (c): $z+v$ has the same derivative $F(z(t))=F(z(t)+v)$, and $z(t)+v\in W+v=W$. □

**Lemma 4.4** (Bounds and Lipschitz constants on compact sets)**.** *Let $F$ be $C^1$ on the open set $W$ and $K\subset W$ compact. Then $F$ is bounded on $K$ and there is $L\geq 0$ with $\|F(x)-F(y)\|\leq L\|x-y\|$ for $x,y\in K$.*
*Source: Standard.*

*Proof.* **Step S4.2: Via Mathlib.** Alternatively: $F$ is locally Lipschitz at every point of $W$ ((E3)), hence locally Lipschitz on $K$, and `LocallyLipschitzOn.exists_lipschitzOnWith_of_compact` ((E3)) gives $L$. The remaining steps are the explanatory proof of that fact.

**Step S4.3: Bounded.** $F$ is continuous on $W \supset K$, so $\|F\| \le M$ on the compact $K$ ((E5)).

**Step S4.4: Contradiction hypothesis.** If no $L$ works, for every $k \in \mathbb{N}$, $k \ge 1$, there are $x_k \ne y_k$ in $K$ with $\|F(x_k) - F(y_k)\| > k\|x_k - y_k\|$; then $\|x_k - y_k\| < 2M/k \to 0$.

**Step S4.5: Compactness.** A subsequence $x_{k_j} \to p \in K$, hence also $y_{k_j} \to p$. By (E3) there are $L_p$ and a neighbourhood $N_p$ of $p$ on which $F$ is $L_p$-Lipschitz. For large $j$ both points lie in $N_p$ and $k_j > L_p$, which contradicts Step S4.4. □

**Definition 4.5** (Uniform data)**.** Let $F$ be $C^1$ on the open set $W$. A set $\Phi \subset E$ has *uniform data* $(\delta, M, L)$, with $\delta > 0$, $M, L \ge 0$, if for every $x \in \Phi$: $\overline{B}(x, \delta) \subset W$, $\|F\| \le M$ on $\overline{B}(x, \delta)$ and $F$ is $L$-Lipschitz on $\overline{B}(x, \delta)$. (Balls are convex, so by (E4) a bound $\|DF\| \le L$ on $\overline{B}(x, \delta) \subset W$ gives the Lipschitz clause.)

**Lemma 4.6** (Compact sets have uniform data)**.** *Let $F$ be $C^1$ on the open set $W$ and $K \subset W$ compact. Then $K$ has uniform data $(\delta, M, L)$ for some $\delta > 0$, and $K_\delta = \{y : \exists x \in K,\ \|y - x\| \le \delta\} \subset W$ is compact.*

*Proof.* **Step S4.6: Choice of $\delta$.** If $K = \emptyset$ put $\delta = 1$: then $K_\delta = \emptyset$ and all assertions are vacuous (no infDist of the empty set is used). If $W = E$ put $\delta = 1$. Otherwise $K \ne \emptyset$ and $A = E \setminus W$ is closed and nonempty, so $x \mapsto \operatorname{dist}(x, A)$ is continuous and positive on $K$ ((E5)); it has a minimum $2\delta > 0$ on the compact nonempty $K$, and then $K_\delta \subset W$. $K_\delta$ is the image of the compact $K \times \overline{B}(0, \delta)$ under $(x, w) \mapsto x + w$, hence compact.

**Step S4.7: Data.** Apply Lemma 4.4 to $K_\delta$: $\|F\| \le M$ and $F$ is $L$-Lipschitz on $K_\delta \supset \overline{B}(x, \delta)$ for $x \in K$. □

**Lemma 4.7** (Uniform data for angle-periodic fields)**.** *Let $E = \mathbb{R}^n \times \mathbb{R}^n$ (max norm), $U \subset \mathbb{R}^n$ open, $W = \mathbb{R}^n \times U$, $F$ $C^1$ on $W$ with $F(\theta + k, I) = F(\theta, I)$ for $(\theta, I) \in W$, $k \in \mathbb{Z}^n$. If $K \subset U$ is compact, then $\Phi = \mathbb{R}^n \times K$ has uniform data $(\delta, M, L)$ for some $\delta \in (0, 1]$.*

*Proof.* **Step S4.8: A compact cell.** If $K = \emptyset$ then $\Phi = \emptyset$ and any $\delta, M, L$ work. Otherwise choose $\delta \in (0, 1]$ with $K_\delta = \{I : \exists J \in K,\ |I - J| \le \delta\} \subset U$ (as in Step S4.6, using the nonempty closed set $\mathbb{R}^n \setminus U$, or $\delta = 1$ if $U = \mathbb{R}^n$). $C = [-1, 2]^n \times K_\delta$ is compact and contained in $W$; let $M, L_C$ be the constants of Lemma 4.4 for $C$.

**Step S4.9: Translating a ball into the cell.** Let $x = (\theta, I) \in \Phi$, $k = (\lfloor \theta_i \rfloor)_i \in \mathbb{Z}^n$ and $x' = x - (k, 0) \in [0, 1)^n \times K$. Since the norm is the max norm and $\delta \le 1$, $\overline{B}(x', \delta) \subset C$, and $\overline{B}(x, \delta) = \overline{B}(x', \delta) + (k, 0) \subset W$. By periodicity $F(y + (k, 0)) = F(y)$ on $\overline{B}(x', \delta)$, so $\|F\| \le M$ and $F$ is $L_C$-Lipschitz on $\overline{B}(x, \delta)$. □

## 4.3. **Local existence.**

**Lemma 4.8** (A priori short-time confinement)**.** *Let $J$ be an interval, $t_0 \in J$, $\alpha : \mathbb{R} \to E$ with $\alpha(t_0) = x$ and $\alpha'(t) = F(\alpha(t))$ for all $t \in J$ (derivative two-sided; $F$ is only assumed to be a function), and suppose $\|F\| \le M$ on $\overline{B}(x, \delta)$ with $M > 0$. Then $\|\alpha(t) - x\| \le M|t - t_0|$ for all $t \in J$ with $|t - t_0| < \delta/M$; in particular $\alpha(t) \in \overline{B}(x, \delta)$.*
*Source: Standard.*

*Proof.* **Step S4.10: Forward bootstrap.** Let $t \in J$, $t_0 < t < t_0 + \delta/M$ (the case $t < t_0$ follows by Lemma 4.3(b) and the case $t = t_0$ is trivial). Let $A = \{s \in [t_0, t] : \|\alpha(u) - x\| \le M(u - t_0)\ \forall u \in [t_0, s]\}$ and $s^* = \sup A$. $t_0 \in A$ and $A$ is closed by continuity of $\alpha$ on $J$, so $s^* \in A$.

**Step S4.11: Extension.** If $s^* < t$, then $\|\alpha(s^*) - x\| \le M(s^* - t_0) < \delta$, so by continuity there is $\eta > 0$, $s^* + \eta \le t$, with $\alpha([s^*, s^* + \eta]) \subset \overline{B}(x, \delta)$. On this interval $\|\alpha'\| = \|F(\alpha)\| \le M$, so by (E4) $\|\alpha(u) - \alpha(s^*)\| \le M(u - s^*)$ and hence $\|\alpha(u) - x\| \le M(u - t_0)$ there, contradicting the definition of $s^*$. So $s^* = t$. □

**Proposition 4.9** (Local existence with uniform time)**.** *Let $F$ be $C^1$ on the open set $W$ and $\Phi$ have uniform data $(\delta, M, L)$; put $M' = \max(M, 1)$ and $\varepsilon = \delta/M'$. Then for every $x \in \Phi$ and every $t_0 \in \mathbb{R}$ there is $\alpha : \mathbb{R} \to E$ with $\alpha(t_0) = x$, $\|\alpha(t) - x\| < \delta$ for all $t \in (t_0 - \varepsilon, t_0 + \varepsilon)$ and $\mathtt{IsSolIn}\, F\, W\, (t_0 - \varepsilon, t_0 + \varepsilon)\, \alpha$.*
*Source: Standard (Picard–Lindelöf).*

*Proof.* **Step S4.12: Picard–Lindelöf.** Apply (E1) with $a = \delta$, bound $M'$, Lipschitz constant $L$ and $T = \varepsilon$ ($M'T = \delta \le a$): there is $\alpha$ with $\alpha(t_0) = x$ and derivative $F(\alpha(t))$ within $[t_0 - \varepsilon, t_0 + \varepsilon]$ at each point of it. At points $t$ of the open interval this is a two-sided derivative, since the closed interval is a neighbourhood of $t$.

**Step S4.13: Staying in $W$.** By Lemma 4.8 on $J = (t_0 - \varepsilon, t_0 + \varepsilon)$, $\|\alpha(t) - x\| \le M'|t - t_0| < \delta$, so $\alpha(t) \in \overline{B}(x, \delta) \subset W$. □

**Theorem 4.10** (Local existence, both time directions)**.** *Let $F$ be $C^1$ on the open set $W$, $x_0 \in W$, $t_0 \in \mathbb{R}$. There are $\varepsilon > 0$ and $\alpha : \mathbb{R} \to E$ with $\alpha(t_0) = x_0$ and $\mathtt{IsSolIn}\, F\, W\, (t_0 - \varepsilon, t_0 + \varepsilon)\, \alpha$. Moreover $\varepsilon$ may be taken uniform on any compact $K \subset W$ (for all $x_0 \in K$).*
*Source: Standard.*

*Proof.* **Step S4.14: Reduction.** $\{x_0\}$ (resp. $K$) is compact; Lemma 4.6 gives uniform data and Proposition 4.9 the claim. □

### 4.4. Uniqueness on every interval.

**Theorem 4.11** (Uniqueness on arbitrary intervals)**.** *Let $F$ be $C^1$ on the open set $W$, $J \subset \mathbb{R}$ an interval (of any kind: open, closed, half-open, unbounded, or a point), $t_0 \in J$, and $y_1, y_2 : \mathbb{R} \to E$ with $\mathtt{IsSolIn}\, F\, W\, J\, y_i$ and $y_1(t_0) = y_2(t_0)$. Then $y_1 = y_2$ on $J$. No confinement, boundedness or smallness of either solution is assumed. In particular, if $y_1, y_2$ are solutions on intervals $J_1, J_2$ and agree at one point of $J_1 \cap J_2$, they agree on $J_1 \cap J_2$ (an interval), and two solutions which agree at one time of $J$ agree at all times of $J$.*
*Source: Standard; used for BM11 Theorem 2.1 (uniqueness) and in the maximal-solution argument of BM11 p. 13.*

*Proof.* **Step S4.15: A compact piece.** Let $t_1 \in J$, $t_1 > t_0$ ($t_1 < t_0$ is the mirror image, using the left-derivative form of (E2); $t_1 = t_0$ is the hypothesis). Then $[t_0, t_1] \subset J$, each $y_i$ is continuous on it, so $K = y_1([t_0, t_1]) \cup y_2([t_0, t_1])$ is compact ((E5)) and $K \subset W$.

**Step S4.16: Lipschitz on $K$.** Lemma 4.4 gives $L$ with $F$ $L$-Lipschitz on $K$.

**Step S4.17: Apply (E2).** Both $y_i$ are continuous on $[t_0, t_1]$, have right derivative $F(y_i(t))$ on $[t_0, t_1)$ (a two-sided derivative gives a right derivative) and take values in $K$; so $y_1(t_1) = y_2(t_1)$. □

### 4.5. Maximal solutions and continuation.

**Lemma 4.12** (Extension to an open interval)**.** *Let $F$ be $C^1$ on the open set $W$, $J \neq \emptyset$ an interval and $\mathtt{IsSolIn}\, F\, W\, J\, y$. Then there are an open interval $J' \supset J$ and $y'$ with $y' = y$ on $J$ and $\mathtt{IsSolIn}\, F\, W\, J'\, y'$.*

*Proof.* **Step S4.18: Right end.** If $J$ has a largest element $b$, apply Theorem 4.10 at $x_0 = y(b) \in W$, $t_0 = b$: $\alpha$ on $(b-\varepsilon, b+\varepsilon)$. Put $y_1(s) = y(s)$ for $s \le b$ and $y_1(s) = \alpha(s)$ for $s > b$. Then $y_1$ is a solution on $J_1 = J \cup (b, b+\varepsilon)$: at $t \in J$, $t < b$, and at $t \in (b, b+\varepsilon)$ it agrees with $y$, resp. $\alpha$, near $t$; at $t = b$ it has derivative $F(y(b))$ within $(-\infty, b]$ (it equals $y$ there) and within $[b, \infty)$ (it equals $\alpha$ there, as $\alpha(b) = y(b)$), so by (E6) it has the two-sided derivative. If $J$ has no largest element, $J_1 = J$, $y_1 = y$.

**Step S4.19: Left end.** Apply Step S4.18 to the reversed solution $y_1(-\cdot)$ (Lemma 4.3(b)). The result has no endpoint, so is an open interval. □

**Theorem 4.13** (Maximal solution)**.** *Let $F$ be $C^1$ on the open set $W$ and $x_0 \in W$. There are $T_- \in [-\infty, 0)$, $T_+ \in (0, +\infty]$ and $z : \mathbb{R} \to E$ with $z(0) = x_0$ and* `IsSolIn` *$F\, W\, (T_-, T_+)\, z$, such that every solution $y$ with $y(0) = x_0$ on an interval $J \ni 0$ satisfies $J \subset (T_-, T_+)$ and $y = z$ on $J$. Then $(T_-, T_+)$ is the* maximal interval *and $z|_{(T_-,T_+)}$ the* maximal solution*; both are unique. For initial time $t_0$ instead of $0$ use Lemma 4.3(a).*
*Source: BM11 p. 13 ("maximal time of existence"); standard.*

*Proof.* **Step S4.20: The union.** Let $\mathcal{J}$ be the set of open intervals $J \ni 0$ carrying a solution $y_J$ with $y_J(0) = x_0$; $\mathcal{J} \neq \emptyset$ by Theorem 4.10. The union $I$ of $\mathcal{J}$ is an open interval $(T_-, T_+) \ni 0$ (a union of intervals containing 0).

**Step S4.21: Gluing.** For $t \in I$ choose $J \in \mathcal{J}$ with $t \in J$ and set $z(t) = y_J(t)$ (and $z(t) = x_0$ outside $I$). This is independent of $J$: for $J, J' \in \mathcal{J}$, $y_J = y_{J'}$ on $J \cap J' \ni 0$ by Theorem 4.11. Each $t \in I$ has an open neighbourhood $J \in \mathcal{J}$ on which $z = y_J$, so $z$ is a solution on $I$ (local nature, (iii) above).

**Step S4.22: Maximality.** If $y$ is a solution on an interval $J \ni 0$ with $y(0) = x_0$, Lemma 4.12 gives an open $J' \supset J$ and a solution $y'$ extending $y$, so $J' \in \mathcal{J}$, $J \subset J' \subset I$, and $y = y' = z$ on $J$ (Theorem 4.11). Uniqueness of the interval $(T_-, T_+)$, and of $z$ as a function on it (equality on this interval of any two representatives; values outside it are arbitrary), is immediate from the maximality property applied twice. □

**Theorem 4.14** (Continuation criterion, uniform form)**.** *Let $F$ be $C^1$ on the open set $W$, $x_0 \in W$, $(T_-, T_+)$ and $z$ the maximal interval and solution of Theorem 4.13. Let $\Phi \subset E$ have uniform data $(\delta, M, L)$ (Definition 4.5). If $T_+ < \infty$ there is no $t_* \in [0, T_+)$ with $z(t) \in \Phi$ for all $t \in [t_*, T_+)$. Symmetrically, if $T_- > -\infty$ there is no $t_* \in (T_-, 0]$ with $z(t) \in \Phi$ for all $t \in (T_-, t_*]$. Equivalently: if the solution stays in $\Phi$ near the right (left) end of its maximal interval, that end is $+\infty$ ($-\infty$).*
*Source: Standard; BM11 p. 13 ("this solution is defined for all time"), Pöschel 1993 p. 198 ("$T = T_*$").*

*Proof.* **Step S4.23: Set-up.** Suppose $T_+ < \infty$ and $z(t) \in \Phi$ on $[t_*, T_+)$. Let $\varepsilon = \delta / \max(M, 1)$ from Proposition 4.9. Choose $t_1 \in [t_*, T_+)$ with $t_1 > T_+ - \varepsilon$.

**Step S4.24: A solution through $z(t_1)$.** Proposition 4.9 with $x = z(t_1) \in \Phi$ and $t_0 = t_1$ gives $\alpha$ on $(t_1 - \varepsilon, t_1 + \varepsilon)$ with $\alpha(t_1) = z(t_1)$. By Theorem 4.11 on $(t_1 - \varepsilon, t_1 + \varepsilon) \cap (T_-, T_+)$, $\alpha = z$ there.

**Step S4.25: Gluing past $T_+$.** Put $y = z$ on $(T_-, T_+)$ and $y = \alpha$ on $[T_+, t_1 + \varepsilon)$; then $y = \alpha$ on $(t_1 - \varepsilon, t_1 + \varepsilon)$ and $y = z$ on $(T_-, T_+)$, two open sets covering $(T_-, t_1 + \varepsilon)$ on each of which $y$ is a solution. So $y$ is a solution on $(T_-, t_1 + \varepsilon) \ni 0$ with $y(0) = x_0$, and $t_1 + \varepsilon > T_+$ contradicts Theorem 4.13. The left end follows by Lemma 4.3(b). □

**Corollary 4.15** (Compact continuation)**.** *In the setting of Theorem 4.14, if $K \subset W$ is compact and $z(t) \in K$ for all $t \in [t_*, T_+)$ (some $t_* \in [0, T_+)$), then $T_+ = +\infty$; likewise on the left.*

*Proof.* Lemma 4.6 gives uniform data for $K$; apply Theorem 4.14. □

**Corollary 4.16** (Closed sets, periodic fields)**.** *Let $E = \mathbb{R}^n \times \mathbb{R}^n$, $U \subset \mathbb{R}^n$ open, $W = \mathbb{R}^n \times U$, $F$ $C^1$ on $W$ with $F(\theta + k, I) = F(\theta, I)$ for $k \in \mathbb{Z}^n$. If the maximal solution through $x_0 \in W$ satisfies $z(t)_2 \in K$ for $t \in [t_*, T_+)$ with $K \subset U$ compact, then $T_+ = +\infty$ (likewise on the left). The angle components $z(t)_1$ may be unbounded.*

*Proof.* Lemma 4.7 gives uniform data for $\Phi = \mathbb{R}^n \times K$; apply Theorem 4.14. □

*Remark* 4.17 (What the closed-set variant really needs). Staying in a closed non-compact set $\Phi \subset W$ does *not* suffice in general (the solution of $\dot{x} = x^2$ on $E = \mathbb{R}$, $W = \mathbb{R}$, $x_0 = 1$, $x(t) = 1/(1-t)$, stays in the closed set $[1, \infty)$ for $0 \le t < 1$ and blows up at $T_+ = 1$). What suffices is the uniform data $(\delta, M, L)$ of Definition 4.5: a $\delta$-neighbourhood of $\Phi$ inside $W$ on which $F$ is bounded and Lipschitz, uniformly. This holds for compact sets, for angle-periodic fields over a compact action set (Lemma 4.7), and for the Newtonian regions bounded away from collisions with bounded velocities (Corollary 4.27).

**Theorem 4.18** (Confinement glue)**.** *Let $F$ be $C^1$ on the open set $W$, $x_0 \in W$, $\tau > 0$, and $\Phi \subset E$ a set with uniform data (e.g. compact $\Phi \subset W$, or as in Lemma 4.7). Assume the* a priori confinement*: every solution $y$ on an open interval $J \ni 0$ with $y(0) = x_0$ satisfies $y(t) \in \Phi$ for all $t \in J$ with $|t| \le \tau$. Then the maximal interval contains $[-\tau, \tau]$ in its interior, so $z$ is a solution on $[-\tau, \tau]$ with $z(t) \in \Phi$ for $|t| \le \tau$, and every solution on $[-L, L]$ $(0 \le L \le \tau)$ with initial value $x_0$ equals $z$ there.*
*Source: BM11 p. 13 (the two-case argument "Therefore this solution is defined for all time"); Pöschel 1993 p. 198.*

*Proof.* **Step S4.26: No exit before $\tau$.** Suppose $T_+ \le \tau$. The maximal solution $z$ on $(T_-, T_+)$ satisfies, for $t \in [0, T_+)$ $(|t| < \tau)$, $z(t) \in \Phi$ by confinement. Theorem 4.14 gives $T_+ = \infty$, a contradiction. So $T_+ > \tau$, and likewise $T_- > -\tau$.

**Step S4.27: Conclusion.** $[-\tau, \tau] \subset (T_-, T_+)$ so $z$ is a solution on $[-\tau, \tau]$ and $z(t) \in \Phi$ for $|t| \le \tau$ by confinement applied to $z$ itself. Uniqueness is Theorem 4.11. □

4.6. **First exit from an open set; barriers.**

**Lemma 4.19** (First exit)**.** *Let* `IsSolIn` $F\, W\, J\, z$ *with $J$ an interval, $0 \in J$, $V \subset E$ open and $z(0) \in V$. Put $A = \{t \in J : t \ge 0,\ z(t) \notin V\}$. Then exactly one of the following holds.*

*(i) $A = \emptyset$: $z(t) \in V$ for all $t \in J$, $t \ge 0$.*

*(ii) $A \neq \emptyset$. Then $t^* = \min A$ exists, $t^* > 0$, $t^* \in J$, $z(t) \in V$ for $t \in [0, t^*)$ and $z(t^*) \in \partial V = \overline{V} \setminus V$. The number $t^*$ is the* first exit time *and equals $\sup\{t \in J : t \ge 0,\ z([0,t]) \subset V\}$; in particular $t^*$ is finite and lies in $J$ ($t^*$ is "inside the solution's interval" and the solution is defined, continuous, at $t^*$).*

*The same holds backwards in time (replace $z(t)$ by $z(-t)$, Lemma 4.3(b)). Only continuity of $z$ on $J$ is used; $F$ and $W$ play no role.*
*Source: Standard; used for BM11 p. 13 ("smallest time $t_*$ such that $\omega(t_*) \in R_K$") and Pöschel 1993 p. 198–199 (stopping-time arguments).*

*Proof.* **Step S4.28: Closedness.** $z$ is continuous on $J$ (consequence (ii)), $E \setminus V$ is closed, so $A$ is closed in $J \cap [0, \infty)$.

**Step S4.29: The minimum.** If $A \neq \emptyset$, $t^* = \inf A \in [0, \infty)$; $t^* \in J$ since $J$ is an interval containing 0 and a point of $A$ that is $\ge t^*$. By closedness $t^* \in A$, i.e. $z(t^*) \notin V$. As $z(0) \in V$, $t^* > 0$.

**Step S4.30: Boundary point.** For $t \in [0, t^*)$, $t \notin A$, so $z(t) \in V$. By continuity $z(t^*) = \lim_{t \uparrow t^*} z(t) \in \overline{V}$, and $z(t^*) \notin V$; as $V$ is open, $\overline{V} \setminus V = \partial V$.

**Step S4.31: The supremum formula.** $z([0,t]) \subset V$ iff $t < t^*$ (for $t \in J$, $t \ge 0$), so the supremum is $t^*$. □

**Definition 4.20** (Invariant sets)**.** $S \subset E$ is *invariant* for $(F, W)$ if for every interval $J'$ and every $y$ with `IsSolIn` $F\,W\,J'\,y$: if $y(t_1) \in S$ for one $t_1 \in J'$ then $y(t) \in S$ for all $t \in J'$. (Equilibria $\{x : F(x) = 0\}$ are invariant by Theorem 4.11; so is any union of such sets and any set $\{G = c\}$ for a first integral $G$.)

**Corollary 4.21** (Barrier)**.** *Let $F$ be $C^1$ on the open set $W$, $V$ open, `IsSolIn` $F\,W\,J\,z$ with $J$ an interval, $0 \in J$, $z(0) \in V$. Let $G \subset E$ with $z(J) \subset G$, and suppose $\partial V \cap G \subset S$ where $S$ is invariant and $z(0) \notin S$ (e.g. $G$ an energy level, Theorem 4.23, and $\partial V \cap G$ a union of invariant sets not meeting the initial point). Then $z(t) \in V$ for all $t \in J$ (both time directions).*

*Proof.* **Step S4.32: Forward.** If not, Lemma 4.19(ii) gives $t^* \in J$ with $z(t^*) \in \partial V \cap G \subset S$; invariance gives $z(0) \in S$, a contradiction.

**Step S4.33: Backward.** Apply Step S4.32 to $t \mapsto z(-t)$, a solution of $-F$ on $-J$; $G$, $\partial V$ are unchanged and $S$ is invariant for $-F$ as well (reversal maps solutions of $F$ to solutions of $-F$ and back). □

4.7. **Hamiltonian layer.** Phase space $P = \mathbb{R}^n \times \mathbb{R}^n$, points $z = (\theta, I)$, max norm. $X_H$ is the frozen `KAM.hamiltonianVectorField`, Definition 1.3 (convention of [K:`bp:p0:hamiltonian`]): $X_H(z) = \big((\partial_{I_i} H(z))_i, (-\partial_{\theta_i} H(z))_i\big)$, with $\partial_{I_i} H(z) = DH(z)(0, e_i)$ and $\partial_{\theta_i} H(z) = DH(z)(e_i, 0)$.

**Lemma 4.22** (Regularity of $X_H$)**.** *Let $W \subset P$ be open and $H : P \to \mathbb{R}$ be $C^{k+1}$ on $W$, $k \geq 0$. Then $X_H$ is $C^k$ on $W$; for $k \geq 1$ it is $C^1$ (hence locally Lipschitz) on $W$, and $H$ is differentiable at every point of $W$.*
*Source: Standard.*

*Proof.* **Step S4.34: Derivative is $C^k$.** $DH : W \to \mathcal{L}(P, \mathbb{R})$ is $C^k$ on the open $W$ ((E6), `ContDiffOn.fderiv_of_isOpen`).

**Step S4.35: Components.** Each coordinate of $X_H$ is $\pm$ the composition of $DH$ with the continuous linear evaluation $\ell \mapsto \ell(0, e_i)$ or $\ell \mapsto \ell(e_i, 0)$; hence $C^k$, and a map into $\mathbb{R}^n \times \mathbb{R}^n$ with $C^k$ coordinates is $C^k$. □

**Theorem 4.23** (Energy conservation)**.** *Let $H : P \to \mathbb{R}$, $J \subset \mathbb{R}$ an interval and $z : \mathbb{R} \to P$ such that for every $t \in J$, $H$ is differentiable at $z(t)$ and $z$ has derivative $X_H(z(t))$ at $t$. Then $t \mapsto H(z(t))$ is constant on $J$. (No regularity of $H$ beyond the stated pointwise differentiability, and no confinement.)*
*Source: BM11 p. 13 and Pöschel 1993 p. 198 ("energy conservation", "$\Delta H = H(I(t), \theta(t))|_0^T$ vanishes"); standard.*

*Proof.* **Step S4.36: Chain rule.** For $t \in J$, $\frac{d}{dt} H(z(t)) = DH(z(t))\big[X_H(z(t))\big]$ ((E6)).

**Step S4.37: Cancellation.** Write $z = z(t)$, $\ell = DH(z)$ and $X_H(z) = (a, b)$ with $a_i = \ell(0, e_i)$, $b_i = -\ell(e_i, 0)$. By linearity, $\ell(a, b) = \sum_i a_i\, \ell(e_i, 0) + \sum_i b_i\, \ell(0, e_i) = \sum_i \ell(0, e_i)\ell(e_i, 0) - \sum_i \ell(e_i, 0)\ell(0, e_i) = 0$.

**Step S4.38: Constancy.** Let $s < t$ in $J$. $H \circ z$ is continuous on $[s, t] \subset J$ and has derivative 0 at every point, so $H(z(t)) = H(z(s))$ ((E6)). □

**Lemma 4.24** (Periodicity in the angles passes to solutions)**.** *Let $W \subset P$ be open with $W + (k, 0) = W$ for all $k \in \mathbb{Z}^n$, and $H : P \to \mathbb{R}$ with $H(\theta + k, I) = H(\theta, I)$ for $(\theta, I) \in W$, $k \in \mathbb{Z}^n$. Then $X_H(\theta + k, I) = X_H(\theta, I)$ on $W$. Consequently, if `IsSolIn` $X_H\,W\,J\,z$ then `IsSolIn` $X_H\,W\,J\,(z + (k, 0))$ for every $k \in \mathbb{Z}^n$, and $H \circ (z + (k, 0)) = H \circ z$.*
*Source: Standard.*

*Proof.* **Step S4.39: Field.** Let $\tau_k(x) = x + (k, 0)$. On the open set $W$, $H \circ \tau_k = H$, so $DH = D(H \circ \tau_k) = DH \circ \tau_k$ at each point of $W$ (the derivative of a translate; both sides are 0 if $H$ is not differentiable, by the definition of `fderiv`). Evaluating on $(0, e_i)$ and $(e_i, 0)$ gives $X_H \circ \tau_k = X_H$ on $W$.

**Step S4.40: Solutions.** Lemma 4.3(c) with $v = (k, 0)$ applies to $F = X_H$. □

4.8. **The frozen solution predicates are instances.**

**Proposition 4.25** (Track N and track I Hamiltonian predicates)**.** *Let $H : P \to \mathbb{R}$, $U \subset \mathbb{R}^n$ open, $W \subset P$ open with $\mathbb{R}^n \times U \subset W$, and $H$ be $C^2$ on $W$. Put $W_U = \mathbb{R}^n \times U$. For every $J \subset \mathbb{R}$ and $z : \mathbb{R} \to P$,*

$$\texttt{IsSolutionOn}\, H\, U\, J\, z \iff \texttt{IsSolIn}\, X_H\, W_U\, J\, z \iff \texttt{IsHamiltonianSolutionOn}\, H\, U\, J\, z,$$

*and $X_H$ is $C^1$ on $W_U$. (The two frozen Hamiltonian predicates have the same formula up to the letter naming the action set, and under the regularity hypotheses both are equivalent to `IsSolIn` as shown below, `KAM.Nekhoroshev.IsSolutionOn` and `KAM.Iso.IsHamiltonianSolutionOn`.)*
*Frozen Lean declaration: `KAM.Nekhoroshev.IsSolutionOn`, `KAM.Iso.IsHamiltonianSolutionOn`*
*Source: Ours (bookkeeping).*

*Proof.* **Step S4.41: Membership.** $z(t)_2 \in U \iff z(t) \in W_U$ by definition of $W_U$.

**Step S4.42: Differentiability is free.** For $z(t) \in W_U \subset W$, $H$ is $C^2$ on the open $W$, hence differentiable at $z(t)$ (Lemma 4.22); so the clause `DifferentiableAt` $\mathbb{R}\, H\, (z\, t)$ is implied by $z(t) \in W_U$, and the two predicates coincide. Smoothness of $X_H$: Lemma 4.22, $k = 1$, restricted to $W_U \subset W$. □

**Proposition 4.26** (Newtonian predicates)**.** *Let $N \geq 2$, $G \in \mathbb{R}$, `mass` : $Fin\, N \to \mathbb{R}$. Let $E_N = (Fin\, N \to \mathbb{R}^2) \times (Fin\, N \to \mathbb{R}^2)$ (max norm), $\ell =$ `planeLength`,*

$$W_N = \{(q, v) : \ell(q_j - q_i) > 0 \ \forall i \neq j\}, \qquad F_N(q, v) = \Big(v,\ \big(\textstyle\sum_{j \neq i} \frac{G\, \mathrm{mass}_j}{\ell(q_j - q_i)^3}(q_j - q_i)\big)_i\Big).$$

*Then $W_N$ is open, $F_N$ is $C^\infty$ on $W_N$, and for $Q, V : \mathbb{R} \to (Fin\, N \to \mathbb{R}^2)$ with $x(t) = (Q(t), V(t))$: `IsNewtonSolution` $G\, \mathrm{mass}\, Q\, V \iff$ `IsSolIn` $F_N\, W_N\, \mathbb{R}\, x$. Likewise for `IsFixedSolution` $G\, m_0\, m\, \varepsilon$ with $W_N' = W_N \cap \{\ell(q_i) > 0 \ \forall i\}$ and the field $F'(q, v) = (v, (-\frac{Gm_0}{\ell(q_i)^3} q_i + \varepsilon \sum_{j \neq i} \frac{Gm_j}{\ell(q_j - q_i)^3}(q_j - q_i))_i)$, and for `IsRestrictedSolutionOn` $\mu\, J$ with $W_\mu = \{(q, p) :$ `CollisionFree` $\mu\, (q, p)\}$ and the total extension $F_\mu$ of `restrictedVectorField` $\mu$ by the same formula (with the values off $W_\mu$ irrelevant): `IsRestrictedSolutionOn` $\mu\, J\, z \iff$ `IsSolIn` $F_\mu\, W_\mu\, J\, z$. In all three cases $F$ is $C^\infty$ on the open set $W$ and the general layer applies with $J = \mathbb{R}$ (resp. the given $J$).*
*Frozen Lean declaration: `KAM.Planetary.IsNewtonSolution`, `KAM.Planetary.IsFixedSolution`, `KAM.Iso.IsRestrictedSolutionOn`*
*Source: Ours (bookkeeping); Fejoz Section 6.1, Newton equations as in the frozen text.*

*Proof.* **Step S4.43: Open domain and smoothness.** $\ell(w)^2 =$ `planeSq`$(w) = w_0^2 + w_1^2$ is a polynomial, so $\{\ell > 0\} = \{$`planeSq` $> 0\}$ is open, and $\ell = \sqrt{\cdot}$ is $C^\infty$ where `planeSq` $> 0$. Hence $\ell(q_j - q_i)^{-3}$ is $C^\infty$ on $W_N$ and $F_N$ is a finite sum of products of $C^\infty$ maps. Same for $F'$ and $F_\mu$ (`radius` $= \sqrt{\texttt{radiusSq}}$ with the shifts by $\texttt{primary}_{1,2}\mu$).

**Step S4.44: Derivatives of pairs.** A map into a finite product has derivative $(a, b)$ at $t$ iff each component has the corresponding derivative (`hasDerivAt_pi`, `HasDerivAt.prodMk` and converse). The first block of the frozen definition (distance positivity) is membership in $W_N$; the second block says $Q$ has derivative $V(t)$; the third says $V$ has derivative the force. Together they say exactly that $x$ has derivative $F_N(x(t))$ at $t$. □

**Corollary 4.27** (Newtonian continuation)**.** *In the setting of Proposition 4.26, let $x$ be the maximal solution through $x_0 \in W_N$ (Theorem 4.13), $\rho > 0$, $\bar{V} \geq 0$ and*

$$\Phi_{\rho,\bar{V}} = \{(q,v) : \ \ell(q_j - q_i) \geq \rho \ (i \neq j), \ |v| \leq \bar{V}\}.$$

*If $T_+ < \infty$ then for every $\rho, \bar{V}$ and every $t_* \in [0, T_+)$ there is $t \in [t_*, T_+)$ with $x(t) \notin \Phi_{\rho,\bar{V}}$ (a finite-time singularity needs near-collisions or unbounded velocities; positions may be unbounded). The same for $F'$ with the additional condition $\ell(q_i) \geq \rho$, and for $F_\mu$ with $\ell \geq \rho$ from both primaries and $|q|, |p| \leq \bar{V}$.*

*Proof.* **Step S4.45: Uniform data.** Let $\delta = \rho/8$. If $|y - x| \leq \delta$ (max norm) with $x \in \Phi_{\rho,\bar{V}}$ then each $\ell(q'_j - q'_i) \geq \rho - 2\sqrt{2}\,\delta \geq \rho/2$, so $y \in W_N$ and $\overline{B}(x,\delta) \subset W_N$. On this ball $\ell \geq \rho/2$ and $|v'| \leq \bar{V} + \delta$; the entries of $F_N$ and $DF_N$ are polynomials in $v'$, the differences $q'_j - q'_i$ (Euclidean length $\leq \ell(q'_j - q'_i)$) and the inverse powers $\ell^{-3}, \ell^{-5}$ ($D(w/\ell^3) = \ell^{-3}I - 3\ell^{-5}ww^\top$), hence bounded by a constant $C(N, G, \mathrm{mass}, \rho, \bar{V})$ *independent of the position of $x$*. By (E4) on the convex ball, $F_N$ is $C$-Lipschitz there. So $\Phi_{\rho,\bar{V}}$ has uniform data.

**Step S4.46: Apply.** Theorem 4.14 gives the claim; the other two fields are identical in structure (for $F_\mu$ the extra terms are linear in $(q,p)$ and bounded on $|q|, |p| \leq \bar{V}$). □

4.9. **The track-N statements.** Throughout, $R, s > 0$, $h : \mathbb{C}^n \to \mathbb{C}$ with `IsAnalyticIntegrable` $R\,s\,h$, $\varepsilon \geq 0$ and $f : \mathbb{C}^n \times \mathbb{C}^n \to \mathbb{C}$ with `IsAnalyticPerturbation` $R\,s\,\varepsilon\,f$ (Definitions 1.11, 1.12), $H = H_{h,f}$ (Definition 1.4), $W_s = \mathbb{R}^n \times B(0, R+s) \subset P$ (open; $B(0,R+s) = \mathcal{B}_s(R)$ by Remark 1.9), and $U \subset B(0,R+s)$ open (in N-HEAD-C, $U = B(0,R)$).

**Lemma 4.28** (Input from Part 1: $H$ is smooth on $W_s$)**.** *$H_{h,f}$ is $C^\infty$ (in particular $C^2$) on the open set $W_s$, $X_H = X_{h+f}$ on $W_s \cap (\mathbb{R}^n \times \mathbb{R}^n)$, and $H(\theta + k, I) = H(\theta, I)$ for $(\theta, I) \in W_s$, $k \in \mathbb{Z}^n$.*
*Source: Part 1, Lemma 2.3(b)–(d) (real smoothness on $W_s$, field $X_H = X_{h+f} \circ \iota$, periodicity); ingredients* [K:`bp:p1:holomorphic-regularity`], [K:`bp:p1:hamiltonian-real`].

*Proof.* **Step S4.47: Citation.** (b), (c), (d) of Lemma 2.3 are exactly the three claims. For orientation, the argument is as follows.

**Step S4.48: Regularity.** The function $G(\theta, I) = h(I) + f(\theta, I)$ is holomorphic on the open set $D_s(R) \supset W_s$ (real points $(\theta, I)$ with $I \in \mathcal{B}_s(R) = B(0,R)_s \cap \mathbb{R}^n$ lie in $T_s \times B(0,R)_s$), hence $C^\infty$ in the real sense ([K:`bp:p1:holomorphic-regularity`]); it is real on $W_s$ by Definition 1.11(iv) and 1.12(iii). So $H = \mathrm{Re}\, G = G$ on $W_s$ is the restriction of $C^\infty$ map to a real subspace: $C^\infty$ on $W_s$; $X_H = X_G$ at real points is [K:`bp:p1:hamiltonian-real`]. (Packaged as Lemma 2.3.)

**Step S4.49: Periodicity.** $f(\theta + k, I) = f(\theta, I)$ for real $\theta$, $I \in \mathcal{B}_s(R)$, and $h$ depends on $I$ only. □

**Theorem 4.29** ($X_H$ is $C^1$, hence locally Lipschitz)**.** *$X_{H_{h,f}}$ is $C^1$ on $W_s \supset \mathbb{R}^n \times B(0,R)$; consequently it is Lipschitz on every compact subset of $W_s$, and on every set $\mathbb{R}^n \times K$ with $K \subset B(0, R+s)$ compact there are $\delta, M, L$ as in Lemma 4.7.*
*Source: BM11 p. 13 (implicit in "the solution"); standard.*

*Proof.* **Step S4.50: Smoothness.** Lemma 4.28 and Lemma 4.22 ($k = 1$).

**Step S4.51: Lipschitz.** Lemma 4.4 on compact sets, Lemma 4.7 (with $U = B(0, R+s)$, using the periodicity of $X_H$ from Lemma 4.24 and Lemma 4.28) for the cylinders. □

**Theorem 4.30** (Local existence and uniqueness for `IsSolutionOn`)**.** *For every $t_0 \in \mathbb{R}$ and $z_0 \in \mathbb{R}^n \times U$ there are $\epsilon > 0$ and $y : \mathbb{R} \to P$ with $y(t_0) = z_0$ and* `IsSolutionOn` $H\,U\,(t_0 - \epsilon, t_0 + \epsilon)\,y$ *(both time directions). If $J$ is any interval, $t_0 \in J$ and $y_1, y_2$ satisfy* `IsSolutionOn` $H\,U\,J\,y_i$ *with $y_1(t_0) = y_2(t_0)$, then $y_1 = y_2$ on $J$.*
*Source: BM11 p. 13 and Theorem 2.1 p. 5; Picard–Lindelöf (external fact (E1)).*

*Proof.* **Step S4.52: Reduction.** Proposition 4.25 (with $W = W_s$, $H$ $C^2$ there by Lemma 4.28) turns $\texttt{IsSolutionOn}\, H\, U$ into $\texttt{IsSolIn}\, X_H\, (\mathbb{R}^n \times U)$ with $X_H$ $C^1$ on $\mathbb{R}^n \times U$.

**Step S4.53: Existence, uniqueness.** Theorem 4.10 and Theorem 4.11. □

**Theorem 4.31** (Energy conservation for $\texttt{IsSolutionOn}$)**.** *If $\texttt{IsSolutionOn}\, H\, U\, J\, z$ for an interval $J$ (here $H$ may be any function $P \to \mathbb{R}$ and $U$ any set), then $H(z(t)) = H(z(t'))$ for $t, t' \in J$. In particular, for $H = H_{h,f}$ and $J = [0, T]$ the quantity $\Delta H = H(z(t))|_0^T$ of Pöschel's Lemma 1 vanishes.*
*Source: Pöschel 1993 p. 198 ("by energy conservation ... $\Delta H$ vanishes"; page read); BM11 p. 13.*

*Proof.* **Step S4.54: Direct.** The definition gives differentiability of $H$ at every $z(t)$ and $z'(t) = X_H(z(t))$; Theorem 4.23. □

**Theorem 4.32** (Maximal solutions and continuation)**.** *Let $z_0 = (\theta_0, I_0)$ with $I_0 \in U$. There is a unique maximal interval $(T_-, T_+) \ni 0$ and a solution $z$ with $z(0) = z_0$ and $\texttt{IsSolutionOn}\, H\, U\, (T_-, T_+)\, z$, such that every $y$ with $y(0) = z_0$ and $\texttt{IsSolutionOn}\, H\, U\, J\, y$ on an interval $J \ni 0$ has $J \subset (T_-, T_+)$ and $y = z$ on $J$ (Theorem 4.13 through Proposition 4.25). If $K \subset U$ is compact and $z(t)_2 \in K$ for all $t \in [t_*, T_+)$ for some $t_* \in [0, T_+)$, then $T_+ = +\infty$; likewise on the left. The angle components $z(t)_1$ may be unbounded: no obstruction arises because $X_H$ is 1-periodic in $\theta$.*
*Source: BM11 p. 13 ("From this estimate we can deduce that the solution $I(t)$ belongs to some compact ball ... this solution is defined for all time"); Pöschel 1993 p. 198 ("$T = T_*$").*

*Proof.* **Step S4.55: General statements.** By Proposition 4.25, $\texttt{IsSolutionOn}\, H\, U\, J\, y \iff \texttt{IsSolIn}\, X_H\, (\mathbb{R}^n \times U)\, J\, y$ and $X_H$ is $C^1$ on $W = \mathbb{R}^n \times U$; maximality is Theorem 4.13.

**Step S4.56: Periodic cylinder.** $X_H$ is 1-periodic in $\theta$ on $W$ (Lemmas 4.24, 4.28). Corollary 4.16 yields $T_+ = \infty$. □

**Theorem 4.33** (Uniqueness on closed intervals)**.** *For $L \geq 0$ and $y_1, y_2$ with $y_1(0) = y_2(0)$ and $\texttt{IsSolutionOn}\, H\, U\, [-L, L]\, y_i$ one has $y_1 = y_2$ on $[-L, L]$. In particular a solution on $[-L, L]$ agrees with every solution on a larger interval containing $[-L, L]$ and the same initial value (clause (d) of Theorem 1.19, where $U = B(0, R)$, $L \in [0, \tau]$, $y$ arbitrary and $z$ the exhibited solution). No confinement of $y$ is assumed: Gronwall is replaced by Theorem 4.11, whose Lipschitz constant comes from the compact set $y_1([-L, L]) \cup y_2([-L, L])$.*
*Source: Standard (Gronwall); clause (d) of the frozen N-HEAD-C.*

*Proof.* **Step S4.57: Apply.** $J = [-L, L]$, $t_0 = 0$ in Theorem 4.30 (uniqueness part). □

**Theorem 4.34** (Confinement glue)**.** *Let $\theta_0 \in \mathbb{R}^n$, $I_0 \in U$, $\tau > 0$ and $\rho > 0$ with $\overline{B}(I_0, \rho) \subset U$ (e.g. $|I_0| < R/2$, $\rho \leq R/4$, $U = B(0, R)$). Assume that for every open interval $J \ni 0$ and every $y$ with $y(0) = (\theta_0, I_0)$ and $\texttt{IsSolutionOn}\, H\, U\, J\, y$ one has $|y(t)_2 - I_0| \leq \rho$ for all $t \in J$ with $|t| \leq \tau$. Then there is $z$ with $z(0) = (\theta_0, I_0)$, $\texttt{IsSolutionOn}\, H\, U\, [-\tau, \tau]\, z$ and $|z(t)_2 - I_0| \leq \rho$ for $|t| \leq \tau$; and for every $L \in [0, \tau]$ and every $y$ with $y(0) = z(0)$ and $\texttt{IsSolutionOn}\, H\, U\, [-L, L]\, y$, $y = z$ on $[-L, L]$. This is the assembly of clauses (a)–(d) of Theorem 1.19 from an a priori estimate (Part 10); $\rho = C\varepsilon^{1/(2n)}$ there.*
*Source: BM11 p. 13 (two cases, "Therefore this solution is defined for all time"); Pöschel 1993 p. 198.*

*Proof.* **Step S4.58: Compact confinement set.** $K = \overline{B}(I_0, \rho)$ is compact and $\subset U$; $\Phi = \mathbb{R}^n \times K$ has uniform data by Lemma 4.7 (with $W = \mathbb{R}^n \times U$, $F = X_H$ periodic by Lemma 4.24).

**Step S4.59: General glue.** By Proposition 4.25 the hypothesis reads: every general solution on an open $J \ni 0$ with this initial value stays in $\Phi$ for $|t| \leq \tau$. Theorem 4.18 gives $z$ on $[-\tau, \tau]$ in $\Phi$, i.e. $|z(t)_2 - I_0| \leq \rho$, and uniqueness on $[-L, L]$ (it is also Theorem 4.33). □

4.10. **Label index for tracks I and P.**

| | |
|---|---|
| `bpN:p3:gen-sol`, `gen-symmetry` | predicate; time shift, reversal, space translation |
| `gen-lip`, `gen-uniform`, `gen-uniform-compact`, `gen-uniform-periodic` | Lipschitz and uniform data |
| `gen-short`, `gen-uniform-exist`, `gen-local` | local existence, uniform time |
| `gen-uniqueness` | uniqueness on all intervals, no confinement |
| `gen-extend`, `gen-maximal` | maximal solutions |
| `gen-continuation`, `-compact`, `-periodic`, `gen-confinement` | continuation, closed sets with uniform bounds, confinement glue |
| `gen-exit`, `gen-invariant`, `gen-barrier` | first exit, barrier |
| `ham-smooth`, `ham-energy`, `ham-periodic` | Hamiltonian layer |
| `ham-reduction`, `newton-reduction`, `newton-continuation` | frozen predicates of I, N, P |
| `holo-input`, `lipschitz`, `local-exist`, `energy`, `continuation`, `uniqueness-closed`, `confinement-glue` | track N specializations |

## 5. Weighted Fourier norms and Cauchy estimates (shared by tracks N, P, K)

This part is Appendix B and Appendix C of Pöschel 1993 (pp. 213–215) in the frozen conventions. It is shared infrastructure: track P (averaging, two-scale KAM) cites its results as [N:`bpN:p4:...`]. The statements are therefore made for functions $u(\theta, I)$ with $\theta \in \mathbb{T}^n$ and $I$ in a neighbourhood of an *arbitrary* set $D \subset \mathbb{C}^m$, with $m$ independent of $n$; only the Poisson bracket (Lemma 5.7) needs $m = n$. All norms on $\mathbb{R}^N$, $\mathbb{C}^N$ are max norms; no Euclidean norm occurs in this part.
Normalisation. Angles have period 1, characters are $\chi_k(\theta) = e^{2\pi i k\cdot\theta}$, and the strip width $s$ is the *geometric* width, $T_s = \{\theta \in \mathbb{C}^n : |\operatorname{Im}\theta|_\infty < s\}$ ([K:`bp:p0:domains`]); the weight is $e^{2\pi|k|_1 s}$, that is, the coefficient radius is $2\pi s$ as in `docs/fourier-conventions.md`. With this choice every formula of Pöschel's Appendix B holds *verbatim*, with no factor $2\pi$ (Remark 5.11); for the $\theta$-derivative the factor $2\pi$ of $\partial_{\theta_j}\chi_k = 2\pi i k_j \chi_k$ is absorbed by the weight. Widths are $r$ (actions) and $s$ (angles); the *losses* are $\rho$ (actions) and $\sigma$ (angles), as in Pöschel's $(\rho, \sigma)$. (The brief's $|u|_{D,r,\sigma}$ is $|u|_{D,r,s}$ here.)

5.1. **External facts.** Besides [K:`bp:p1:ext-cauchy`], [K:`bp:p1:ext-integral`], [K:`bp:p1:ext-fourier`], [K:`bp:p1:ext-mvt`], the proofs use:

(E1) *Holomorphic dependence on a parameter.* Let $Q \subset \mathbb{R}^p$ be a compact box, $U \subset \mathbb{C}^q$ open, $G : Q \times U \to \mathbb{C}^d$ continuous, holomorphic in $I \in U$ for each $x \in Q$, with $D_I G$ continuous on $Q \times U$. Then $g(I) = \int_Q G(x, I)\,dx$ is holomorphic on $U$ and $Dg(I) = \int_Q D_I G(x, I)\,dx$. (Mathlib: `hasFDerivAt_integral_of_dominated_of_fderiv_le` with a bound on a compact polydisc.) Exists.

(E2) *Uniform limits.* If $K_j$ are holomorphic on an open $U \subset \mathbb{C}^q$ and $K_j \to K$ uniformly on $U$, then $K$ is holomorphic on $U$, and $DK_j \to DK$ uniformly on $\{z : \bar{P}(z, \eta) \subset U\}$ for each $\eta > 0$ (the derivatives by Lemma 5.8 applied to $K_j - K_l$; holomorphy by [K:`bp:p1:uniform-limits`]). Exists in this form modulo that lemma.

(E3) *Series and integrals.* Tonelli for nonnegative double series and for integrals of nonnegative functions; the Cauchy product of two absolutely convergent series; termwise integration of a uniformly convergent series over a box ([K:`bp:p1:ext-integral`]). Exists.

5.2. **Domains, function class, norm.**

**Definition 5.1** (Domains)**.** Let $m, n \geq 0$, $D \subset \mathbb{C}^m$ any subset, $r, s > 0$. Put $V_r D = \{I \in \mathbb{C}^m : \exists I' \in D,\ |I - I'|_\infty < r\}$ and $\mathcal{D}_{r,s} = T_s \times V_r D \subset \mathbb{C}^n \times \mathbb{C}^m$ (Pöschel's $V_{r,s}$). For $z = (\theta, I)$ let

$\bar P(z;a,b)=\{(\theta',I'):|\theta'-\theta|_\infty\le a,|I'-I|_\infty\le b\}$. Then: (a) $V_rD$ and $\mathcal D_{r,s}$ are open, and $\mathcal D_{r',s'}\subset\mathcal D_{r,s}$ if $r'\le r$, $s'\le s$. (b) If $0<r'<r$, $0<s'<s$ and $z\in\mathcal D_{r',s'}$, then $\bar P(z;s-s',r-r')\subset\mathcal D_{r,s}$.
Source: Pos93 p. 213 ("analytic on complex neighbourhoods of a fixed domain $D\times\mathbb T^n$"); ours (max norms).

*Proof.* **Step S5.1: Openness and monotonicity.** $V_rD=\bigcup_{I'\in D}P(I',r)$ is a union of open polydiscs, and $T_s$ is open; monotonicity is clear.

**Step S5.2: Polydisc.** Let $|\theta'-\theta|_\infty\le s-s'$ and $|I'-I|_\infty\le r-r'$. Then $|\operatorname{Im}\theta_j'|\le|\operatorname{Im}\theta_j|+s-s'<s$; and with $I_0\in D$, $|I-I_0|<r'$: $|I'-I_0|<r'+(r-r')=r$. □

**Definition 5.2** (The class $\mathcal O(r,s)$ and the weighted norm)**.** $\mathcal O(r,s)$ is the set of $u:\mathcal D_{r,s}\to\mathbb C$ that are holomorphic (complex differentiable on the open set $\mathcal D_{r,s}\subset\mathbb C^{n+m}$) and 1-periodic in each $\theta_j$. For $u\in\mathcal O(r,s)$ and $k\in\mathbb Z^n$, $I\in V_rD$, put

$$\hat u_k(I)=\int_{[0,1]^n}u(x,I)\,e^{-2\pi ik\cdot x}\,dx,\qquad |u|_{I,s}=\sum_{k\in\mathbb Z^n}|\hat u_k(I)|\,e^{2\pi|k|_1 s},\qquad |u|_{r,s}=\sup_{I\in V_rD}|u|_{I,s}\in[0,\infty].$$

(This is $|u|_{D,r,s}$; the integrand is continuous because $\mathbb R^n\subset T_s$.) For a vector- or matrix-valued $u=(u_a)_a$ (components in $\mathcal O(r,s)$) the coefficients are taken componentwise and $|\hat u_k|_\infty=\max_a|\hat u_{a,k}|$ replaces $|\hat u_k|$ in $|u|_{I,s}$. If $|u|_{r,s}<\infty$ we say $u$ has *finite norm*.
Source: Pos93 p. 213 (definition of $|u|_{I,s}$, $|u|_{r,s}$), in the frozen convention of `docs/fourier-conventions.md` ("Bridge to strip sup norms": weight $e^{2\pi|k|_1 s}$).

Monotonicity: $|u|_{r',s'}\le|u|_{r,s}$ for $r'\le r$, $s'\le s$ (restriction to $\mathcal D_{r',s'}$; the coefficients at $I\in V_{r'}D$ are unchanged). For a vector, $\max_a|u_a|_{r,s}\le|u|_{r,s}\le\sum_a|u_a|_{r,s}$.

**Lemma 5.3** (Coefficients, derivatives, series)**.** *Let $u\in\mathcal O(r,s)$.*

- *(a) Each $\hat u_k$ is holomorphic on $V_rD$.*
- *(b) $\partial_{\theta_j}u$, $\partial_{I_i}u\in\mathcal O(r,s)$, $\widehat{(\partial_{\theta_j}u)}_k=2\pi ik_j\hat u_k$, and $\widehat{(\partial_{I_i}u)}_k=\partial_{I_i}\hat u_k$.*
- *(c) (Series) Let $a_k:V_rD\to\mathbb C$ ($k\in\mathbb Z^n$) be holomorphic with $S:=\sup_{I\in V_rD}\sum_k|a_k(I)|e^{2\pi|k|_1 s}<\infty$. Then $v=\sum_k a_k\chi_k$ converges absolutely and uniformly on $\mathcal D_{r,s'}$ for each $s'<s$, $v\in\mathcal O(r,s)$, $\hat v_k=a_k$ and $|v|_{r,s}=S$.*
- *(d) (Representation) If $|u|_{r,s}<\infty$ then $u(\theta,I)=\sum_k\hat u_k(I)\chi_k(\theta)$ on $\mathcal D_{r,s}$, and $|\hat u_k(I)|e^{2\pi|k|_1 s}\le|u|_{r,s}$.*

*Source: Standard; the one-variable ingredients are* [K:`bp:p1:fourier-basic`], [K:`bp:p1:summation`].

*Proof.* **Step S5.3: (a).** Fix $I_0\in V_rD$ and $\eta>0$ with $\bar P(I_0,\eta)\subset V_rD$. The integrand $G(x,I)=u(x,I)e^{-2\pi ik\cdot x}$ is continuous on $[0,1]^n\times P(I_0,\eta)$, holomorphic in $I$, and $D_IG$ is continuous there, because $Du$ is holomorphic ([K:`bp:p1:holomorphic-regularity`]) hence continuous. (E1) gives holomorphy of $\hat u_k$ near $I_0$.

**Step S5.4: (b).** The derivatives are holomorphic ([K:`bp:p1:holomorphic-regularity`]) and periodic (differentiate the identity $u(\theta+e_j,I)=u(\theta,I)$). The formula for $\theta_j$ is [K:`bp:p1:fourier-basic`](a) at fixed $I$ (the restriction of $u(\cdot,I)$ to $\mathbb R^n$ is continuous, periodic and $C^1$). The formula for $I_i$ is the derivative formula of (E1) as in Step S5.3.

**Step S5.5: (c).** Each hypothesis $|a_k(I)|e^{2\pi|k|_1 s}\le S$ gives, on $\mathcal D_{r,s'}$, $|a_k(I)\chi_k(\theta)|\le|a_k(I)|e^{2\pi|k|_1 s'}\le Se^{-2\pi|k|_1(s-s')}$, which is summable ([K:`bp:p1:exp-sums`](b)). By the Weierstrass test the series converges uniformly on $\mathcal D_{r,s'}$, and each term is holomorphic, so $v$ is holomorphic there ((E2)); as $s'<s$ is arbitrary, $v$ is holomorphic on $\mathcal D_{r,s}$. It is periodic since each $\chi_k$ is. Integrating termwise ((E3)) against $\chi_{-k}$ over $[0,1]^n$ gives $\hat v_k=a_k$ (as in the coefficient step of [K:`bp:p1:summation`]); the norm identity is then the definition.

**Step S5.6: (d).** The bound is a single term of the sum $|u|_{I,s} \le |u|_{r,s}$. Fix $I$. The function $v = \sum_k \hat{u}_k(I)\chi_k$ is holomorphic and 1-periodic on $T_s$ with coefficients $\hat{u}_k(I)$ ([K:bp:p1:summation], first part), and $u(\cdot, I) - v$ is holomorphic, 1-periodic, with all coefficients zero, so $u(\cdot, I) = v$ by [K:bp:p1:fourier-basic](c). □

5.3. **Lemma B.1: sup norm versus weighted norm.**

**Lemma 5.4** (B.1)**.** *Let $u \in \mathcal{O}(r, s)$ (scalar, vector or matrix valued;* $\sup|u|$ *is the sup of the max norm of the values).*

*(a)* $\sup_{\mathcal{D}_{r,s}} |u| \le |u|_{r,s}$.
*(b) If $\sigma > 0$ and $|u| \le M$ on $\mathcal{D}_{r,s+\sigma}$ (so $u \in \mathcal{O}(r, s+\sigma)$), then $|u|_{r,s} \le \coth(\pi\sigma)^n M$.*

*In Pöschel's form: $|u|^{\sup}_{r,s} \le |u|_{r,s} \le \coth^n \sigma_P \, |u|^{\sup}_{r,s+2\sigma_P}$ with $\sigma_P = \pi\sigma$ (the coefficient radius grows by $2\sigma_P = 2\pi\sigma$); see Remark 5.11.*
*Source: Pos93 p. 213, Lemma B.1; frozen-convention form:* [K:bp:p1:summation], [K:bp:p1:decay], [K:bp:p1:exp-sums]*(b); this is Lemma 2.9 with the $I$-dependence made explicit.*

*Proof.* **Step S5.7: (a).** If $|u|_{r,s} = \infty$ there is nothing to prove. Otherwise Lemma 5.3(d) gives, for $(\theta, I) \in \mathcal{D}_{r,s}$, $|u(\theta, I)| \le \sum_k |\hat{u}_k(I)||\chi_k(\theta)| \le \sum_k |\hat{u}_k(I)| e^{2\pi|k|_1 s}$, using $|\chi_k(\theta)| \le e^{2\pi|k|_1 |\mathrm{Im}\,\theta|_\infty}$ ([K:bp:p0:domains]). For vector values apply this to each component and note $|\hat{u}_{a,k}| \le |\hat{u}_k|_\infty$.

**Step S5.8: (b).** Fix $I \in V_r D$. The function $u(\cdot, I)$ is holomorphic, periodic and bounded by $M$ on $T_{s+\sigma}$, so [K:bp:p1:decay] gives $|\hat{u}_k(I)|_\infty \le e^{-2\pi|k|_1(s+\sigma)} M$ (limit $\rho' \uparrow s+\sigma$), and

$$|u|_{I,s} \le M \sum_k e^{-2\pi|k|_1\sigma} = \coth(\pi\sigma)^n M \qquad ([\text{K:bp:p1:exp-sums}](b),\ x = 2\pi\sigma).$$

Take the supremum over $I$. □

5.4. **Lemma B.2: algebra property.**

**Lemma 5.5** (B.2)**.** *Let $u, v \in \mathcal{O}(r, s)$ be scalar with finite norms. Then $uv \in \mathcal{O}(r, s)$, $\widehat{(uv)}_k(I) = \sum_l \hat{u}_{k-l}(I)\hat{v}_l(I)$ (absolutely convergent), and $|uv|_{I,s} \le |u|_{I,s}\,|v|_{I,s}$ for every $I \in V_r D$; hence $|uv|_{r,s} \le |u|_{r,s}|v|_{r,s}$, and $|u^j|_{r,s} \le |u|^j_{r,s}$.*
*Source: Pos93 p. 213, Lemma B.2. (The coefficient-level inequality is proved in Lean: `docs/fourier-conventions.md`, "Verified coefficient multiplication", `KAM/Fourier/Convolution.lean`; its identification with the product of functions is the remaining bridge.)*

*Proof.* **Step S5.9: Product is in the class.** $uv$ is holomorphic and periodic.

**Step S5.10: Coefficients.** Fix $I$. By Lemma 5.3(d), at real $x$, $u(x, I) = \sum_k \hat{u}_k \chi_k(x)$ and $v(x, I) = \sum_l \hat{v}_l \chi_l(x)$, both absolutely convergent ((E3)). The Cauchy product gives $u(x, I)v(x, I) = \sum_k c_k \chi_k(x)$ with $c_k = \sum_l \hat{u}_{k-l}\hat{v}_l$, and $\sum_k |c_k| \le |u|_{I,0}|v|_{I,0} < \infty$, so the convergence is uniform in $x$ and termwise integration gives $\widehat{(uv)}_k(I) = c_k$.

**Step S5.11: Norm.** Since $|k|_1 \le |k-l|_1 + |l|_1$, $|uv|_{I,s} = \sum_k |c_k| e^{2\pi|k|_1 s} \le \sum_k \sum_l |\hat{u}_{k-l}| e^{2\pi|k-l|_1 s} |\hat{v}_l| e^{2\pi|l|_1 s} = |u|_{I,s}|v|_{I,s}$ (Tonelli, with the bijection $(k, l) \mapsto (k-l, l)$). Take the supremum over $I$; induct for $u^j$. □

5.5. **Lemma B.3: Cauchy estimates in the weighted norm.**

**Lemma 5.6** (B.3)**.** *Let $u \in \mathcal{O}(r_0, s_0)$ have finite norm, $0 < r < r_0$, $0 < s < s_0$. Then*

*(a) (pointwise in $I$) for every $I \in V_{r_0}D$:* $\displaystyle\sum_{j=1}^{n}|\partial_{\theta_j}u|_{I,s} \le \frac{1}{e(s_0-s)}|u|_{I,s_0}$*; consequently* $\displaystyle\sup_{I\in V_{r_0}D}\sum_j|\partial_{\theta_j}u|_{I,s} \le \frac{|u|_{r_0,s_0}}{e(s_0-s)}$ *and, for each single $j$ (global norm),* $|\partial_{\theta_j}u|_{r_0,s} \le \frac{1}{e(s_0-s)}|u|_{r_0,s_0}$. The sum over $j$ of the global norms $\sum_j\sup_I|\partial_{\theta_j}u|_{I,s}$ is not bounded by this constant *(for $D$ disconnected it can be larger by a factor up to $n$; with the norm $\sharp$ of Lemma 5.10 it is bounded);*

*(b)* $\displaystyle\max_{1\le i\le m}|\partial_{I_i}u|_{r,s_0} \le \frac{1}{r_0-r}|u|_{r_0,s_0}$.

*In particular (taking $r_0 = r+\rho$, $s_0 = s+\sigma$):* $\sup_{I\in V_rD}\sum_j|\partial_{\theta_j}u|_{I,s-\sigma} \le (e\sigma)^{-1}|u|_{r,s}$, $|\partial_{\theta_j}u|_{r,s-\sigma} \le (e\sigma)^{-1}|u|_{r,s}$ *for each $j$, and* $\max_i|\partial_{I_i}u|_{r-\rho,s} \le \rho^{-1}|u|_{r,s}$ *for $0<\sigma<s$, $0<\rho<r$. The same holds for vector- and matrix-valued $u$, componentwise (and with the pointwise sum over the $\theta$-index outside, never the sum of global norms).*

*Source: Pos93 pp. 213–214, Lemma B.3 (with $s$ geometric; Pöschel's $1/(e\sigma)$ is the same number, Remark 5.11).*

*Proof.* **Step S5.12: (a): coefficients.** Fix $I \in V_{r_0}D$. By Lemma 5.3(b), $\sum_j|\partial_{\theta_j}u|_{I,s} = \sum_k\sum_j 2\pi|k_j||\hat u_k|e^{2\pi|k|_1 s} = \sum_k 2\pi|k|_1 e^{-2\pi|k|_1(s_0-s)}|\hat u_k|e^{2\pi|k|_1 s_0}$. By [K:bp:p1:exp-sums](a) with $\tau = 1$, $a = 2\pi(s_0-s)$: $2\pi|k|_1e^{-2\pi|k|_1(s_0-s)} \le 2\pi/(e\cdot 2\pi(s_0-s))$. Hence the sum is $\le (e(s_0-s))^{-1}|u|_{I,s_0}$ for this $I$. The two global consequences follow by taking the supremum over $I$ of the left side (resp. of a single summand, which is at most the sum) and using $|u|_{I,s_0} \le |u|_{r_0,s_0}$.

**Step S5.13: (b): Cauchy integral.** Fix $I \in V_rD$, $i$, and put $\rho' = r_0 - r$. For $|\zeta| \le \rho'$ the point $I+\zeta e_i$ lies in $V_{r_0}D$: if $I_0 \in D$ with $|I-I_0| < r$ then $|I+\zeta e_i - I_0| < r+\rho' = r_0$. By Lemma 5.3(a),(b) and [K:bp:p1:ext-cauchy], $\partial_{I_i}\hat u_k(I) = \frac{1}{2\pi i}\oint_{|\zeta|=\rho'}\hat u_k(I+\zeta e_i)\zeta^{-2}\,d\zeta$, so $|\partial_{I_i}\hat u_k(I)| \le \frac{1}{2\pi}\oint|\hat u_k(I+\zeta e_i)|\rho'^{-2}|d\zeta|$. Multiply by $e^{2\pi|k|_1 s_0}$, sum over $k$ and use Tonelli ((E3)): $|\partial_{I_i}u|_{I,s_0} \le \frac{1}{2\pi}\oint\rho'^{-2}|u|_{I+\zeta e_i,s_0}|d\zeta| \le \rho'^{-1}|u|_{r_0,s_0}$. The vector case is componentwise, the matrix case likewise. □

5.6. **Lemma B.4: the Poisson bracket.** For $m=n$ and $u,v$ holomorphic on an open subset of $\mathbb{C}^n\times\mathbb{C}^n$, $\{u,v\} = \sum_{i=1}^n\big(\partial_{\theta_i}u\,\partial_{I_i}v - \partial_{I_i}u\,\partial_{\theta_i}v\big)$ (Definition 6.2; $\{u,v\} = Du\cdot X_v$ for the frozen field $X_v = (\partial_I v, -\partial_\theta v)$).

**Lemma 5.7** (B.4)**.** *Let $m=n$, $u\in\mathcal{O}(r_0,s_0)$, $v\in\mathcal{O}(r,s)$ with finite norms, $0<\rho<r$, $0<\sigma<s$, $r-\rho<r_0$, $s-\sigma<s_0$. Then $\{u,v\}\in\mathcal{O}(r-\rho,s-\sigma)$ and*

$$|\{u,v\}|_{r-\rho,s-\sigma} \le \frac{1}{e}\Big(\frac{1}{(r_0-r+\rho)\,\sigma} + \frac{1}{(s_0-s+\sigma)\,\rho}\Big)|u|_{r_0,s_0}|v|_{r,s}.$$

*Source: Pos93 p. 214, Lemma B.4 (same formula, with Pöschel's $(r_0,s_0,r,s,\rho,\sigma)$ read as our geometric quantities).*

*Proof.* Write $\{u,v\} = \sum_i\partial_{\theta_i}u\,\partial_{I_i}v - \sum_i\partial_{I_i}u\,\partial_{\theta_i}v$, all terms restricted to $\mathcal{D}_{r-\rho,s-\sigma}\subset\mathcal{D}_{r_0,s_0}\cap\mathcal{D}_{r,s}$ (monotonicity).

**Step S5.14: Pointwise product estimate.** Fix $I\in V_{r-\rho}D$ and write $|\cdot|_{I,\cdot} = |\cdot|_{I,s-\sigma}$, $t = s-\sigma$. By Lemma 5.5 (pointwise in $I$) and the triangle inequality for $|\cdot|_{I,t}$ (coefficients of a sum are sums), $|\{u,v\}|_{I,t} \le \sum_i|\partial_{\theta_i}u|_{I,t}|\partial_{I_i}v|_{I,t} + \sum_i|\partial_{I_i}u|_{I,t}|\partial_{\theta_i}v|_{I,t} \le \Big(\sum_i|\partial_{\theta_i}u|_{I,t}\Big)\max_i|\partial_{I_i}v|_{I,t} + \max_i|\partial_{I_i}u|_{I,t}\Big(\sum_i|\partial_{\theta_i}v|_{I,t}\Big)$. The angular sums are kept *pointwise in $I$*.

**Step S5.15: Four derivative bounds.** By Lemma 5.6, with (a) pointwise and (b) global, and monotonicity of the norm in both widths: $\sum_i|\partial_{\theta_i}u|_{I,t}\le\frac{|u|_{I,s_0}}{e(s_0-s+\sigma)}\le\frac{|u|_{r_0,s_0}}{e(s_0-s+\sigma)}$ ((a) with $s\to s-\sigma$; $I\in V_{r_0}D$ since $r-\rho<r_0$); $\max_i|\partial_{I_i}u|_{I,t}\le\frac{|u|_{r_0,s_0}}{r_0-r+\rho}$ ((b) with $r\to r-\rho$); $\sum_i|\partial_{\theta_i}v|_{I,t}\le\frac{|v|_{I,s}}{e\sigma}\le\frac{|v|_{r,s}}{e\sigma}$ ((a), pointwise) and $\max_i|\partial_{I_i}v|_{I,t}\le\frac{|v|_{r,s}}{\rho}$ ((b)). Inserting the four bounds in the previous display and taking the supremum over $I\in V_{r-\rho}D$ *last* gives the claim; the constant is unchanged and no factor $n$ appears. □

5.7. **Lemma C.1: the general Cauchy inequality.**

**Lemma 5.8** (C.1)**.** *Let $A,B$ be finite-dimensional complex normed spaces, $v\in A$, $R>0$, and $G:P_A(v,R)=\{a:|a-v|_A<R\}\to B$ holomorphic with $|G|_B\le M$ on this ball. Then the operator norm satisfies $|DG(v)|_{B,A}\le M/R$.*
*Source: Pos93 p. 215, Lemma C.1 ("analytic maps of Banach spaces"; only the finite-dimensional case is needed).*

*Proof.* **Step S5.16: Reduction to one variable.** Let $u\in A$, $u\neq0$. $f(\zeta)=G(v+\zeta u)$ is holomorphic on $|\zeta|<R/|u|_A$ with $|f|\le M$ and $f'(0)=DG(v)u$.

**Step S5.17: Cauchy.** For every $\varrho<R/|u|_A$, [K:`bp:p1:ext-cauchy`] gives $|f'(0)|\le M/\varrho$; let $\varrho\uparrow R/|u|_A$: $|DG(v)u|_B\le M|u|_A/R$. □

This also gives (E2)'s derivative statement: apply it to $K_j-K_l$ on the ball of radius $\eta$.

5.8. **Projections and truncation.**

**Lemma 5.9** ($P_S$, $T_K$, averages)**.** *Let $u\in\mathcal{O}(r,s)$ have finite norm and $S\subset\mathbb{Z}^n$. Put $P_Su=\sum_{k\in S}\hat u_k\chi_k$. Then $P_Su\in\mathcal{O}(r,s)$, $\widehat{(P_Su)}_k=\hat u_k$ for $k\in S$ and $0$ otherwise, and $|P_Su|_{r,s}\le|u|_{r,s}$. In particular, for $K\ge0$ the truncation $T_Ku=P_{\{|k|_1\le K\}}u$ (a trigonometric polynomial in $\theta$) and the average $\langle u\rangle(I)=\hat u_0(I)=P_{\{0\}}u$ (independent of $\theta$) satisfy $|T_Ku|_{r,s}\le|u|_{r,s}$, $|\langle u\rangle|_{r,s}\le|u|_{r,s}$, and*

$$u-T_Ku=P_{\{|k|_1>K\}}u,\qquad |u-T_Ku|_{r,s-\sigma}\le e^{-2\pi K\sigma}|u|_{r,s}\qquad(0<\sigma<s).$$

*If $\Lambda\subset\mathbb{Z}^n$ is any set (e.g. a lattice, Pos93 p. 192) and $u$ is said to be in $\Lambda$-normal form when $u=P_\Lambda u$, then $P_\Lambda u$, $u-P_\Lambda u$ are in the class with norms $\le|u|_{r,s}$. For vector-valued $u$ everything is componentwise.*
*Source: Pos93 p. 192 (definition of $T_K$, $P_\Lambda$) and p. 212 ("$|f-T_Kf|_{r,s-\sigma}\le e^{-K\sigma}|f|_{r,s}$"), geometric form.*

*Proof.* **Step S5.18: Class and coefficients.** The coefficient family $a_k=\hat u_k\mathbf{1}_S(k)$ consists of holomorphic functions (Lemma 5.3(a)) with $\sup_I\sum_k|a_k|e^{2\pi|k|_1s}\le|u|_{r,s}<\infty$; Lemma 5.3(c) gives all claims including $|P_Su|_{r,s}\le|u|_{r,s}$.

**Step S5.19: Tail.** For $|k|_1>K$: $e^{2\pi|k|_1(s-\sigma)}=e^{-2\pi|k|_1\sigma}e^{2\pi|k|_1s}\le e^{-2\pi K\sigma}e^{2\pi|k|_1s}$. Sum over $k$ with $|k|_1>K$ and take the supremum over $I$. Also $u=\sum_k\hat u_k\chi_k$ (Lemma 5.3(d)) gives $u-T_Ku=P_{\{|k|_1>K\}}u$. □

5.9. **The stronger norm ♯ (sum of coefficient suprema).** Track P's averaging uses $\|f\|_{r,\Sigma}=\sum_k\sup_I|f_k(I)|e^{\Sigma|k|_1}$. With $s=\Sigma/(2\pi)$ this is, in our notation,

$$|u|^\sharp_{r,s}=\sum_{k\in\mathbb{Z}^n}\ \sup_{I\in V_rD}|\hat u_k(I)|_\infty\,e^{2\pi|k|_1s}\ \in[0,\infty].$$

Always $|u|_{r,s}\le|u|^\sharp_{r,s}$; the reverse fails by an unbounded factor on disconnected $D$ (on $M$ disjoint action components carry one distinct mode each: $|\cdot|$, $=1$, $|\cdot|^\sharp=M$), so the two norms are *not* interchangeable and a bound for one does not give a bound for the other. *Width conversion for track P:* every width and loss of P (including a non-loss width $\bar s$) is converted by $s=\Sigma/(2\pi)$,

$\sigma = \Delta\Sigma/(2\pi)$; the action losses are unchanged. Then the angular Cauchy constant $1/(e\sigma)$ reads $2\pi/(e\,\Delta\Sigma)$, the tail factor $e^{-2\pi K\sigma}$ reads $e^{-K\Delta\Sigma}$, and the composition parameter $\eta = 2|\phi|/(\rho\sigma)$ reads $4\pi|\phi|^\sharp/(\rho\,\Delta\Sigma)$.

**Lemma 5.10** (Variants for $\sharp$)**.** *All of Lemma 5.4(a),(b), Lemma 5.5, Lemma 5.6, Lemma 5.7 and Lemma 5.9 hold with $|\cdot|$, replaced by $|\cdot|^\sharp$, with the same constants. In Lemma 5.6(a) the stronger* global *form holds: $\sum_j |\partial_{\theta_j} u|^\sharp_{r_0,s} \le |u|^\sharp_{r_0,s_0}/(e(s_0-s))$. For finite-norm $u$ the series $\sum_k \hat u_k \chi_k$ of Lemma 5.3(c) may be taken with $S^\sharp := \sum_k \sup_I |a_k| e^{2\pi|k|_1 s} < \infty$ (the hypothesis is then stronger, and implies the one used there).*

*Source: Ours; the blueprint's reading of Pos93 App. B for the norm of track P's two-scale iteration.*

*Proof.* In every proof above the supremum over $I$ is taken of a sum over $k$; for $\sharp$ it is taken inside, coefficient by coefficient, and each step survives.

**Step S5.20: B.1.** (a) $|u(\theta,I)| \le \sum_k |\hat u_k(I)| e^{2\pi|k|_1 s} \le |u|^\sharp_{r,s}$. (b) the decay $\sup_I |\hat u_k(I)|_\infty \le e^{-2\pi|k|_1(s+\sigma)} M$ is uniform in $I$, so $|u|^\sharp_{r,s} \le \coth(\pi\sigma)^n M$.

**Step S5.21: B.2.** $\sup_I |\sum_l \hat u_{k-l}(I)\hat v_l(I)| \le \sum_l \sup_I |\hat u_{k-l}| \sup_I |\hat v_l|$, and then the weights as in Step S5.11.

**Step S5.22: B.3.** (a): $\sum_j \sup_I |2\pi k_j \hat u_k(I)| = 2\pi|k|_1 \sup_I |\hat u_k|$ (the sum over $j$ is of the coefficient factors $|k_j|$ only), then Step S5.12 coefficientwise. (b): $\sup_{I\in V_r D} |\partial_{I_i}\hat u_k(I)| \le (r_0 - r)^{-1} \sup_{I'\in V_{r_0}D} |\hat u_k(I')|$ by the Cauchy integral of Step S5.13; then sum over $k$.

**Step S5.23: B.4.** The convolution bound of the previous steps gives, coefficientwise in the suprema, $|\{u,v\}|^\sharp_{r,s} \le \sum_i |\partial_{\theta_i} u|^\sharp_{r,s} |\partial_{I_i} v|^\sharp_{r,s} + \dots$, and the global forms of (a),(b) bound the four factors as in Step S5.15.

**Step S5.24: Projections.** $|P_S u|^\sharp_{r,s} = \sum_{k\in S} \sup_I |\hat u_k| e^{2\pi|k|_1 s} \le |u|^\sharp_{r,s}$ and the tail estimate is the same. □

*Remark* 5.11 (Dictionary with Pöschel's normalisation and with Part 1). Pöschel works with $2\pi$-periodic angles $\varphi$, characters $e^{ik\cdot\varphi}$, weight $e^{|k|_1 s_P}$ and losses $(\rho_P, \sigma_P)$. Under the symplectic scaling $\varphi = 2\pi\theta$, $I_P = I/2\pi$ (so $d\varphi \wedge dI_P = d\theta \wedge dI$; time and the values of Hamiltonians are unchanged) his objects become ours with $s_P = 2\pi s$, $\sigma_P = 2\pi\sigma$, $r_P = r/2\pi$, $\rho_P = \rho/2\pi$. Then $e^{|k|_1 s_P} = e^{2\pi|k|_1 s}$ and $\rho_P\sigma_P = \rho\sigma$; the quantities in which Lemmas B.3–B.5 and the Iterative Lemma are written (for instance $|\phi|/(\rho\sigma)$ and the norm $|W\cdot|$) are therefore the same numbers. The conversion of Part 1, $\tilde H = 2\pi H(\varphi/2\pi, I)$ with $I$ unchanged (Lemma 2.8), gives the same smallness ratio: $\tilde\phi/(\rho\sigma_P) = 2\pi\phi/(\rho\cdot 2\pi\sigma) = \phi/(\rho\sigma)$. So a condition "$|\phi| < \rho\sigma/2$" means the same in all three conventions, and the $2\pi$-conversion does not affect the constants of the Iterative Lemma beyond the scalings of Lemma 2.8(b) (that bookkeeping is Part 6's). Where `docs/fourier-conventions.md` writes the derivative loss as $2\pi/(e\delta)$ for coefficient radius $s$, Lemma 5.6(a) is the same statement for the geometric width: a coefficient-radius loss $\delta = 2\pi\sigma$ gives $2\pi/(e\cdot 2\pi\sigma) = 1/(e\sigma)$.

## 6. Canonical transformations: time-one maps and the Lie transform (shared by tracks N, P)

This part is the canonical-transformation toolkit of Pöschel 1993, Appendix A (pp. 211–213: Taylor's formula, the flow $X^t_\phi$ of a small Hamiltonian field, its displacement and derivative bounds) and Lemma B.5 (pp. 214–215), in the frozen conventions of Part 5 (period 1, geometric widths, max norms). Track P cites it as [N:`bpN:p5:...`]. Generating functions $\phi$ live on general domains $\mathcal{D}_{r,s} = T_s \times V_r D$ with $D \subset \mathbb{C}^n$ arbitrary (Definition 5.1); $m = n$ because a Hamiltonian structure is needed. The ODE existence, uniqueness and continuation used are those of Part 4 (general layer); the weighted-norm estimates are those of Part 5.

*Remark* 6.1 (Decision for Lemma B.5: the Lie series, not the sup-norm fallback). **Route.** `PLAN.md` left open whether $|u \circ X_\phi^1| \le (1-2|\phi|/(\rho\sigma))^{-1}|u|$ is proved by Pöschel's convergent series $u \circ X_\phi^1 = \sum_h \frac{1}{h!}\operatorname{ad}_\phi^h u$ or by a sup-norm estimate of the displaced function and a conversion back with Lemma 5.4(b). *We take the series*, for three reasons. (1) The Normal Form Lemma iterates $N \approx Ks/(12\ln 2)$ times and every step feeds the *weighted* norm of $f_+$ into the next step. The fallback re-enters the weighted norm through $\coth(\pi\sigma)^n$ in every step; this factor exceeds 1 by a fixed amount, so the contraction $\varepsilon_{i+1} \le \frac14\varepsilon_i$ of the Iterative Lemma would be destroyed and not merely perturbed in constants. (2) The fallback needs $u \circ X_\phi^1$ to be holomorphic and periodic *before* one can speak of its Fourier coefficients, that is, holomorphic dependence of the flow on the initial point (Lemma 6.4(b)), a statement not in Mathlib. The series route gets holomorphy of $u \circ X_\phi^1$ for free (it is a norm-convergent series of holomorphic functions); the dependence on initial data is then needed only for the claims on the map $\Phi = X_\phi^1$ itself (Proposition 6.13). (3) The ingredients of the series route are already labelled results: the bracket estimate (Lemma 5.7), a one-line induction (Lemma 6.9), a Taylor remainder along a solution of $\dot z = X_\phi(z)$ (Part 4 and Lemma 6.8) and a geometric series. **What is lost.** The equality $u \circ X_\phi^1 = \sum_h \frac{1}{h!}\operatorname{ad}_\phi^h u$ is proved here for $\eta := 2|\phi|/(\rho\sigma) \le \frac12$ instead of Pöschel's $\eta < 1$ (Theorem 6.10); the bound itself is $(1-\eta)^{-1}$ as in the source. For $\eta < 1$ the equality also holds, by real-analyticity of $t \mapsto u(X_\phi^t z)$ and the identity theorem, but real-analytic dependence on time is another external fact; since the frozen headline only asserts the existence of constants, Part 6 simply uses $|\phi| \le \rho\sigma/4$ in place of $|\phi| < \rho\sigma/2$.

## 6.1. **Setting.**

**Definition 6.2** (Hamiltonian field, bracket, scaled norm)**.** Points of $\mathbb{C}^n \times \mathbb{C}^n$ are $z = (\theta, I)$ (angles first, as in the frozen field); $J = \begin{pmatrix} 0 & \mathbf{1} \\ -\mathbf{1} & 0 \end{pmatrix}$, so $J^2 = -\mathbf{1}$, $J^\top = -J$. For $u$ holomorphic on an open set let $\nabla u = (\partial_\theta u, \partial_I u)^\top$,

$$X_u = J\nabla u = (\partial_I u, -\partial_\theta u), \qquad \{u,v\} = \sum_{i=1}^n \big(\partial_{\theta_i} u\,\partial_{I_i} v - \partial_{I_i} u\,\partial_{\theta_i} v\big) = Du\cdot X_v = \nabla u^\top J \nabla v = -\{v,u\}.$$

$X_u$ is the frozen Hamiltonian field (`KAM.hamiltonianVectorField`, Definition 1.3; for a real-valued function on real points see [K:`bp:p1:hamiltonian-real`]); the bracket is normalised so that $\frac{d}{dt}u(X_\phi^t(z)) = \{u,\phi\}(X_\phi^t(z))$ (Lemma 6.7). For $\rho,\sigma > 0$ the *scaled norm* on $\mathbb{C}^n \times \mathbb{C}^n$ is $\nu_{\rho,\sigma}(a,b) = \max(|a|_\infty/\sigma, |b|_\infty/\rho)$ (Pöschel's $|W\cdot|$, $W = \operatorname{diag}(\sigma^{-1},\rho^{-1})$ in angle-first order); it is a complex norm, and its closed ball of radius $c$ about $z = (\theta, I)$ is $\bar P(z; c\sigma, c\rho)$. Iterated brackets: $\operatorname{ad}_\phi^0 u = u$, $\operatorname{ad}_\phi^j u = \{\operatorname{ad}_\phi^{j-1} u, \phi\}$.

Source: Pos93 pp. 211–212 (flow $X_\phi^t$, $W$), p. 214 (ad).

Source gap: We did not locate the printed definition of the bracket's sign in pp. 211–215; only the two-term structure (Lemma B.4) and $\{\phi,h\} = \omega\cdot\partial_\theta\phi$ for $h = h(I)$ (p. 211, homological equation, divisors $ik\cdot\omega$) are used. With the convention above $\{\phi,h\} = \partial_\theta\phi\cdot\partial_I h$, which is that equation.

The bracket of holomorphic periodic functions is holomorphic and periodic ([K:`bp:p1:holomorphic-regularity`]); $\operatorname{ad}_\phi^j u$ is holomorphic on the common domain of $u$ and $\phi$; $\{u,v\} = 0$ if $u, v$ depend only on $I$.

**Lemma 6.3** (Size of the field)**.** *Let $\phi \in \mathcal{O}(r,s)$ have finite norm, $0 < \rho < r$, $0 < \sigma < s$. On $\mathcal{D}_{r-\rho,s-\sigma}$: $|\partial_{I_i}\phi| \le |\phi|_{r,s}/\rho$, $|\partial_{\theta_i}\phi| \le |\phi|_{r,s}/(e\sigma)$, and*

$$\nu_{\rho,\sigma}(X_\phi) = \max\Big(\frac{\max_i|\partial_{I_i}\phi|}{\sigma}, \frac{\max_i|\partial_{\theta_i}\phi|}{\rho}\Big) \le \frac{|\phi|_{r,s}}{\rho\sigma}.$$

*Source: Pos93 p. 212 ("$|\phi_I|_\infty \le q\delta/(\alpha\rho)$, $|WX_\phi| \le q\delta/(\alpha\rho\sigma)$").*

*Proof.* **Step S6.1: Derivatives.** Lemma 5.6 (with $r_0 = r$, $s_0 = s$ and the target widths $r-\rho$, $s-\sigma$) gives, for each single index, $|\partial_{I_i}\phi|_{r-\rho,s} \le \rho^{-1}|\phi|_{r,s}$ and $|\partial_{\theta_j}\phi|_{r,s-\sigma} \le (e\sigma)^{-1}|\phi|_{r,s}$ (the individual global angular estimate, not a sum of global norms). Restrict to $\mathcal{D}_{r-\rho,s-\sigma}$ (monotonicity), and apply Lemma 5.4(a) to each derivative.

**Step S6.2: The field.** Insert into the definition of $\nu_{\rho,\sigma}$: $\frac{1}{\sigma}\cdot\frac{|\phi|}{\rho} = \frac{|\phi|}{\rho\sigma}$ and $\frac{1}{\rho}\cdot\frac{|\phi|}{e\sigma} \le \frac{|\phi|}{\rho\sigma}$. □

## 6.2. **The flow of a small holomorphic vector field.**

**Lemma 6.4** (Flow of a small holomorphic field)**.** *Let $\nu$ be a complex norm on $\mathbb{C}^N$, $\Omega \subset \mathbb{C}^N$ open, $F : \Omega \to \mathbb{C}^N$ holomorphic with $\nu(F) \le \varepsilon$ on $\Omega$ and $0 \le \varepsilon \le \frac{1}{2}$. For $c > 0$ let $\Omega^c = \{z : \bar{B}_\nu(z, c) \subset \Omega\}$ (an open set).*

- *(a) (Existence, uniqueness, displacement) For $z \in \Omega^{1/2}$ there is $\gamma_z : \mathbb{R} \to \mathbb{C}^N$ with* `IsSolIn` $F\,\Omega\,[-1,1]\,\gamma_z$ *and $\gamma_z(0) = z$; any solution in $\Omega$ through $z$ on an interval $J \ni 0$ equals $\gamma_z$ on $J \cap [-1,1]$. Put $X^t(z) = \gamma_z(t)$ ($|t| \le 1$). Then $\nu(X^t(z) - z) \le |t|\,\varepsilon$.*
- *(b) (Regularity) $(t,z) \mapsto X^t(z)$ is continuous on $[-1,1]\times\Omega^{1/2}$; for each $t$ the map $X^t : \Omega^{1/2} \to \Omega$ is holomorphic; $(t,z) \mapsto D_z X^t(z)$ is continuous, and $Y(t) = D_z X^t(z)$ satisfies $Y(t) = \mathbf{1} + \int_0^t DF(X^\tau z)Y(\tau)\,d\tau$.*
- *(c) (Inverse, derivative) For $z \in \Omega^1$: $X^1(z) \in \Omega^{1/2}$ and $X^{-1}(X^1 z) = z$; and the $\nu$-operator norm satisfies $|DX^t(z) - \mathbf{1}|_\nu \le 2\varepsilon$ for $|t| \le 1$. In the same way $X^1(X^{-1}z) = z$.*
- *(d) (Translations) If $\tau \in \mathbb{C}^N$, $\Omega + \tau = \Omega$ and $F(w+\tau) = F(w)$ on $\Omega$, then $\Omega^{1/2} + \tau = \Omega^{1/2}$ and $X^t(z+\tau) = X^t(z) + \tau$.*
- *(e) (Reality) Let $N = 2n$ and suppose $F(x) \in \mathbb{R}^N$ for $x \in \Omega \cap \mathbb{R}^N$. Then $X^t(x) \in \mathbb{R}^N$ for $x \in \Omega^{1/2} \cap \mathbb{R}^N$.*

*Source: Pos93 p. 212 ("$X^t_\phi$ maps $V_{r-\frac{3}{2}\rho,s-\frac{3}{2}\sigma}$ into $V_{r-\rho,s-\sigma}$ for $0 \le t \le 1$ and satisfies $|W(X^t_\phi - \mathrm{id})| \le \ldots$"; the derivative estimate by Lemma C.1). The general-$\nu$ formulation and the holomorphy in (b) are ours (Pöschel asserts "real analytic").*

*Proof.* Regard $\mathbb{C}^N$ as a real normed space of dimension $2N$ with norm $\nu$, extend $F$ by 0 outside $\Omega$ (Remark 4.1); $F$ is $C^1$ on $\Omega$ ([K:`bp:p1:holomorphic-regularity`], restriction of scalars).

**Step S6.3: (a) The maximal solution stays in a compact set.** Let $z \in \Omega^{1/2}$, $(T_-, T_+)$ and $\gamma$ the maximal interval and solution (Theorem 4.13). For $t \in [0, T_+)$, $\nu(\gamma'(t)) \le \varepsilon$, so $\nu(\gamma(t) - z) \le \varepsilon t$ ([K:`bp:p1:ext-mvt`] in one variable). If $T_+ \le 1$, then $\gamma(t) \in K = \bar{B}_\nu(z, \varepsilon) \subset \bar{B}_\nu(z, \frac{1}{2}) \subset \Omega$ for $t \in [0, T_+)$, with $K$ compact, so $T_+ = \infty$ by Corollary 4.15, a contradiction. Hence $T_+ > 1$; symmetrically (time reversal, Lemma 4.3(b)) $T_- > -1$. Uniqueness on an interval $J \ni 0$ is Theorem 4.11; the displacement bound was shown.

**Step S6.4: (b) Picard iterates.** Fix $z_0 \in \Omega^{1/2}$. As $\bar{B}_\nu(z_0, \frac{1}{2})$ is compact in the open $\Omega$, there is $\eta > 0$ with $\bar{B}_\nu(z_0, \frac{1}{2} + \eta) \subset \Omega$. Let $\Lambda = \sup\{|DF(w)|_\nu : w \in \bar{B}_\nu(z_0, \frac{1}{2} + \eta)\} < \infty$; $F$ is $\Lambda$-Lipschitz on this convex ball ([K:`bp:p1:ext-mvt`]). For $(t,z) \in [-1,1] \times B_\nu(z_0, \eta)$ put $y_0(t,z) = z$, $y_{k+1}(t,z) = z + \int_0^t F(y_k(\tau,z))\,d\tau$. By induction on $k$: $\nu(y_k(t,z) - z) \le |t|\varepsilon$ (so $y_k(t,z) \in \bar{B}_\nu(z,\varepsilon) \subset \bar{B}_\nu(z_0, \frac{1}{2} + \eta) \subset \Omega$, and the integral is defined); $y_k$ is continuous; $y_k(t,\cdot)$ is holomorphic; $D_z y_k$ is continuous in $(t,z)$. The last two follow from (E1) applied to $t\int_0^1 F(y_k(t\tau', z))\,d\tau'$, whose $z$-derivative is $t\int_0^1 DF(y_k)D_z y_k\,d\tau'$.

**Step S6.5: (b) Convergence and identification.** $\nu(y_{k+1} - y_k)(t,z) \le \varepsilon\Lambda^k|t|^{k+1}/(k+1)!$ by induction, so $y_k \to y_\infty$ uniformly on $[-1,1] \times B_\nu(z_0, \eta)$. The limit is continuous and satisfies $y_\infty(t,z) = z + \int_0^t F(y_\infty)\,d\tau$ ($F$ is $\Lambda$-Lipschitz), so $y_\infty(\cdot, z)$ is $C^1$ with derivative $F(y_\infty)$, i.e. a solution in $\Omega$ on $[-1,1]$ through $z$, and $y_\infty(t,z) = X^t(z)$ by (a). Hence $X^t$ is continuous in $(t,z)$ and, by (E2), holomorphic on $B_\nu(z_0,\eta)$ with $D_z y_k \to D_z X^t$ uniformly on $[-1,1] \times B_\nu(z_0, \eta/2)$, so

$D_z X^t(z)$ is jointly continuous. As $z_0$ was arbitrary, (b) holds except for the integral equation. Differentiating $X^t(w) = w + \int_0^t F(X^\tau w)\,d\tau$ in $w$ under the integral ((E1); the integrand's $w$-derivative $DF(X^\tau w)D_w X^\tau(w)$ is jointly continuous) gives the equation for $Y$.

**Step S6.6: (c).** Let $z \in \Omega^1$, $w = X^1(z)$. Then $\nu(w-z) \le \varepsilon \le \frac{1}{2}$, so $\bar B_\nu(w, \frac12) \subset \bar B_\nu(z,1) \subset \Omega$, i.e. $w \in \Omega^{1/2}$. The map $t \mapsto \gamma_z(1+t)$ is a solution on $[-1,0]$ through $w$, so it equals $\gamma_w$ there (Theorem 4.11) and $X^{-1}(w) = \gamma_z(0) = z$. The statement for $X^1 \circ X^{-1}$ is this one for $-F$ (Lemma 4.3(b)). For the derivative: $B_\nu(z, \frac12) \subset \Omega^{1/2}$ (as above), $G = X^t - \mathrm{id}$ is holomorphic there by (b) with $\nu(G) \le \varepsilon$ by (a); Lemma 5.8 with $R = \frac12$ gives $|DG(z)|_\nu \le 2\varepsilon$.

**Step S6.7: (d),(e).** (d): $\Omega^{1/2} + \tau = \Omega^{1/2}$ since the $\nu$-balls translate; $t \mapsto \gamma_z(t) + \tau$ is a solution through $z+\tau$ (Lemma 4.3(c)), so equals $\gamma_{z+\tau}$ by uniqueness. (e): apply (a) to $E = \mathbb{R}^N$, $W = \Omega \cap \mathbb{R}^N$, the real field $F|_W$ (it is $C^1$ and real by hypothesis, [K:`bp:p1:real-restriction`]) and the compact set $\bar B_\nu(x,\varepsilon) \cap \mathbb{R}^N \subset W$: this gives a real solution on $[-1,1]$ through $x$, which is also a solution in $\Omega$, hence equals $\gamma_x$. □

**Lemma 6.5** (Symplectic flow)**.** *In Lemma 6.4 let $N = 2n$, $\Omega \subset \mathbb{C}^n \times \mathbb{C}^n$, and $F = X_\phi$ with $\phi$ holomorphic on $\Omega$. Then for $z \in \Omega^{1/2}$, $|t| \le 1$, $Y = DX^t(z)$ satisfies*

$$Y^\top J Y = J, \qquad Y J Y^\top = J, \qquad \det Y = \pm 1$$

*($Y^\top$ the plain transpose of the complex matrix).*
*Source: Standard (Hamiltonian flows are symplectic; used implicitly by Pos93 pp. 211–212, "canonical transformation $\Phi$").*

*Proof.* **Step S6.8: Variational equation.** By Lemma 6.4(b), $Y \in C^1$ with $Y' = DF(X^t z)Y$, $Y(0) = \mathbf{1}$.

**Step S6.9: The field.** $F = J\nabla\phi$, so $DF = JS$ with $S = D\nabla\phi$ the Hessian, symmetric ([K:`bp:p1:ext-schwarz`]). Then $DF^\top J = S^\top J^\top J = S$ and $J\,DF = J^2 S = -S$, so $DF^\top J + J\,DF = 0$.

**Step S6.10: Conservation.** $\frac{d}{dt}(Y^\top J Y) = Y^\top(DF^\top J + J\,DF)Y = 0$, so $Y^\top J Y = J$ for all $t$. Taking determinants, $\det(Y)^2 = 1$; $Y$ is invertible and inverting $Y^\top J Y = J$ gives $Y^{-1}J^{-1}Y^{-\top} = J^{-1}$, that is, $J = Y J Y^\top$ ($J^{-1} = -J$). □

**Corollary 6.6** (Canonical maps: bracket invariance and conjugation)**.** *In the setting of Lemma 6.5, let $z \in \Omega^{1/2}$, $\Phi = X^t$ ($|t| \le 1$), and $u, v, H$ holomorphic on a neighbourhood of $\Phi(z)$. Then $u \circ \Phi$ is holomorphic near $z$ and*

- *(a) $\{u\circ\Phi, v\circ\Phi\}(z) = \{u,v\}(\Phi(z))$;*
- *(b) $D\Phi(z)\,X_{H\circ\Phi}(z) = X_H(\Phi(z))$;*
- *(c) if $y : J \to \Omega^{1/2}$ satisfies $\dot y = X_{H\circ\Phi}(y)$ with $H$ holomorphic on a neighbourhood of $\Phi(y(J))$, then $x = \Phi \circ y$ satisfies $\dot x = X_H(x)$.*

*Source: Standard.*

*Proof.* **Step S6.11: Chain rule.** $\Phi$ is holomorphic (Lemma 6.4(b)) and $\nabla(u\circ\Phi) = D\Phi^\top(\nabla u)\circ\Phi$.

**Step S6.12: (a).** $\{u\circ\Phi, v\circ\Phi\} = \nabla u^\top\, Y J Y^\top\, \nabla v = \nabla u^\top J \nabla v$ at $\Phi(z)$ by Lemma 6.5.

**Step S6.13: (b),(c).** $D\Phi\, X_{H\circ\Phi} = Y\,J\,Y^\top(\nabla H)\circ\Phi = J(\nabla H)\circ\Phi = X_H \circ \Phi$; (c) is the chain rule $\dot x = D\Phi(y)\dot y = D\Phi(y) X_{H\circ\Phi}(y)$. □

For real data: if $\phi$ is real on real points then so is $F = X_\phi$ (Step S6.7), $\Phi$ maps real points to real points, its derivative at a real point is a real matrix ([K:`bp:p1:real-restriction`]), and (a)–(c) hold for real $u, v, H$ with the real restriction of $X$ ([K:`bp:p1:hamiltonian-real`]).

6.3. **Taylor's formula along the flow.**

**Lemma 6.7** (Derivative along the flow)**.** *Let $\Omega \subset \mathbb{C}^n \times \mathbb{C}^n$, $\phi$, $F = X_\phi$, $\varepsilon \le \frac{1}{2}$ as in Lemma 6.5; let $\Omega_1 \subset \Omega$ be open and $u$ holomorphic on $\Omega_1$, and $z \in \Omega^{1/2}$ with $X^t(z) \in \Omega_1$ for $|t| \le 1$. Then $g(t) = u(X^t(z))$ is $C^\infty$ on $[-1,1]$ and $g^{(j)}(t) = (\mathrm{ad}_\phi^j u)(X^t(z))$ for all $j \ge 0$ (each $\mathrm{ad}_\phi^j u$ is holomorphic on $\Omega_1$).*

*Source: Pos93 p. 211 ("Taylor's formula" for $h \circ X_\phi^t$); standard.*

*Proof.* **Step S6.14: One derivative.** For $v$ holomorphic on $\Omega_1$, $v$ is real-differentiable with derivative $Dv$, and $\gamma = X^\cdot(z)$ has derivative $X_\phi(\gamma)$, so $\frac{d}{dt} v(\gamma(t)) = Dv(\gamma) \cdot X_\phi(\gamma) = \{v, \phi\}(\gamma(t))$ (Definition 6.2), which is continuous in $t$.

**Step S6.15: Induction.** Apply the previous step to $v = \mathrm{ad}_\phi^j u$, holomorphic on $\Omega_1$ (derivatives of holomorphic functions, [K:`bp:p1:holomorphic-regularity`]). □

**Lemma 6.8** (Taylor with integral remainder along the flow)**.** *In the setting of Lemma 6.7, for every $N \ge 1$,*

$$u(X^1 z) = \sum_{j=0}^{N-1} \frac{1}{j!} (\mathrm{ad}_\phi^j u)(z) + R_N(z),$$

$$R_N(z) = \int_0^1 \frac{(1-t)^{N-1}}{(N-1)!} (\mathrm{ad}_\phi^N u)(X^t z)\, dt, \qquad |R_N(z)| \le \frac{1}{N!} \max_{0 \le t \le 1} |(\mathrm{ad}_\phi^N u)(X^t z)|.$$

*In particular ($N = 1$ for $f_0$, $N = 2$ for $h$), if $h, f_0$ are holomorphic on $\Omega_1$ and the trajectory of $z$ stays in $\Omega_1$,*

$$(h + f_0)(X^1 z) = h(z) + \{h, \phi\}(z) + f_0(z) + \int_0^1 \big\{(1-t)\{h,\phi\} + f_0,\ \phi\big\}(X^t z)\, dt. \tag{1}$$

*Source: Pos93 p. 211, displayed formula for $H_0 \circ \Phi$.*

*Proof.* **Step S6.16: Taylor.** Apply the Taylor formula with integral remainder for $C^N$ functions of one real variable (external fact: induction on $N$ with one integration by parts per step, [K:`bp:p1:ext-integral`]) to $g(t) = u(X^t z)$ at 0, evaluated at 1, using $g^{(j)}(0) = (\mathrm{ad}_\phi^j u)(z)$ and $g^{(N)}(t) = (\mathrm{ad}_\phi^N u)(X^t z)$ (Lemma 6.7). The bound uses $\int_0^1 (1-t)^{N-1}/(N-1)!\, dt = 1/N!$.

**Step S6.17: The displayed formula.** Apply the case $N = 2$ to $h$ ($\mathrm{ad}^2 h = \{\{h,\phi\},\phi\}$) and the case $N = 1$ to $f_0$ and add: the remainders are $\int_0^1 (1-t)\{\{h,\phi\},\phi\}(X^t z)\, dt + \int_0^1 \{f_0, \phi\}(X^t z)\, dt$, and brackets are bilinear. □

6.4. **Lemma B.5: composition in the weighted norm.**

**Lemma 6.9** (Iterated brackets)**.** *Let $\phi \in \mathcal{O}(r', s')$, $u \in \mathcal{O}(r, s)$ have finite norms, $0 < \rho < r$, $0 < \sigma < s$, $r + \rho \le r'$, $s + \sigma \le s'$, and $\eta = 2|\phi|_{r',s'}/(\rho\sigma)$. Then for every $h \ge 1$,*

$$|\mathrm{ad}_\phi^h u|_{r-\rho, s-\sigma} \le \Big(\frac{2h^2}{e\rho\sigma}\Big)^h \frac{h!}{(2h)!} |\phi|_{r',s'}^h |u|_{r,s}, \qquad \text{hence} \qquad \frac{1}{h!} |\mathrm{ad}_\phi^h u|_{r-\rho,s-\sigma} \le \Big(\frac{e\eta}{4}\Big)^h |u|_{r,s}.$$

*Source: Pos93 p. 215 (the displayed estimates for $|\mathrm{ad}_\phi^h u|$ with $\tilde\rho = \rho/h$, $\tilde\sigma = \sigma/h$).*

*Proof.* **Step S6.18: Setup.** Fix $h \ge 1$, $\kappa = \rho/h$, $\lambda = \sigma/h$, $r_i = r - i\kappa$, $s_i = s - i\lambda$ ($0 \le i \le h$), so $r_h = r - \rho$, $s_h = s - \sigma$. By induction each $\mathrm{ad}_\phi^i u \in \mathcal{O}(r_i, s_i)$.

**Step S6.19: One step.** Let $1 \le i \le h$. Since $\mathrm{ad}_\phi^i u = \{\mathrm{ad}_\phi^{i-1} u, \phi\} = -\{\phi, \mathrm{ad}_\phi^{i-1} u\}$, Lemma 5.7 with $(u, v; r_0, s_0; r, s; \rho, \sigma) \to (\phi, \mathrm{ad}_\phi^{i-1} u; r', s'; r_{i-1}, s_{i-1}; \kappa, \lambda)$ applies (note $\kappa < r_{i-1}$, $r_i < r'$, and

likewise for $s$) and gives $|\mathrm{ad}_\phi^i u|_{r_i,s_i} \le \frac{1}{e}\Big(\frac{1}{(r'-r_{i-1}+\kappa)\lambda}+\frac{1}{(s'-s_{i-1}+\lambda)\kappa}\Big)|\phi|_{r',s'}|\mathrm{ad}_\phi^{i-1}u|_{r_{i-1},s_{i-1}}$. Now $r'-r_{i-1}+\kappa=(r'-r)+i\kappa\ge\rho+i\kappa=(h+i)\kappa$ and likewise $s'-s_{i-1}+\lambda\ge(h+i)\lambda$; so the bracket is $\le\frac{2}{e(h+i)\kappa\lambda}$.

**Step S6.20: Product.** Multiplying over $i=1,\dots,h$ and using $\prod_{i=1}^h(h+i)=(2h)!/h!$ and $\kappa\lambda=\rho\sigma/h^2$ gives the first estimate. For the second, $(2h)!\ge(2h/e)^{2h}$ (from $e^N\ge N^N/N!$), so $\frac{1}{h!}|\mathrm{ad}_\phi^h u|\le\frac{(\eta h^2/e)^h}{(2h)!}|u|\le\big(\eta\frac{h^2}{e}\cdot\frac{e^2}{4h^2}\big)^h|u|=(\frac{e\eta}{4})^h|u|$. □

**Theorem 6.10** (Lemma B.5: $|u\circ X_\phi^1|\le(1-\eta)^{-1}|u|$)**.** *Let $D\subset\mathbb{C}^n$, $0<\rho<r\le r'-\rho$, $0<\sigma<s\le s'-\sigma$, $\phi\in\mathcal{O}(r',s')$ with finite norm and $\eta=2|\phi|_{r',s'}/(\rho\sigma)\le\frac12$. Put $\Omega=\mathcal{D}_{r'-\rho,s'-\sigma}\supset\mathcal{D}_{r,s}$ and $\nu=\nu_{\rho,\sigma}$.*

*(i) $\nu(X_\phi)\le\eta/2\le\frac14$ on $\Omega$; the flow $X^t=X_\phi^t$, $|t|\le1$, of Lemma 6.4 is defined on $\Omega^{1/2}\supset\mathcal{D}_{r-\rho,s-\sigma}$, with $X^t(\mathcal{D}_{r-\rho,s-\sigma})\subset\mathcal{D}_{r,s}$.*

*(ii) For every $u\in\mathcal{O}(r,s)$ with finite norm, $\sum_{h\ge0}\frac{1}{h!}\mathrm{ad}_\phi^h u$ converges in $|\cdot|_{r-\rho,s-\sigma}$ to a function $S\in\mathcal{O}(r-\rho,s-\sigma)$ and $S=u\circ X_\phi^1$ on $\mathcal{D}_{r-\rho,s-\sigma}$. In particular $u\circ X_\phi^1\in\mathcal{O}(r-\rho,s-\sigma)$ and*

$$|u\circ X_\phi^1|_{r-\rho,s-\sigma}\le\Big(1-\frac{e\eta}{4}\Big)^{-1}|u|_{r,s}\le\Big(1-\frac{2}{\rho\sigma}|\phi|_{r',s'}\Big)^{-1}|u|_{r,s}.$$

*(iii) The same holds for $X_\phi^t$, $0\le t\le1$ (and $-1\le t\le0$), with $\eta$ replaced by $|t|\eta$.*

*(iv) If $\phi$ and $u$ are real on real points, $u\circ X_\phi^1$ is real on real points of $\mathcal{D}_{r-\rho,s-\sigma}$.*

*Source: Pos93 pp. 214–215, Lemma B.5 (with $|\phi|_{r_0,s_0}<\rho\sigma/2$ there).*
*Source gap: Pöschel writes "Consider the Lie series expansion $u\circ X_\phi^1=\sum\frac{1}{h!}\mathrm{ad}_\phi^h u$" and estimates the series; the equality with the composition is not proved on p. 214. It is proved here for $\eta\le\frac12$; see Remark 6.1. The remark that the smallness condition implies $X_\phi^t:V_{r-\rho,s-\sigma}\to V_{r,s}$ is (i).*

*Proof.* **Step S6.21: The field and the domain.** Since $r\le r'-\rho$, $s\le s'-\sigma$: $\mathcal{D}_{r,s}\subset\Omega$. Lemma 6.3 (for $\phi\in\mathcal{O}(r',s')$ and losses $\rho,\sigma$) gives $\nu(X_\phi)\le|\phi|/(\rho\sigma)=\eta/2=:\varepsilon\le\frac14$ on $\Omega$. If $z\in\mathcal{D}_{r-\rho,s-\sigma}$, then $\bar B_\nu(z,\frac12)=\bar P(z;\sigma/2,\rho/2)\subset\mathcal{D}_{r-\rho/2,s-\sigma/2}\subset\mathcal{D}_{r,s}\subset\Omega$ (Definition 5.1(b)), so $z\in\Omega^{1/2}$. By Lemma 6.4(a), $X^t(z)\in\bar P(z;\varepsilon\sigma,\varepsilon\rho)\subset\mathcal{D}_{r-(1-\varepsilon)\rho,s-(1-\varepsilon)\sigma}\subset\mathcal{D}_{r,s}$. This is (i).

**Step S6.22: The series.** By Lemma 6.9 (applicable: $r+\rho\le r'$, $s+\sigma\le s'$), $\frac{1}{h!}\mathrm{ad}_\phi^h u\in\mathcal{O}(r-\rho,s-\sigma)$ has norm $\le(e\eta/4)^h|u|_{r,s}$, and $e\eta/4<1$. The sum of norms is $\le(1-e\eta/4)^{-1}|u|_{r,s}$. By Lemma 5.4(a) the partial sums converge uniformly on $\mathcal{D}_{r-\rho,s-\sigma}$, so $S$ is holomorphic ((E2)) and periodic, and termwise integration ((E3)) shows that its Fourier coefficients are the sums of the coefficients; the triangle inequality in $\sum_k|\hat S_k(I)|e^{2\pi|k|_1(s-\sigma)}$ gives $|S|_{r-\rho,s-\sigma}\le(1-e\eta/4)^{-1}|u|_{r,s}$; and $1-e\eta/4\ge1-\eta$.

**Step S6.23: The remainder tends to 0.** Let $z\in\mathcal{D}_{r-\rho,s-\sigma}$, $t\in[0,1]$, $w=X^t(z)$. By Step S6.21, $w\in\mathcal{D}_{r-\hat\rho,s-\hat\sigma}$ with $\hat\rho=(1-\varepsilon)\rho$, $\hat\sigma=(1-\varepsilon)\sigma$, $0<\hat\rho<r$, $r+\hat\rho\le r'$, and likewise for $s$. Lemma 6.9 with losses $(\hat\rho,\hat\sigma)$ and $\hat\eta=2|\phi|/(\hat\rho\hat\sigma)=\eta/(1-\varepsilon)^2\le\frac{16}{9}\eta\le\frac89$ gives $|\mathrm{ad}_\phi^N u|_{r-\hat\rho,s-\hat\sigma}\le N!(e\hat\eta/4)^N|u|_{r,s}$, and $e\hat\eta/4\le\frac{2e}{9}<1$. By Lemma 5.4(a), $|(\mathrm{ad}_\phi^N u)(X^tz)|\le N!(e\hat\eta/4)^N|u|_{r,s}$. Lemma 6.8 (with $\Omega_1=\mathcal{D}_{r,s}$, $\Omega_1$-trajectory by Step S6.21) gives $|R_N(z)|\le(e\hat\eta/4)^N|u|_{r,s}\to0$. Hence $u(X^1z)=\lim_N\sum_{j<N}\frac{1}{j!}(\mathrm{ad}_\phi^j u)(z)=S(z)$. This proves (ii).

**Step S6.24: (iii), (iv).** $X_\phi^t=X_{t\phi}^1$ (the field $X_{t\phi}=tX_\phi$ and $s\mapsto X_\phi^{ts}(z)$ solves it), and $|t\phi|=|t||\phi|$, so (iii) is (ii) for $t\phi$. (iv): Lemma 6.4(e) with $F=X_\phi$ real on real points (a real-valued $\phi$ on real points has real partials there). □

**Proposition 6.11** (Lie transform with first-order remainder, in the weighted norm)**.** *In the setting of Theorem 6.10, let $h\in\mathcal{O}(r,s)$ (any norm, possibly $\infty$) with $w:=\{h,\phi\}\in\mathcal{O}(r,s)$ of finite norm,*

*and $f_0 \in \mathcal{O}(r,s)$ with finite norm. Then on $\mathcal{D}_{r-\rho,s-\sigma}$*

$$(h+f_0)\circ X_\phi^1 = h+w+f_0+\mathcal{R}, \qquad \mathcal{R} = \sum_{j\geq 1}\Big(\frac{1}{(j+1)!}\operatorname{ad}_\phi^j w + \frac{1}{j!}\operatorname{ad}_\phi^j f_0\Big) = \int_0^1 \{(1-t)w+f_0,\phi\}\circ X_\phi^t\,dt,$$

*the series converges in $|\cdot|_{r-\rho,s-\sigma}$, $\mathcal{R}\in\mathcal{O}(r-\rho,s-\sigma)$, and*

$$|\mathcal{R}|_{r-\rho,s-\sigma} \leq \sum_{j\geq 1}\Big(\frac{e\eta}{4}\Big)^j\Big(\frac{|w|_{r,s}}{j+1}+|f_0|_{r,s}\Big) \leq \frac{\eta}{1-\eta}\Big(\tfrac{1}{2}|w|_{r,s}+|f_0|_{r,s}\Big).$$

*The integral form is Lemma 6.8, (1); the pointwise identity requires no bound on $h$ itself.*
*Source: Pos93 p. 211 (the decomposition $H_0\circ\Phi = h+\{h,\phi\}+f_0+\int\ldots$) and p. 212–213 (estimate of $f_+$); the series form of the remainder is ours.*

*Proof.* **Step S6.25: Derivatives of $h$.** Since $\operatorname{ad}_\phi^1 h = w$, $\operatorname{ad}_\phi^j h = \operatorname{ad}_\phi^{j-1} w$ $(j\geq 1)$. In Lemma 6.7 (with $\Omega_1 = \mathcal{D}_{r,s}$) $g(t) = h(X^t z)$ has $g^{(j)}(t) = (\operatorname{ad}_\phi^{j-1} w)(X^t z)$; Taylor's formula to order $N$ gives $h(X^1 z) = h(z) + \sum_{i=0}^{N-2}\frac{1}{(i+1)!}(\operatorname{ad}_\phi^i w)(z) + R'_N$, with $|R'_N| \leq \frac{1}{N!}\max_t|(\operatorname{ad}_\phi^{N-1} w)(X^t z)|$. By Step S6.23 (Lemma 6.9 with $u=w$ and the losses $\hat\rho,\hat\sigma$) this is $\leq \frac{(N-1)!}{N!}(e\hat\eta/4)^{N-1}|w|_{r,s}\to 0$. So $h\circ X^1 = h + \sum_{i\geq 0}\frac{1}{(i+1)!}\operatorname{ad}_\phi^i w$ pointwise, the term $i=0$ being $w$.

**Step S6.26: Derivatives of $f_0$.** By Theorem 6.10(ii), $f_0\circ X^1 = \sum_{j\geq 0}\frac{1}{j!}\operatorname{ad}_\phi^j f_0$.

**Step S6.27: Sum and bound.** Adding gives the first identity with $\mathcal{R}$ as stated (terms $j=0$ are $w$, $f_0$). Lemma 6.9 bounds $\frac{1}{j!}|\operatorname{ad}_\phi^j w| \leq (e\eta/4)^j|w|$, hence $\frac{1}{(j+1)!}|\operatorname{ad}_\phi^j w| \leq \frac{1}{j+1}(e\eta/4)^j|w|$, and similarly for $f_0$. The geometric sum with $\frac{1}{j+1}\leq\frac{1}{2}$ and $e\eta/4\leq\eta$ gives the last bound. Convergence in the norm and membership of $\mathcal{O}(r-\rho,s-\sigma)$ are as in Step S6.22.

**Step S6.28: Integral form.** Equality with the integral is (1), because $\{h,\phi\}=w$; the trajectory stays in $\mathcal{D}_{r,s}$ by Step S6.21. □

**Corollary 6.12** (Integral remainders in the weighted norm)**.** *In the setting of Theorem 6.10 let $v_t\in\mathcal{O}(r,s)$ $(0\leq t\leq 1)$ have finite norms with $V=\sup_t|v_t|_{r,s}<\infty$, and suppose $(t,z)\mapsto v_t(z)$ and $(t,z)\mapsto Dv_t(z)$ are continuous on $[0,1]\times\mathcal{D}_{r,s}$. Then $G(z)=\int_0^1 v_t(X_\phi^t z)\,dt$ lies in $\mathcal{O}(r-\rho,s-\sigma)$ and $|G|_{r-\rho,s-\sigma}\leq(1-\eta)^{-1}V$.*
*Source: Pos93 p. 212–213 ("$|u\circ X_\phi^t|\leq(1-2|\phi|/\rho\sigma)^{-1}|u|$ uniformly for $0\leq t\leq 1$", applied inside the integral).*

*Proof.* **Step S6.29: Integrand.** $g_t = v_t\circ X^t \in \mathcal{O}(r-\rho,s-\sigma)$ with $|g_t|\leq(1-t\eta)^{-1}V\leq(1-\eta)^{-1}V$ (Theorem 6.10(iii)). The map $(t,z)\mapsto g_t(z)$ is continuous and $D_z g_t = Dv_t(X^t z)D_z X^t(z)$ is continuous (Lemma 6.4(b)).

**Step S6.30: Holomorphy and coefficients.** By (E1), $G$ is holomorphic; it is periodic. Fubini on $[0,1]\times[0,1]^n$ gives $\hat G_k(I) = \int_0^1 (\hat{g_t})_k(I)\,dt$, so by Tonelli $|G|_{I,s-\sigma}\leq\int_0^1|g_t|_{I,s-\sigma}\,dt\leq(1-\eta)^{-1}V$. □

## 6.5. The canonical map $\Phi = X_\phi^1$ on weighted domains.

**Proposition 6.13** (Iterative-Lemma package for $\Phi$)**.** *Let $D\subset\mathbb{C}^n$, $\phi\in\mathcal{O}(r,s)$ with finite norm, $0<\rho<r/2$, $0<\sigma<s/2$, and $\varepsilon := |\phi|_{r,s}/(\rho\sigma)\leq\frac{1}{2}$. Put $\Omega=\mathcal{D}_{r-\rho,s-\sigma}$, $\nu=\nu_{\rho,\sigma}$, $\Phi = X_\phi^1$, $\Psi = X_\phi^{-1}$ (Lemma 6.4). Then:*

- *(a) $\nu(X_\phi)\leq\varepsilon$ on $\Omega$; $\Phi,\Psi$ are defined and holomorphic on $\mathcal{D}_{r-\frac{3}{2}\rho,s-\frac{3}{2}\sigma}\subset\Omega^{1/2}$ and map it into $\Omega$. For every $t\in[-1,1]$, $\nu(X^t-\mathrm{id})\leq|t|\varepsilon$; explicitly $|\Phi_\theta-\theta|_\infty\leq|\phi|_{r,s}/\rho$ and $|\Phi_I-I|_\infty\leq|\phi|_{r,s}/(e\sigma)$, and the same for $\Psi$.*

*(b)* $\Phi(\theta+k,I)=\Phi(\theta,I)+(k,0)$ *for* $k\in\mathbb{Z}^n$*; if* $\phi$ *is real on real points,* $\Phi$ *maps real points to real points.*
*(c) (Derivative) on* $\mathcal{D}_{r-2\rho,s-2\sigma}$*:* $|D\Phi-\mathbf{1}|_\nu\le 2\varepsilon$*, i.e.* $|W(D\Phi-\mathbf{1})W^{-1}|\le 2|\phi|_{r,s}/(\rho\sigma)$ *for* $W=\mathrm{diag}(\sigma^{-1},\rho^{-1})$*.*
*(d) (Canonical)* $(D\Phi)^\top J\,D\Phi=J$*, and Corollary 6.6 holds for* $\Phi$*.*
*(e) (Inverse)* $\Psi\circ\Phi=\mathrm{id}=\Phi\circ\Psi$ *on* $\mathcal{D}_{r-2\rho,s-2\sigma}$*, and* $\Phi(\mathcal{D}_{r-2\rho,s-2\sigma})\subset\mathcal{D}_{r-\frac{3}{2}\rho,s-\frac{3}{2}\sigma}$*.*

*Source: Pos93 p. 212 ("Equivalently,* $X_\phi$ *satisfies* $|WX_\phi|_{\mathcal{P}}\le\frac{q\delta}{\alpha\rho\sigma}\le\frac12$ *…* $X_\phi^t$ *maps* $V_{r-\frac32\rho,s-\frac32\sigma}$ *into* $V_{r-\rho,s-\sigma}$ *…* $|W(DX_\phi^t-I_{2n})W^{-1}|\le 2\frac{q\delta}{\alpha\rho\sigma}$ *uniformly on* $V_{r-2\rho,s-2\sigma}$*"), with* $|\phi|_{r,s}$ *in place of* $q\delta/\alpha$*.*

*Proof.* **Step S6.31: Field.** Lemma 6.3 gives $\nu(X_\phi)\le\varepsilon\le\frac12$ on $\Omega$, and $\Omega$ is open. For $z\in\mathcal{D}_{r-c\rho,s-c\sigma}$ $(c\ge\frac32)$ the ball $\bar B_\nu(z,c-1)=\bar P(z;(c-1)\sigma,(c-1)\rho)$ lies in $\Omega$ (Definition 5.1(b)): so $\mathcal{D}_{r-\frac32\rho,s-\frac32\sigma}\subset\Omega^{1/2}$ and $\mathcal{D}_{r-2\rho,s-2\sigma}\subset\Omega^1$.

**Step S6.32: (a),(b).** Lemma 6.4(a),(b),(d),(e) (with $\tau=(k,0)$: $\Omega+\tau=\Omega$ and $X_\phi$ is periodic, as $\phi$ is). $X^t(z)\in\bar P(z;\varepsilon\sigma,\varepsilon\rho)\subset\mathcal{D}_{r-\rho,s-\sigma}$ for $z\in\mathcal{D}_{r-\frac32\rho,s-\frac32\sigma}$ because $\varepsilon\le\frac12$. The explicit component bounds are $|X^t_\theta-\theta|\le|t|\sup|\partial_I\phi|\le|t||\phi|/\rho$ and $|X^t_I-I|\le|t|\sup|\partial_\theta\phi|\le|t||\phi|/(e\sigma)$ (mean value inequality along the trajectory and Lemma 6.3). For $\Psi=X_\phi^{-1}=X^1_{-\phi}$ the same with $-\phi$.

**Step S6.33: (c)–(e).** (c) is Lemma 6.4(c) with $\Omega^1$ and Step S6.31. (d) is Lemma 6.5 and Corollary 6.6. (e): Lemma 6.4(c) for $z\in\Omega^1$ gives $\Psi(\Phi z)=z$ and $\Phi(\Psi z)=z$; and $\Phi(z)\in\bar B_\nu(z,\varepsilon)\subset\mathcal{D}_{r-\frac32\rho,s-\frac32\sigma}$ for $z\in\mathcal{D}_{r-2\rho,s-2\sigma}$ $(\varepsilon\le\frac12)$. □

6.6. **Variants for the norm ♯.**

**Proposition 6.14** (Composition in the norm ♯)**.** *Lemma 6.3, Lemma 6.9, Theorem 6.10(i)–(iv), Proposition 6.11, Corollary 6.12 and Proposition 6.13 hold with every weighted norm* $|\cdot|$ *replaced by* $|\cdot|^\sharp$ *(Lemma 5.10), with the same constants and the same hypothesis* $\eta=2|\phi|^\sharp_{r',s'}/(\rho\sigma)\le\frac12$*. For the consumer with coefficient radius* $\Sigma=2\pi s$ *and loss* $\Delta\Sigma=2\pi\sigma$ *the parameter is* $\eta=4\pi|\phi|^\sharp/(\rho\,\Delta\Sigma)$ *(Section 5.9).*

*Proof.* **Step S6.34: Where the norm enters.** The geometry (flow, symplecticity, Taylor, displacement) uses the norm only through sup bounds $|u(z)|\le|u|^\sharp$ (Lemma 5.10, B.1(a)), which hold for ♯ as for the pointwise norm; the vector-field bound Lemma 6.3 uses the individual derivative estimates, valid for ♯.

**Step S6.35: Series.** Lemma 6.9 uses only the bracket estimate; Theorem 6.10 Step S6.22 uses the triangle inequality, valid for ♯ since $\sup_I|\sum_h c_k^h(I)|\le\sum_h\sup_I|c_k^h|$. The integral corollary uses $\sup_I\int_0^1|(\hat g_t)_k|\,dt\le\int_0^1\sup_I|(\hat g_t)_k|\,dt$ and Tonelli. □

*Remark* 6.15 (Use by Part 6 and track P; what is not provided). Part 6 applies Proposition 6.11 with $h$ the integrable part, $w=\{h,\phi\}=g_+-f_0$ by the homological equation (so $|w|$ is controlled by $|\chi|$ and $|T_Kf|$, not by $|h|$), $f_0=g+T_Kf$, and Theorem 6.10 for $(f-T_Kf)\circ X^1_\phi$, with the losses $(\rho,\sigma)$ from $(r,s)$ to $(r-\rho,s-\sigma)$ (so $\phi$ lives on $(r+\rho,s+\sigma)$, or one re-indexes as Pöschel: $\phi\in\mathcal{O}(r,s)$, $u\in\mathcal{O}(r-\rho,s-\sigma)$, result on $\mathcal{D}_{r-2\rho,s-2\sigma}$). The smallness used is $\eta=2|\phi|/(\rho\sigma)\le\frac12$; the displacement and derivative bounds of Proposition 6.13 need only $\varepsilon\le\frac12$. The sup-norm *fallback* of `PLAN.md` is not used. Not proved here: the stability transfer (Part 7) and the estimate of $|W(\Phi_1\circ\cdots\circ\Phi_N-\mathrm{id})|$ for compositions of the maps of successive steps, which is Part 6's (the displacement bounds of (a) add, since each $\Phi_i$ maps into the domain of $\Phi_{i+1}$ by the choice of losses).

## 7. The homological equation, the Iterative Lemma and the Normal Form Lemma (shared by tracks N, P)

This part is Section 2 of Pöschel 1993 (the Normal Form Lemma, pp. 192–196, with the Iterative Lemma, p. 193) and Appendix A (pp. 211–213), in the frozen conventions of Part 5: angles of period 1, weight $e^{2\pi|k|_1 s}$, geometric widths, max norms on $\mathbb{R}^n, \mathbb{C}^n$, and the Lie transform of Part 6. It is shared infrastructure: track P derives CP10's averaging lemma ($\alpha_1 = \alpha_2$, $\Lambda = \{0\}$) from Theorem 7.6, which is therefore stated for an arbitrary set $\Lambda \subset \mathbb{Z}^n$ (including $\{0\}$), arbitrary $n$, and an arbitrary domain $D \subset \mathbb{C}^n$, with all constants explicit. Nothing but Parts 5 and 6 is used.
Notation. $Z_K = \{k \in \mathbb{Z}^n : |k|_1 \le K\}$ for real $K \ge 0$ (the $\ell^1$ ball: it is the cut-off of $T_K$, Lemma 5.9); $\omega = \partial_I h$ is the frequency; $p > 1$ and $q = p/(p-1)$, so $p^{-1} + q^{-1} = 1$ (Pöschel takes $q = 9$, $p = \frac{9}{8}$; nothing below depends on that choice); for $\alpha > 0$ and $s > 0$

$$\bar\alpha := 2\pi\alpha, \qquad \kappa := 2\pi K s.$$

Two dictionaries with Pöschel's normalisation are used, one at a time (Remark 5.11). (i) *Symplectic scaling* $\varphi = 2\pi\theta$, $I_P = I/2\pi$: time and the values of Hamiltonians, generators and remainders are unchanged; $\omega_P = 2\pi\omega$, so $\alpha_P = 2\pi\alpha = \bar\alpha$; $r_P = r/2\pi$, so $\alpha_P r_P = \alpha r$ (not $\bar\alpha r$); $s_P = 2\pi s$, $\sigma_P = 2\pi\sigma$, $\rho_P = \rho/2\pi$, so $\rho_P\sigma_P = \rho\sigma$ and $e^{|k|_1 s_P} = e^{2\pi|k|_1 s}$; the Hessian-Lipschitz constant scales as $L_P = (2\pi)^2 L$ and the divisor-radius condition $KLr \le \alpha/p$ is the same condition in both conventions. (ii) *Fixed-action Hamiltonian scaling* (Part 1, Lemma 2.8) $\tilde H = 2\pi H(\varphi/2\pi, I)$: the action variable and $r$ are unchanged, but values of Hamiltonians, generators and norms are multiplied by $2\pi$, $\tilde\alpha = \bar\alpha$, and $\rho\tilde\sigma = 2\pi\rho\sigma$; the ratio $|\phi|/(\rho\sigma)$ is unchanged. No blanket substitution "$\alpha \to \bar\alpha$ with $r$ unchanged" is claimed. The quantities $\bar\alpha$ and $\kappa = 2\pi K s$ are Pöschel's $\alpha$ and $Ks$; the arithmetic below is redone in our variables and does not depend on either dictionary.
Lattice cut-offs (finding N06). Only the $\ell^1$ ball $Z_K$ occurs here, and $|k \cdot v| \le |k|_1 |v|_\infty$ is all that is used about it. Nonresonance of the geometry of resonances (Part 8) is obtained on the Euclidean ball $\{\|k\|_2 \le K_g\}$; since $Z_K \subset \{\|k\|_2 \le K\}$ ($\|k\|_2 \le |k|_1$), it holds on $Z_K$ as soon as $K = K_a \le K_g$. The factor $\sqrt{n}$ enters only through the radius condition $KLr \le \alpha/p$ with $L = \sqrt{n}M$ against the covering radius of Part 8, which holds iff $K_g \ge \sqrt{n} K_a$ (Part 9, Remark 9.1, Corollary 9.16); it is carried by the caller. The results below are insensitive to the choice of $K$.

### 7.1. External facts.

(E1) *Numerics.* $2/e < 1$; $e^{-3\ln 2} = \frac{1}{8}$; $e^2 < 8$, hence $\ln 2 > \frac{2}{3}$ and $12 \ln 2 > 8$; $\sum_{i \ge 1} 4^{1-i} = \frac{4}{3}$; $(1 - \frac{1}{2^7})^{-1} = \frac{128}{127}$, $(1 - \frac{1}{8})^{-1} = \frac{8}{7}$. Exists.
(E2) *Chain rule for holomorphic maps.* Compositions of holomorphic maps are holomorphic and $D(F \circ G) = DF \circ G \cdot DG$. Exists.

Besides these, [K:bp:p1:ext-mvt] (mean value inequality) and the external facts of Parts 5 and 6 are used.

### 7.2. Setting and the divisor bound.

**Definition 7.1** (Resonant setting)**.** Let $D \subset \mathbb{C}^n$, $r > 0$, $K \ge 0$, $\alpha > 0$, $L \ge 0$, $\Lambda \subset \mathbb{Z}^n$, and let $h$ be holomorphic on $V_r D$ (Definition 5.1), regarded as a $\theta$-independent function on $\mathcal{D}_{r,s}$ for every $s > 0$; $\omega = \partial_I h : V_r D \to \mathbb{C}^n$. We say that $(D, r, K, \alpha, L, \Lambda, h)$ satisfies

(N1) ($(\alpha, K)$-*nonresonant modulo* $\Lambda$) $|k \cdot \omega(I)| \ge \alpha$ for all $I \in D$ and $k \in Z_K \setminus \Lambda$ (Pöschel p. 192);
(N2) (*frequency Lipschitz bound*) $|\omega(I) - \omega(I_0)|_\infty \le L|I - I_0|_\infty$ whenever $I_0 \in D$, $I \in V_r D$, $|I - I_0|_\infty < r$.

For $\Lambda = \{0\}$ and $\Lambda = \mathbb{Z}^n$ these are the completely nonresonant and the trivial cases. If $0 \notin \Lambda$ then (N1) forces $D = \emptyset$ (take $k = 0$), which is harmless. For $g = \sum_{j \in J} g_j$ ($J$ finite, $g_j \in \mathcal{O}(r_j, s_j)$ with

finite norm, $r_j \ge r$, $s_j \ge s$) and $0 < \rho < r$, $0 < \sigma < s$ put

$$|g|_* := \sum_{j\in J} \frac{1}{2}\Big(\frac{\rho}{r_j - r + \rho} + \frac{\sigma}{s_j - s + \sigma}\Big)|g_j|_{r_j,s_j} \qquad \text{(depends on the decomposition).}$$

If $J = \{1\}$, $g_1 = g \in \mathcal{O}(r,s)$ then $|g|_* = |g|_{r,s}$.
Source: Pos93 p. 192 (nonresonance), p. 193 ($|g|_*$), p. 211–212 ("on the domain $D$ we have $|k\cdot\omega(I)| \ge \alpha \ldots$ by the mean value theorem and our assumptions on $Q$ and $r$"): the Lipschitz bound is stated as hypothesis (N2) instead of via $\sup|Q|$, in max norms.

**Lemma 7.2** (Lipschitz bound from the Hessian)**.** *If $\omega = \partial_I h$ is holomorphic on $V_rD$ and the operator norm of $D\omega = \partial_I^2 h$ (max norm on both sides) is $\le L$ on $V_rD$, then (N2) holds. In particular $L = nM_C$ if every entry of the Hessian has modulus $\le M_C$ (Lemma 2.5).*
*Source: Pos93 p. 212 (mean value theorem); standard.*

*Proof.* **Step S7.1: Segment.** For $I_0 \in D$, $I \in V_rD$ with $|I - I_0|_\infty < r$ the segment $[I_0, I]$ lies in the polydisc $P(I_0, r) \subset V_rD$ (convex, Definition 5.1). [K:`bp:p1:ext-mvt`] gives $|\omega(I) - \omega(I_0)|_\infty \le L|I - I_0|_\infty$. □

**Lemma 7.3** (Divisor bound)**.** *Assume (N1), (N2) and $KLr \le \alpha/p$. Then $|k\cdot\omega(I)| \ge \alpha - KLr \ge \alpha/q$ for all $I \in V_rD$ and $k \in Z_K \setminus \Lambda$.*
*Source: Pos93 p. 211–212 ("$|k\cdot\omega(I)| \ge \alpha - MKr \ge (1 - \frac{1}{p})\alpha = \alpha/q$").*

*Proof.* **Step S7.2: Move to $D$.** Let $I \in V_rD$ and choose $I_0 \in D$ with $|I - I_0|_\infty < r$. Then $|k\cdot\omega(I)| \ge |k\cdot\omega(I_0)| - |k|_1|\omega(I) - \omega(I_0)|_\infty \ge \alpha - KLr$ by (N1), (N2). Finally $\alpha - \alpha/p = \alpha/q$. □

## 7.3. **The homological equation.**

**Proposition 7.4** (Homological equation)**.** *Assume (N1), (N2) and $KLr \le \alpha/p$. Let $s > 0$, $f \in \mathcal{O}(r,s)$ with $\delta = |f|_{r,s} < \infty$, and define*

$$\xi = P_\Lambda T_K f, \qquad \phi = \sum_{k\in Z_K\setminus\Lambda} \frac{\hat f_k(I)}{2\pi i\, k\cdot\omega(I)}\,\chi_k(\theta), \qquad \chi_k(\theta) = e^{2\pi i k\cdot\theta}.$$

*Then $\xi, \phi \in \mathcal{O}(r,s)$ are finite sums of characters (trigonometric polynomials of degree $\le K$ in $\theta$), and*

*(a) $\{\phi, h\} + \xi = T_K f$ on $\mathcal{D}_{r,s}$, where $\{\phi, h\} = \partial_\theta\phi\cdot\omega$; $P_\Lambda\phi = 0$, $P_\Lambda\xi = \xi$, $\xi = P_{\Lambda\cap Z_K} f$;*
*(b) $|\xi|_{r,s} \le \delta$, $|T_K f|_{r,s} \le \delta$, and $|\phi|_{r,s} \le q\delta/\bar\alpha = q\delta/(2\pi\alpha)$;*
*(c) if $\Lambda = -\Lambda$, $f$ is real on real points and $\omega(I) \in \mathbb{R}^n$ for real $I \in V_rD$, then $\xi$ and $\phi$ are real on real points.*

*Source: Pos93 p. 211–212 ("$\xi = P_\Lambda T_K f$, $\phi = L^{-1}(T_K f - P_\Lambda T_K f)$, $\phi_k = f_k/(ik\cdot\omega(I)) \ldots |\xi|_{r,s} \le \delta$, $|\phi|_{r,s} \le q\delta/\alpha$"); the $2\pi$ in the divisor is the period-1 convention, so that the bound reads $q\delta/\bar\alpha$ as in the source.*

*Proof.* **Step S7.3: Well defined and holomorphic.** For $k \in Z_K \setminus \Lambda$, $k\cdot\omega$ is holomorphic on $V_rD$ and has modulus $\ge \alpha/q > 0$ (Lemma 7.3), so $\hat\phi_k := \hat f_k/(2\pi i k\cdot\omega)$ is holomorphic (Lemma 5.3(a)). The sets $Z_K \cap \Lambda$ and $Z_K \setminus \Lambda$ are finite. Hence $\phi, \xi$ are finite sums of terms $a_k(I)\chi_k(\theta)$ with $a_k$ holomorphic and $\sup_I \sum|a_k|e^{2\pi|k|_1 s} < \infty$, so they lie in $\mathcal{O}(r,s)$ with the stated coefficients (Lemma 5.3(c)); $\xi = P_\Lambda T_K f = P_{\Lambda\cap Z_K} f$ by Lemma 5.9.

**Step S7.4: Norms.** $|\hat\phi_k(I)| \le \frac{q}{2\pi\alpha}|\hat f_k(I)|$ by Lemma 7.3; multiplying by $e^{2\pi|k|_1 s}$, summing over $k \in Z_K \setminus \Lambda$ and taking the supremum over $I$ gives $|\phi|_{r,s} \le q\delta/\bar\alpha$. The bounds for $\xi$ and $T_K f$ are Lemma 5.9.

**Step S7.5: The equation.** By Lemma 5.3(b), $\partial_{\theta_j}\phi$ has coefficients $2\pi i k_j \hat\phi_k$; since $h$ is independent of $\theta$, $\{\phi,h\} = \sum_j \partial_{\theta_j}\phi\,\omega_j$, and multiplication by the $\theta$-independent $\omega_j(I)$ multiplies each coefficient by $\omega_j(I)$ (integrate against $\chi_{-k}$). So $\widehat{\{\phi,h\}}_k = 2\pi i(k\cdot\omega)\hat\phi_k = \hat f_k$ for $k \in Z_K \setminus \Lambda$ and 0 otherwise. Adding $\hat\xi_k = \hat f_k \mathbf{1}_{Z_K\cap\Lambda}(k)$ gives the coefficients of $T_K f$; coefficients determine the function (Lemma 5.3(d)).

**Step S7.6: Reality.** If $f$ is real on real points, $\overline{\hat f_k(I)} = \hat f_{-k}(I)$ for real $I$ (conjugate the integral defining $\hat f_k$). If $\omega(I)$ is real and $k \notin \Lambda$ with $-k \notin \Lambda$, then $\overline{\hat\phi_k} = \hat f_{-k}/\overline{2\pi i k\cdot\omega} = -\hat f_{-k}/(2\pi i k\cdot\omega) = \hat\phi_{-k}$; the set $Z_K\setminus\Lambda$ is symmetric. A finite sum $\sum a_k\chi_k$ with $\bar a_k = a_{-k}$ is real at real $\theta$. The same for $\xi$. □

**Lemma 7.5** (Bracket with a decomposed $g$)**.** *Let $\phi \in \mathcal{O}(r,s)$ and $g = \sum_{j\in J} g_j$ as in Definition 7.1, $0<\rho<r$, $0<\sigma<s$. Then $\{g,\phi\} \in \mathcal{O}(r-\rho, s-\sigma)$ and $|\{g,\phi\}|_{r-\rho,s-\sigma} \le \dfrac{2}{e\rho\sigma}|g|_*|\phi|_{r,s} \le \dfrac{2}{\rho\sigma}|g|_*|\phi|_{r,s}$.*

*Source: Pos93 p. 212 ("$|\{g,\phi\}|_{r-\rho,s-\sigma} \le \frac{q}{\alpha\rho\sigma}|g|_*|f|_{r,s}$"), with the bracket estimate of Lemma 5.7.*

*Proof.* **Step S7.7: One term.** Lemma 5.7 with $(u,v;r_0,s_0)\to(g_j,\phi;r_j,s_j)$ (admissible: $r-\rho<r_j$, $s-\sigma<s_j$) gives $|\{g_j,\phi\}|_{r-\rho,s-\sigma} \le \frac{1}{e}\big(\frac{1}{(r_j-r+\rho)\sigma}+\frac{1}{(s_j-s+\sigma)\rho}\big)|g_j||\phi| = \frac{2}{e\rho\sigma}\cdot\frac12\big(\frac{\rho}{r_j-r+\rho}+\frac{\sigma}{s_j-s+\sigma}\big)|g_j||\phi|$.

**Step S7.8: Sum.** The bracket is bilinear and the norm satisfies the triangle inequality; sum over $j$ and use (E1) ($e>2$). □

### 7.4. The Iterative Lemma.

**Theorem 7.6** (Iterative Lemma)**.** *Let $(D,r,K,\alpha,L,\Lambda,h)$ satisfy (N1), (N2) with $KLr \le \alpha/p$ ($p>1$, $q=p/(p-1)$), let $s>0$, $0<\rho$, $0<\sigma$ with $2\rho<r$, $2\sigma<s$, and put $\bar\alpha = 2\pi\alpha$. Let $H = h+g+f$ on $\mathcal{D}_{r,s}$, where $g=\sum_{j\in J} g_j$ is as in Definition 7.1 (any finite decomposition), $f\in\mathcal{O}(r,s)$, and*

$$\delta := |f|_{r,s} \le \frac{\bar\alpha\,\rho\sigma}{4q}, \qquad \text{equivalently} \qquad u := \frac{q\delta}{\bar\alpha\rho\sigma} \le \frac14.$$

*Then there is $\phi\in\mathcal{O}(r,s)$ (the one of Proposition 7.4, so $|\phi|_{r,s}\le q\delta/\bar\alpha$) such that the time-one map $\Phi = X_\phi^1$ of Part 6 satisfies:*

- *(a) $\Phi$ is defined and holomorphic on $\mathcal{D}_{r-\frac32\rho,s-\frac32\sigma}$ with values in $\mathcal{D}_{r-\rho,s-\sigma}\subset\mathcal{D}_{r,s}$; it is canonical: $(D\Phi)^\top J\,D\Phi = J$, $\Phi(\theta+k,I) = \Phi(\theta,I)+(k,0)$ ($k\in\mathbb{Z}^n$), and $\Phi$ is injective on $\mathcal{D}_{r-2\rho,s-2\sigma}$ with left inverse $X_\phi^{-1}$ (Proposition 6.13(d),(e),(b));*
- *(b) (displacement) with $\nu=\nu_{\rho,\sigma}$ of Definition 6.2: $\nu(\Phi-\mathrm{id})\le u$ on $\mathcal{D}_{r-\frac32\rho,s-\frac32\sigma}$, and $|\Phi_\theta-\theta|_\infty \le \dfrac{q\delta}{\bar\alpha\rho}$, $|\Phi_I - I|_\infty \le \dfrac{q\delta}{e\,\bar\alpha\sigma}$; on $\mathcal{D}_{r-2\rho,s-2\sigma}$: $|D\Phi-\mathbf{1}|_\nu \le 2u$;*
- *(c) (normal form) $H\circ\Phi = h+g_+ + f_+$ on $\mathcal{D}_{r-2\rho,s-2\sigma}$ with*

$$g_+ = g + P_\Lambda T_K f, \qquad |g_+ - g|_{r,s}\le\delta, \qquad f_+\in\mathcal{O}(r-2\rho,s-2\sigma);$$

- *(d) (estimate of $f_+$)*

$$|f_+|_{r-2\rho,s-2\sigma} \le \Big(1-\frac{2q\delta}{\bar\alpha\rho\sigma}\Big)^{-1}\Big(\frac{2q}{e\,\bar\alpha\rho\sigma}\big(|g|_*+\delta\big)+e^{-2\pi K\sigma}\Big)\delta,$$

  *with $|g|_*$ computed with these $\rho,\sigma$;*
- *(e) (reality) if $\Lambda=-\Lambda$, and $h,g,f$ are real on real points, then $\phi$, $g_+$, $f_+$ are real on real points and $\Phi$ maps real points to real points.*

*Source: Pos93 p. 193 (Iterative Lemma; the hypothesis there is $|f|_{r,s} < \alpha\rho\sigma/2q$, the estimate $|f_+| \le (1 - \frac{2q\delta}{\alpha\rho\sigma})^{-1}(\frac{|g|_*+|f|}{\alpha\rho\sigma/q} + e^{-K\sigma})|f|$) and Appendix A, pp. 211–213. Pöschel's $\alpha, \sigma, |W(\Phi - \mathrm{id})| \le \frac{q\delta}{\alpha\rho\sigma}$ correspond to $\bar\alpha, 2\pi\sigma$ and $u$ (Remark 5.11).* Deviation*: the smallness is $|\phi| \le \rho\sigma/4$ (so $\eta = 2|\phi|/(\rho\sigma) \le \frac12$) instead of $< \rho\sigma/2$, because Theorem 6.10 proves the Lie series identity only for $\eta \le \frac12$ (Remark 6.1); the estimate has the extra factor $2/e < 1$ from Lemma 5.7. Constants only change.*

*Proof.* **Step S7.9: The generating function.** By Proposition 7.4, $\phi, \xi$ exist with $\{\phi, h\} + \xi = T_K f$, $|\xi|, |T_K f| \le \delta$, $|\phi|_{r,s} \le q\delta/\bar\alpha$. Put $\varepsilon := |\phi|_{r,s}/(\rho\sigma) \le u \le \frac14$ and $\eta := 2|\phi|_{r,s}/(\rho\sigma) \le 2u \le \frac12$.

**Step S7.10: Items (a), (b).** Proposition 6.13 (with $\rho < r/2$, $\sigma < s/2$, $\varepsilon \le \frac12$) gives (a), the displacement bounds of (b) with $|\phi|/(\rho\sigma) \le u$, $|\phi|/\rho \le q\delta/(\bar\alpha\rho)$, $|\phi|/(e\sigma) \le q\delta/(e\bar\alpha\sigma)$, and $|D\Phi - \mathbf{1}|_\nu \le 2\varepsilon \le 2u$.

**Step S7.11: Taylor identity.** Put $f_0 = g + T_K f \in \mathcal{O}(r - \rho, s - \sigma)$ (finite norm: $g = \sum g_j$ and monotonicity, $|g|_{r,s} \le \sum |g_j|$), $w = \{h, \phi\} = -\{\phi, h\} = \xi - T_K f$ (Proposition 7.4(a), antisymmetry, Definition 6.2); $w$ has finite norm. Apply Proposition 6.11 with the data $(r, s; r', s') \to (r - \rho, s - \sigma; r, s)$ (admissible: $\rho < r - \rho$ iff $2\rho < r$, $r - \rho \le r' - \rho$; likewise for $s$; and $\eta \le \frac12$): on $\mathcal{D}_{r-2\rho,s-2\sigma}$, $(h+f_0)\circ\Phi = h+w+f_0+\mathcal{R}$ with $\mathcal{R} = \int_0^1 \{(1-t)w+f_0, \phi\}\circ X_\phi^t\, dt$. Here $h+w+f_0 = h+g+\xi = h+g_+$, and $(1-t)w + f_0 = g + f_t$ with $f_t := (1-t)\xi + tT_K f$.

**Step S7.12: Decomposition of $H \circ \Phi$.** $H = (h + f_0) + (f - T_K f)$ with $f - T_K f \in \mathcal{O}(r, s)$, so on $\mathcal{D}_{r-2\rho,s-2\sigma}$

$$H \circ \Phi = h + g_+ + f_+, \qquad f_+ := \mathcal{R} + (f - T_K f) \circ \Phi.$$

Both terms are holomorphic and periodic (Theorem 6.10(ii), Corollary 6.12), so $f_+ \in \mathcal{O}(r-2\rho, s-2\sigma)$ once the norm below is finite. This is (c); $|g_+ - g| = |\xi| \le \delta$.

**Step S7.13: The integrand.** $v_t := \{g + f_t, \phi\} \in \mathcal{O}(r - \rho, s - \sigma)$ is affine in $t$, so $(t, z) \mapsto v_t(z)$ and $Dv_t(z)$ are continuous. By Lemma 7.5 and Lemma 5.7 (the single-term decomposition $f_t$, $|f_t| \le \delta$ since $|\xi|, |T_K f| \le \delta$),

$$|v_t|_{r-\rho,s-\sigma} \le \frac{2}{e\rho\sigma}\big(|g|_* + \delta\big)|\phi|_{r,s} \le \frac{2q}{e\,\bar\alpha\rho\sigma}\big(|g|_* + \delta\big)\delta =: V.$$

**Step S7.14: The two terms.** Corollary 6.12 (data as in Step S7.11) gives $|\mathcal{R}|_{r-2\rho,s-2\sigma} \le (1 - \eta)^{-1}V$. By Lemma 5.9 and monotonicity, $|f - T_K f|_{r-\rho,s-\sigma} \le |f - T_K f|_{r,s-\sigma} \le e^{-2\pi K\sigma}\delta$, and Theorem 6.10(ii) (same data) gives $|(f - T_K f) \circ \Phi|_{r-2\rho,s-2\sigma} \le (1 - \eta)^{-1}e^{-2\pi K\sigma}\delta$. Add and use $(1 - \eta)^{-1} \le (1 - 2q\delta/(\bar\alpha\rho\sigma))^{-1}$ ($|\phi| \le q\delta/\bar\alpha$). This is (d).

**Step S7.15: Reality.** If $\Lambda = -\Lambda$ and $h, g, f$ are real, then $\omega$ is real at real points ($h$ is real-valued and holomorphic on the open set $V_r D$, so its partial derivatives at real points of $V_r D$ are real), and Proposition 7.4(c) gives $\phi, \xi$ real; $\Phi$ preserves real points (Proposition 6.13(b)); so $g_+ = g + \xi$ is real and $f_+ = H \circ \Phi - h - g_+$ is real on real points of $\mathcal{D}_{r-2\rho,s-2\sigma}$. □

**Corollary 7.7** (Iterative Lemma, direct divisor form)**.** *Let $D \subset \mathbb{C}^n$, $r > 0$, $K \ge 0$, $\alpha > 0$, $\Lambda \subset \mathbb{Z}^n$, $p > 1$, $q = p/(p-1)$, $h$ holomorphic on $V_r D$ with $\omega = \partial_I h$, and assume, in place of (N1), (N2) and $KLr \le \alpha/p$, the single hypothesis*

$$\text{(N-dir)} \qquad |k \cdot \omega(I)| \ge \frac{\alpha}{q} \quad \textit{for all } I \in V_r D,\ k \in Z_K \setminus \Lambda.$$

*Then Proposition 7.4, Theorem 7.6 (all of (a)–(e), with the same constants and the same hypotheses on $s, \rho, \sigma, g, f, \delta$) and Remark 7.8 hold verbatim.*

*Source: Ours (observation); track P,* [P:bpP:p2:direct]*, records the same remark. Pos93 pp. 211–212 uses the divisor bound only on the whole neighbourhood.*

*Proof.* **Step S7.16: Where the old hypotheses enter.** In the proofs of Proposition 7.4 and Theorem 7.6 the hypotheses (N1), (N2) and $KLr \le \alpha/p$ are used only through Lemma 7.3, i.e. through the bound $|k\cdot\omega(I)| \ge \alpha/q$ on $V_rD$, $k \in Z_K \setminus \Lambda$ (Steps S7.3, S7.4; Steps S7.9–S7.15 use only the conclusions of the homological equation, Parts 5, 6, and the holomorphy and (for reality) the realness of $h$ on $V_rD$, which is part of the data). (N-dir) is exactly that bound.

**Step S7.17: Conclusion.** Replace Lemma 7.3 by (N-dir) in those two steps; nothing else changes. In particular (N-dir) follows from the original hypotheses (Lemma 7.3), so the corollary generalises the theorem. Only the Iterative Lemma is claimed in direct form; the Normal Form Lemma keeps its hypotheses (its Step S7.19 uses $KLr \le \alpha/p$ only to invoke the Iterative Lemma and in the displacement constant, which was not re-examined). □

*Remark* 7.8 (Special cases for track P). (1) *First step, $g = 0$:* $|g|_* = 0$ and (d) reads $|f_+| \le (1 - \frac{2q\delta}{\bar\alpha\rho\sigma})^{-1}\big(\frac{2q\delta}{e\bar\alpha\rho\sigma} + e^{-2\pi K\sigma}\big)\delta$. (2) *Averaging, $\Lambda = \{0\}$:* $P_\Lambda T_K f = \hat f_0 = \langle f\rangle$, the $\theta$-average; (N1) is $|k\cdot\omega(I)| \ge \alpha$ for $0 < |k|_1 \le K$; the transformed Hamiltonian is $h + g + \langle f\rangle + f_+$. (3) *Constants:* no constant depends on $f$ or on $H$ beyond $\delta$, $|g|_*$; the data are $(n, r, s, \rho, \sigma, K, \alpha, L, p)$. The hypotheses involve only $\delta$ through $u \le \frac14$.

## 7.5. Composition of the steps.

**Lemma 7.9** (Telescoping). *Let $U_0 \supset U_1 \supset \cdots \supset U_{m+1}$ be subsets of $\mathbb{C}^{2n}$ and $\Phi_i : U_{i+1} \to U_i$ $(0 \le i \le m)$ maps. Put $\Psi = \Phi_0 \circ \cdots \circ \Phi_m : U_{m+1} \to U_0$. Then for every $z \in U_{m+1}$ and every seminorm $\mu$ on $\mathbb{C}^{2n}$ (e.g. $|\Pi_I \cdot|_\infty$, $|\Pi_\theta \cdot|_\infty$), $\mu(\Psi(z) - z) \le \sum_{i=0}^m \sup_{U_{i+1}} \mu(\Phi_i - \mathrm{id})$. If every $\Phi_i$ is holomorphic and $(D\Phi_i)^\top J D\Phi_i = J$, then $\Psi$ is holomorphic, $(D\Psi)^\top J D\Psi = J$; if every $\Phi_i$ is injective, so is $\Psi$.*

*Proof.* **Step S7.18: Telescoping.** Put $z_{m+1} = z$, $z_i = \Phi_i(z_{i+1}) \in U_i$ (so $z_{i+1} \in U_{i+1}$ for the next step because $U_{i+1} \subset U_i$). Then $\Psi(z) - z = \sum_{i=0}^m (z_i - z_{i+1})$ and $z_i - z_{i+1} = (\Phi_i - \mathrm{id})(z_{i+1})$; apply the triangle inequality. The chain rule (E2) gives $D\Psi = \prod D\Phi_i(z_{i+1})$ and $(D\Psi)^\top J D\Psi = J$ by induction; injectivity is clear. □

## 7.6. The Normal Form Lemma.

**Theorem 7.10** (Normal Form Lemma). *Let $(D, r, K, \alpha, L, \Lambda, h)$ satisfy (N1), (N2), $K > 0$, and $KLr \le \alpha/p$ $(p > 1,\ q = p/(p-1))$. Let $s > 0$ and $H = h + f$ with $f \in \mathcal{O}(r, s)$ of finite norm $\varepsilon := |f|_{r,s}$. Assume*

$$\text{(i)}\quad \varepsilon \le \frac{1}{2^7 q}\cdot\frac{\alpha r}{K}, \qquad\qquad \text{(ii)}\quad \kappa = 2\pi K s \ge 6.$$

*Put $r_* = r/2$, $s_* = s/6$, $c = 2qK/(\alpha r)$, $g_0 = P_\Lambda T_K f$. Then there are*

- *a holomorphic, injective map $\Psi : \mathcal{D}_{r_*,s_*} \to \mathcal{D}_{r,s}$ with $(D\Psi)^\top J D\Psi = J$ and $\Psi(\theta + k, I) = \Psi(\theta, I) + (k, 0)$ $(k \in \mathbb{Z}^n)$;*
- *$g, f_* \in \mathcal{O}(r_*, s_*)$, with $g$ a trigonometric polynomial in $\theta$ with frequencies in $\Lambda \cap Z_K$ (so $P_\Lambda g = g$: $h + g + f_*$ is in $\Lambda$-resonant normal form up to $f_*$),*

*such that $H \circ \Psi = h + g + f_*$ on $\mathcal{D}_{r_*,s_*}$ and*

*(a) $|g - g_0|_{r_*,s_*} \le 16c\,\varepsilon^2$ $\quad$ $(g = g_0$ if $N = 0$ (Step S7.21) or in the second case of Step S7.20);*
*(b) $|f_*|_{r_*,s_*} \le e^{-2\pi K s/6}\varepsilon = e^{-\kappa/6}\varepsilon$;*
*(c) $|\Pi_I\Psi - I|_\infty \le qK\varepsilon/\alpha$ $\big(= \frac12 c\, r\varepsilon\big)$ and $|\Pi_\theta\Psi - \theta|_\infty \le 10\,q\varepsilon/(2\pi\alpha r)$ on $\mathcal{D}_{r_*,s_*}$;*
*(d) if $\Lambda = -\Lambda$ and $h, f$ are real on real points, then $g$, $f_*$ are real on real points and $\Psi$ maps real points to real points.*

*The data on which $\Psi, g, f_*$ are constructed are $(h, f, K, \alpha, L, \Lambda, p)$ and $(r, s)$; $r_*, s_*, c$ depend only on $(r, s, K, \alpha, q)$.*

*Source: Pos93 p. 192 (Normal Form Lemma) and pp. 193–196 (proof). Changes: (1) frozen conventions ($\bar\alpha = 2\pi\alpha$, $\kappa = 2\pi Ks$ are his $\alpha$, $Ks$; max norms); (2) $|\Pi_I\Psi - I| \le qK\varepsilon/\alpha$ in the max norm instead of $\|\Pi_I\Psi - I\| \le c\varepsilon r = 2qK\varepsilon/\alpha$ (Euclidean); (3) the smallness $\eta \le \frac12$ of Theorem 6.10 is built into Theorem 7.6; every numerical step below was recomputed (Remark 7.13).*
*Source gap: Pöschel's text p. 194 ends the second case with "$\le e^{Ks/6}\varepsilon$"; the correct (and used) bound is $e^{-Ks/6}\varepsilon$, as the first line of that display shows.*

*Proof.* Throughout $\bar\alpha = 2\pi\alpha$ and $\kappa = 2\pi Ks$; note $\bar\alpha s = \alpha\kappa/K$ and $K \ge 3/(\pi s)$. *Indexing:* $V_i = \mathcal{D}_{r_i,s_i}$, $f_0 := f$, $g_{-1} := 0$; step $i$ applies Theorem 7.6 to $H_i = h + g_{i-1} + f_i$ on $V_i$ with losses $(\rho_i, \sigma_i)$, producing $\Phi_i : V_{i+1} \to V_i$, $g_i = g_{i-1} + P_\Lambda T_K f_i$, $f_{i+1}$ on $V_{i+1}$, $r_{i+1} = r_i - 2\rho_i$, $s_{i+1} = s_i - 2\sigma_i$; $\varepsilon_i := |f_i|_{V_i}$, $\varepsilon_0 = \varepsilon$. ($V_{i+1}$ stands for $\mathcal{D}_{r_i-2\rho_i, s_i-2\sigma_i}$, which is where $\Phi_i$ is injective and $f_{i+1}$ lives; $\Phi_i$ maps it into $\mathcal{D}_{r_i-\frac32\rho_i, s_i-\frac32\sigma_i} \subset V_i$.)

**Step S7.19: Step 0: hypotheses of the Iterative Lemma.** Let $\rho_0 = r/8$, $\sigma_0 = s/6$, so $r_1 = \frac34 r$, $s_1 = \frac23 s$, $\rho_0\sigma_0 = rs/48$. Then $2\rho_0 < r$, $2\sigma_0 < s$, $KLr \le \alpha/p$. From (i), (ii): $\varepsilon \le \frac{\alpha r}{2^7 qK} \le \frac{\alpha r}{2^7 q}\cdot\frac{\pi s}{3} = \frac{\pi\alpha rs}{384q} = \frac{\bar\alpha\rho_0\sigma_0}{16q}$. Hence $y_0 := \frac{q\varepsilon}{\bar\alpha\rho_0\sigma_0} \le \frac{1}{16}$ and the hypothesis $u \le \frac14$ of Theorem 7.6 holds. Also $\kappa/6 = 2\pi K\sigma_0 \ge 1$.

**Step S7.20: Step 0: two cases.** Apply Theorem 7.6(d) with $g = 0$ ($|g|_* = 0$), $\delta = \varepsilon$: $|f_1| \le (1-2y_0)^{-1}\big(\frac{2}{e}y_0 + e^{-\kappa/6}\big)\varepsilon$, with $(1-2y_0)^{-1} \le \frac87$.

*Case A: $e^{-\kappa/6} \le y_0$.* Then $|f_1| \le \frac87\big(\frac2e + 1\big)y_0\varepsilon \le \frac{16}{7}y_0\varepsilon$, so

$$\varepsilon_1 \le \frac{16}{7}\cdot\frac{48\,q\varepsilon^2}{\bar\alpha rs} \le \frac{2^7 q\varepsilon^2}{\bar\alpha rs} \le \frac{2^7 q\varepsilon}{\bar\alpha rs}\cdot\frac{\alpha r}{2^7 qK}\,\varepsilon = \frac{\varepsilon}{2\pi Ks} = \frac{\varepsilon}{\kappa} \le \frac{\varepsilon}{6} \le \frac{\varepsilon}{4}. \tag{6.1}$$

*Case B: $e^{-\kappa/6} > y_0$.* Repeat the step with twice the losses $\rho' = r/4$, $\sigma' = s/3$ (so $2\rho' < r$, $2\sigma' < s$, $\bar\alpha\rho'\sigma' = 4\bar\alpha\rho_0\sigma_0$, the hypothesis $u' = y_0/4 \le \frac14$ holds). Then $|f_1|_{r/2,s/3} \le (1 - y_0/2)^{-1}\big(\frac{2}{e}\cdot\frac{y_0}{4} + e^{-2\kappa/6}\big)\varepsilon$; as $e^{-2\kappa/6} \le e^{-1}e^{-\kappa/6}$ and $y_0 < e^{-\kappa/6}$, this is $\le \frac{32}{31}\big(\frac{1}{2e} + \frac1e\big)e^{-\kappa/6}\varepsilon \le e^{-\kappa/6}\varepsilon$. We stop: $\Psi = \Phi_0'$ on $\mathcal{D}_{r/2,s/3} \supset \mathcal{D}_{r_*,s_*}$, $g = g_0$, $f_* = f_1$ restricted to $V_{r_*,s_*}$. Items (a), (b) hold; items (c), (d) in Step S7.24. From now on *Case A*.

**Step S7.21: Number of steps.** Let $N = \lfloor \kappa/(12\ln 2)\rfloor \ge 0$, so $N \le \frac{\kappa}{12\ln 2} < N+1$ and $\kappa \ge 12\ln 2\, N > 8N$ ((E1)). If $N = 0$ stop after Step 0: $\Psi = \Phi_0$, $g = g_0$, $f_* = f_1$ on $V_1 \supset \mathcal{D}_{r_*,s_*}$ and $|f_*| \le \varepsilon_1 \le \varepsilon/4 \le e^{-\kappa/6}\varepsilon$, because $\kappa < 12\ln 2$ gives $e^{-\kappa/6} > e^{-2\ln 2} = \frac14$. Let $N \ge 1$ and, for $1 \le i \le N$,

$$\rho_i = \rho = \frac{r}{8N}, \quad \sigma_i = \sigma = \frac{s}{4N}, \qquad r_i = \tfrac34 r - 2(i-1)\rho, \quad s_i = \tfrac23 s - 2(i-1)\sigma;$$

then $r_{N+1} = \frac34 r - \frac{r}{4} = r_*$ and $s_{N+1} = \frac23 s - \frac{s}{2} = s_*$. Note $2\pi K\sigma = \kappa/(4N) \ge 3\ln 2$, i.e. $e^{-2\pi K\sigma} \le \frac18$, and $\rho\sigma = rs/(32N^2)$. Smallness constants: from (i), $\kappa > 8N$ and $K = \kappa/(2\pi s)$,

$$\varepsilon \le \frac{\alpha r\cdot 2\pi s}{2^7 q\kappa} \le \frac{\bar\alpha rs}{2^{10}qN}, \qquad \text{hence by (6.1)} \quad \varepsilon_1 \le \frac{2^7 q}{\bar\alpha rs}\Big(\frac{\bar\alpha rs}{2^{10}qN}\Big)^2 = \frac{\bar\alpha rs}{2^{13}qN^2} = \frac{\bar\alpha\rho\sigma}{2^8 q}. \tag{6.2}$$

**Step S7.22: Induction.** *Claim:* for $1 \le i \le N$, step $i$ is admissible and $\varepsilon_{i+1} \le \varepsilon_i/4$. Hence $\varepsilon_i \le 4^{1-i}\varepsilon_1$ for $1 \le i \le N+1$, and $|g_i - g_{i-1}|_{V_i} = |P_\Lambda T_K f_i| \le \varepsilon_i$ (Lemma 5.9). Induct on $i$; $\varepsilon_1 \le \varepsilon/4$ is (6.1) (so for $i = 1$ the claim "$\varepsilon_i \le 4^{1-i}\varepsilon_1$" is trivial). Let $i \le N$ and suppose $\varepsilon_j \le 4^{1-j}\varepsilon_1$ for $1 \le j \le i$. *Admissibility.* $r_i \ge r_{N+1} = r/2 > 2\rho$; $s_i \ge s_N = s_* + 2\sigma = \frac{s}{6} + \frac{s}{2N} > 2\sigma$; $KLr_i \le KLr \le \alpha/p$; and $\varepsilon_i \le \varepsilon_1 \le \bar\alpha\rho\sigma/(2^8 q) \le \bar\alpha\rho\sigma/(4q)$ by (6.2). *Size of $|g_{i-1}|_*$* (computed with $(r_i, s_i, \rho, \sigma)$ in Definition 7.1). Decompose $g_{i-1} = g_0 + \sum_{j=1}^{i-1}(g_j - g_{j-1})$, the term $j \ge 1$ living on $V_j$ with $r_j - r_i = 2(i-j)\rho \ge 2\rho$, $s_j - s_i \ge 2\sigma$, so its weight is $\le \frac12(\frac13 + \frac13) \le \frac12$; the term

$g_0 = P_\Lambda T_K f$ lives on $V_0 = \mathcal{D}_{r,s}$ with norm $\le \varepsilon$ and $r - r_i \ge r - r_1 = \frac{r}{4}$, $s - s_i \ge \frac{s}{3}$, so its weight is $\le \frac{1}{2}\big(\frac{r/8N}{r/4} + \frac{s/4N}{s/3}\big) = \frac{5}{8N} \le \frac{1}{N}$. Thus

$$|g_{i-1}|_* \le \tfrac{1}{2}(\varepsilon_{i-1} + \cdots + \varepsilon_1) + \tfrac{1}{N}\varepsilon \le \tfrac{1}{2}\cdot\tfrac{4}{3}\varepsilon_1 + \tfrac{\varepsilon}{N} \le \varepsilon_1 + \frac{\varepsilon}{N} \le \frac{\bar\alpha\rho\sigma}{2^8 q} + \frac{\bar\alpha\rho\sigma}{2^5 q} \le \frac{\bar\alpha\rho\sigma}{2^4 q},$$

using (6.2) and $\varepsilon/N \le \bar\alpha rs/(2^{10}qN^2) = 2^5\bar\alpha\rho\sigma/(2^{10}q)$. *The step.* Theorem 7.6(d) with $\delta = \varepsilon_i$: $\frac{2q}{\bar\alpha\rho\sigma}\varepsilon_i \le 2^{-7}$, so $(1-2^{-7})^{-1} = \frac{128}{127}$, and $\frac{2q}{e\bar\alpha\rho\sigma}(|g_{i-1}|_* + \varepsilon_i) \le \frac{2}{e}(\frac{1}{16} + \frac{1}{2^8}) \le \frac{1}{16} + \frac{1}{2^8}$, $e^{-2\pi K\sigma} \le \frac{1}{8}$:

$$\varepsilon_{i+1} \le \frac{128}{127}\Big(\frac{1}{2^4} + \frac{1}{2^8} + \frac{1}{8}\Big)\varepsilon_i \le \frac{128}{127}\cdot 0.1915\,\varepsilon_i < \frac{1}{4}\varepsilon_i.$$

**Step S7.23: Conclusion of the construction.** Put $\Psi = \Phi_0\circ\cdots\circ\Phi_N$, $g = g_N = g_0 + \sum_{i=1}^N P_\Lambda T_K f_i$, $f_* = f_{N+1}$, all restricted to $V_{N+1} = \mathcal{D}_{r_*,s_*}$. Since $H_{i+1} = H_i\circ\Phi_i$, $H\circ\Psi = H_{N+1} = h + g + f_*$. $g$ is a finite sum of trigonometric polynomials with frequencies in $\Lambda \cap Z_K$. Lemma 7.9 with $U_i = V_i$ gives the symplectic, periodic (each $\Phi_i$ is, Theorem 7.6(a)) and injective $\Psi$. (b): $|f_*| = \varepsilon_{N+1} \le 4^{-N}\varepsilon_1 \le 4^{-N-1}\varepsilon \le e^{-\kappa/6}\varepsilon$, because $N + 1 > \kappa/(12\ln 2)$ gives $4^{-(N+1)} = e^{-(N+1)2\ln 2} < e^{-\kappa/6}$. (a): $g - g_0 = \sum_{i=1}^N P_\Lambda T_K f_i$, so $|g - g_0| \le \frac{4}{3}\varepsilon_1 \le \frac{4}{3}\cdot\frac{2^7 q\varepsilon^2}{\bar\alpha rs}$ by (6.1); $\bar\alpha s = \alpha\kappa/K$ and $\kappa > 8N \ge 8$ give $\le \frac{4}{3}\cdot\frac{2^7}{8}\cdot\frac{qK\varepsilon^2}{\alpha r} \le 32\frac{qK\varepsilon^2}{\alpha r} = 16c\varepsilon^2$.

**Step S7.24: (c) Displacement.** By Lemma 7.9 and Theorem 7.6(b) ($|\Phi_I - I| \le q\delta/(e\bar\alpha\sigma)$, $|\Phi_\theta - \theta| \le q\delta/(\bar\alpha\rho)$, $\delta = \varepsilon_i$): in Case A

$$|\Pi_I\Psi - I|_\infty \le \frac{q\varepsilon}{e\bar\alpha\sigma_0} + \sum_{i=1}^N \frac{q\varepsilon_i}{e\bar\alpha\sigma} \le \frac{6q\varepsilon}{e\bar\alpha s} + \frac{q}{e\bar\alpha\sigma}\cdot\frac{4}{3}\varepsilon_1.$$

Now $1/(\bar\alpha s) = K/(\alpha\kappa) \le K/(6\alpha)$; $1/(\bar\alpha\sigma) = 4NK/(\alpha\kappa) \le 4K/(12\ln 2\,\alpha)$ because $N \le \kappa/(12\ln 2)$, and $\varepsilon_1 \le \varepsilon/4$; so the right side is at most $\frac{qK\varepsilon}{e\alpha}\big(1 + \frac{4}{12\ln 2}\cdot\frac{4}{3}\cdot\frac{1}{4}\big) \le \frac{qK\varepsilon}{e\alpha}\cdot 1.17 \le \frac{qK\varepsilon}{\alpha}$. Similarly $|\Pi_\theta\Psi - \theta|_\infty \le \frac{q\varepsilon}{\bar\alpha\rho_0} + \frac{q}{\bar\alpha\rho}\cdot\frac{4}{3}\varepsilon_1 = \frac{8q\varepsilon}{\bar\alpha r} + \frac{8Nq}{\bar\alpha r}\cdot\frac{4}{3}\cdot\frac{\varepsilon}{\kappa} \le \frac{q\varepsilon}{\bar\alpha r}\big(8 + \frac{32}{3\cdot 12\ln 2}\big) \le \frac{10\,q\varepsilon}{\bar\alpha r}$ (using $\varepsilon_1 \le \varepsilon/\kappa$ from (6.1), $N/\kappa \le 1/(12\ln 2)$). In Case B, $\Psi = \Phi_0'$ with losses $(2\rho_0, 2\sigma_0)$ and $\delta = \varepsilon$, and the bounds are smaller by the factor $\frac{1}{2}$.

**Step S7.25: (d) Reality.** Each step preserves reality (Theorem 7.6(e)); $g_i$, $f_i$, $H_i$ stay real on real points and so do the $\Phi_i$; compositions of real-point-preserving maps preserve real points. □

**Corollary 7.11** (Consequence for the action variable)**.** *In Theorem 7.10, if $(\theta', I') = \Psi(\theta, I)$ with $(\theta, I) \in \mathcal{D}_{r_*,s_*}$ then $|I' - I|_\infty \le qK\varepsilon/\alpha$; in particular if $\varepsilon \le \alpha r/(2^7 qK)$ (hypothesis (i)) this is $\le r/2^7$.*
*Source: Pos93 p. 196, last display.*

*Proof.* Theorem 7.10(c) and hypothesis (i). □

7.7. **The case $2\pi Ks < 6$.**

**Proposition 7.12** (No transformation for small $Ks$)**.** *Let $H = h + f$ on $\mathcal{D}_{r,s}$, $f \in \mathcal{O}(r,s)$ with finite norm $\varepsilon$ and $\kappa = 2\pi Ks < 6$. Then, with $\Psi = \mathrm{id}$, $g = 0$, $f_* = f$, one has $H\circ\Psi = h + g + f_*$ on $\mathcal{D}_{r,s}$ and*

$$|\Psi - \mathrm{id}| = 0, \qquad |f_*|_{r,s} = \varepsilon \le e\cdot e^{-\kappa/6}\varepsilon,$$

*and $g = 0 = P_\Lambda g$ is in $\Lambda$-normal form for every $\Lambda$. (No nonresonance, convexity or smallness hypothesis is used.)*
*Source: Pos93 p. 197, first line ("For $Ks \le 6$ the same reasoning applies in original coordinates, letting $g = 0$ and $f_* = f_\varepsilon$").*
*Source gap: The source does not say what is claimed about $|f_*|$ in this case. With $f_* = f$, the bound $|f_*| \le e^{-\kappa/6}\varepsilon$ of the Normal Form Lemma is* false *for $\kappa < 6$ (take $\Psi = \mathrm{id}$, $f \ne 0$, $K \to 0$); what holds is $|f_*| \le e^{1-\kappa/6}\varepsilon$, since $\kappa/6 < 1$. This helper is not needed for the stability bound: the source's p. 197 proof is*

*Proposition 1 directly in the original coordinates (no Normal Form Lemma bound is asserted for the identity map), and Part 7 (Remark 8.12) proves it that way on the full strip with the original time* $\frac{sr}{5\varepsilon}e^{Ks/6}$ *and no loss of the factor* $e^{-1}$*. The trivial statement is recorded here once for citation.*

*Proof.* **Step S7.26: Trivial.** Identity map, $g = 0$; $|f_*| = \varepsilon$ and $1 \le e^{1-\kappa/6}$ for $\kappa \le 6$. □

### 7.8. **Deviations, errata, uses.**

*Remark* 7.13 (Deviations from Pöschel and bookkeeping). (1) *Lie-series range.* Part 6 proves $u \circ X_\phi^1 = \sum_h \frac{1}{h!}\operatorname{ad}_\phi^h u$ for $\eta = 2|\phi|/(\rho\sigma) \le \frac{1}{2}$ only. Hence the Iterative Lemma assumes $|\phi| \le \rho\sigma/4$, which is implied by $\delta \le \bar\alpha\rho\sigma/(4q)$ (Pöschel: $\delta < \alpha\rho\sigma/2q$). All hypotheses on $\varepsilon$ in the Normal Form Lemma are already stronger than this: $\varepsilon_0 \le \bar\alpha\rho_0\sigma_0/(16q)$ and $\varepsilon_i \le \bar\alpha\rho\sigma/(2^8 q)$ (Steps S7.19, S7.22). The numerical constants $2^7$, $4^{1-i}$, $N = \lfloor\kappa/(12\ln 2)\rfloor$, $r_* = r/2$, $s_* = s/6$, $16c\varepsilon^2$ are therefore *the same as Pöschel's* (re-derived above), and the number of iteration steps is unchanged.

(2) *Constants gained.* The factor $2/e < 1$ in the estimate of $f_+$ (from $\frac{1}{e} < \frac{1}{2}$ in Lemma 5.7); where the source's arithmetic is reused we simply bound $2/e \le 1$.

(3) *Norms.* Max norms on actions and angles replace his Euclidean/sup norms; the divisor bound uses $|k\cdot v| \le |k|_1|v|_\infty$, which is why $Z_K$ is the $\ell^1$ ball and hypothesis (N2) is stated in $|\cdot|_\infty$. The Euclidean bound $\|\Pi_I\Psi - I\| \le c\varepsilon r$ of the source follows from the max-norm statement of Theorem 7.10(c) only with a factor $\sqrt{n}$.

(4) *Not proved in the source, proved here:* the $\theta$-displacement of $\Psi$ and the injectivity of $\Psi$ (from Proposition 6.13(e)); the $|W_0(\Psi - \mathrm{id})|$ bound of p. 195 ("usual telescoping arguments") is replaced by the componentwise bound of Theorem 7.10(c).
Source gap: Pöschel's last display of p. 195 asserts $|W_0(D\Psi - I)W_0^{-1}| \le 2q\varepsilon_0/(\alpha\rho_0\sigma_0)$ by telescoping; the later steps contribute $\sum_{i\ge1} q\varepsilon_i/(\bar\alpha\rho\sigma)$ in the scaled norm $\nu_{\rho,\sigma}$, which is not obviously $\lesssim q\varepsilon_0/(\bar\alpha\rho_0\sigma_0)$ for large $N$ since $\rho\sigma/(\rho_0\sigma_0) = 3/(2N^2)$. It is not used by the stability estimates (only $\Pi_I\Psi$ is), so it is not reproduced.

(5) *The second case of the first step* (his "twice the step size") is Step S7.20, Case B, and ends with the correct sign of the exponent (see the
Source gap:
of Theorem 7.10).

(6) *Domains.* The generating function lives on $\mathcal{D}_{r,s}$, the map $\Phi$ on $\mathcal{D}_{r-\frac{3}{2}\rho,s-\frac{3}{2}\sigma}$ and the new remainder on $\mathcal{D}_{r-2\rho,s-2\sigma}$; hence the chain $V_{i+1} = \mathcal{D}_{r_i-2\rho_i,s_i-2\sigma_i}$ and $\Psi$ on $\mathcal{D}_{r/2,s/6}$ as in the source.

*Remark* 7.14 (Consumers). Part 7 uses Theorem 7.10, Corollary 7.11 and Proposition 7.12; Part 9 fixes $q = 9$, $p = \frac{9}{8}$ and, with Part 8, chooses $K = K_a$ with $K_g = \sqrt{n}K_a$ (so $K_a \le K_g$ transfers nonresonance from the Euclidean ball to $Z_{K_a}$, and $K_g \ge \sqrt{n}K_a$ gives the radius condition; Part 9, Remark 9.1) and the domain $D$. The Cauchy estimates entering through Lemma 5.7 are those of Lemma 5.6 in its corrected pointwise-in-$I$ form (findings N45-01/02); this part never sums global derivative norms over the angle index. Track P (Part 2 of its plan) takes Theorem 7.6 with $\Lambda = \{0\}$, Remark 7.8(2), the elementary constant bookkeeping being its own.

## 8. Stability estimates: nonresonant, convex, quasi-convex and resonant (shared by tracks N, I)

This part is Section 3 of Pöschel 1993 (pp. 196–200): the nonresonant Proposition 1, the two energy estimates near a resonance (Lemma 1 for $m$-convex $h$, Lemma 2 for $\ell, m$-quasi-convex $h$) and the Resonant Stability Lemma (Proposition 2), which carries these estimates from normal-form coordinates back to the original ones. The idea is that in normal form the action of an orbit moves only through the non-resonant remainder $f_*$ (exponentially small) and through the resonant part

$g$, whose gradient lies in the span of $\Lambda$. Energy conservation and convexity of $h$ transverse to that span then bound the motion in the resonant directions by $\sqrt{\zeta/m}$, without any further small-divisor analysis; the span itself is handled because $D$ is close to the exact resonance space.
Conventions. Everything is in the frozen conventions of Parts 5–7: angles of period 1, $\chi_k(\theta) = e^{2\pi i k\cdot\theta}$, geometric strip width $s$, weight $e^{2\pi|k|_1 s}$, max norms $|\cdot| = |\cdot|_\infty$ on actions, $\kappa = 2\pi K s$, $\bar\alpha = 2\pi\alpha$. The convexity argument needs an inner product: there the Euclidean norm $\|\cdot\|_2$ on $\mathbb{R}^n$ is used, with the comparisons of Lemma 2.1 ($|x| \le \|x\|_2 \le \sqrt{n}|x|$). Time and the values of Hamiltonians are those of the frozen system $\dot\theta = \partial_I H$, $\dot I = -\partial_\theta H$ with period-1 angles; no rescaling to Pöschel's $2\pi$-normalisation is needed, because Part 5 (Remark 5.11) shows that the quantities $s\,r$, $s\,m\,r^2/\omega$, $|\phi|/(\rho\sigma)$ are the same numbers in both normalisations (checked below where they occur, Remark 8.21). We assume $n \ge 1$.

### 8.1. **External facts.**

(E1) *Banach fixed point theorem.* A contraction of a nonempty closed subset of $\mathbb{R}^N$ into itself has a fixed point (Mathlib `ContractingWith.exists_fixedPoint`). Exists.
(E2) *Taylor, Lagrange form, order one.* If $\varphi : [0,1] \to \mathbb{R}$ is $C^2$, then $\varphi(1) = \varphi(0) + \varphi'(0) + \frac{1}{2}\varphi''(\xi)$ for some $\xi \in (0,1)$ (Mathlib `taylor_mean_remainder_lagrange`). Exists.
(E3) *Numerics.* $e > 2.7$, so $\frac{1}{12e} < 0.0307$, $\frac{6}{5e} < 0.442$, $\frac{15e}{192} > \frac{1}{5}$; Young's inequality $ab \le \frac{a^2}{2c} + \frac{cb^2}{2}$ ($c > 0$); $xe^{-x} \le e^{-1}$ for $x \ge 0$. Exists.
(E4) *Orthogonal projections.* For a subspace $E \subset \mathbb{R}^n$ with orthogonal projection $\Pi$ and $\Pi^\perp = 1 - \Pi$: $\Pi$ is self-adjoint, $\|\Pi v\|_2 \le \|v\|_2$, $\langle u, \Pi v\rangle = \langle \Pi u, v\rangle$, and $\|\Pi v\|_2 = \operatorname{dist}_2(v, E^\perp)$. Exists.

Track-K and earlier-part results used: [K:`bp:p1:ext-mvt`], [K:`bp:p1:holomorphic-regularity`], [K:`bp:p1:hamiltonian-real`], [K:`bp:p1:real-restriction`]; (E3), Lemma 2.1, Lemma 2.7; Part 4 (Definition 4.2, Lemma 4.19, Theorems 4.11, 4.13, Corollary 4.16, Lemma 4.22, Theorem 4.23); Part 5 (Definition 5.1, 5.2, Lemmas 5.3, 5.4, 5.6, 5.8, 5.9); Lemma 6.5; Part 7 (Definition 7.1, Lemma 7.2, Theorem 7.10, Proposition 7.12).

### 8.2. **Real systems on a real neighbourhood of $D$.**

**Definition 8.1** (Real neighbourhoods, real systems, orbits)**.** Let $D \subset \mathbb{R}^n$ and $\varrho, s > 0$. Put $U_\varrho D = V_\varrho D \cap \mathbb{R}^n = \{I \in \mathbb{R}^n : \exists J \in D,\ |I - J| < \varrho\}$ (Definition 5.1); it is open, and $W_\varrho := \mathbb{R}^n \times U_\varrho D$ is open in $\mathbb{R}^{2n}$ and equals $\mathcal{D}_{\varrho,s} \cap \mathbb{R}^{2n}$ (real $\theta$ lies in $T_s$). For $0 < \varrho \le r$, $W_\varrho \subset W_r$. $\mathcal{O}(r,s)^{\mathbb{R}}$ denotes the set of $G \in \mathcal{O}(r,s)$ with $G(x) \in \mathbb{R}$ for $x \in W_r$ ("real on real points"). For such $G$ the *real Hamiltonian* is $G^{\mathbb{R}}(x) = G(x)$ for $x \in W_r$ and $G^{\mathbb{R}}(x) = 0$ otherwise (a total function $\mathbb{R}^{2n} \to \mathbb{R}$; its values off $W_r$ never matter, Remark 4.1). A *$G$-orbit in $W_\varrho$ on $J$* ($J \subset \mathbb{R}$ an interval, $0 < \varrho \le r$) is a $z : \mathbb{R} \to \mathbb{R}^{2n}$ with $\mathtt{IsSolIn}\, X_{G^{\mathbb{R}}}\, W_\varrho\, J\, z$ (Definition 4.2). We write $z = (\theta, I)$ and $I(t) = z(t)_2$. For $h$ holomorphic on $V_r D$ and real on $U_r D$: $h_{\mathbb{R}} = h|_{U_r D}$, $\omega = \nabla h_{\mathbb{R}}$ (real, $= \partial_I h$), $Q(I) = D^2 h_{\mathbb{R}}(I)$ (symmetric).
Source: Ours (bookkeeping for [K:`bp:p1:hamiltonian-real`] and Part 4); Pöschel's $U_r D$, $V_r D$ (p. 189) in max norm.

**Lemma 8.2** (Holomorphic real systems)**.** *Let $G \in \mathcal{O}(r,s)^{\mathbb{R}}$, $0 < \varrho \le r$. Then (a) $G^{\mathbb{R}}$ is $C^\infty$ on $W_r$ and $G^{\mathbb{R}}(\theta + k, I) = G^{\mathbb{R}}(\theta, I)$; (b) at $x \in W_r$, $G^{\mathbb{R}}$ is differentiable and $X_{G^{\mathbb{R}}}(x) = X_G(x) = \big((\partial_{I_i} G(x))_i, (-\partial_{\theta_i} G(x))_i\big) \in \mathbb{R}^{2n}$ (complex partials at the real point); (c) $X_{G^{\mathbb{R}}}$ is $C^1$ on $W_r$ and 1-periodic in $\theta$; (d) Theorems 4.23, 4.11, 4.13, Corollary 4.16 and Lemma 4.19 apply to $F = X_{G^{\mathbb{R}}}$ on the open set $W_\varrho$; in particular $G^{\mathbb{R}}$ is constant along every $G$-orbit on every interval (energy conservation).*
*Source: Ours;* [K:`bp:p1:hamiltonian-real`], [K:`bp:p1:holomorphic-regularity`]*; the special case $G = h + f$ of N-HEAD-C is Lemma 2.3.*

*Proof.* **Step S8.1: Smoothness.** $G$ is holomorphic on $\mathcal{D}_{r,s}$, hence $C^\infty$ in the real sense ([K:bp:p1:holomorphic-regularity], (E3)); $W_r = \iota^{-1}(\mathcal{D}_{r,s})$ is open and $G^{\mathbb{R}} = G \circ \iota$ on it, so $G^{\mathbb{R}} \in C^\infty(W_r)$, with $D^k G^{\mathbb{R}} = D^k G \circ \iota$ on real vectors. Periodicity is the periodicity in $\mathcal{O}(r,s)$.

**Step S8.2: Field.** [K:bp:p1:hamiltonian-real] with $V = \mathcal{D}_{r,s}$ (open, $G$ holomorphic, real at the real points of $V$ since $V \cap \mathbb{R}^{2n} = W_r$): $G^{\mathbb{R}}$ agrees with $\operatorname{Re} G$ on the neighbourhood $W_r$ of each real point of $V$, so $X_{G^{\mathbb{R}}} = X_G$ there. The field is $C^1$ by Lemma 4.22, periodic by (a) and Lemma 4.24.

**Step S8.3: (d).** Lemma 4.22 ($G^{\mathbb{R}} \in C^2$ on the open set $W_\varrho$); the cited results need only $F$ $C^1$ on the open set $W_\varrho$, periodicity (for the corollary) and, for energy, differentiability of $G^{\mathbb{R}}$ at the points of the orbit. □

**Lemma 8.3** (Derivative bound at real points)**.** *Let $u \in \mathcal{O}(\varrho, s)$ have finite norm. Then for all $\theta \in \mathbb{R}^n$, $I \in U_\varrho D$: $\sum_{j=1}^n |\partial_{\theta_j} u(\theta, I)| \le |u|_{\varrho,s}/(e\,s)$; in particular $\|\partial_\theta u(\theta, I)\|_2 \le |u|_{\varrho,s}/(es)$. Also $|u(x)| \le |u|_{\varrho,s}$ for $x \in \mathcal{D}_{\varrho,s}$.*
*Source: Pos93 p. 198 ("by the Cauchy estimate of Lemma B.3": $\sup\|\partial_\theta f_*\| \le \frac{1}{es}|f_*|_{r,s}$); the limit $s' \downarrow 0$ in Lemma 5.6(a) is ours.*

*Proof.* **Step S8.4: Strip loss.** For $0 < s' < s$ and fixed $I$, Step S5.12 gives $\sum_j |\partial_{\theta_j} u|_{I,s'} \le |u|_{I,s}/(e(s-s'))$ (Lemma 5.6(a)). By the pointwise bound in Step S5.7 (Lemma 5.4(a)), $|v(\theta, I)| \le |v|_{I,s'}$ for $\theta \in T_{s'} \supset \mathbb{R}^n$.

**Step S8.5: Limit.** Let $s' \downarrow 0$; $\|v\|_2 \le \sum_j |v_j|$ (Lemma 2.1). The last claim is Lemma 5.4(a). □

**Lemma 8.4** (Resonant gradients lie in the span of $\Lambda$)**.** *Let $\Lambda \subset \mathbb{Z}^n$, $\Pi = \Pi_\Lambda$ the Euclidean orthogonal projection onto $\operatorname{span}_{\mathbb{R}} \Lambda$, $\Pi^\perp = 1 - \Pi$. If $g \in \mathcal{O}(r,s)^{\mathbb{R}}$ has finite norm and $P_\Lambda g = g$, then $\partial_\theta g(\theta, I) \in \operatorname{ran} \Pi$, i.e. $\Pi^\perp \partial_\theta g = 0$, for all $(\theta, I) \in W_r$.*
*Source: Pos93 p. 198 ("the vector $Q\Delta I$ is given by the integral ... of $\partial_\theta f_*$ only, since $g$ is in $\Lambda$-resonant normal form").*

*Proof.* **Step S8.6: Coefficients.** The components of $u = \Pi^\perp \partial_\theta g$ are finite combinations of $\partial_{\theta_j} g \in \mathcal{O}(r,s)$ with coefficients $\hat{u}_k = 2\pi i\, \Pi^\perp k\, \hat{g}_k$ (Lemma 5.3(b)). For $k \in \Lambda$, $\Pi^\perp k = 0$; for $k \notin \Lambda$, $\hat{g}_k = 0$ because $g = P_\Lambda g$ (Lemma 5.9). So $\hat{u}_k = 0$ for all $k$.

**Step S8.7: Vanishing.** For $s' < s$, $u$ has finite norm at width $s'$ (Lemma 5.6(a)), so $u = \sum_k \hat{u}_k \chi_k = 0$ on $\mathcal{D}_{r,s'}$ (Lemma 5.3(d)); real points lie in it. □

## 8.3. Transfer through a symplectic change of variables.

**Lemma 8.5** (Conjugation of orbits)**.** *Let $H \in \mathcal{O}(r,s)^{\mathbb{R}}$, $\tilde{H} \in \mathcal{O}(\tilde{r}, \tilde{s})^{\mathbb{R}}$, and $\Psi : \mathcal{D}_{\tilde{r},\tilde{s}} \to \mathcal{D}_{r,s}$ holomorphic, real at real points, with $(D\Psi)^\top J D\Psi = J$ and $\tilde{H} = H \circ \Psi$. If $\tilde{z}$ is an $\tilde{H}$-orbit in $W_{\tilde{r}}$ on $J$, then $z = \Psi \circ \tilde{z}$ is an $H$-orbit in $W_r$ on $J$.*
*Source: Standard; Corollary 6.6(c) for a single flow map, here for an arbitrary symplectic $\Psi$.*

*Proof.* **Step S8.8: Real points.** $\Psi$ maps $W_{\tilde{r}} = \mathcal{D}_{\tilde{r},\tilde{s}} \cap \mathbb{R}^{2n}$ into $\mathcal{D}_{r,s} \cap \mathbb{R}^{2n} = W_r$; the real derivative $Y = D\Psi(x)$ at $x \in W_{\tilde{r}}$ is the (real) restriction of the complex one ([K:bp:p1:real-restriction]), and $Y^\top J Y = J$; hence $\det Y = \pm 1$ and $Y J Y^\top = J$ (Step S6.10).

**Step S8.9: Chain rule.** $\tilde{H}^{\mathbb{R}} = H^{\mathbb{R}} \circ \Psi$ on $W_{\tilde{r}}$, so $\nabla \tilde{H}^{\mathbb{R}}(x) = Y^\top \nabla H^{\mathbb{R}}(\Psi x)$ and, with $X_u = J\nabla u$, $\frac{d}{dt}\Psi(\tilde{z}) = Y\tilde{z}' = Y J Y^\top \nabla H^{\mathbb{R}}(\Psi\tilde{z}) = X_{H^{\mathbb{R}}}(\Psi\tilde{z})$ for $t \in J$ (Lemma 8.2(b)); $z(t) \in W_r$ by Step S8.8. □

**Lemma 8.6** (Real preimages of a near-identity map)**.** *Let $D \subset \mathbb{R}^n$, $\tilde{r}, \tilde{s} > 0$, $\Psi : \mathcal{D}_{\tilde{r},\tilde{s}} \to \mathbb{C}^{2n}$ holomorphic, real at real points, $G = \Psi - \mathrm{id}$, with $|G_\theta| \le d_\theta$, $|G_I| \le d_I$ on $\mathcal{D}_{\tilde{r},\tilde{s}}$. Suppose there are*

*$0 < \varrho' \le \tilde r$, $0 < \sigma' \le 2\tilde s$ with $\kappa_* := \max(d_\theta/\sigma', d_I/\varrho') \le \frac14$. Then for every $y = (\theta_y, I_y) \in \mathbb{R}^n \times D$ there is $x = (\theta_x, I_x) \in W_{\tilde r}$ with $\Psi(x) = y$, $|\theta_x - \theta_y| \le d_\theta$, $|I_x - I_y| \le d_I$.*
*Source: Ours. Pöschel p. 196 and p. 200 assert ("$\Psi^{-1}$ takes any given initial position in $D \times \mathbb{T}^n$ to some initial position in $U_\rho D \times \mathbb{T}^n$") without proof; a left inverse (Proposition 6.13(e)) does not give this.*

*Proof.* Let $\nu(a,b) = \max(|a|/\sigma', |b|/\varrho')$ on $\mathbb{C}^{2n}$ (the scaled norm of Definition 6.2), so $\nu(G) \le \kappa_*$.

**Step S8.10: A closed convex set.** $X = \{x \in \mathbb{R}^{2n} : \nu(x - y) \le \kappa_*\}$ is closed, convex, nonempty. For $x \in X$: $|I_x - I_y| \le \varrho'/4 < \tilde r$ and $I_y \in D$, so $x \in W_{\tilde r}$.

**Step S8.11: Self-map.** $T(x) = y - G(x)$ is real ($\Psi$ real at real points) and $\nu(T(x) - y) = \nu(G(x)) \le \kappa_*$: $T : X \to X$.

**Step S8.12: Contraction.** For $x \in X$ the complex $\nu$-ball $\{\nu(z - x) < \frac12\}$ lies in $\mathcal{D}_{\tilde r, \tilde s}$: $|\operatorname{Im}\theta'| \le |\theta' - \theta_x| < \sigma'/2 \le \tilde s$ and $|I' - I_y| < \varrho'/2 + \varrho'/4 < \tilde r$. On it $G$ is holomorphic with $\nu(G) \le \kappa_*$, so Lemma 5.8 ($R = \frac12$) gives $|DG(x)|_\nu \le 2\kappa_* \le \frac12$. The real derivative of $G|_{\mathbb{R}^{2n}}$ is the restriction ((E3)); [K:bp:p1:ext-mvt] on the convex set $X$ gives $\nu(T(x) - T(x')) \le \frac12 \nu(x - x')$.

**Step S8.13: Fixed point.** (E1) gives $x = y - G(x)$, i.e. $\Psi(x) = x + G(x) = y$, and $y - x = G(x)$ gives the displacement bounds. □

**Corollary 8.7** (Preimages for the Normal Form Lemma)**.** *In Theorem 7.10 with $\Lambda = -\Lambda$, $h, f$ real on real points, $\kappa \ge 6$: every $y \in \mathbb{R}^n \times D$ is $\Psi(x)$ for some $x \in W_{r/2}$ with $|I_x - I_y| \le qK\varepsilon/\alpha$ ($\le r/2^7$).*

*Proof.* **Step S8.14: Apply Lemma 8.6.** $\Psi$ is real at real points (Theorem 7.10(d)); $\tilde r = r/2$, $\tilde s = s/6$, $d_I = qK\varepsilon/\alpha \le r/2^7$ (hypothesis (i)), $d_\theta = 10q\varepsilon/(2\pi\alpha r)$ (item (c)). Take $\varrho' = r/2$, $\sigma' = s/3 = 2\tilde s$: $d_I/\varrho' \le \frac{1}{64}$ and, by (i), $d_\theta/\sigma' = \frac{30q\varepsilon}{2\pi\alpha r s} \le \frac{30}{2\pi 2^7 K s} = \frac{30}{2^7\kappa} \le \frac{30}{768} < \frac14$. □

**Lemma 8.8** (Transfer of an a priori confinement through a symplectic map)**.** *Let $D \subset \mathbb{R}^n$, $H \in \mathcal{O}(r,s)^{\mathbb{R}}$, $\tilde H \in \mathcal{O}(\tilde r, \tilde s)^{\mathbb{R}}$, $\Psi : \mathcal{D}_{\tilde r,\tilde s} \to \mathcal{D}_{r,s}$ as in Lemma 8.5, and $|\cdot|_*$ a norm on $\mathbb{R}^n$ with $|\cdot| \le |\cdot|_*$. Assume*

- *(T1) $|\Pi_I \Psi(x) - I_x|_* \le d_*$ for all $x \in W_{\tilde r}$;*
- *(T2) every $y \in \mathbb{R}^n \times D$ equals $\Psi(x)$ for some $x \in W_{\tilde r}$;*
- *(T3) (a priori confinement in the new variables) there are $\varrho > 0$ with $d_* + \varrho < \tilde r$ and $T : D \to (0, \infty]$ such that for every $I_y \in D$, every interval $\tilde J \ni 0$ and every $\tilde H$-orbit $\tilde z$ in $W_{\tilde r}$ on $\tilde J$ with $|\tilde I(0) - I_y|_* \le d_*$: $|\tilde I(t) - \tilde I(0)|_* \le \varrho$ for all $t \in \tilde J$, $|t| \le T(I_y)$.*

*Then for every interval $J \ni 0$ and every $H$-orbit $z$ in $W_r$ on $J$ with $I_y := I(0) \in D$:*

$$|I(t) - I(0)|_* \le \varrho + 2d_* \qquad (t \in J,\ |t| \le T(I_y)).$$

*Source: Ours; this is the transfer step of Pöschel pp. 196, 200 ("Going back to original coordinates"), in stopped form so that Part 10 can connect it with existence (Theorem 4.34).*

*Proof.* **Step S8.15: A normal-form orbit.** By (T2) choose $x \in W_{\tilde r}$ with $\Psi(x) = z(0)$; by (T1) $|I_x - I_y|_* \le d_*$. Let $\tilde z$ with maximal interval $(T_-, T_+)$ be the maximal solution of $X_{\tilde H^{\mathbb{R}}}$ in $W_{\tilde r}$ through $x$ (Lemma 8.2(d), Theorem 4.13).

**Step S8.16: It lives long enough.** (T3) applies on $(T_-, T_+)$. The set $K = \{I : |I - I_x|_* \le \varrho\}$ is compact and, since $|I - I_y| \le |I - I_y|_* \le \varrho + d_* < \tilde r$ and $I_y \in D$, contained in $U_{\tilde r} D$. If some $t \ge 0$ with $t \le T(I_y)$ had $t \ge T_+$, then $T_+ < \infty$ and $\tilde I(t') \in K$ for all $t' \in [0, T_+)$, so $T_+ = \infty$ by Corollary 4.16 (the field is periodic, Lemma 8.2(c)): a contradiction. Likewise for $t \le 0$. Thus $\{|t| \le T(I_y)\} \subset (T_-, T_+)$.

**Step S8.17: Identification.** $\Psi \circ \tilde z$ is an $H$-orbit in $W_r$ on $(T_-, T_+)$ (Lemma 8.5) with value $y$ at 0, as is $z$ on $J$; Theorem 4.11 (with $F = X_{H^{\mathbb{R}}}$, $W = W_r$) gives $z = \Psi \circ \tilde z$ on $J \cap (T_-, T_+)$.

**Step S8.18: Estimate.** For $t \in J$, $|t| \le T(I_y)$: $I(t) - I(0) = (\Pi_I \Psi(\tilde z(t)) - \tilde I(t)) + (\tilde I(t) - \tilde I(0)) + (\tilde I(0) - I_y)$, of size $\le d_* + \varrho + d_*$ by (T1), (T3) and $\tilde I(0) = I_x$. □

**Corollary 8.9** (Stopped form for any extension of the field)**.** *Let $H \in \mathcal{O}(r,s)^{\mathbb{R}}$, $|\cdot|_* \ge |\cdot|$ a norm, $0 < \varrho < r$, $T \in (0,\infty]$ and suppose that every $H$-orbit in $W_r$ on every interval $J \ni 0$ with $I(0) \in D$ satisfies $|I(t) - I(0)|_* \le \varrho$ for $t \in J$, $|t| \le T$. Let $F : \mathbb{R}^{2n} \to \mathbb{R}^{2n}$ be any function with $F = X_{H^{\mathbb{R}}}$ on $W_r$ and $z : \mathbb{R} \to \mathbb{R}^{2n}$, $J \ni 0$ an interval such that $\forall t \in J$, `HasDerivAt` $z\,(F(z\,t))\,t$ (the genuine two-sided derivative of the total curve $z : \mathbb{R} \to \mathbb{R}^{2n}$ at every time of $J$, endpoints included; no differentiability of $z$ is asserted at times outside $J$), and $z(0) \in \mathbb{R}^n \times D$. Then $|I(t) - I(0)|_* \le \varrho$ for $t \in J$, $|t| \le T$; in particular $z(t) \in W_r$ there.*

*Proof.* **Step S8.19: The stopped interval.** $\{t \in J : z(t) \in W_r\}$ is relatively open in $J$ and contains 0; let $J'$ be its connected component through 0. $z$ is an $H$-orbit in $W_r$ on $J'$, so the bound holds on $J' \cap [-T,T]$.

**Step S8.20: No exit.** If $t_e \in J \cap [-T,T]$ were an endpoint of $J'$ not in $J'$, continuity would give $|I(t_e) - I(0)|_* \le \varrho$, hence $|I(t_e) - I(0)| < r$ and $I(t_e) \in U_r D$ ($I(0) \in D$), so $t_e$ would belong to the component: contradiction. □

**Lemma 8.10** (First exit from a norm ball)**.** *Let $|\cdot|_*$ be a norm on $\mathbb{R}^n$, $J \ni 0$ an interval, $z : J \to \mathbb{R}^{2n}$ continuous, $I_0 = I(0)$, $\varrho > 0$. If $|I(t_1) - I_0|_* > \varrho$ for some $t_1 \in J$, then there is $t^* \in J$ between 0 and $t_1$ with $|I(t^*) - I_0|_* = \varrho$ and $|I(\tau) - I_0|_* \le \varrho$ for all $\tau$ between 0 and $t^*$.*

*Proof.* **Step S8.21: Exit set.** Apply Lemma 4.19 to $V = \mathbb{R}^n \times \{|I - I_0|_* < \varrho\}$ (open; $z(0) \in V$), forward if $t_1 > 0$, and to $t \mapsto z(-t)$ if $t_1 < 0$ (only continuity is used): $t^*$ exists, $z(\tau) \in V$ before it and $z(t^*) \in \overline{V} \setminus V$, i.e. $|I(t^*) - I_0|_* \le \varrho$ and $\not< \varrho$. □

8.4. **Proposition 1: nonresonant stability.**

**Proposition 8.11** (Nonresonant stability estimate)**.** *Let $n \ge 1$, $D \subset \mathbb{R}^n$, $r, s, K, \alpha > 0$, $L \ge 0$, $p > 1$, $q = p/(p-1)$, $\kappa = 2\pi K s$. Let $h$ be holomorphic on $V_r D$, real on $U_r D$, $f \in \mathcal{O}(r,s)^{\mathbb{R}}$ with $\varepsilon := |f|_{r,s} < \infty$, and $H = h + f$. Assume*

*(i) (*completely $\alpha, K$-nonresonant*) $|k \cdot \omega(I)| \ge \alpha$ for $I \in D$ and $k \in \mathbb{Z}^n$, $0 < |k|_1 \le K$;*
*(ii) the Lipschitz bound (N2) for $\omega$ with constant $L$, and $KLr \le \alpha/p$;*
*(iii) $\varepsilon \le \dfrac{\alpha r}{2^7 q K}$.*

*Put $T_{\mathrm{nr}} = \dfrac{s\,r}{5\varepsilon} e^{\kappa/6}$ ($= +\infty$ if $\varepsilon = 0$). Then for every interval $J \ni 0$ and every $H$-orbit $z$ in $W_r$ on $J$ with $I(0) \in D$:*

$$|I(t) - I(0)| \le \tfrac{r}{2} \ \ (\le r) \qquad \text{for all } t \in J,\ |t| \le T_{\mathrm{nr}}.$$

*No convexity is used. The constants depend only on $(r, s, K, \alpha, p)$ and $\varepsilon$.*
*Source: Pos93 p. 196 (Proposition 1: conclusion $\|I(t) - I_0\| \le r$ for $|t| \le \frac{sr}{5\varepsilon} e^{Ks/6}$; hypotheses $\varepsilon \le \frac{1}{2^7 q} \frac{\alpha r}{K}$, $r \le \min(\alpha/(pMK), r_0)$). Deviations: the conclusion is $r/2$ (the proof gives it); the case $\kappa < 6$ (his one line p. 197) is done with $\Psi = \mathrm{id}$ and no loss of width, so the factor $e$ of Proposition 7.12 is absorbed (Remark 8.12); the transfer to the original coordinates includes the surjectivity of $\Psi$, which the source does not prove.*

*Proof.* Let $\tilde H = h + g + f_*$ be the normal form below.

**Step S8.22: Case A, $\kappa \ge 6$: data.** Theorem 7.10 with $\Lambda = \{0\}$ ($\Lambda \cap Z_K = \{0\}$) applies: (N1) is (i), (N2) and $KLr \le \alpha/p$ are (ii), $\varepsilon \le \alpha r/(2^7 qK)$ is (iii). It gives $\Psi : \mathcal{D}_{r/2, s/6} \to \mathcal{D}_{r,s}$ holomorphic, symplectic, real at real points, $H \circ \Psi = h + g + f_*$ with $g$ independent of $\theta$, $f_* \in \mathcal{O}(r/2, s/6)^{\mathbb{R}}$, $\eta := |f_*|_{r/2, s/6} \le e^{-\kappa/6}\varepsilon$, and $|\Pi_I \Psi - I| \le d := qK\varepsilon/\alpha \le r/2^7$. Put $\tilde r = r/2$, $\tilde s = s/6$, $\varrho = \frac{15}{32} r$, $d_* = d$, $|\cdot|_* = |\cdot|$, $T = T_{\mathrm{nr}}$. (T1) is the displacement bound, (T2) is Corollary 8.7; $d_* + \varrho \le \frac{61}{128} r < \tilde r$.

**Step S8.23: Case A: (T3).** Let $\tilde z$ be an $\tilde H$-orbit in $W_{\tilde r}$ on $\tilde J$, $|\tilde I(0)-I_y|\le d$, $I_y\in D$, and suppose $|\tilde I(t_1)-\tilde I(0)|>\varrho$ for some $t_1\in\tilde J$, $|t_1|\le T_{\rm nr}$. Lemma 8.10 gives $t^*$, $|t^*|\le|t_1|$, $|\tilde I(t^*)-\tilde I(0)|=\varrho$, with $\tilde I(\tau)$ in the closed ball $B$ of radius $\varrho$ about $\tilde I(0)$ for $\tau$ between 0 and $t^*$; $B\subset U_{\tilde r}D$ as $\varrho+d<\tilde r$. On this stretch $\dot{\tilde I}=-\partial_\theta\tilde H^{\mathbb R}=-\partial_\theta f_*(\tilde z)$ ($h$ and $g$ do not depend on $\theta$; Lemma 8.2(b)), so by Lemma 8.3 $|\dot{\tilde I}|\le\eta/(e\tilde s)=6\eta/(es)$ and, by the mean value inequality ([K:bp:p1:ext-mvt]),

$$\varrho=|\tilde I(t^*)-\tilde I(0)|\le\frac{6\eta|t^*|}{es}\le\frac{6e^{-\kappa/6}\varepsilon}{es}\cdot\frac{sr\,e^{\kappa/6}}{5\varepsilon}=\frac{6}{5e}r<0.442\,r<\tfrac{15}{32}r=\varrho,$$

a contradiction ((E3); if $\varepsilon=0$ then $\eta=0$ and the left side is 0). So (T3) holds with $T(I_y)=T_{\rm nr}$.

**Step S8.24: Case A: conclusion.** Lemma 8.8: $|I(t)-I(0)|\le\frac{15}{32}r+\frac{2}{2^7}r=\frac{31}{64}r\le\frac r2$.

**Step S8.25: Case B, $\kappa<6$.** Take $\Psi=\mathrm{id}$ on $\mathcal D_{r,s}$, $\tilde H=H$, $g=0$, $f_*=f$ (Proposition 7.12), $\tilde r=r$, $\tilde s=s$, $d_*=0$, $\varrho=r/2$: (T1), (T2) are trivial. For (T3): as in Step S8.23, with $\tilde I(\tau)$ in the ball of radius $r/2$ about $\tilde I(0)\in D$ ($\subset U_rD$), $|\dot{\tilde I}|\le\varepsilon/(es)$ and $|\tilde I(t^*)-\tilde I(0)|\le\frac{\varepsilon}{es}\cdot\frac{sr\,e^{\kappa/6}}{5\varepsilon}<\frac r5<\frac r2$ (as $e^{\kappa/6}<e$): contradiction. Lemma 8.8 gives $|I(t)-I(0)|\le r/2$. □

*Remark* 8.12 (The factor $e$ in the nonresonant case). Proposition 7.12 gives only $|f_*|_{r,s}\le e\cdot e^{-\kappa/6}\varepsilon$ for $\kappa<6$. Used with the widths $(r/2,s/6)$ of the case $\kappa\ge6$ it would give the coefficient $31/384<\frac15$ in the time bound. Case B avoids this by not losing width ($\tilde s=s$, $\eta=\varepsilon$): the available time is $\frac{es\,r}{2\varepsilon}>\frac{sr}{5\varepsilon}e^{\kappa/6}$ for all $\kappa<6$, so the stated $T_{\rm nr}$ is attained with room to spare, and no factor $e$ is lost in the final statement.

8.5. **The energy argument.**

**Definition 8.13** (Resonance data, convexity, quasi-convexity)**.** Let $D\subset\mathbb R^n$, $\Lambda\subset\mathbb Z^n$, $\Pi=\Pi_\Lambda$ (Lemma 8.4), $R_\Lambda=\{\omega\in\mathbb R^n:k\cdot\omega=0\ \forall k\in\Lambda\}=\ker\Pi$. *$D$ is $\delta$-close to exact $\Lambda$-resonances* if $\|\Pi\omega(I)\|_2=\mathrm{dist}_2(\omega(I),R_\Lambda)\le\delta$ for all $I\in D$. For $h$ as in Definition 8.1 and $U\subset U_rD$: *$h$ is $m$-convex on $U$* if $Q(I)[\xi,\xi]\ge m\|\xi\|_2^2$ for $I\in U$, $\xi\in\mathbb R^n$; *$\ell,m$-quasi-convex on $U$* if for all $I\in U$ and $\xi\in\mathbb R^n$: $|\langle\omega(I),\xi\rangle|>\ell\|\xi\|_2$ or $Q(I)[\xi,\xi]\ge m\|\xi\|_2^2$ (equivalently $|\langle\omega,\xi\rangle|\le\ell\|\xi\|_2\Rightarrow Q[\xi,\xi]\ge m\|\xi\|_2^2$, Lemma 2.7(b); never full positive definiteness, review N05). $M\ge m$ denotes a bound $\|Q(I)\|_{2\to2}\le M$ on $U_rD$ (the normalisation $m\le M$ is always achievable by enlarging $M$).
Source: Pos93 pp. 190, 197.

**Lemma 8.14** (Energy–convexity inequality (reusable core))**.** *Let $U\subset\mathbb R^n$ be open, $h_{\mathbb R}\in C^2(U)$, $g,f\in C^1(\mathbb R^n\times U;\mathbb R)$, $H^{\mathbb R}=h_{\mathbb R}(I)+g+f$ on $\mathbb R^n\times U$ (any values elsewhere), $\Pi$ an orthogonal projection of $(\mathbb R^n,\|\cdot\|_2)$ with $\Pi^\perp\partial_\theta g\equiv0$. Let $J\ni0$ be an interval, $z$ with $\mathtt{IsSolIn}\,X_{H^{\mathbb R}}(\mathbb R^n\times U)\,J\,z$, $t_1\in J$, $I_0=I(0)$, $\Delta I=I(t_1)-I_0$, $I_s=I_0+s\Delta I$, $\omega_0=\nabla h_{\mathbb R}(I_0)$, and assume $I_s\in U$ for $s\in[0,1]$. Then*

*(a) (energy) $h_{\mathbb R}(I(t_1))-h_{\mathbb R}(I_0)=-\big[(g+f)(z(t_1))-(g+f)(z(0))\big]$;*
*(b) (resonant directions) $\|\Pi^\perp\Delta I\|_2\le|t_1|\sup_\tau\|\partial_\theta f(z(\tau))\|_2$, $\tau$ between 0 and $t_1$;*
*(c) (Taylor) $h_{\mathbb R}(I(t_1))-h_{\mathbb R}(I_0)-\langle\omega_0,\Delta I\rangle=\frac12Q(I_\xi)[\Delta I,\Delta I]$ for some $\xi\in(0,1)$;*
*(d) if $Q(I_s)[\Delta I,\Delta I]\ge m\|\Delta I\|_2^2$ for all $s\in[0,1]$, then*

$$\frac m2\|\Delta I\|_2^2\le\big|(g+f)(z(t_1))-(g+f)(z(0))\big|+\|\Pi\omega_0\|_2\|\Delta I\|_2+\|\omega_0\|_2\|\Pi^\perp\Delta I\|_2.$$

*Source: Pos93 p. 198 ("Following Lochak … expanding around $I_0$", $\Delta h=\langle\omega_0,\Delta I\rangle+\int_0^1(1-s)\langle Q(I(s))\Delta I,\Delta I\rangle ds$, energy conservation, the vector $Q\Delta I$ …), with the Lagrange remainder instead of the integral one. This is the part wanted by track I (energy confinement).*

*Proof.* **Step S8.26: Energy.** $H^{\mathbb R}$ is $C^1$ on the open set $\mathbb R^n\times U$, hence differentiable along the orbit, and Theorem 4.23 on the interval between 0 and $t_1$ gives $H^{\mathbb R}(z(t_1))=H^{\mathbb R}(z(0))$; this is (a).

**Step S8.27: Resonant projection.** $\dot I = -\partial_\theta H^{\mathbb{R}}(z) = -\partial_\theta g - \partial_\theta f$ ($h_{\mathbb{R}}$ is $\theta$-independent). Hence $t \mapsto \Pi^\perp I(t)$ has derivative $-\Pi^\perp \partial_\theta f(z(t))$, of norm $\le \|\partial_\theta f(z(t))\|_2$; the mean value inequality ([K:bp:p1:ext-mvt], $\Pi^\perp$ linear) gives (b).

**Step S8.28: Taylor and combination.** (c) is (E2) for $\varphi(s) = h_{\mathbb{R}}(I_s)$ ($\varphi'(0) = \langle \omega_0, \Delta I\rangle$, $\varphi'' = Q(I_s)[\Delta I, \Delta I]$). Under the hypothesis of (d), (c) gives $\frac{m}{2}\|\Delta I\|^2 \le h_{\mathbb{R}}(I(t_1)) - h_{\mathbb{R}}(I_0) - \langle \omega_0, \Delta I\rangle$; insert (a), and $\langle \omega_0, \Delta I\rangle = \langle \Pi\omega_0, \Delta I\rangle + \langle \omega_0, \Pi^\perp \Delta I\rangle$ ((E4)) with Cauchy–Schwarz. □

The resonant setting (RS).. $n \ge 1$, $D \subset \mathbb{R}^n$, $r, s > 0$, $\Lambda \subset \mathbb{Z}^n$, $h$ holomorphic on $V_r D$ and real on $U_r D$, $0 < m \le M$ with $\|Q(I)\|_{2\to 2} \le M$ on $U_r D$, $g, f_* \in \mathcal{O}(r,s)^{\mathbb{R}}$ of finite norm with $P_\Lambda g = g$, $\zeta \ge |g|_{r,s} + |f_*|_{r,s}$, $\eta = |f_*|_{r,s}$, $H = h + g + f_*$, $\delta > 0$ with $D$ $\delta$-close to exact $\Lambda$-resonances, $\delta \le mr/4$, $\zeta \le mr^2/32$, and $\rho := \delta/(4M)$ $\big(\le r/16\big)$. An initial action $I_0$ is *admissible* if $\|I_0 - J_0\|_2 < \rho$ for some $J_0 \in D$.

**Lemma 8.15** (Admissible initial actions)**.** *In (RS), for admissible $I_0$ (with $J_0$): (a) the closed ball $\bar B = \{I : \|I - I_0\|_2 \le r - \rho\} \subset U_r D$; (b) $\|\Pi\omega(I)\|_2 \le \frac{5}{4}\delta + M\|I - I_0\|_2$ for $I \in \bar B$; in particular $\|\Pi\omega(I_0)\|_2 \le \frac{5}{4}\delta$.*

*Proof.* **Step S8.29: Ball.** $|I - J_0| \le \|I - J_0\|_2 < (r-\rho) + \rho = r$, so $I \in U_r D$; likewise the segment $[J_0, I_0] \subset U_r D$.

**Step S8.30: Closeness.** By the mean value inequality for $\omega$ in $\|\cdot\|_2$ ($D\omega = Q$, $\|Q\| \le M$, convex segments inside $U_r D$): $\|\omega(I_0) - \omega(J_0)\|_2 \le M\rho = \delta/4$, so $\|\Pi\omega(I_0)\| \le \delta + \delta/4$; and $\|\Pi\omega(I)\| \le \|\Pi\omega(I_0)\| + M\|I - I_0\|_2$. □

**Lemma 8.16** (No exit under convexity along the segment)**.** *In (RS) let $z$ be an $H$-orbit in $W_r$ on $J \ni 0$, $I_0 = I(0)$ admissible, and $t_1 \in J \setminus \{0\}$ with*

*(i) $\|I(\tau) - I_0\|_2 \le r - \rho$ for $\tau$ between $0$ and $t_1$, and $\|I(t_1) - I_0\|_2 = r - \rho$;*
*(ii) $Q(I_s)[\Delta I, \Delta I] \ge m\|\Delta I\|_2^2$ for all $s \in [0,1]$ ($\Delta I = I(t_1) - I_0$, $I_s = I_0 + s\Delta I$);*
*(iii) $|t_1|\,\|\omega(I_0)\|_2\,\eta \le s\,m\,r^2/12$.*

*Then a contradiction arises (so these three cannot hold together).*
*Source: Pos93 p. 198 (end of the proof of Lemma 1): $\frac{m}{3}\|\Delta I\|^2 \le 2\zeta + \frac{5}{2m}\delta^2 + T\|\omega_0\|\frac{1}{es}|f_*| \le \frac{mr^2}{4} < \frac{m}{3}(r-\rho)^2$; all numbers recomputed.*

*Proof.* Put $\Delta = \|\Delta I\|_2 = r - \rho \ge \frac{15}{16} r$.

**Step S8.31: Apply the core.** $U = U_r D$, $\bar B \subset U$ (Lemma 8.15) is convex so $I_s \in U$; $g, f_*$ are $C^\infty$ on $W_r$ (Lemma 8.2), $\Pi^\perp \partial_\theta g = 0$ (Lemma 8.4), the orbit is a solution of $X_{H^{\mathbb{R}}}$ in $\mathbb{R}^n \times U$. Lemma 8.14(d) with (ii): $\frac{m}{2}\Delta^2 \le A + B + C$, $A = |\Delta(g + f_*)|$, $B = \|\Pi\omega_0\|\Delta$, $C = \|\omega_0\|\|\Pi^\perp \Delta I\|$.

**Step S8.32: Three terms.** $A \le 2(|g|_{r,s} + |f_*|_{r,s}) \le 2\zeta \le \frac{mr^2}{16}$ (Lemma 8.3, last claim). $B \le \frac{5}{4}\delta\Delta \le \frac{5}{2m}\delta^2 + \frac{m}{6}\Delta^2$ (Lemma 8.15, Young), and $\frac{5}{2m}\delta^2 \le \frac{5mr^2}{32}$. By Lemma 8.14(b), (i) and Lemma 8.3, $C \le \|\omega_0\||t_1|\frac{\eta}{es} \le \frac{mr^2}{12e} < 0.0307\, mr^2$ by (iii).

**Step S8.33: Contradiction.** Hence $\frac{m}{3}\Delta^2 \le mr^2(\frac{1}{16} + \frac{5}{32} + \frac{1}{12e}) < \frac{mr^2}{4}$, while $\frac{m}{3}\Delta^2 \ge \frac{m}{3}\cdot\frac{225}{256}r^2 > 0.29\, mr^2$ ((E3)). □

8.6. **Lemma 1 and Lemma 2.**

**Lemma 8.17** (Lemma 1: stability for $m$-convex $h$ in normal form)**.** *In (RS) let $h$ be $m$-convex on $U_r D$. Then for every interval $J \ni 0$ and every $H$-orbit $z$ in $W_r$ on $J$ with $I_0 = I(0)$ admissible:*

$$\|I(t) - I_0\|_2 \le r - \rho \qquad (t \in J,\ |t| \le T_*), \qquad T_* = \frac{s\,m\,r^2}{12\,\|\omega(I_0)\|_2\,\eta},$$

*with $T_* = +\infty$ if $\|\omega(I_0)\|_2\eta = 0$. (In particular for $\Lambda = \mathbb{Z}^n$, $f_* = 0$: all $t \in J$.)*

*Source: Pos93 p. 197 (Lemma 1: $\delta \le mr/4$, $\zeta \le mr^2/32$, $\rho = \delta/4M$, $T_* = \frac{s}{12\|\omega(I_0)\|}\frac{mr^2}{\eta}$). Statement in stopped form ("as long as the action stays in $W_r$": the orbit is an orbit in $W_r$ on $J$); with exit analysis by Lemma 8.10.*

*Proof.* **Step S8.34: Exit.** Suppose $\|I(t_1) - I_0\|_2 > r - \rho$ for some $t_1 \in J$, $|t_1| \le T_*$. Lemma 8.10 ($|\cdot|_* = \|\cdot\|_2$) gives $t^* \in J$, $|t^*| \le |t_1|$, with (i) of Lemma 8.16 for $t^*$.

**Step S8.35: Hypotheses (ii), (iii).** $I_s \in \bar{B} \subset U_rD$ (Lemma 8.15), so (ii) is $m$-convexity; (iii) is $|t^*| \le T_*$. Lemma 8.16 gives the contradiction. □

**Lemma 8.18** (Lemma 2: stability for $\ell, m$-quasi-convex $h$ in normal form)**.** *In (RS) let $h$ be $\ell, m$-quasi-convex on $U_rD$ and assume*

$$r \le 2\ell/m \qquad \text{and} \qquad Mr \le \ell/4 \tag{R}$$

*(the second is an* added *hypothesis, Remark 8.20(1)). For admissible $I_0$ let $\Omega = \max\{\|\omega(I)\|_2 : \|I - I_0\|_2 \le r - \rho\}$. Then for every $H$-orbit $z$ in $W_r$ on $J \ni 0$ with $I(0) = I_0$ admissible,*

$$\|I(t) - I_0\|_2 \le r - \rho \quad (t \in J,\ |t| \le T_*), \qquad T_* = \frac{s\, m\, r^2}{12\,\Omega\,\eta}\ (= +\infty \text{ if } \Omega\eta = 0).$$

*Source: Pos93 p. 199 (Lemma 2: $r \le 2\ell/m$, $\Omega = \sup_{\|I-I_0\|\le r-\rho}\|\omega(I)\|$, proof by contradiction at the first exit time). The hypothesis $Mr \le \ell/4$ repairs a gap of the proof (Remark 8.20(1)).*

*Proof.* **Step S8.36: Exit.** As in Lemma 8.17, a violation yields $t^*$, $|t^*| \le T_*$ with (i) of Lemma 8.16. Put $\Delta I$, $I_s$ as there, $\Delta = r - \rho \ge \frac{15}{16}r$; $I_s \in \bar{B}$ and $\|\omega(I_s)\| \le \Omega$.

**Step S8.37: (iii).** $\|\omega(I_0)\| \le \Omega$, so $|t^*|\|\omega(I_0)\|\eta \le |t^*|\Omega\eta \le smr^2/12$.

**Step S8.38: The cone condition along the segment.** Fix $s' \in [0,1]$. Then $|\langle \omega(I_{s'}), \Delta I\rangle| \le \|\Pi\omega(I_{s'})\|\Delta + \|\omega(I_{s'})\|\|\Pi^{\perp}\Delta I\|$. By Lemma 8.15(b) ($\|I_{s'} - I_0\| \le \Delta \le r$) the first term is $\le (\frac{5}{4}\delta + Mr)\Delta \le (\frac{5}{16}mr + Mr)\Delta$. By Lemma 8.14(b), Lemma 8.3 and Step S8.37 the second is $\le \Omega|t^*|\frac{\eta}{es} \le \frac{mr^2}{12e} \le \frac{16}{15}\cdot\frac{mr}{12e}\Delta < 0.0328\, mr\,\Delta$. With $mr \le 2\ell$ and $Mr \le \ell/4$:

$$|\langle \omega(I_{s'}), \Delta I\rangle| \le \big(0.3453 \cdot 2\ell + \tfrac{\ell}{4}\big)\Delta < 0.941\,\ell\,\|\Delta I\|_2.$$

**Step S8.39: Convexity along the segment.** Since $|\langle \omega(I_{s'}), \Delta I\rangle| \le \ell\|\Delta I\|_2$, the first alternative of quasi-convexity at $I_{s'} \in U_rD$ fails for $\xi = \Delta I$, so $Q(I_{s'})[\Delta I, \Delta I] \ge m\|\Delta I\|^2$: (ii) of Lemma 8.16 holds. That lemma gives the contradiction. □

## 8.7. **Proposition 2: the Resonant Stability Lemma.**

**Proposition 8.19** (Resonant Stability Lemma)**.** *Let $n \ge 1$, $D \subset \mathbb{R}^n$, $r, s, K, \alpha > 0$, $1 < p \le \frac{3}{2}$ and $q = p/(p-1)$ ($\ge 3$), $\kappa = 2\pi Ks$. Let $\Lambda \subset \mathbb{Z}^n$ be a subgroup, $0 < m \le M$, $\ell, \delta > 0$, $h$ holomorphic on $V_rD$ and real on $U_rD$ with*

$$\|\partial_I^2 h(I)\|_{2\to 2} \le M \quad \text{for all } I \in V_rD \text{ (complex Hessian, operator norm on } (\mathbb{C}^n, \|\cdot\|_2)),$$

*$h$ $\ell, m$-quasi-convex on $U_rD$, $f \in \mathcal{O}(r,s)^{\mathbb{R}}$ with $\varepsilon = |f|_{r,s} < \infty$, $H = h + f$. Assume:*

- *(a) $D$ is $\alpha, K$-nonresonant modulo $\Lambda$: $|k\cdot\omega(I)| \ge \alpha$ for $I \in D$, $k \in Z_K \setminus \Lambda$, and $\delta$-close to exact $\Lambda$-resonances;*
- *(b) $\varepsilon \le \frac{p}{q}\cdot\frac{mr^2}{2^7}$ and $\delta \le \frac{mr}{8}$;*
- *(c) $r \le \min\Big(\frac{\alpha}{p\sqrt{n}\,MK},\ \frac{4\ell}{m}\Big)$ and (added) $Mr \le \ell/2$.*

*For $I_0 \in D$ let $\Omega(I_0) = \max\{\|\omega(I)\|_2 : \|I - I_0\|_2 \le r/2\}$ and*

$$T(I_0) = \frac{s\,m\,r^2}{288\,\Omega(I_0)\,\varepsilon}\,e^{\kappa/6} \qquad (= +\infty \text{ if } \Omega\varepsilon = 0).$$

*Then for every interval $J \ni 0$ and every $H$-orbit $z$ in $W_r$ on $J$ with $I_0 = I(0) \in D$,*

$$\|I(t) - I_0\|_2 \le \tfrac{r}{2} \ (\text{so } |I(t) - I_0| \le r) \qquad (t \in J,\ |t| \le T(I_0)),$$

*and if $\Lambda = \mathbb{Z}^n$ this holds for all $t \in J$.*

*Source: Pos93 pp. 199–200 (Proposition 2: $\varepsilon \le \frac{p}{q}\frac{mr^2}{2^7}$, $\delta \le \frac{mr}{8}$, $r \le \min(\frac{\alpha}{pMK}, \frac{4\ell}{m}, r_0)$, $q \ge 3$; $\|I(t) - I_0\| \le r$ for $|t| \le \frac{s}{288\Omega}\frac{mr^2}{\varepsilon}e^{Ks/6}$; $\Lambda = \mathbb{Z}^n$: all $t$). Deviations: $\sqrt{n}$ in the first bound of (c) (Euclidean Hessian bound $M$ against the max-norm Lipschitz bound $L = \sqrt{n}M$ of Part 7); added hypothesis $Mr \le \ell/2$ and the radius $r/2$ for $\kappa < 6$ (Remark 8.20(1),(2)); $\Omega$ taken over the radius $r/2$ (the proof confines to $r/2$).*

*Proof.* **Step S8.40: Reductions.** Replace $\delta$ by $\delta' = mr/8 \ge \delta$ ($D$ is $\delta'$-close). Then $\rho' := \delta'/(4M) = \frac{mr}{32M}$. Since $\frac{p}{q} = \frac{1}{q-1} \le \frac{1}{2}$, (b) gives $\varepsilon \le mr^2/2^8$. By (c) and $m \le M$: $L := \sqrt{n}M$ satisfies $KLr \le \alpha/p$, and $\varepsilon \le \frac{pmr^2}{q2^7} \le \frac{pMr^2}{q2^7} \le \frac{\alpha r}{2^7 qK}$. The complex Hessian bound gives (N2) with $L = \sqrt{n}M$ (Lemma 7.2: $|Qv|_\infty \le \|Qv\|_2 \le M\|v\|_2 \le \sqrt{n}M|v|_\infty$). Put $\tilde r = r/2$; $M\tilde r \le \ell/4$, $\tilde r \le 2\ell/m$, $\delta' = m\tilde r/4$.

**Step S8.41: Three cases.** (A) $\Lambda \ne \mathbb{Z}^n$, $\kappa \ge 6$: Theorem 7.10 (conditions (N1), (N2), $K > 0$, (i), (ii) hold; $\Lambda = -\Lambda$, reality) gives $\Psi : \mathcal{D}_{r/2,s/6} \to \mathcal{D}_{r,s}$ with $H \circ \Psi = h + g + f_*$, $P_\Lambda g = g$, $\tilde s = s/6$, $\eta \le e^{-\kappa/6}\varepsilon$ and $|g|_{r,s}$-type bound: $|g|_{r/2,s/6} \le |g_0|_{r,s} + 16c\varepsilon^2$, where $16c\varepsilon^2 = \frac{32qK\varepsilon^2}{\alpha r} \le \frac{\varepsilon}{4}$ by (i), so $|g|_{r/2,s/6} + \eta \le \frac{5}{4}\varepsilon + \frac{\varepsilon}{e} < 2\varepsilon =: \zeta$. (B) $\Lambda \ne \mathbb{Z}^n$, $\kappa < 6$: $\Psi = \mathrm{id}$ on $\mathcal{D}_{r/2,s}$, $g = 0$, $f_* = f$, $\tilde s = s$, $\eta = \varepsilon$, $\zeta = \varepsilon$. (C) $\Lambda = \mathbb{Z}^n$: $\Psi = \mathrm{id}$, $\tilde s = s$, $g = f$ ($P_\Lambda g = g$), $f_* = 0$, $\eta = 0$, $\zeta = \varepsilon$. In all cases $\tilde H = h + g + f_*$ on $\mathcal{D}_{\tilde r,\tilde s}$ is in (RS) with $(r,s) \to (\tilde r, \tilde s)$ and $\delta'$: $\delta' = m\tilde r/4$, and $\zeta \le 2\varepsilon \le mr^2/128 = m\tilde r^2/32$.

**Step S8.42: (T1), (T2).** Case A: $\|\Pi_I\Psi - I\|_2 \le \sqrt{n}\frac{qK\varepsilon}{\alpha} \le \frac{q\varepsilon}{pMr} \le \frac{mr}{2^7 M} = \frac{\rho'}{4} =: d_*$ (Theorem 7.10(c), $\alpha \ge p\sqrt{n}MKr$, (b)); (T2) is Corollary 8.7. Cases B, C: $d_* = 0$, trivial. Norm $|\cdot|_* = \|\cdot\|_2$; $\varrho := \tilde r - \rho'$; then $d_* + \varrho < \tilde r$.

**Step S8.43: (T3).** Let $\tilde z$ be an $\tilde H$-orbit in $W_{\tilde r}$, $\|\tilde I(0) - I_y\|_2 \le d_* < \rho'$, $I_y \in D$: $\tilde I(0)$ is admissible. Lemma 8.18 (hypotheses (R) at radius $\tilde r$) gives $\|\tilde I(t) - \tilde I(0)\|_2 \le \varrho$ for $|t| \le \tilde T_* = \frac{\tilde s\, m\tilde r^2}{12\tilde\Omega\eta}$, $\tilde\Omega = \max_{\|I - \tilde I(0)\|_2 \le \varrho}\|\omega\|_2$. Since $\varrho + d_* \le \tilde r - \rho' + \rho'/4$, the ball lies in $\{\|I - I_y\|_2 \le r/2\}$ so $\tilde\Omega \le \Omega(I_y)$. Case A: $\tilde T_* \ge \frac{(s/6)(mr^2/4)}{12\Omega\, e^{-\kappa/6}\varepsilon} = T(I_y)$. Case B: $\tilde T_* = \frac{smr^2}{48\Omega\varepsilon} \ge T(I_y)$ because $e^{\kappa/6} < e < 6$. Case C: $\tilde T_* = \infty$.

**Step S8.44: Conclusion.** Lemma 8.8: $\|I(t) - I_0\|_2 \le \varrho + 2d_* \le \tilde r - \rho' + \rho'/2 < \frac{r}{2}$ for $|t| \le T(I_0)$ ($t \in J$), and all $t$ in case C. Finally $|\cdot| \le \|\cdot\|_2$. □

## 8.8. **Errata, deviations, dictionary, consumers.**

*Remark* 8.20 (Suspected defects in Pos93 (to the ledger)). (1) *p. 199, proof of Lemma 2*: "$|\langle\omega(I(s)), P\Delta I\rangle| \le \frac{5}{4}\delta\|\Delta I\|$". The closeness $\frac{5}{4}\delta$ was derived (p. 198) only for $U_\rho D$, but $I(s)$ ranges over the ball of radius $r - \rho$ about $I_0$, where the mean value theorem gives only $\|P\omega(I(s))\| \le \frac{5}{4}\delta + M\|I(s) - I_0\|$. Without a bound on $Mr$ the cone condition $|\langle\omega(I(s)), \Delta I\rangle| \le \ell\|\Delta I\|$ need not hold, so Lemma 2 (and Proposition 2, which uses it) is not proved as printed. Repair: the hypothesis $Mr \le \ell/4$ (Lemma 8.18), resp. $Mr \le \ell/2$ in Proposition 8.19. In Theorem 1 (p. 190) one has $r \le r_0/(AK)$, $r_0 \le 4\ell/m$, $A = 11M/m$, so $Mr \le \frac{4\ell}{11K}$, and Part 9 must check $Mr \le \ell/2$ there: $K \ge 1$ suffices only for the blocks $\Lambda \ne \mathbb{Z}^n$; on the block $\mathbb{Z}^n$ ($r = r_0$) it needs $r_0 \le \ell/(2M)$ (Part 10, Remark 10.9(1)). The lemma may be true without it; we do not claim so.

(2) *p. 200, Proposition 2, case $Ks < 6$*: "we simply apply those two lemmata to the given hamiltonian $H$ with $g = 0$ and $f_* = f_\varepsilon$"; Lemma 2 needs $r \le 2\ell/m$ while only $r \le 4\ell/m$ is assumed. *Classification (review N7-02):* the scan says only to apply the two lemmas to $H$ with $g = 0$; it states no radius for this application (the preceding proof had already introduced a half-radius), so this is an *omitted radius choice*, not a confirmed false application. We supply it explicitly: apply them at radius $r/2$ (case B).

(3) *p. 196, Proposition 1 and p. 200*: "$\Psi^{-1}$ takes any given initial position in $D \times \mathbb{T}^n$ to some initial position in $U_\rho D \times \mathbb{T}^n$" needs surjectivity of $\Psi$ onto a neighbourhood of $D \times \mathbb{T}^n$; not argued (Lemma 8.6 supplies it by a contraction argument using the displacement bounds of Theorem 7.10(c)).

(4) *p. 197, first line*: "For $Ks \le 6 \ldots f_* = f_\varepsilon$". *Allegation rejected (review N7-03):* the bound $|f_*| \le e^{-Ks/6}\varepsilon$ is false for $f_* = f \ne 0$ (Part 7, Proposition 7.12), but p. 197 never asserts it; it proves Proposition 1 directly in the original coordinates. Not a source defect; our direct small-$Ks$ argument is Remark 8.12.

(5) Misprint-free checks: the numbers $\frac{m}{3}\|\Delta I\|^2 \le \frac{mr^2}{4} < \frac{m}{3}(r-\rho)^2$ (p. 198) and $T_*$, $\frac{s}{288\Omega}$ (p. 200) were recomputed and agree; the transfer constants in Proposition 8.19 are sharper than printed ($\Psi$-displacement $qK\varepsilon/\alpha$ instead of $2qK\varepsilon/\alpha$ in the max norm).

*Remark* 8.21 (Dictionary with Pöschel's normalisation). With $\varphi = 2\pi\theta$, $I_P = I/2\pi$ (Remark 5.11): $s_P = 2\pi s$, $r_P = r/2\pi$, $\omega_P = 2\pi\omega$, $m_P = (2\pi)^2 m$, $M_P = (2\pi)^2 M$, $\ell_P = 2\pi\ell$, $\delta_P = 2\pi\delta$ (distance of the frequency to a resonance subspace scales like $\omega$; review N7-01), $\rho_P = \rho/2\pi$, while the Hamiltonian and remainder sizes $\eta, \varepsilon, \zeta$ are unchanged (the substitution is symplectic), up to the Euclidean/max conventions. Then $s_P r_P = sr$, $s_P m_P r_P^2/\omega_P = s\,m\,r^2/\omega$, so $T_*$, $T_{\mathrm{nr}}$ and $T$ have the printed form; and $\partial_\varphi f_* = \partial_\theta f_*/2\pi$ makes $|\partial_\theta f_*| \le \eta/(es)$ the same estimate as his $|\partial_\theta f_*| \le \eta/(es_P)$.

*Remark* 8.22 (Consumers). Part 9 (Theorem 1 of Pöschel) takes $q = 9$, $p = \frac{9}{8}$ ($\frac{p}{q} = \frac{1}{8}$, $q \ge 3$), $\Lambda$ a maximal $K$-lattice, $D$ a block of Part 8 (with $K = K_a$ and $K_g = \sqrt{n}K_a$: $K_a \le K_g$ transfers nonresonance from the Euclidean ball to $Z_{K_a}$ and $K_g \ge \sqrt{n}K_a$ is the compatibility with the radius condition $r \le \alpha/(p\sqrt{n}MK_a)$ of Proposition 8.19; Part 9, Remark 9.1, Corollary 9.16), and uses Proposition 8.11 for $\Lambda = \{0\}$ and Proposition 8.19 otherwise. The added hypothesis $Mr \le \ell/2$ holds for every block with $K \ge 1$ *other than the block* $\mathbb{Z}^n$, where $r = r_0$ and it requires $r_0 \le \ell/(2M)$ (Part 10, Remark 10.9(1); in Part 2 this is the entry $\ell/(2B_c)$ of $r_0$). With Part 1: $M = B_c$ (Lemma 2.5), $\ell$ as defined in Lemma 2.7 and the curvature parameter $m = \mu$ of its clause (b) (two distinct constants); the Hessian bound on $V_r D$ requires $V_r D \subset B(0,R)_s$ (Lemma 2.2(c)), and the quasi-convexity conversion, stated on $B(0,R)$, requires $U_r D \subset B(0,R)$, to be proved at the import (it holds for the max-norm tube $U_{r_0}P \subset B(0, 5R/8)$, Lemma 2.10(c), Lemma 2.2(c)); containment of the complex tube in the analytic domain alone does not give it. Part 10 connects the stopped form with existence through Corollary 8.9 (take $F = X_{H_{h,f}}$, which equals $X_{H^{\mathbb{R}}}$ on $W_r$ when $U_r D \subset \mathcal{B}_s(R)$) and Theorem 4.34. Track I takes Lemma 8.14 and Lemmas 8.16, 8.17, 8.18, which use no property of $h, g, f_*$ beyond the stated ones.

## 9. Geometry of resonances (shared by tracks N and P)

This part is Section 4 of Pöschel 1993 (pp. 200–203): the volume $|\Lambda|$ of an integer lattice, maximal $K$-lattices, resonance zones and blocks, the covering $\mathbb{R}^n = B_\Theta \cup B_1^* \cup \cdots \cup B_n^*$, the Geometric Lemma and the Covering Lemma. It is pure linear algebra and lattice geometry: no Hamiltonian, no regularity and no holomorphy enters, the dimension $n \ge 1$ and the sublattices of $\mathbb{Z}^n$ are arbitrary, and the frequency map is only a map $\omega : P \to \mathbb{R}^n$. Track P may take every statement below as it stands.

Conventions. Throughout this part $\|\cdot\|$ is the *Euclidean* norm on $\mathbb{R}^n$ ($= |\cdot|_2$ of Part 2), $x \cdot y$ the dot product, $|k|_1 = \sum|k_i|$ and $|x|_\infty = \max|x_i|$; matrices are real, $A^\top$ is the transpose. The

geometry needs orthogonal projections, hence the Euclidean norm; Part 2, Lemma 2.1 relates it to the max norm and to $|k|_1$. Since a lattice cut-off appears in two roles, we write $K$ (or $K_g$) for the *geometric* cut-off, a bound on $\|k\|$, and $K_a$ for the *analytic* cut-off of Parts 5–8, a bound on $|k|_1$, with $Z_{K_a} = \{k \in \mathbb{Z}^n : |k|_1 \le K_a\}$ (Part 7). The geometric ball is $\mathcal{E}_K = \{k \in \mathbb{Z}^n : \|k\| \le K\}$ (a finite set). The compatibility of the two cut-offs is Corollary 9.16 (finding N06 of the review of Part 0).

*Remark* 9.1 (Direction of the cut-off compatibility: correction to Parts 7 and 8). (Historical.) An earlier draft of Parts 7 and 8 (remarks on cut-offs and "consumers") stated that the analytic cut-off must satisfy $K_a \ge \sqrt{n}K_g$; the caller remarks of both parts have since been corrected to $K_g \ge \sqrt{n}K_a$. The earlier inequality was the wrong way round for what Part 8 needs. The nonresonance *conclusion* of the Geometric Lemma holds on the Euclidean ball $\mathcal{E}_{K_g}$, and Part 7 needs it on the $\ell^1$ ball $Z_{K_a}$, which is contained in $\mathcal{E}_{K_a}$ (not in $\mathcal{E}_{K_g}$ unless $K_a \le K_g$). So one needs $K_a \le K_g$. The factor $\sqrt{n}$ enters in the other place: the radius condition $r \le \alpha/(p\sqrt{n}\,MK_a)$ of Part 8 (Proposition 8.19(c)) is implied by the covering radius $r_\Lambda = \alpha_\Lambda/(pMK_g)$ exactly when $K_g \ge \sqrt{n}K_a$. Hence the correct compatibility is $K_g \ge \sqrt{n}\,K_a$ (Corollary 9.16), and Part 9 should take, e.g., $K_g = \sqrt{n}K_a$ (with $K_a \ge 1/\sqrt{n}$ so that $K_g \ge 1$). With $K_a \ge \sqrt{n}K_g > K_g$ the claim "$Z_{K_a} \setminus \Lambda$ is nonresonant on the blocks" would not follow. Example: $n = 2$, $K_g = 2$, $K_a = 3 \ge \sqrt{2}K_g$, $\omega = (1, -2)$. Every nonzero $k \in \mathcal{E}_2$ has $|k \cdot \omega| \ge 1$ (all 12 vectors checked), but $k = (2, 1)$ has $|k|_1 = 3 \le K_a$, $\|k\| = \sqrt{5} > 2$ and $k \cdot \omega = 0$. (Taking $\lambda_1 = \frac{1}{2}$, the point $\omega$ lies in $B_\Theta$ for a Geometric-Lemma family.)

### 9.1. **External facts.**

(E1) *Subgroups of $\mathbb{Z}^n$.* Every subgroup of $\mathbb{Z}^n$ is a free $\mathbb{Z}$-module of rank $d \le n$ (Mathlib `Submodule.basisOfPid`, applied to the standard basis `Pi.basisFun`; `LinearAlgebra/FreeModule/PID.lean`). Exists.

(E2) *Gram matrices.* If $A \in \mathbb{R}^{n\times d}$ has injective $x \mapsto Ax$ then $A^\top A$ is positive definite and $\det(A^\top A) > 0$ (Mathlib `Matrix.PosDef.conjTranspose_mul_self` and `Matrix.PosDef.det_pos`; `LinearAlgebra/Matrix/PosDef.lean`); $\det(MN) = \det M \det N$, $\det M^\top = \det M$ (`Matrix.det_mul`, `Matrix.det_transpose`). Exists.

(E3) *Schur complement.* For $G \in \mathbb{R}^{d\times d}$ invertible, $\det\left(\begin{smallmatrix} G & b \\ c^\top & \gamma \end{smallmatrix}\right) = \det G\,(\gamma - c^\top G^{-1} b)$ (Mathlib `Matrix.det_fromBlocks`$_{11}$; `LinearAlgebra/Matrix/SchurComplement.lean`). Exists.

(E4) *Unimodular matrices.* A matrix $U \in \mathbb{Z}^{d\times d}$ is invertible over $\mathbb{Z}$ iff $\det U \in \{\pm 1\}$ (`Matrix.isUnit_iff_isUnit_det`, `Int.isUnit_iff`); the change-of-basis matrix between two bases of a free module is invertible (`Basis.invertibleToMatrix`). A nonzero integer has modulus $\ge 1$. Exists.

(E5) *Independence over $\mathbb{Q}$ and $\mathbb{R}$.* Columns of an integer matrix that are $\mathbb{Z}$-linearly independent are $\mathbb{Q}$-independent (`LinearIndependent.iff_fractionRing`, `RingTheory/Localization/Module.lean`); $\mathbb{Q}$-independence of the $d$ columns of a rational matrix is the nonvanishing of one $d \times d$ minor (`Matrix.rank_eq_finrank_span_cols` and rank by minors), which does not depend on the field, so they are $\mathbb{R}$-independent. Exists (the second half as a short lemma over the field extension).

(E6) *Elementary facts.* Cauchy–Schwarz $|x \cdot y| \le \|x\|\|y\|$; a finite nonempty partially ordered set has a maximal element; finite sets of vectors in $\mathbb{Z}^n$ (the ball $\mathcal{E}_K$); a finite spanning set of a vector space contains a basis. Exists.

Part 2 (Lemma 2.1) is used for the norm comparisons; no track-K result is used.

### 9.2. **Lattices, dimension, volume.**

**Definition 9.2** (Lattices)**.** A *lattice* is a subgroup $\Lambda \le \mathbb{Z}^n$. Its *span* is $W_\Lambda = \operatorname{span}_{\mathbb{R}} \Lambda \subset \mathbb{R}^n$ and its *dimension* is $d = \dim \Lambda := \dim_{\mathbb{R}} W_\Lambda$. The trivial lattice is $\Theta = \{0\}$ ($d = 0$). The *resonance space* of $\Lambda$ is $R_\Lambda = \{\omega \in \mathbb{R}^n : k \cdot \omega = 0\ \forall k \in \Lambda\} = W_\Lambda^\perp$, of dimension $n - d$.
Source: Pos93 pp. 200–201.

**Lemma 9.3** (Bases and unimodular change)**.** *Let $\Lambda \le \mathbb{Z}^n$.*

*(a) $\Lambda$ has a $\mathbb{Z}$-basis $a_1, \dots, a_d$ with $d \le n$; every $\mathbb{Z}$-basis is $\mathbb{R}$-linearly independent, hence $\dim_{\mathbb{R}} W_\Lambda = d = \operatorname{rank}_{\mathbb{Z}} \Lambda$. For a basis put $A = (a_1 | \dots | a_d) \in \mathbb{Z}^{n\times d}$ ("basis matrix").*

*(b) If $\tilde{A} = AU$ is another basis matrix then $U \in \mathbb{Z}^{d\times d}$ and $\det U = \pm 1$; conversely $AU$ is a basis matrix if $U \in \mathbb{Z}^{d\times d}$, $\det U = \pm 1$.*

*(c) If $\Gamma \le \Lambda$ with $\dim \Gamma = \dim \Lambda = d$ and basis matrices $\tilde{A}$ of $\Gamma$ and $A$ of $\Lambda$, then $\tilde{A} = AN$ with $N \in \mathbb{Z}^{d\times d}$ and $\det N \neq 0$.*

*Source: Pos93 p. 201 ("there is a unimodular $d \times d$-matrix $U$ such that $\tilde{A} = AU$").*

*Proof.* **Step S9.1: Basis.** (a): (E1) gives a $\mathbb{Z}$-basis with $d \le n$ elements; (E5) gives $\mathbb{R}$-independence.

**Step S9.2: Unimodular.** (b): each column of $\tilde{A}$ lies in $\Lambda$, so $\tilde{A} = AU$ with an integer matrix $U$ (coordinates in the basis $A$) and likewise $A = \tilde{A}\tilde{U}$; then $A = AU\tilde{U}$ gives $U\tilde{U} = 1$ by injectivity of $A$ ((E2)), so $\det U \det \tilde{U} = 1$ with both integers, i.e. $\det U = \pm 1$ ((E4)). Conversely $AU$ has columns in $\Lambda$, and $U^{-1} \in \mathbb{Z}^{d\times d}$ shows they generate $\Lambda$ and are independent.

**Step S9.3: Sublattice.** (c): $\tilde{A} = AN$ as in (b). If $Nx = 0$ then $\tilde{A}x = ANx = 0$, so $x = 0$ ($\tilde{A}$ injective by (a)); thus $N$ is injective, $\det N \neq 0$. □

**Definition 9.4** (Volume of a lattice)**.** For $\Lambda \neq \Theta$ with basis matrix $A$ put $|\Lambda| := \sqrt{\det(A^\top A)}$; $|\Theta| := 1$. (The radicand is $> 0$ by (E2) and Lemma 9.3(a); geometrically $|\Lambda|$ is the $d$-dimensional volume of the parallelepiped spanned by a basis.)
Source: Pos93 p. 200–201.

**Lemma 9.5** (Orthogonal projection onto a span)**.** *Let $A \in \mathbb{R}^{n\times d}$ ($d \ge 1$) have independent columns, $W$ their span, $G = A^\top A$ (invertible, (E2)) and $\Pi_W = AG^{-1}A^\top$, $Q_W = 1 - \Pi_W$. Then:*

*(a) $\Pi_W^\top = \Pi_W = \Pi_W^2$, $\Pi_W v \in W$ for all $v$, $\Pi_W v = v$ for $v \in W$, and $Q_W v \perp W$. Hence $\|v\|^2 = \|\Pi_W v\|^2 + \|Q_W v\|^2$, $\|\Pi_W v\| \le \|v\|$, $\|Q_W v\| \le \|v\|$. For $d = 0$ put $\Pi_{\{0\}} = 0$.*

*(b) $\Pi_W v$ is the unique $w \in W$ with $v - w \perp W$.*

*(c) $\operatorname{dist}_2(\omega, W^\perp) = \|\Pi_W \omega\|$ for all $\omega \in \mathbb{R}^n$ (the minimum is attained at $Q_W\omega$).*

*(d) If $W \subset W'$ are two such spans, then $\Pi_{W'}\Pi_W = \Pi_W \Pi_{W'} = \Pi_W$.*

*Consequently $R_\Lambda = W_\Lambda^\perp$ and $\operatorname{dist}_2(\omega, R_\Lambda) = \|\Pi_{W_\Lambda}\omega\|$; write $\Pi_\Lambda = \Pi_{W_\Lambda}$, $Q_\Lambda = Q_{W_\Lambda}$.*
*Source: Standard (Pos93 p. 202 uses these properties silently).*

*Proof.* **Step S9.4: Algebra.** $\Pi_W^\top = AG^{-\top}A^\top = \Pi_W$ ($G$ symmetric), $\Pi_W^2 = AG^{-1}(A^\top A)G^{-1}A^\top = \Pi_W$, $\Pi_W A = A$, so $\Pi_W v = v$ on $W$; $A^\top Q_W = A^\top - GG^{-1}A^\top = 0$ gives $Q_W v \perp W$. Pythagoras follows from $\Pi_W v \cdot Q_W v = v \cdot \Pi_W Q_W v = 0$.

**Step S9.5: Uniqueness, distance.** (b): if $w, w' \in W$ both have $v - w, v - w' \perp W$ then $w - w' \in W \cap W^\perp = 0$. (c): for $u \in W^\perp$, $\|\omega - u\|^2 = \|\Pi_W\omega\|^2 + \|Q_W\omega - u\|^2$ (as $\Pi_W\omega \in W \perp Q_W\omega - u$), with equality for $u = Q_W\omega \in W^\perp$.

**Step S9.6: Nested spans.** (d): $\Pi_W v \in W \subset W'$ so $\Pi_{W'}\Pi_W v = \Pi_W v$; transposing gives $\Pi_W \Pi_{W'} = \Pi_W$. □

**Lemma 9.6** (Volume)**.** *Let $n \ge 1$ and let $\Lambda, \Gamma \le \mathbb{Z}^n$ be lattices.*

*(a) (independence of the basis) $\det(A^\top A)$ does not depend on the basis matrix $A$ of $\Lambda$; so $|\Lambda|$ is well defined.*

*(b) $|\Lambda|^2 = \det(A^\top A)$ is a positive integer; hence $|\Lambda| \ge 1$.*

*(c) If $\Gamma \le \Lambda$ and $\dim \Gamma = \dim \Lambda$ then $|\Gamma| = |\det N||\Lambda| \ge |\Lambda|$, $N$ as in Lemma 9.3(c).*

*(d) $|\mathbb{Z}g| = \|g\|$ for $g \in \mathbb{Z}^n \setminus \{0\}$; $|\mathbb{Z}^n| = 1$.*

*(e) (adding a vector) If $k \in \mathbb{Z}^n \setminus W_\Lambda$, then $\Lambda + \mathbb{Z}k$ has dimension $\dim \Lambda + 1$ and $|\Lambda + \mathbb{Z}k| = |\Lambda|\,\|Q_\Lambda k\|$, with $Q_\Lambda k \neq 0$.*

*(f)* (Hadamard) *If $v_1, \dots, v_d \in \mathbb{Z}^n$ are $\mathbb{R}$-independent and generate $\Gamma$, then $|\Gamma| \le \prod_i \|v_i\|$.*

*Source: Pos93 p. 201 (a), p. 202 (the inequalities "the volume of $\Lambda_+$ is not bigger than the volume of $\Lambda$ times the length of the component of $k$ orthogonal to $\Lambda$"; (c),(e) make this precise).*

*Proof.* **Step S9.7: (a),(b).** If $\tilde{A} = AU$, $\det \tilde{A}^\top \tilde{A} = \det U^\top \det(A^\top A) \det U = (\det U)^2 \det A^\top A = \det A^\top A$ by Lemma 9.3(b) and (E2). The matrix $A^\top A$ is an integer matrix, so its determinant is an integer, $> 0$ by (E2). For $\Lambda = \Theta$, $|\Theta| = 1$.

**Step S9.8: (c).** $\tilde{A}^\top \tilde{A} = N^\top (A^\top A) N$, so $|\Gamma|^2 = (\det N)^2 |\Lambda|^2$ with $\det N \in \mathbb{Z} \setminus \{0\}$ (Lemma 9.3(c)), $|\det N| \ge 1$. (For $\Lambda = \Theta$ there is nothing to prove.)

**Step S9.9: (d).** $A = g$ gives $\sqrt{g^\top g} = \|g\|$; $A = 1$ gives 1.

**Step S9.10: (e).** By Lemma 9.5(a), $k \notin W_\Lambda$ means $Q_\Lambda k \neq 0$ (else $k = \Pi_\Lambda k \in W_\Lambda$). With $A$ a basis matrix of $\Lambda$ the columns of $A' = (A|k)$ generate $\Lambda + \mathbb{Z}k$ and are $\mathbb{R}$-independent, hence form a basis, $\dim = d + 1$. Then $A'^\top A' = \left(\begin{smallmatrix} G & A^\top k \\ k^\top A & k^\top k \end{smallmatrix}\right)$, $G = A^\top A$, and (E3) gives $\det A'^\top A' = \det G\,(k^\top k - k^\top A G^{-1} A^\top k) = \det G\,(\|k\|^2 - k \cdot \Pi_\Lambda k) = \det G\,\|Q_\Lambda k\|^2$, since $k \cdot \Pi_\Lambda k = \|\Pi_\Lambda k\|^2$ (Lemma 9.5(a)) and Pythagoras. For $\Lambda = \Theta$ this reads $|\mathbb{Z}k| = \|k\|$ by (d).

**Step S9.11: (f).** Induction on $d$ with (e): for $\Gamma_j = \langle v_1, \dots, v_j \rangle$, $|\Gamma_j| = |\Gamma_{j-1}| \|Q_{\Gamma_{j-1}} v_j\| \le |\Gamma_{j-1}| \|v_j\|$ (Lemma 9.5(a)), starting from $|\Theta| = 1$. □

*Remark* 9.7 (Totality). Every square root and division in this part is guarded: $|\Lambda| > 0$ (Lemma 9.6(b)); $G^{-1}$ exists by (E2); $\|Q_\Lambda k\| > 0$ when divided by (Lemma 9.6(e)); the radicands $\|\Pi_+ \omega\|^2 - \|\Pi \omega\|^2$ and $A^2 - 1$ in Lemma 9.13 are shown to be $\ge 0$ (indeed $> 0$) before the square root is taken. Nothing is evaluated outside its domain.

### 9.3. Maximal $K$-lattices, zones, blocks, covering of frequency space.

**Definition 9.8** ($K$-lattices)**.** Let $K \ge 0$. A lattice $\Lambda$ is a *$K$-lattice* if it is generated (as a group) by $\Lambda \cap \mathcal{E}_K$. It is a *maximal $K$-lattice* if, moreover, every $K$-lattice $\Lambda' \supset \Lambda$ with $\dim \Lambda' = \dim \Lambda$ equals $\Lambda$. The family of maximal $K$-lattices is $\mathbb{M}_K$; $\mathbb{M}_K^d$ denotes those of dimension $d$.

Source: Pos93 p. 200 ("lattices generated by vectors in $\mathbb{Z}_K^n$ and not properly contained in any other lattice of the same dimension"). The Euclidean ball is the one used on p. 202 ("it suffices to assume $\|k\| \le K$"). If one reads "any other lattice" as an arbitrary subgroup of $\mathbb{Z}^n$ the maximal ones are the saturated ones $W \cap \mathbb{Z}^n$; these belong to our $\mathbb{M}_K$ (Lemma 9.9(a)), and all statements below hold for the larger family $\mathbb{M}_K$.

**Lemma 9.9** (Maximal $K$-lattices)**.** *Let $K \ge 0$ and $\Lambda$ a $K$-lattice.*

- *(a) $\Lambda \in \mathbb{M}_K$ iff $\mathcal{E}_K \cap W_\Lambda \subset \Lambda$ (i.e. every integer vector of norm $\le K$ in the real span of $\Lambda$ lies in $\Lambda$). In particular $\Theta \in \mathbb{M}_K$, and a saturated $K$-lattice is maximal.*
- *(b) Every $K$-lattice $\Gamma$ is contained in some $\Lambda \in \mathbb{M}_K$ with $\dim \Lambda = \dim \Gamma$. In particular, for $k \in \mathcal{E}_K \setminus \{0\}$ there is $\Lambda \in \mathbb{M}_K^1$ with $k \in \Lambda$.*
- *(c) If $\Lambda \in \mathbb{M}_K$ and $k \in \mathcal{E}_K \setminus \Lambda$, then $k \notin W_\Lambda$, so $\dim \Lambda < n$ and $\Lambda + \mathbb{Z}k$ is a $K$-lattice of dimension $\dim \Lambda + 1$.*
- *(d) If $\Lambda \in \mathbb{M}_K$ and $\dim \Lambda = n$ then $\mathcal{E}_K \subset \Lambda$; if $K \ge 1$ then $\Lambda = \mathbb{Z}^n$, and $\mathbb{Z}^n \in \mathbb{M}_K$. If $K < 1$ then $\mathbb{M}_K = \{\Theta\}$.*
- *(e) If $\Lambda \in \mathbb{M}_K$ has dimension $d$ then $1 \le |\Lambda| \le K^d$.*

*Source: Pos93 pp. 200–202 (the properties used in the proofs of the Geometric Lemma and of the optimality remark); the verification is ours.*

*Proof.* **Step S9.12: Characterisation.** ($\Rightarrow$) Let $k \in \mathcal{E}_K \cap W_\Lambda$. Then $\Lambda + \mathbb{Z}k$ is generated by $(\Lambda \cap \mathcal{E}_K) \cup \{k\} \subset \mathcal{E}_K$, so it is a $K$-lattice; $k \in W_\Lambda$ gives $W_{\Lambda + \mathbb{Z}k} = W_\Lambda$, same dimension; maximality gives $\Lambda + \mathbb{Z}k = \Lambda$, $k \in \Lambda$. ($\Leftarrow$) Let $\Lambda' \supset \Lambda$ be a $K$-lattice of the same dimension. Then $W_\Lambda \subset W_{\Lambda'}$

and the dimensions agree, so $W_{\Lambda'} = W_\Lambda$; $\Lambda'$ is generated by $\Lambda' \cap \mathcal{E}_K \subset \mathcal{E}_K \cap W_\Lambda \subset \Lambda$, so $\Lambda' \subset \Lambda$. Saturated: $\Lambda = W_\Lambda \cap \mathbb{Z}^n \supset \mathcal{E}_K \cap W_\Lambda$. $\Theta$: $W = \{0\}$.

**Step S9.13: Existence.** (b): the set $\mathcal{S}$ of $K$-lattices $\Lambda \supset \Gamma$ with $\dim \Lambda = \dim \Gamma$ is finite (each is generated by a subset of the finite set $\mathcal{E}_K$), nonempty and ordered by inclusion; take an inclusion-maximal element $\Lambda$. Any $K$-lattice $\Lambda' \supset \Lambda$ of the same dimension is in $\mathcal{S}$, hence equals $\Lambda$. For $k \neq 0$ apply this to $\Gamma = \mathbb{Z}k$ ($K$-lattice of dimension 1).

**Step S9.14: (c).** By (a), $k \in \mathcal{E}_K \cap W_\Lambda$ would give $k \in \Lambda$; so $k \notin W_\Lambda$, and $\dim \Lambda + 1 = \dim(W_\Lambda + \mathbb{R}k) \leq n$. $\Lambda + \mathbb{Z}k$ is a $K$-lattice as in Step S9.12, of dimension $\dim \Lambda + 1$.

**Step S9.15: (d).** $W_\Lambda = \mathbb{R}^n$, so $\mathcal{E}_K \subset \Lambda$ by (a). If $K \geq 1$ then $e_i \in \mathcal{E}_K$, so $\Lambda = \mathbb{Z}^n$; and $\mathbb{Z}^n$ is generated by $\mathcal{E}_K \cap \mathbb{Z}^n$ and is saturated, so it is in $\mathbb{M}_K$ by (a). If $K < 1$: $\mathcal{E}_K = \{0\}$, so the only $K$-lattice is $\Theta$.

**Step S9.16: (e).** The lower bound is Lemma 9.6(b). $\Lambda \cap \mathcal{E}_K$ spans $W_\Lambda$ ($\dim d$), so it contains $d$ independent vectors $v_1, \dots, v_d$ ((E6)); they generate $\Gamma \leq \Lambda$ with $\dim \Gamma = d$. By Lemma 9.6(c),(f): $|\Lambda| \leq |\Gamma| \leq \prod \|v_i\| \leq K^d$. For $d = 0$ both sides are 1. □

**Definition 9.10** (Zones, blocks)**.** Fix $K \geq 0$ and positive numbers $\lambda_1, \dots, \lambda_n$. For $\Lambda \in \mathbb{M}_K$ of dimension $d \geq 1$ put

$$\delta_\Lambda = \frac{\lambda_d}{|\Lambda|}, \qquad Z_\Lambda = \{\omega \in \mathbb{R}^n : \operatorname{dist}_2(\omega, R_\Lambda) < \delta_\Lambda\} = \{\omega : \|\Pi_\Lambda \omega\| < \delta_\Lambda\}$$

(Lemma 9.5(c)); $Z_\Theta = \mathbb{R}^n$. Put $Z_d^* = \bigcup_{\Lambda \in \mathbb{M}_K^d} Z_\Lambda$ for $1 \leq d \leq n$ and $Z_{n+1}^* = \emptyset$, and

$$B_\Lambda = Z_\Lambda \setminus Z_{d+1}^* \quad (\Lambda \in \mathbb{M}_K^d,\ d \geq 1), \qquad B_\Theta = \mathbb{R}^n \setminus Z_1^*, \qquad B_d^* = \bigcup_{\Lambda \in \mathbb{M}_K^d} B_\Lambda \ (1 \leq d \leq n).$$

(The zones depend on $K$ and on $\lambda = (\lambda_1, \dots, \lambda_n)$; $Z_d^*$ is empty if $\mathbb{M}_K^d$ is.)
Source: Pos93 p. 201.

**Proposition 9.11** (Covering of frequency space)**.** *For every $K \geq 0$ and $\lambda_1, \dots, \lambda_n > 0$: $\mathbb{R}^n = B_\Theta \cup B_1^* \cup \dots \cup B_n^*$, i.e. every $\omega \in \mathbb{R}^n$ lies in $B_\Lambda$ for some $\Lambda \in \mathbb{M}_K$ (a finite family).*
*Source: Pos93 p. 201 (the chain $\mathbb{R}^n = Z_\Theta = B_\Theta \cup Z_1^* = B_\Theta \cup B_1^* \cup Z_2^* = \dots$).*

*Proof.* **Step S9.17: Finite family.** $\mathbb{M}_K$ consists of subgroups generated by subsets of the finite set $\mathcal{E}_K$, so it is finite.

**Step S9.18: Top dimension.** Fix $\omega$. Let $\mathcal{D} = \{d \in \{1, \dots, n\} : \omega \in Z_d^*\}$. If $\mathcal{D} = \emptyset$ then $\omega \notin Z_1^*$, so $\omega \in B_\Theta$. Otherwise let $d = \max \mathcal{D}$ and choose $\Lambda \in \mathbb{M}_K^d$ with $\omega \in Z_\Lambda$. Since $d+1 \notin \mathcal{D}$ (or $d = n$, $Z_{n+1}^* = \emptyset$), $\omega \notin Z_{d+1}^*$, so $\omega \in Z_\Lambda \setminus Z_{d+1}^* = B_\Lambda \subset B_d^*$. □

## 9.4. **The Geometric Lemma.**

**Lemma 9.12** (A numerical inequality)**.** *If $A \geq \sqrt{2}$ then $A^2 - 1 \geq 1 > 0$ and $\sqrt{A^2-1} - 1 \geq A - \sqrt{2}$.*
*Source: Pos93 p. 202 ("since $\sqrt{A^2-1} - 1 \geq A - \sqrt{2} \geq E$").*

*Proof.* $A^2 - 1 \geq 1$ is clear. Put $c = \sqrt{2} - 1 \in (0, 1)$; $A - c \geq 1 > 0$. Then $(A-c)^2 \leq A^2 - 1 \iff -2cA + c^2 \leq -1 \iff A \geq \frac{1+c^2}{2c} = \frac{4-2\sqrt{2}}{2(\sqrt{2}-1)} = \sqrt{2}$ (using $c^2 = 3 - 2\sqrt{2}$). Both sides being nonnegative, $\sqrt{A^2-1} \geq A - c$. □

**Lemma 9.13** (Geometric Lemma)**.** *Let $n \geq 1$, $K > 0$, $E > 0$, $A \geq E + \sqrt{2}$, $\lambda_1, \dots, \lambda_n > 0$ with*

$$\frac{\lambda_{d+1}}{\lambda_d} \geq AK \qquad (1 \leq d < n),$$

*and let $\mathbb{M}_K$, $\delta_\Lambda$, $B_\Lambda$ be as in Definition 9.10.*

(a) *($B_\Theta$) For $\omega \in B_\Theta$ and $k \in \mathbb{Z}^n$ with $0 < \|k\| \le K$: $|k \cdot \omega| \ge \lambda_1$.*
(b) *($B_\Lambda$) Let $\Lambda \in \mathbb{M}_K$ with $d = \dim \Lambda \ge 1$, $\omega \in B_\Lambda$ and $k \in \mathbb{Z}^n \setminus \Lambda$ with $\|k\| \le K$. Then $|k \cdot \omega| \ge E\,K\,\delta_\Lambda$. Also $\operatorname{dist}_2(\omega, R_\Lambda) < \delta_\Lambda$. (For $d = n$ there is no such $k$, Lemma 9.9(d).)*

*So $B_\Theta$ is completely $K$-nonresonant with constant $\alpha_\Theta = \lambda_1$, and $B_\Lambda$ is $K$-nonresonant modulo $\Lambda$ with constant $\alpha_\Lambda \ge EK\delta_\Lambda$, in the Euclidean cut-off $\|k\| \le K$.*
*Source: Pos93 pp. 201–202 (Geometric Lemma and its proof). Deviation: the proof is organised with the explicit formulas of Lemma 9.14, which replace the chain "$k \cdot \omega = Qk \cdot QP_+\omega + Pk \cdot P\omega \ldots$" (same content, with all projections made explicit); all inequalities have been recomputed and agree.*

**Lemma 9.14** (The projection formula)**.** *Let $W \subset \mathbb{R}^n$ be the span of a basis matrix of a lattice $\Lambda$ (or $W = \{0\}$), $k \in \mathbb{R}^n \setminus W$, $W_+ = W + \mathbb{R}k$, $\Pi = \Pi_W$, $Q = Q_W$, $\Pi_+ = \Pi_{W_+}$. Then $Qk \ne 0$ and for every $\omega \in \mathbb{R}^n$*

$$\Pi_+\omega - \Pi\omega = \frac{Qk \cdot \omega}{\|Qk\|^2}\, Qk, \qquad k \cdot \omega = \Pi k \cdot \Pi\omega + Qk \cdot \omega, \qquad |Qk \cdot \omega| = \|Qk\|\,\|\Pi_+\omega - \Pi\omega\|,$$

*and $\|\Pi_+\omega\|^2 = \|\Pi\omega\|^2 + \|\Pi_+\omega - \Pi\omega\|^2$.*

*Proof.* $Qk \ne 0$ as in Lemma 9.6(e). Put $c = Qk \cdot \omega / \|Qk\|^2$ and $v = \Pi\omega + c\,Qk$. Now $Qk = k - \Pi k \in W_+$, so $v \in W_+$; $W_+$ is the span of $d+1$ independent vectors (a basis of $W$ and $k$), so Lemma 9.5 applies to it. Further $\omega - v = Q\omega - c\,Qk \perp W$ (as $Q\omega, Qk \perp W$) and $(\omega - v) \cdot Qk = Q\omega \cdot Qk - c\|Qk\|^2 = \omega \cdot Qk - c\|Qk\|^2 = 0$ (since $Q\omega \cdot Qk = \omega \cdot Q^2 k$, $Q^2 = Q$). As $W_+ = W \oplus \mathbb{R}Qk$, $\omega - v \perp W_+$, and Lemma 9.5(b) gives $\Pi_+\omega = v$. The first formula follows. Next, $k = \Pi k + Qk$ and $\Pi k \cdot \omega = \Pi k \cdot \Pi\omega$ (symmetry of $\Pi$, $\Pi^2 = \Pi$) give the second. The third: $\|\Pi_+\omega - \Pi\omega\| = |c|\|Qk\| = |Qk \cdot \omega| / \|Qk\|$. The last: $\Pi\omega \in W \perp Qk$ (Pythagoras). □

*Proof of Lemma 9.13.* **Step S9.19: (a): reduction to dimension one.** Let $\omega \in B_\Theta$ and $0 < \|k\| \le K$. By Lemma 9.9(b) there is $\Lambda_k \in \mathbb{M}^1_K$ containing $k$; $\Lambda_k = \mathbb{Z}g$ with $g \in \mathbb{Z}^n \setminus \{0\}$ (a rank-one subgroup of $\mathbb{Z}^n$, Lemma 9.3(a)), $k = jg$ with $j \in \mathbb{Z} \setminus \{0\}$, so $\|g\| \le \|k\|$ and $|\Lambda_k| = \|g\|$ (Lemma 9.6(d)). Hence $\delta_{\Lambda_k} = \lambda_1 / \|g\| \ge \lambda_1 / \|k\|$.

**Step S9.20: (a): conclusion.** As $\omega \notin Z_1^* \supset Z_{\Lambda_k}$: $\operatorname{dist}_2(\omega, R_{\Lambda_k}) \ge \delta_{\Lambda_k}$. Here $R_{\Lambda_k} = g^\perp = k^\perp$ and $\operatorname{dist}_2(\omega, k^\perp) = |k \cdot \omega| / \|k\|$ (Lemma 9.5(c), $\|\Pi_{\mathbb{R}k}\omega\| = |k \cdot \omega| / \|k\|$). So $|k \cdot \omega| \ge \|k\|\,\delta_{\Lambda_k} \ge \lambda_1$.

**Step S9.21: (b): setting.** Let $\Lambda, d, \omega, k$ be as in (b); write $V = |\Lambda|$, $\lambda = \lambda_d$, $\delta = \delta_\Lambda = \lambda / V$, $\Pi = \Pi_\Lambda$, $Q = Q_\Lambda$. The last assertion is $\omega \in Z_\Lambda$. By Lemma 9.9(c), $k \notin W_\Lambda$, $d < n$, and $\Lambda + \mathbb{Z}k$ is a $K$-lattice of dimension $d+1$; by Lemma 9.9(b) there is $\Lambda_+ \in \mathbb{M}^{d+1}_K$ with $\Lambda_+ \supset \Lambda + \mathbb{Z}k$. Then $W_{\Lambda_+} \supset W_\Lambda + \mathbb{R}k =: W_+$ and both have dimension $d+1$, so $W_{\Lambda_+} = W_+$ and $\Pi_{\Lambda_+} = \Pi_+$ of Lemma 9.14. Put $V_+ = |\Lambda_+|$, $\lambda_+ = \lambda_{d+1}$.

**Step S9.22: Volumes.** By Lemma 9.6(c),(e): $V_+ \le |\Lambda + \mathbb{Z}k| = V\|Qk\|$, i.e. $\|Qk\| \ge V_+ / V$; and $\|Qk\| \le \|k\| \le K$ gives $V_+ \le KV$.

**Step S9.23: The two projections.** As $\omega \in B_\Lambda \subset Z_\Lambda$: $\|\Pi\omega\| < \delta = \lambda / V \le K\lambda / V_+$ (Step S9.22). As $\omega \notin Z^*_{d+1} \supset Z_{\Lambda_+}$: $\|\Pi_+\omega\| \ge \delta_{\Lambda_+} = \lambda_+ / V_+ \ge AK\lambda / V_+$ (hypothesis on the $\lambda$'s). With $A^2 - 1 > 0$ (Lemma 9.12, as $A \ge E + \sqrt{2} > \sqrt{2}$):

$$\|\Pi_+\omega - \Pi\omega\|^2 = \|\Pi_+\omega\|^2 - \|\Pi\omega\|^2 \ge (A^2 - 1)\Big(\frac{K\lambda}{V_+}\Big)^2 \quad \text{(Lemma 9.14)}.$$

Hence the radicand is positive and $\|\Pi_+\omega - \Pi\omega\| \ge \frac{K\lambda}{V_+}\sqrt{A^2 - 1}$.

**Step S9.24: The divisor.** By Lemma 9.14: $k \cdot \omega = \Pi k \cdot \Pi\omega + Qk \cdot \omega$ and $|Qk \cdot \omega| = \|Qk\|\|\Pi_+\omega - \Pi\omega\| \ge \frac{V_+}{V} \cdot \frac{K\lambda}{V_+}\sqrt{A^2 - 1} = K\delta\sqrt{A^2 - 1}$; while $|\Pi k \cdot \Pi\omega| \le \|k\|\|\Pi\omega\| \le K\delta$. So

$$|k \cdot \omega| \ge K\delta\big(\sqrt{A^2 - 1} - 1\big) \ge K\delta(A - \sqrt{2}) \ge EK\delta$$

by Lemma 9.12 and $A - \sqrt{2} \geq E$. □

9.5. **The Covering Lemma in action space.**

**Lemma 9.15** (Covering Lemma)**.** *Let $n \geq 1$; $p, b, r_0, M, m > 0$; $K \geq 1$; $A \geq \frac{pM}{bm} + \sqrt{2}$. Let $P$ be any set and $\omega : P \to \mathbb{R}^n$ any map (no regularity). Put*

$$\lambda_d = \frac{b\,m\,r_0}{A^n K^n} A^d K^d \ (1 \leq d \leq n), \qquad r_\Lambda = \frac{r_0}{A^n K^n} \frac{A^d K^d}{|\Lambda|} \ (d = \dim \Lambda),$$
$$\alpha_\Lambda = pMK\, r_\Lambda, \qquad \delta_\Lambda = b\,m\,r_\Lambda$$

*for $\Lambda \in \mathbb{M}_K$ (for $\Lambda = \Theta$: $r_\Theta = r_0/(AK)^n$, $\alpha_\Theta = pMKr_\Theta$), with the blocks $B_\Lambda$ of Definition 9.10 for these $K, \lambda$, and $D_\Lambda = \{I \in P : \omega(I) \in B_\Lambda\}$. Then:*

- *(a) (*covering*) $P = \bigcup_{\Lambda \in \mathbb{M}_K} D_\Lambda$, a finite union, and the $D_\Lambda$ may be discarded when empty.*
- *(b) (*consistency*, for $\dim \Lambda = d \geq 1$ only) $\delta_\Lambda = \lambda_d/|\Lambda|$, i.e. the number $\delta_\Lambda = bmr_\Lambda$ is the radius of the zone $Z_\Lambda$ of Definition 9.10. For $\Lambda = \Theta$ no such statement is made ($\lambda_0$ is not defined and $Z_\Theta = \mathbb{R}^n$): $\delta_\Theta = bmr_\Theta$ is only an auxiliary positive number.*
- *(c) (*nonresonance*) If $I \in D_\Theta$ and $k \in \mathbb{Z}^n$, $0 < \|k\| \leq K$: $|k \cdot \omega(I)| \geq \lambda_1 \geq \alpha_\Theta$. If $\Lambda \in \mathbb{M}_K$, $\dim \Lambda \geq 1$, $I \in D_\Lambda$, $k \in \mathbb{Z}^n \setminus \Lambda$, $\|k\| \leq K$: $|k \cdot \omega(I)| \geq \alpha_\Lambda$, and $\operatorname{dist}_2(\omega(I), R_\Lambda) < \delta_\Lambda = bmr_\Lambda$.*
- *(d) (*radii*) $r_\Theta = \dfrac{r_0}{(AK)^n} \leq r_\Lambda \leq \dfrac{r_0}{AK}$ for every $\Lambda \in \mathbb{M}_K$ with $1 \leq \dim \Lambda < n$; $r_{\mathbb{Z}^n} = r_0$ (the only lattice of dimension $n$). Hence $r_\Theta \leq r_\Lambda \leq r_0$ for all $\Lambda \in \mathbb{M}_K$.*

*Source: Pos93 p. 203 (Covering Lemma with $\alpha_\Lambda = pMKr_\Lambda$, $\delta_\Lambda = bmr_\Lambda$) and p. 204 ($b = \frac{1}{8}$; bounds $r_\Theta \leq r_\Lambda \leq r_{\mathbb{Z}^n} = r_0$, and $r_\Lambda \leq r_0/(AK)$ otherwise). The parameters $m, M$ enter only through $E = pM/(bm)$ and play no other role, as Pos93 notes. Pos93's pull-back "setting $D_\Lambda = \{I \in P : \omega(I) \in B_\Lambda\}$ and discarding…" is (a).*

*Proof.* **Step S9.25: Parameters of the Geometric Lemma.** $\lambda_{d+1}/\lambda_d = AK$ $(1 \leq d < n)$, $\lambda_d > 0$. Put $E = pM/(bm) > 0$; then $A \geq E + \sqrt{2}$, and $K \geq 1 > 0$. Lemma 9.13 applies. Also, for $\dim \Lambda = d \geq 1$, $\delta_\Lambda = bmr_\Lambda = \frac{bmr_0 A^d K^d}{A^n K^n |\Lambda|} = \lambda_d/|\Lambda|$ (b) (for $\Lambda = \Theta$ only the auxiliary number $\delta_\Theta = bmr_\Theta > 0$ is defined), and $EK\delta_\Lambda = \frac{pM}{bm} K\, bm\, r_\Lambda = pMKr_\Lambda = \alpha_\Lambda$.

**Step S9.26: Covering.** Let $I \in P$. Proposition 9.11 applied to the point $\omega(I) \in \mathbb{R}^n$ gives $\Lambda \in \mathbb{M}_K$ with $\omega(I) \in B_\Lambda$, i.e. $I \in D_\Lambda$. Finiteness: Proposition 9.11.

**Step S9.27: Nonresonance.** For $I \in D_\Lambda$, $\omega(I) \in B_\Lambda$; apply Lemma 9.13(b), (a) to the point $\omega(I)$; the constants are $EK\delta_\Lambda = \alpha_\Lambda$ above and, for $\Theta$, $\lambda_1 = bm\, r_0 AK/(AK)^n = bmA \cdot K\, r_\Theta \geq pMK\, r_\Theta = \alpha_\Theta$ since $bmA \geq pM$.

**Step S9.28: Radii.** $r_\Lambda = r_0(AK)^{d-n}/|\Lambda|$. For $d < n$: $|\Lambda| \geq 1$ and $AK \geq \sqrt{2} > 1$ give $r_\Lambda \leq r_0(AK)^{-1}$. Lower bound: $|\Lambda| \leq K^d$ (Lemma 9.9(e)) gives $r_\Lambda \geq r_0(AK)^{d-n}K^{-d} = r_0 A^d K^{-n} A^{-n} \geq r_0/(AK)^n$ as $A \geq 1$ (for $d = 0$ equality). $d = n$: $\Lambda = \mathbb{Z}^n$ (Lemma 9.9(d); $K \geq 1$), $|\mathbb{Z}^n| = 1$, $r_{\mathbb{Z}^n} = r_0$; and $r_\Lambda \leq r_0/(AK) \leq r_0$ for $1 \leq d < n$, $r_\Theta \leq r_0$. □

**Corollary 9.16** (Euclidean versus $\ell^1$ cut-offs; interface for Parts 7, 8 and 9)**.** *In the setting of Lemma 9.15 write $K = K_g$ for the geometric cut-off, and let $K_a \geq 0$ be an analytic cut-off.*

- *(a) (*inclusions with the dimension factor*) $Z_{K_a} \subset \mathcal{E}_{K_a} \subset Z_{\sqrt{n}K_a}$, i.e. $\|k\| \leq |k|_1 \leq \sqrt{n}\|k\|$ (Lemma 2.1(b)).*
- *(b) If $K_a \leq K_g$, then for every $\Lambda \in \mathbb{M}_{K_g}$ and $I \in D_\Lambda$ the point $I$ is $(\alpha_\Lambda, K_a)$-nonresonant modulo $\Lambda$ in the sense of Definition 7.1(N1): $|k \cdot \omega(I)| \geq \alpha_\Lambda$ for all $k \in Z_{K_a} \setminus \Lambda$ (for $\Lambda = \Theta$, $k \in Z_{K_a} \setminus \{0\}$; the constant is $\alpha_\Theta \leq \lambda_1$). Also $\operatorname{dist}_2(\omega(I), R_\Lambda) < \delta_\Lambda$, i.e. $D_\Lambda$ is $\delta_\Lambda$-close to exact $\Lambda$-resonances (Definition 8.13).*

(c) *If* $K_g \geq \sqrt{n}K_a > 0$ *then* $r_\Lambda = \dfrac{\alpha_\Lambda}{pMK_g} \leq \dfrac{\alpha_\Lambda}{p\sqrt{n}\,MK_a}$, *the radius condition of Proposition 8.19(c); and (b) holds because* $\sqrt{n}K_a \geq K_a$. *A choice satisfying all requirements is* $K_g = \sqrt{n}K_a$ *with* $K_a \geq 1/\sqrt{n}$.

*Source: Ours (review N06 and Remark 9.1); Pos93 p. 202 only says "it suffices to assume* $\|k\| \leq K$ *rather than the stronger* $|k| \leq K$*".*

*Proof.* **Step S9.29: Inclusions.** (a) is Lemma 2.1(b). (b): $k \in Z_{K_a} \setminus \Lambda$ has $\|k\| \leq |k|_1 \leq K_a \leq K_g$, so Lemma 9.15(c) applies; closeness is the last assertion there. (c): $K_g \geq \sqrt{n}K_a$ gives $\frac{1}{pMK_g} \leq \frac{1}{p\sqrt{n}MK_a}$; $\alpha_\Lambda = pMK_g r_\Lambda$ by definition. □

9.6. **Errata, deviations, consumers.**

*Remark* 9.17 (Suspected defects in Pos93 and deviations). (1) *p. 200, definition of maximal* $K$*-lattices*: "lattices that are generated by vectors in $\mathbb{Z}^n_K = \{k : |k| \leq K\}$ and are not properly contained in any other lattice of the same dimension". The proof of the Geometric Lemma (p. 202) uses the Euclidean ball ("it suffices to assume $\|k\| \leq K$") and from maximality only the consequence "$k \in \mathbb{Z}^n$, $\|k\| \leq K$, $k \notin \Lambda \Rightarrow k \notin W_\Lambda$". We use the Euclidean ball and the weaker maximality of Definition 9.8, which is exactly this consequence (Lemma 9.9(a)); the definition is ambiguous as printed ("any other lattice": any subgroup, or any $K$-lattice).

(2) *p. 202, case* $d \geq 1$: "the maximal $K$-lattice $\Lambda_+$ generated by $\Lambda$ and $k$ together has dimension $d+1 \leq n$" and "the volume of $\Lambda_+$ is not bigger than the volume of $\Lambda$ times the length of the component of $k$ orthogonal to $\Lambda$": true, but each needs an argument (Lemma 9.9(b),(c) for existence and dimension, Lemma 9.6(c),(e) for the volume). The bound $\|P\omega\| \leq \lambda/V \leq K\lambda/V_+$ also uses $V_+ \leq KV$, which is not stated (Step S9.22).

(3) *p. 202, case* $B_\Theta$: "let $\Lambda$ be the unique one-dimensional maximal lattice containing $k$. Then $|\Lambda| \leq \|k\|$ …". Uniqueness is true (such a $\Lambda$ is $\mathbb{Z}k_0$, $k_0$ the primitive vector on $\mathbb{R}k$, since $k_0 \in \mathcal{E}_K \cap \mathbb{R}k$) but is not argued and is not needed; we only use existence (Lemma 9.9(b)) and $|\Lambda| = \|g\| \leq \|k\|$ (Step S9.19).

(4) *p. 203, "the covering lemma"* is stated with $A \geq \frac{pM}{bm} + \sqrt{2}$ and uses $\lambda_d = \frac{bmr_0}{A^n K^n} A^d K^d$: all checks (Lemma 9.15) agree, including $\alpha_\Lambda = EK\delta_\Lambda$. The claim "$r_\Theta \leq r_\Lambda \leq r_{\mathbb{Z}^n} = r_0$" (p. 204) needs $|\Lambda| \leq K^d$ and $|\Lambda| \geq 1$ (Lemma 9.9(e)), not argued in the source.

(5) *The* $\sqrt{n}$ *direction* between $\|k\|$ and $|k|_1$ is Remark 9.1 (it concerns Parts 7, 8 and the PLAN, not Pos93).

(6) No misprint in the numbers of pp. 200–204 concerning the geometry was found; $\frac{p}{q}$-type constants belong to Parts 8 and 9.

*Remark* 9.18 (Consumers). Part 9 chooses $q, p, b$, $A = 11M/m \geq pM/(bm) + \sqrt{2}$ (to be checked there with $M = nM_{\mathbb{C}}$), $K = K_g = \sqrt{n}K_a$, applies Lemma 9.15 to $P = B(0,R)$ (or the relevant action ball) and $\omega = \omega_h$, and uses Corollary 9.16 for the hypotheses (a), (c) of Proposition 8.19 and of Proposition 8.11 (with $\Lambda = \Theta$). The radii in Lemma 9.15(d) are the ones used on p. 204 ($r_\Theta \leq r_\Lambda \leq r_0/(AK)$ for $\Lambda \neq \mathbb{Z}^n$, $r_{\mathbb{Z}^n} = r_0$). Track P takes Definitions 9.2–9.10, Lemmas 9.6, 9.9, Proposition 9.11 and Lemma 9.13 for any $n$, any $K$ and any $\lambda_1, \ldots, \lambda_n > 0$ as they stand.

## 10. Pöschel's Theorem 1 as an internal theorem

This part is Section 5 of Pöschel 1993 ("Proof of Theorem 1", pp. 203–204) together with the statement of Theorem 1 (pp. 190–191): it assembles the Normal Form Lemma and the stability estimates (Parts 7, 8) with the covering of action space by resonance blocks (Part 9). The result, Theorem 10.7, is an *a priori* statement about orbits on a real tube around an arbitrary set $P \subset \mathbb{R}^n$; Part 11 connects it with existence (Theorem 4.34). It is stated in the frozen conventions (angles of period 1, weight $e^{2\pi|k|_1 s}$, max norms on actions in $|\cdot|$, Euclidean norm $\|\cdot\|$ where the geometry and

convexity need it), so that its hypotheses are exactly those of Part 8. The conversion to Pöschel's printed form and to the data of Part 2 is Remark 10.10.

Conventions. $V_rD$ is the max-norm tube of Definition 5.1, $U_rD = V_rD \cap \mathbb{R}^n$, $\mathcal{O}(r,s)$ the class of Definition 5.2 (with $D$ understood), $\mathcal{O}(r,s)^{\mathbb{R}}$ and $W_r = \mathbb{R}^n \times U_rD$ as in Definition 8.1; $Z_K = \{k : |k|_1 \le K\}$, $\mathcal{E}_K = \{k : \|k\| \le K\}$, $\mathbb{M}_K$, $D_\Lambda$, $r_\Lambda$, $\alpha_\Lambda$, $\delta_\Lambda$ as in Part 9. As in Part 8, an $H$-orbit is an orbit of the real Hamiltonian $H^{\mathbb{R}} = (h+f)|_{W_r}$ (extended by 0) and $\omega(I) = \nabla h_{\mathbb{R}}(I) \in \mathbb{R}^n$ for real $I$. The three numerical parameters of Pöschel's proof are fixed once and for all:

$$q = 9, \qquad p = \tfrac{9}{8} \ \ (\text{so } q = \tfrac{p}{p-1} \ge 3,\ 1 < p \le \tfrac{3}{2},\ \tfrac{p}{q} = \tfrac{1}{8}), \qquad b = \tfrac{1}{8}.$$

### 10.1. **External facts.**

(E1) *Numerics.* $\sqrt{2} < 2$; $\frac{9}{8}\cdot 8 = 9$; $2^{10}/288 > 3$; $\frac{2^{10}}{5\cdot 8} = 25.6 > 1$; $x^n \ge x^2$ for $x \ge 1$, $n \ge 2$; for $x > 0$, $a = \frac{1}{2n}$: $(x^{-1})^{2n\,a} = x^{-1}$ (`Real.rpow_natCast`, `Real.rpow_mul`). Exists.

(E2) *Maximum on a compact set.* A continuous real function on a nonempty compact subset of $\mathbb{R}^n$ attains its maximum. Exists.

Besides these: Lemma 2.3, Lemma 2.1, [K:`bp:p1:ext-mvt`]; Part 8 (Definition 8.1, Definition 8.13, Propositions 8.11, 8.19, Corollary 8.9); Part 9 (Lemma 9.9, Proposition 9.11, Lemma 9.15, Corollary 9.16).

### 10.2. **Setting and constants.**

**Definition 10.1** (Setting of Theorem 1)**.** A *Theorem-1 setting* consists of $n \ge 2$, a set $P \subset \mathbb{R}^n$, numbers $r_0, s > 0$, $0 < m \le M$, $\ell > 0$, and functions $h$, $f$ such that

(S1) $h$ is holomorphic on $V_{r_0}P$ and real on $U_{r_0}P$; $\|\partial_I^2 h(I)\|_{2\to 2} \le M$ for all $I \in V_{r_0}P$ (operator norm of the complex Hessian on $(\mathbb{C}^n, \|\cdot\|)$);

(S2) $h$ is $\ell, m$-quasi-convex on $U_{r_0}P$ (Definition 8.13): for $I \in U_{r_0}P$, $\xi \in \mathbb{R}^n$, $|\omega(I)\cdot\xi| \le \ell\|\xi\| \Rightarrow Q(I)[\xi,\xi] \ge m\|\xi\|^2$;

(S3) $r_0 \le \dfrac{\ell}{2M}$;

(S4) $f \in \mathcal{O}(r_0, s)^{\mathbb{R}}$ (relative to $D = P$) with $\varepsilon_f := |f|_{r_0,s} < \infty$.

The *constants* are

$$A = \frac{11M}{m}, \qquad \varepsilon_0 = \frac{m\, r_0^2}{2^{10} A^{2n}}, \qquad a = \frac{1}{2n}, \qquad R_0 = \frac{r_0}{A},$$

and for a number $\varepsilon \in (0, \varepsilon_0]$ (the bound for $f$; we always assume $\varepsilon_f \le \varepsilon$)

$$K = \Big(\frac{\varepsilon_0}{\varepsilon}\Big)^a, \qquad K_a = \frac{K}{\sqrt{n}}, \qquad \kappa = 2\pi K_a s, \qquad r_\Theta = \frac{r_0}{(AK)^n}.$$

$W := W_{r_0} = \mathbb{R}^n \times U_{r_0}P$, $H^{\mathbb{R}} = (h+f)|_W$ (and 0 off $W$). For $I_0 \in P$ put $\Omega_0(I_0) = \max\{\|\omega(I)\| : \|I - I_0\| \le R_0\}$ (Lemma 10.4).

Source: Pos93 pp. 189–191 (hypotheses and parameters of Theorem 1: $|f_\varepsilon|_{P,r_0,s_0} \le \varepsilon \le \varepsilon_0 = m r_0^2/(2^{10}A^{2n})$, $A = 11M/m$, $a = \frac{1}{2n}$, $R_0 = r_0/A$) and p. 203 (proof, $q = 9$, $p = \frac{9}{8}$, $b = \frac{1}{8}$); the geometric strip width $s$ is $s_0/2\pi$ of the source. Deviations: (S3) replaces $r_0 \le 4\ell/m$ (Remark 10.9(1)); the analytic cut-off is $K_a = K/\sqrt{n}$ (Remark 10.9(2)); max-norm tubes.

**Lemma 10.2** (Elementary relations)**.** *In a Theorem-1 setting with $0 < \varepsilon \le \varepsilon_0$:*

*(a)* $A \ge 11$, $A \ge \dfrac{pM}{bm} + \sqrt{2}$ *(with $p = \frac{9}{8}$, $b = \frac{1}{8}$),* $R_0 \le r_0/11$, *and* $r_0 \le 4\ell/m$.

*(b)* $K \ge 1$, $K^{2n} = \varepsilon_0/\varepsilon$, $(\varepsilon/\varepsilon_0)^a = 1/K$, *and* $\varepsilon = \dfrac{m\, r_\Theta^2}{2^{10}}$; *moreover* $\varepsilon_0 \le m r_0^2/2^{10}$.

*(c)* $K_a > 0$, $\sqrt{n}K_a = K$ *and* $K_a \le K$.

*Proof.* **Step S10.1: (a).** $M \ge m$ gives $A \ge 11$ and $\frac{pM}{bm} = \frac{9}{8}\cdot 8\frac{M}{m} = 9\frac{M}{m}$, so $A - \frac{pM}{bm} = \frac{2M}{m} \ge 2 > \sqrt{2}$ ((E1)). $R_0 = r_0/A$. Finally $r_0 \le \frac{\ell}{2M} \le \frac{\ell}{2m} \le \frac{4\ell}{m}$ by (S3).

**Step S10.2: (b).** $\varepsilon \le \varepsilon_0$ gives $\varepsilon_0/\varepsilon \ge 1$, so $K \ge 1$; $K^{2n} = (\varepsilon_0/\varepsilon)^{2na} = \varepsilon_0/\varepsilon$ ((E1)), and $(\varepsilon/\varepsilon_0)^a = K^{-1}$. Then $\frac{mr_\Theta^2}{2^{10}} = \frac{mr_0^2}{2^{10}A^{2n}K^{2n}} = \frac{\varepsilon_0}{K^{2n}} = \varepsilon$; $A^{2n} \ge 1$ gives the last claim.

**Step S10.3: (c).** Clear from $K_a = K/\sqrt{n}$, $n \ge 2$. □

**Lemma 10.3** ($m \le M$ is automatic)**.** *Let $n \ge 2$, $P \ne \emptyset$, and (S1), (S2) hold with some $m > 0$ and $M > 0$ (not assuming $m \le M$). Then $m \le M$.*

*Proof.* **Step S10.4: A transverse vector.** Take $I \in P \subset U_{r_0}P$ and $\xi \in \mathbb{R}^n \setminus \{0\}$ with $\omega(I)\cdot\xi = 0$ ($n \ge 2$). Then $|\omega\cdot\xi| = 0 \le \ell\|\xi\|$, so $Q(I)[\xi,\xi] \ge m\|\xi\|^2$ by (S2). The real Hessian is the restriction of the complex one (Lemma 2.3), so $Q(I)[\xi,\xi] \le \|Q(I)\xi\|\|\xi\| \le M\|\xi\|^2$ by (S1). Hence $m \le M$. □

**Lemma 10.4** (The frequency size $\Omega_0$)**.** *In a Theorem-1 setting let $I_0 \in P$ and $\bar B = \{I \in \mathbb{R}^n : \|I - I_0\| \le R_0\}$. Then $\bar B \subset U_{r_0}P$, $\Omega_0(I_0) = \max_{\bar B}\|\omega\|$ exists, and $\|\omega(I_0)\| \le \Omega_0(I_0) \le \|\omega(I_0)\| + MR_0 < \infty$.*

*Proof.* **Step S10.5: Inclusion.** For $I \in \bar B$: $|I - I_0| \le \|I - I_0\| \le R_0 < r_0$ (Lemma 10.2(a), Lemma 2.1), so $I \in V_{r_0}P \cap \mathbb{R}^n$.

**Step S10.6: Max and bounds.** $h_{\mathbb{R}}$ is $C^2$ on the open set $U_{r_0}P$ (Lemma 2.3), so $\omega$ is continuous; $\bar B$ is compact, convex, nonempty, so the maximum exists ((E2)). $I_0 \in \bar B$ gives the lower bound; the mean value inequality ([K:bp:p1:ext-mvt]) with $\|D\omega\| = \|Q\| \le M$ ((S1)) on the convex set $\bar B$ gives $\|\omega(I) - \omega(I_0)\| \le MR_0$. □

10.3. **The hypotheses of Part 8 on every block.** Fix a Theorem-1 setting, $\varepsilon \in (0, \varepsilon_0]$ with $\varepsilon_f \le \varepsilon$, $K, K_a, \kappa$ as in Definition 10.1. Apply Lemma 9.15 with the parameters $p = \frac{9}{8}$, $b = \frac{1}{8}$, $r_0$, $M$, $m$, $K$ ($= K_g$) $\ge 1$, $A = 11M/m$ (admissible by Lemma 10.2(a)), to the set $P$ and the map $\omega|_P$: this defines $\mathbb{M}_K$, and for $\Lambda \in \mathbb{M}_K$ the numbers $r_\Lambda$, $\alpha_\Lambda = pMKr_\Lambda$, $\delta_\Lambda = bmr_\Lambda = \frac{1}{8}mr_\Lambda$ and the blocks $D_\Lambda \subset P$.

**Proposition 10.5** (Each block satisfies the hypotheses of Propositions 8.11, 8.19)**.** *Let $\Lambda \in \mathbb{M}_K$, $I_0 \in D_\Lambda$, and put $D = \{I_0\}$, $r = r_\Lambda$, $\alpha = \alpha_\Lambda$, $L = \sqrt{n}M$. Then:*

- *(a) (*domains*) $r_\Theta \le r \le r_0$ and $r \le r_0/(AK)$ if $\Lambda \ne \mathbb{Z}^n$; $V_rD \subset V_{r_0}P$, $U_rD \subset U_{r_0}P$, $W_r \subset W$; $h$ is holomorphic on $V_rD$, real on $U_rD$, with $\|\partial_I^2 h\|_{2\to 2} \le M$ on $V_rD$, $\ell, m$-quasi-convex on $U_rD$; $f \in \mathcal{O}(r,s)^{\mathbb{R}}$ and $|f|_{r,s} \le \varepsilon$.*
- *(b) (*size of $\varepsilon$*) $\varepsilon \le \dfrac{mr^2}{2^{10}} = \dfrac{p}{q}\cdot\dfrac{mr^2}{2^7}$.*
- *(c) (*radius*) $r = \dfrac{\alpha}{p\sqrt{n}\,MK_a}$, i.e. $K_aLr = \alpha/p$; $r \le \dfrac{4\ell}{m}$; $Mr \le \dfrac{\ell}{2}$.*
- *(d) (*nonresonance*) If $\Lambda = \Theta$: $|k\cdot\omega(I_0)| \ge \alpha_\Theta$ for all $k \in \mathbb{Z}^n$, $0 < |k|_1 \le K_a$. If $\Lambda \ne \Theta$: $|k\cdot\omega(I_0)| \ge \alpha$ for all $k \in Z_{K_a} \setminus \Lambda$ (no such $k$ if $\Lambda = \mathbb{Z}^n$), and $\|\Pi_\Lambda\omega(I_0)\| < \delta_\Lambda \le \frac{1}{8}mr$.*
- *(e) (*Lipschitz bound*) $\omega$ satisfies the condition (N2) on $V_rD$ with constant $L = \sqrt{n}M$.*
- *(f) (*nonresonant smallness*) If $\Lambda = \Theta$: $\varepsilon \le \dfrac{\alpha r}{2^7 qK_a}$.*

*Source: Pos93 pp. 203–204 ("The stability estimates now apply to all blocks simultaneously, if we have $r_0 \le 4\ell/m$ and $\varepsilon \le mr_\Theta^2/2^{10} = \varepsilon_0/K^{2n}$"); every inequality recomputed; (c) carries the added hypothesis of Part 8 (Remark 10.9(1)).*

*Proof.* **Step S10.7: (a): domains.** Lemma 9.15(d): $r_\Theta \le r_\Lambda \le r_0$, and $r_\Lambda \le r_0/(AK)$ for $1 \le \dim\Lambda < n$; also $r_\Theta = r_0/(AK)^n \le r_0/(AK)$ since $AK \ge 1$, $n \ge 1$; only $\Lambda = \mathbb{Z}^n$ has $r_\Lambda = r_0$. $I_0 \in P$ and $r \le r_0$ give $V_rD \subset V_{r_0}P$ and the inclusions of the real tubes and of $W_r \subset W$. All

hypotheses (S1), (S2) are inherited by subsets. For $f$: restriction of $f$ to $\mathcal{D}_{r,s}$ is in $\mathcal{O}(r,s)^{\mathbb{R}}$ and $|f|_{r,s} \le |f|_{r_0,s} \le \varepsilon$ (monotonicity, Part 5).

**Step S10.8: (b).** By Lemma 10.2(b) and $r \ge r_\Theta$: $\varepsilon = mr_\Theta^2/2^{10} \le mr^2/2^{10}$; $\frac{p}{q} \cdot \frac{1}{2^7} = \frac{1}{8\cdot 2^7} = 2^{-10}$.

**Step S10.9: (c).** $\alpha = pMKr$ and $K = \sqrt{n}K_a$ (Lemma 10.2(c)) give $\frac{\alpha}{p\sqrt{n}MK_a} = \frac{pMKr}{p\sqrt{n}MK_a} = r$. By (a) and (S3): $Mr \le Mr_0 \le \ell/2$, and $r \le r_0 \le 4\ell/m$ (Lemma 10.2(a)).

**Step S10.10: (d).** $I_0 \in D_\Lambda$. By Corollary 9.16(b) (applicable since $K_a \le K$, Lemma 10.2(c)): for $k \in Z_{K_a} \setminus \Lambda$ ($k \in Z_{K_a} \setminus \{0\}$ if $\Lambda = \Theta$) $|k\cdot\omega(I_0)| \ge \alpha_\Lambda$, and $\operatorname{dist}_2(\omega(I_0), R_\Lambda) = \|\Pi_\Lambda \omega(I_0)\| < \delta_\Lambda = bmr$ (Lemma 9.5(c)); $\Lambda = \mathbb{Z}^n$: $Z_{K_a} \setminus \mathbb{Z}^n = \emptyset$.

**Step S10.11: (e).** For $v \in \mathbb{C}^n$ and $I \in V_r D$: $|\partial_I^2 h(I)v| \le \|\partial_I^2 h(I)v\| \le M\|v\| \le \sqrt{n}M|v|$ (Lemma 2.1(a)); Lemma 7.2.

**Step S10.12: (f).** By (c), $\frac{\alpha r}{2^7 q K_a} = \frac{pMKr^2}{2^7 qK_a} = \frac{\sqrt{n}\,M\,r^2}{2^{10}} \ge \frac{mr^2}{2^{10}} \ge \varepsilon$ by (b), $M \ge m$, $\frac{p}{q} = \frac{1}{8}$. □

**Lemma 10.6** (Stability on one block, stopped form)**.** *In the situation of Proposition 10.5 let $F : \mathbb{R}^{2n} \to \mathbb{R}^{2n}$ be any function with $F = X_{H^{\mathbb{R}}}$ on $W$, let $J \ni 0$ be an interval and $z = (\theta, I) : \mathbb{R} \to \mathbb{R}^{2n}$ with $z'(t) = F(z(t))$ for $t \in J$ and $I(0) = I_0$. Put $\Omega_\Lambda(I_0) = \max\{\|\omega(I)\| : \|I - I_0\| \le r_\Lambda/2\}$ (finite, $\ge \|\omega(I_0)\|$, as in Lemma 10.4) and*

$$T_\Theta = \frac{s\, r_\Theta}{5\varepsilon}\, e^{\kappa/6}, \qquad T_\Lambda = \frac{s\, m\, r_\Lambda^2}{288\, \Omega_\Lambda(I_0)\, \varepsilon}\, e^{\kappa/6} \ (\Theta \ne \Lambda \ne \mathbb{Z}^n), \qquad T_{\mathbb{Z}^n} = \infty.$$

*Then $z(t) \in W$ for all $t \in J$ with $|t| \le T_\Lambda$ (where $T_\Theta = T_\Lambda$ for $\Lambda = \Theta$), and for those $t$*

$$|I(t) - I_0| \le \tfrac{1}{2} r_\Lambda, \qquad \text{and} \qquad \|I(t) - I_0\| \le \tfrac{1}{2} r_\Lambda \quad \text{if } \Lambda \ne \Theta.$$

*(If $\Omega_\Lambda(I_0) = 0$ then $T_\Lambda = +\infty$ for $\Theta \ne \Lambda \ne \mathbb{Z}^n$, as in Proposition 8.19.)*

*Proof.* Let $D = \{I_0\}$, $r = r_\Lambda$, $\alpha = \alpha_\Lambda$, $W_r = \mathbb{R}^n \times U_r D \subset W$, so that $X_{H^{\mathbb{R}}}$ computed on $W_r$ is the restriction of the one on $W$ ($W_r$ is open and the two real Hamiltonians agree on it).

**Step S10.13: Case $\Lambda = \Theta$.** Proposition 8.11 with $(D, r, s, K, \alpha, L, p) \to (\{I_0\}, r_\Theta, s, K_a, \alpha_\Theta, \sqrt{n}M, \frac{9}{8})$: (i) is Proposition 10.5(d); (ii) is (e) and (c) ($K_a L r = \alpha/p \le \alpha/p$); (iii) is (f). $h$, $f$ as required by (a). It gives $|I(t) - I(0)| \le r_\Theta/2$ for every $H$-orbit in $W_{r_\Theta}$ on every $J' \ni 0$ with $I(0) \in D$ and $|t| \le T_{\mathrm{nr}} = s r_\Theta e^{\kappa/6}/(5\varepsilon)$.

**Step S10.14: Case $\Lambda \ne \Theta$.** Proposition 8.19 with $(D, r, K, \alpha, \ell, m, M, \delta, p) \to (\{I_0\}, r_\Lambda, K_a, \alpha_\Lambda, \ell, m, M, \delta_\Lambda, \frac{9}{8})$ and the subgroup $\Lambda$: the Hessian bound and quasi-convexity are (a); (a) of that proposition is Proposition 10.5(d) (the closeness with $\delta = \delta_\Lambda$); (b) is $\varepsilon \le \frac{p}{q}\frac{mr^2}{2^7}$ from (b) here and $\delta_\Lambda \le \frac{1}{8}mr$ from (d); (c) is Proposition 10.5(c) (first bound with equality, second bound, and the added $Mr \le \ell/2$). Its conclusion: $\|I(t) - I(0)\| \le r/2$ for $|t| \le T(I_0)$ ($= T_\Lambda$), and for all $t$ if $\Lambda = \mathbb{Z}^n$.

**Step S10.15: Stopped form.** In each case the hypothesis of Corollary 8.9 holds for $H$, $D = \{I_0\}$, $\varrho = r_\Lambda/2 < r_\Lambda$, $T = T_\Lambda$ and the norm $|\cdot|_* = |\cdot|$ ($\Lambda = \Theta$) resp. $\|\cdot\|$ (note $|\cdot| \le \|\cdot\|$); $F = X_{H^{\mathbb{R}}}$ on $W_r$. The corollary gives the claims and $z(t) \in W_r \subset W$. □

10.4. **Theorem 1.**

**Theorem 10.7** (Pöschel's Theorem 1, stopped form)**.** *Let $(n, P, r_0, s, m, M, \ell, h, f)$ be a Theorem-1 setting (Definition 10.1), $P \ne \emptyset$, and let $\varepsilon \in (0, \varepsilon_0]$ with $|f|_{r_0,s} \le \varepsilon$. With $a = \frac{1}{2n}$, $A = 11M/m$, $R_0 = r_0/A$, $K = (\varepsilon_0/\varepsilon)^a$, $\Omega_0(I_0)$ as above, let*

$$F : \mathbb{R}^{2n} \to \mathbb{R}^{2n} \quad \text{be any function with } F = X_{H^{\mathbb{R}}} \text{ on } W = \mathbb{R}^n \times U_{r_0} P,$$

*$J \ni 0$ an interval and $z = (\theta, I) : \mathbb{R} \to \mathbb{R}^{2n}$ with $z'(t) = F(z(t))$ for all $t \in J$, and $I_0 := I(0) \in P$. Then:*

*(a) (*nonexceptional initial action*) If $\|\omega(I_0)\| \ge \frac{1}{8}mr_0$, then for all $t \in J$ with*

$$|t| \le T_0 \exp\Big(\frac{\pi s}{3\sqrt{n}}\Big(\frac{\varepsilon_0}{\varepsilon}\Big)^a\Big), \qquad T_0 = \frac{A^2 s}{\Omega_0(I_0)}$$

*one has $z(t) \in W$ and $|I(t) - I_0| \le \frac{1}{2}R_0\big(\varepsilon/\varepsilon_0\big)^a$ $\big(\le R_0(\varepsilon/\varepsilon_0)^a\big)$.*

*(b) (*exceptional initial action*) If $\|\omega(I_0)\| < \frac{1}{8}mr_0$, then for all $t \in J$: $z(t) \in W$ and $\|I(t)-I_0\| \le \frac{1}{2}r_0$.*

*The constants $A, \varepsilon_0, a, R_0$ depend only on $(n, r_0, m, M)$ and $\Omega_0(I_0)$ only on $h, I_0, R_0$; $s$ enters only through $T_0$ and the exponent.*

*Source: Pos93 Theorem 1 (pp. 190–191) with its proof, pp. 203–204. Deviations from the printed statement (details in Remarks 10.9, 10.10): (1) hypothesis $r_0 \le \ell/(2M)$ instead of $r_0 \le 4\ell/m$; (2) the exponent is $\frac{\pi s}{3\sqrt{n}}K$ $(= \frac{2\pi s}{6}\cdot\frac{K}{\sqrt{n}})$ instead of $\frac{s_0}{6}K$ ($s_0 = 2\pi s$ in the source's normalisation); (3) the stability radius is $\frac{1}{2}R_0/K$ in the max norm; the exceptional branch is $\|\omega(I_0)\| < \frac{1}{8}mr_0$ with radius $\frac{1}{2}r_0$ (the source has $\le$ and $r_0$); (4) stopped form for any extension $F$ of the field; (5) all printed constants of Pos93 p. 204 reproduced ($\varepsilon_0$, $A$, $a$, $R_0$, $T_0$) are correct as printed except for these changes.*

*Proof.* **Step S10.16: Parameters.** Lemma 10.2 applies ($K \ge 1$, $K_a = K/\sqrt{n}$), Lemma 10.3 is not needed since $m \le M$ is a standing hypothesis; Lemma 10.4: $\Omega_0(I_0) \ge \|\omega(I_0)\|$. Apply Lemma 9.15 as at the start of §10.3 to $P$ and $\omega|_P$.

**Step S10.17: (b): the block $\mathbb{Z}^n$.** Let $\|\omega(I_0)\| < \frac{1}{8}mr_0 = \delta_{\mathbb{Z}^n}$ ($r_{\mathbb{Z}^n} = r_0$, Lemma 9.15(d)). $\mathbb{Z}^n \in \mathbb{M}_K$ ($K \ge 1$, Lemma 9.9(d)), $W_{\mathbb{Z}^n} = \mathbb{R}^n$, $\Pi_{\mathbb{Z}^n} = 1$, so $\omega(I_0) \in Z_{\mathbb{Z}^n}$; since $Z^*_{n+1} = \emptyset$, $B_{\mathbb{Z}^n} = Z_{\mathbb{Z}^n}$ (Definition 9.10), thus $I_0 \in D_{\mathbb{Z}^n}$. Lemma 10.6 with $T_{\mathbb{Z}^n} = \infty$: $z(t) \in W$ and $\|I(t) - I_0\| \le r_0/2$ for all $t \in J$.

**Step S10.18: (a): choice of a block.** Let $\|\omega(I_0)\| \ge \frac{1}{8}mr_0$. By Lemma 9.15(a) there is $\Lambda \in \mathbb{M}_K$ with $I_0 \in D_\Lambda$. If $\Lambda = \mathbb{Z}^n$ then, as in Step S10.17 (and Lemma 9.15(c)), $\|\omega(I_0)\| = \|\Pi_{\mathbb{Z}^n}\omega(I_0)\| < \delta_{\mathbb{Z}^n} = \frac{1}{8}mr_0$, a contradiction; so $\Lambda \ne \mathbb{Z}^n$. By Lemma 10.6, for $t \in J$, $|t| \le T_\Lambda$: $z(t) \in W$ and $|I(t) - I_0| \le \frac{1}{2}r_\Lambda \le \frac{1}{2}\cdot\frac{r_0}{AK} = \frac{1}{2}R_0K^{-1} = \frac{1}{2}R_0(\varepsilon/\varepsilon_0)^a$ (Proposition 10.5(a), Lemma 10.2(b)).

**Step S10.19: (a): the time, block $\Theta$.** Let $\Lambda = \Theta$. By Lemma 10.2(b), $\varepsilon = mr_\Theta^2/2^{10}$ gives $\frac{r_\Theta}{\varepsilon} = \frac{2^{10}}{m\,r_\Theta} = \frac{2^{10}(AK)^n}{mr_0}$. With $mr_0 \le 8\|\omega(I_0)\| \le 8\Omega_0$ (hypothesis of (a), Lemma 10.4):

$$T_\Theta = \frac{s\,e^{\kappa/6}}{5}\cdot\frac{2^{10}(AK)^n}{mr_0} \ge \frac{2^{10}}{40}\cdot\frac{(AK)^n s}{\Omega_0}e^{\kappa/6} \ge \frac{A^2 s}{\Omega_0}e^{\kappa/6}$$

since $2^{10}/40 = 25.6 \ge 1$ and $(AK)^n \ge (AK)^2 \ge A^2$ ($n \ge 2$, $AK \ge 1$) ((E1)).

**Step S10.20: (a): the time, blocks $\Lambda \ne \Theta, \mathbb{Z}^n$.** Let $d = \dim\Lambda \in [1, n-1]$. Lemma 9.9(e): $|\Lambda| \le K^d$; hence $r_\Lambda = \frac{r_0(AK)^d}{(AK)^n|\Lambda|} \ge r_0A^{d-n}K^{-n}$ and, with $\varepsilon = \varepsilon_0K^{-2n} = \frac{mr_0^2}{2^{10}A^{2n}K^{2n}}$,

$$\frac{mr_\Lambda^2}{\varepsilon} \ge mr_0^2A^{2d-2n}K^{-2n}\cdot\frac{2^{10}A^{2n}K^{2n}}{mr_0^2} = 2^{10}A^{2d} \ge 2^{10}A^2.$$

The ball $\{\|I - I_0\| \le r_\Lambda/2\}$ is contained in $\{\|I - I_0\| \le R_0\}$ because $r_\Lambda/2 \le \frac{r_0}{2AK} \le R_0$ ($K \ge 1$), so $\Omega_\Lambda(I_0) \le \Omega_0(I_0)$, and $\Omega_\Lambda(I_0) \ge \|\omega(I_0)\| \ge \frac{1}{8}mr_0 > 0$. Therefore

$$T_\Lambda = \frac{s}{288\,\Omega_\Lambda}\cdot\frac{mr_\Lambda^2}{\varepsilon}\,e^{\kappa/6} \ge \frac{2^{10}}{288}\cdot\frac{A^2 s}{\Omega_0}e^{\kappa/6} \ge \frac{A^2 s}{\Omega_0}e^{\kappa/6}.$$

**Step S10.21: (a): conclusion.** In both cases $T_\Lambda \ge T_0e^{\kappa/6}$, and $\kappa/6 = \frac{2\pi K_a s}{6} = \frac{\pi s}{3\sqrt{n}}K = \frac{\pi s}{3\sqrt{n}}(\varepsilon_0/\varepsilon)^a$. So every $t \in J$ with $|t| \le T_0\exp(\frac{\pi s}{3\sqrt{n}}(\varepsilon_0/\varepsilon)^a)$ satisfies $|t| \le T_\Lambda$, and Step S10.18 gives (a). □

*Remark* 10.8 (What the stopped form says; interface with Part 11). The orbit $z$ is only assumed to solve $z' = F(z)$ on $J$ with $I(0) \in P$; *nothing* says that it stays in $W$. The conclusion includes $z(t) \in W$ for the times stated, which is exactly what is needed to feed `IsSolutionOn` and Theorem 4.34: take $F = X_{H_{h,f}}$ (the Lean field), which equals $X_{H^{\mathbb{R}}}$ on $W$ whenever $U_{r_0}P \subset \mathcal{B}_s(R)$ (Lemma 2.3), $J$ an open interval and $z$ a solution in $B(0,R)$. The orbit's angles $\theta_0 \in \mathbb{R}^n$ are arbitrary; all bounds are on $I$.

## 10.5. **Deviations, errata, dictionary.**

*Remark* 10.9 (Suspected defects in Pos93 and repairs (to the ledger)). (1) *p. 190 (Theorem 1) and p. 204, the radius condition.* Printed: "$r_0 \le 4l/m$", and p. 204: "The stability estimates now apply to all blocks simultaneously, if we have $r_0 \le 4l/m$ and $\varepsilon \le \dots$". Proposition 8.19 (Lemma 2, p. 199) needs in addition $Mr \le \ell/2$ on every block (Remark 8.20(1)). For the blocks $\Lambda \neq \mathbb{Z}^n$ one has $r_\Lambda \le r_0/(AK)$, hence $Mr_\Lambda \le \frac{4\ell}{11K} \le \frac{\ell}{2}$ from $r_0 \le 4\ell/m$ alone; *but the block $D_{\mathbb{Z}^n}$ has $r_{\mathbb{Z}^n} = r_0$*, and $r_0 \le 4\ell/m$ does not give $Mr_0 \le \ell/2$ (the ratio $M/m$ is unbounded). (That $K \ge 1$ suffices is true only for $\Lambda \neq \mathbb{Z}^n$; Part 8 has already been qualified accordingly, commit `176161c`.) The zone radius cannot be shrunk on that block, since $\delta_{\mathbb{Z}^n} = \frac{1}{8}mr_0$ is tied to $r_0$ and Lemma 2 needs $\delta \le \frac{1}{8}mr$. *Corrected hypothesis:* $r_0 \le \ell/(2M)$ ((S3)), which implies $r_0 \le 4\ell/m$ (Lemma 10.2(a)) and is used in Proposition 10.5(c). Only $r_0$, hence $R_0, \varepsilon_0, C$, change in the frozen headline, where only the existence of constants is asserted; Part 2 (Lemma 2.10) already has the entry $\ell/(2B_c)$ in the minimum defining $r_0$ (applied in commit `176161c`; Remark 10.10). This item is the inherited p. 199 proof gap together with this route's additional radius premise (a strengthening of the source hypothesis), not a numerical misprint.

(2) *Norm adaptation (not a Pos93 misprint): p. 192, p. 202 ("it suffices to assume $\|k\| \le K$") and the exponent.* The nonresonance of the Geometric Lemma is on the Euclidean ball $\mathcal{E}_K$, the Normal Form Lemma is for $Z_K = \{|k|_1 \le K\}$ with the Lipschitz constant of $\omega$ in the max norm. Part 9 (Remark 9.1) shows that the compatible choice is $K_g = \sqrt{n}K_a$, so the exponential in the time bound is $e^{2\pi K_a s/6} = e^{\frac{\pi s}{3\sqrt{n}}K}$, i.e. the printed $e^{Ks/6}$ loses a factor $\sqrt{n}$ in the exponent *in this blueprint's max/$\ell^1$ framework*: a disclosed adaptation of the norms and of the chosen Lipschitz estimate, not an error in the source. This only changes the constant $c$ of the frozen headline ($c \to c/\sqrt{n}$). (In Pöschel's own Euclidean framework the factor does not arise; our tubes and the cut-off are in the max/$\ell^1$ norms.)

(3) *p. 204, "$A = 11\frac{M}{m} \ge \frac{pM}{bm} + \sqrt{2}$".* True and not a misprint: it uses $M \ge m$, which is automatically available for $n \ge 2$ (Lemma 10.3, an elementary justification supplied here) though not stated by the source. Likewise the covering uses $|\Lambda| \le K^d$ and $|\Lambda| \ge 1$ (Part 9, Remark 9.17(4)).

(4) *p. 191, exceptional branch.* Printed: "except when $\|\omega(I_0)\| \le mr_0/8$ in which case $\|I(t) - I_0\| \le r_0$ for all $t$". Proved: for $\|\omega(I_0)\| < mr_0/8$, radius $r_0/2$; the boundary case $\|\omega(I_0)\| = mr_0/8$ lies in a block $\Lambda \neq \mathbb{Z}^n$ and satisfies the nonexceptional estimate (a). Not a defect; the statement is slightly sharper.

(5) *p. 204, time bound.* The chain "$\frac{s}{288\Omega}\frac{mr_\Lambda^2}{\varepsilon} \ge A^2\frac{s}{\Omega_0}$" and "$\frac{sr_\Theta}{5\varepsilon} \ge \frac{A^nK^ns}{\Omega_0} \ge A^2\frac{s}{\Omega_0}$" were recomputed (Steps S10.19, S10.20); they hold with room ($\frac{2^{10}}{288} > 3$, 25.6), and use $\Omega_\Lambda \le \Omega_0$ (the stability ball $\|I - I_0\| \le r_\Lambda/2$ lies in $\|I - I_0\| \le R_0$), not stated. The printed $e^{Ks/6}$ factor was never written in the chain (it multiplies both sides); the time bound of the printed theorem, $T_0\exp(\frac{s_0}{6}(\varepsilon_0/\varepsilon)^a)$, is recovered with the $\sqrt{n}$ change of (2).

(6) *Misprints.* No numerical misprint was found on the audited assembly pages pp. 190–191, 203–204 (items (2), (3) are a norm adaptation and an elementary justification, and (1) a proof gap with its strengthening, none a misprint); $\varepsilon_0 = mr_0^2/(2^{10}A^{2n})$, $R_0 = r_0/A$,

$a = \frac{1}{2n}$, $T_0 = A^2 s_0/\Omega_0$ are correct as printed (in his normalisation), and the equality $\varepsilon = m r_\Theta^2/2^{10} = \varepsilon_0/K^{2n}$ is exact for $K = (\varepsilon_0/\varepsilon)^a$.

*Remark* 10.10 (Dictionary with Pöschel's printed normalisation and with Part 2). The substitution of Lemma 2.8 ($\tilde H = 2\pi H(\varphi/2\pi, I)$, same actions and time) multiplies $m, M, \ell, \omega, \varepsilon$ by $2\pi$ and the strip width by $2\pi$ ($s_0 = 2\pi s$), and leaves $r_0$, $A = 11M/m$, $\ell/M$, $a$, $R_0$, $m r_0^2/\varepsilon$, $\varepsilon/\varepsilon_0$, $A^2 s/\Omega_0$ unchanged; so Theorem 10.7 *is* Pöschel's Theorem 1 for $\tilde H$ with $s_0/6 \to s_0/(6\sqrt{n})$ in the exponent, $r_0 \le \ell/(2M)$, $\frac12 R_0$ in place of $R_0$. For Part 11, with the data of Lemma 2.10 ($m \to \mu$, $M \to B_c$, $\ell$, $P = \{|J| \le R/2\}$, strip width $s/2$, $\varepsilon_f \le F_{\mathrm{fr}}\varepsilon$ with $F_{\mathrm{fr}} = \coth^n(\pi s/4)$, Lemma 2.9):

(i) $r_0$ must satisfy (S3): replace the entry $2\ell/\mu$ by $\ell/(2B_c)$ (it is smaller, since $B_c \ge 2\mu$); all other entries of the minimum remain, (this correction, and the $\sqrt{n}$ in $c$ of (ii), are already applied in Part 2, commit `176161c`), and then $r_0 \le 4\ell/\mu$, $r_0 \le 4\omega_{\min}/\mu$ and the tube inclusions of Lemma 2.2(c) hold. The max-norm tube of Definition 5.1 satisfies $V_{r_0}P \subset B(0,R)_s$ directly ($|I-J|_\infty < r_0 \le s/2$).

(ii) The exponent constant is $c = \frac{\pi s}{6\sqrt{n}}\big(\varepsilon_0^{\mathrm{fr}}/F_{\mathrm{fr}}\big)^a$ with $\varepsilon_0^{\mathrm{fr}} = \mu r_0^2/(2^{10}A^{2n})$ (this *equals* the current Part 2 $c$, since $\varepsilon_0^{\mathrm{fr}}/F_{\mathrm{fr}} = \varepsilon_*/F$ and $\pi s = s_0$; it is not to be divided by $\sqrt{n}$ again), the radius constant is $C = \frac{r_0}{2A}(F_{\mathrm{fr}}/\varepsilon_0^{\mathrm{fr}})^a$ (or $\frac{r_0}{A}(\dots)^a$, cruder), and the time prefactor is $T_0 = A^2\frac{s}{2}/\Omega_0$ with $\Omega_0 \le \sqrt{n}M$-type bounds as in Lemma 2.10(g).

(iii) The exceptional branch is empty: $\frac18\mu r_0 \le \frac18\mu \cdot \frac{4\omega_{\min}}{\mu} = \frac12\omega_{\min} < \omega_{\min} \le \|\omega(I_0)\|_2$ for $I_0 \in B(0, R/2)$ (Lemma 2.7(c)).

*Remark* 10.11 (Consumers and totality). Part 11 uses Theorem 10.7(a) only (the branch (b) is empty by (iii)) together with Remark 10.8. Every division and root is guarded: $\Omega_0(I_0) \ge \frac18 m r_0 > 0$ in (a) (Lemma 10.4); $\varepsilon > 0$ for $\varepsilon^{\pm a}$ (the case $\varepsilon = 0$ is N-ZERO, Part 3); $\Omega_\Lambda(I_0) > 0$ in Step S10.20. The hypothesis $n \ge 2$ is used in Lemma 10.3 and in $(AK)^n \ge A^2$; the frozen $n \ge 3$ is stronger. Not used: Theorem 1* (p. 191, $\mu \in [0,1]$), as planned.

## 11. Assembly of N-HEAD-C

This part proves the frozen headline Theorem 1.19 (N-HEAD-C). It adds no analysis: it chooses the four constants $C, T, c, \varepsilon_0$ from the seven data $(n, R, s, M, m, M_{\mathbb{C}}, \omega_{\min})$ by Lemma 2.10 (§11.2), verifies one by one the hypotheses of Pöschel's Theorem 1 in the stopped form of Part 10 (Theorem 10.7) for every $(h, f, \varepsilon)$ satisfying (H1)–(H7) (§11.3), converts its conclusion into an a-priori confinement estimate for the frozen solution predicate (§11.4), and then obtains existence and uniqueness from the ODE layer of Part 4 (§11.5, §11.6). Throughout $a = \frac{1}{2n}$, $|\cdot|$ is the max norm, $\|\cdot\|$ the Euclidean norm, $\coth^n(\cdot) = (\coth(\cdot))^n$. *Two radii are kept apart:* the solution lives in $B(0,R)$ (the set in `IsSolutionOn`), the hypotheses (H4), (H5) are on $\mathcal{B}_s(R) = B(0, R+s)$, (H3) on $B(0,R)$; the tube $U_{r_0}P$ below lies in $B(0, 5R/8)$.

### 11.1. External facts.

(E1) *Real powers.* For $a > 0$, $x \mapsto x^a$ is monotone on $[0,\infty)$; $(xy)^a = x^a y^a$ for $x, y \ge 0$; $(x^{2n})^a = x$ for $x \ge 0$, $a = \frac{1}{2n}$, $n \ge 1$; $x^{-a} = 1/x^a$ for $x > 0$; exp is monotone. (`Real.rpow_le_rpow`, `Real.mul_rpow`, `Real.rpow_natCast`, `Real.rpow_mul`, `Real.rpow_neg`, `Real.exp_le_exp`.) Exists.

Besides this: Lemmas 2.10, 2.2, 2.3, 2.4, 2.5, 2.7, 2.9, 2.1; Theorem 10.7, Definition 10.1, Lemma 10.4, Remark 10.8; Theorems 4.30, 4.32, 4.33, 4.34; Part 3 (N-ZERO).

### 11.2. The choice of constants.

**Definition 11.1** (Choice of constants and of the Theorem-1 data)**.** Let $n \ge 3$ and $R, s, M, m, M_{\mathbb{C}}, \omega_{\min} > 0$ (the seven data). Let $q, B_c, \mu, \ell, r_0, A, s_0, \sigma, F, \varepsilon_*, C, c, T, \varepsilon_0$ be the numbers defined, in this order, in Lemma 2.10 (with $r_0 = \min\big(\frac{s}{2}, \frac{R}{8}, \frac{\ell}{2B_c}, \frac{4\omega_{\min}}{\mu}\big)$, $C = \frac{r_0}{A}(F/\varepsilon_*)^a$,

$c = \frac{s_0}{6\sqrt{n}}(\varepsilon_*/F)^a$, $T = \frac{A^2 s_0}{4\pi n M}$, $\varepsilon_0 = \min(\varepsilon_*/F, (R/4C)^{2n})$). The *Theorem-1 data* derived from them are

$$P = \{J \in \mathbb{R}^n : |J| \le R/2\}, \quad s_1 = \tfrac{s}{2}, \quad F_1 = \coth^n\!\big(\tfrac{\pi s}{4}\big), \quad \varepsilon_1 = \frac{\mu\, r_0^2}{2^{10} A^{2n}}, \quad R_0 = \frac{r_0}{A},$$

and, for $\varepsilon > 0$, $\varepsilon' = F_1 \varepsilon$. Here $(\mu, B_c)$ play the roles of Part 10's $(m, M)$, $s_1$ is its geometric strip width, $\varepsilon_1$ its $\varepsilon_0$.

Source: Ours; `docs/tracks/N/sources.md`, "Uniform conversion"; the bookkeeping between Pöschel's $2\pi$-normalisation and the frozen one is Remark 10.10.

**Lemma 11.2** (Properties of the choice)**.** *In the situation of Definition 11.1:*

*(a)* (uniformity) $C, T, c, \varepsilon_0 > 0$, *and they, like all intermediate numbers, depend only on* $(n, R, s, M, m, M_{\mathbb{C}}, \omega_{\min})$ *— not on* $h, f, \varepsilon, \theta_0, I_0$, *nor on the bound in Definition 1.11(iii); and* $C\varepsilon_0^a \le R/4$.

*(b)* (radius conditions) $0 < \mu \le B_c$; $A = 11 B_c/\mu$ *(so* $A \ge 22$*);* $r_0 \le \min(\frac{s}{2}, \frac{R}{8})$, $r_0 \le \frac{\ell}{2B_c}$, $r_0 \le \frac{4\omega_{\min}}{\mu}$; $R_0 = r_0/A$.

*(c)* (dictionary) $\varepsilon_* = 2\pi\varepsilon_1$, $F = 2\pi F_1$. *Consequently* $\varepsilon' \le \varepsilon_1 \iff \varepsilon \le \varepsilon_*/F$; *in particular* $0 < \varepsilon \le \varepsilon_0$ *gives* $0 < \varepsilon' \le \varepsilon_1$; *and for* $\varepsilon > 0$

$$\tfrac{1}{2} R_0 \Big(\frac{\varepsilon'}{\varepsilon_1}\Big)^a = \tfrac{1}{2}\, C\, \varepsilon^a, \qquad \frac{\pi s_1}{3\sqrt{n}} \Big(\frac{\varepsilon_1}{\varepsilon'}\Big)^a = c\, \varepsilon^{-a}.$$

*(d)* (size) $0 < \varepsilon \le \varepsilon_0$ *implies* $C\varepsilon^a \le R/4$.

*Source: Lemma 2.10(a),(b); ours for (c),(d).*

*Proof.* **Step S11.1: Items (a), (b).** Lemma 2.10(a), (b) (positivity, dependence, order of choice, $C\varepsilon_0^a \le R/4$, $r_0 \le \ell/(2B_c)$, $A \ge 22$); the other bounds on $r_0$ are the entries of the minimum. $\mu = q/2 \le q \le B_c$ as $B_c = nM_{\mathbb{C}} + q$ (Step S2.31).

**Step S11.2: Dictionary.** $\varepsilon_* = 2\pi\mu r_0^2/(2^{10}A^{2n}) = 2\pi\varepsilon_1$ and $F = 2\pi \coth^n(\pi s/4) = 2\pi F_1$ by definition. Hence $\varepsilon' = F_1\varepsilon \le \varepsilon_1 \iff 2\pi F_1 \varepsilon \le 2\pi\varepsilon_1 \iff F\varepsilon \le \varepsilon_*$. $\varepsilon \le \varepsilon_0 \le \varepsilon_*/F$ gives the first consequence.

**Step S11.3: Radius and exponent.** $\varepsilon'/\varepsilon_1 = F_1\varepsilon/\varepsilon_1 = F\varepsilon/\varepsilon_*$ (cancel $2\pi$). So $\frac{1}{2}R_0(\varepsilon'/\varepsilon_1)^a = \frac{1}{2}\frac{r_0}{A}(F/\varepsilon_*)^a\varepsilon^a = \frac{1}{2}C\varepsilon^a$ ((E1)). And $\varepsilon_1/\varepsilon' = \varepsilon_*/(F\varepsilon)$, $\frac{\pi s_1}{3\sqrt{n}} = \frac{\pi s}{6\sqrt{n}} = \frac{s_0}{6\sqrt{n}}$ ($s_0 = \pi s$), so the left side is $\frac{s_0}{6\sqrt{n}}(\varepsilon_*/F)^a\varepsilon^{-a} = c\varepsilon^{-a}$ ((E1)).

**Step S11.4: Size.** Monotonicity of $x \mapsto x^a$ ((E1)): $C\varepsilon^a \le C\varepsilon_0^a \le R/4$ by (a). □

11.3. **Verification of the hypotheses of Theorem 10.7.**

**Proposition 11.3** (Hypotheses of Theorem 1 hold, with the exceptional branch empty)**.** *Let the seven data and the constants of Definition 11.1 be given, and let* $h, f, \varepsilon$ *satisfy (H1)–(H7) of Theorem 1.19; write* $V = V_{r_0}P = \{I \in \mathbb{C}^n : \exists J \in P,\ |I - J| < r_0\}$ *(max-norm tube, Definition 5.1) and* $U = V \cap \mathbb{R}^n$. *Then:*

*(a)* (tubes) $V \subset B(0,R)_s$ *and* $U \subset B(0, 5R/8) \subset B(0,R) \subset \mathcal{B}_s(R)$; $W := \mathbb{R}^n \times U \subset W_s = \mathbb{R}^n \times \mathcal{B}_s(R)$; $P \ne \emptyset$ *and* $B(0, R/2) \subset P$.

*(b) ((S1))* $h$ *is holomorphic on* $V$, *real on* $U$, *and the operator norm of the complex Hessian for* $\|\cdot\|$ *is* $\le B_c$ *on* $V$.

*(c) ((S2))* $h$ *is* $\ell, \mu$*-quasi-convex on* $U$ *(Definition 8.13); and* $0 < \mu \le B_c$.

*(d) ((S3))* $r_0 \le \ell/(2B_c)$.

*(e) ((S4))* $f|_{T_{s_1}\times V} \in \mathcal{O}(r_0, s_1)^{\mathbb{R}}$ *(relative to* $D = P$*) and* $|f|_{r_0,s_1} \le \varepsilon'$, *with* $0 < \varepsilon' \le \varepsilon_1$. *Hence* $(n, P, r_0, s_1, \mu, B_c, \ell, h, f|_{T_{s_1}\times V})$ *is a Theorem-1 setting whose constants are* $A$, $\varepsilon_1$, $a$, $R_0$ *of Definition 11.1.*

*(f)* (exceptional branch is empty) *For $I_0 \in B(0,R/2)$: $\|\omega_h(I_0)\| \geq \omega_{\min} > \frac{1}{8}\mu r_0$.*

*(g)* (frequency size and time) *For $I_0 \in B(0,R/2)$ the number $\Omega_0(I_0)$ of Definition 10.1 satisfies $0 < \omega_{\min} \leq \Omega_0(I_0) \leq \sqrt{n}M$, hence $T_0 := A^2 s_1/\Omega_0(I_0) \geq 2T$.*

*(h)* (the field) *$H_{h,f} = (h+f)|_W$ on $W$ (real valued), and $X_{H_{h,f}} = X_{H^{\mathbb{R}}}$ on $W$, where $H^{\mathbb{R}} = (h+f)|_W$ is the real Hamiltonian of Part 10; so every $F$ with $F = X_{H_{h,f}}$ on $W$ is admissible in Theorem 10.7. (Here $X_{H_{h,f}}$ is the frozen field of Definition 1.3, a total function on $\mathbb{R}^n \times \mathbb{R}^n$.)*

*Source: Ours; every conversion is Lemma 2.10(c)–(g) re-proved for the max-norm tubes of Parts 5, 10 (Lemma 2.10 now also states them for the max-norm tubes, with the Euclidean tubes as a consequence; the proofs below nevertheless re-verify each clause directly, using only facts that hold on all of $B(0,R)_s$, resp. $B(0,R)$). Pos93 pp. 190–191 for the hypotheses.*

*Proof.* **Step S11.5: (a): tubes.** Let $I \in V$, $J \in P$, $|I-J| < r_0$. Then $|J| \leq R/2 < R$ and $|I-J| < r_0 \leq s/2 < s$, so $I \in B(0,R)_s$ (Definition 1.1). If $I$ is real then $|I| < R/2 + r_0 \leq R/2 + R/8 = 5R/8 < R < R+s$, and $B(0,R+s) = \mathcal{B}_s(R)$ (Lemma 2.2(a)). $0 \in P$, and $B(0,R/2) \subset P$ by definition.

**Step S11.6: (b): $h$ and its Hessian.** By (H1), Definition 1.11(i), (iv): $h$ is holomorphic on $B(0,R)_s \supset V$, and $\operatorname{Im} h = 0$ on $\mathcal{B}_s(R) \supset U$, i.e. $h$ is real on $U$. By (H2) and Lemma 2.5, on $B(0,R)_s \supset V$ the operator norm of $Q(I) = (\partial_i\partial_j h(I))$ for the Euclidean norm is $\leq nM_{\mathbb{C}} \leq B_c$.

**Step S11.7: (c), (d): quasi-convexity and radius.** $U \subset B(0,R)$ by (a), so Lemma 2.7(b) (valid at every point of $B(0,R)$ under (H1)–(H4), with $q, B_c, \mu, \ell$ as in Definition 11.1) gives, for $I \in U$ and $\xi \in \mathbb{R}^n$, $|\omega_h(I)\cdot\xi| \leq \ell\|\xi\| \Rightarrow D^2 h_{\mathbb{R}}(I)[\xi,\xi] \geq \mu\|\xi\|^2$. The symmetric matrix $D^2 h_{\mathbb{R}}(I)$ is the real Hessian $Q$ of $h_{\mathbb{R}} = h|_U$ and $\omega_h = \nabla h_{\mathbb{R}}$ (Lemma 2.3(c)), so this is the definition of $\ell,\mu$-quasi-convexity. $\mu \leq B_c$ is Lemma 11.2(b), and (d) is the same lemma, (b).

**Step S11.8: (e): the perturbation.** Put $\rho = \frac{3s}{4}$, $\delta = \frac{s}{4}$, so $T_{3s/4} \subset T_s$ and $\rho - \delta = s_1$. *Class.* By (H6)(i), $f$ is holomorphic on $D_s(R) \supset T_{s_1} \times V$ (as $V \subset B(0,R)_s$, (a)); 1-periodicity in each $\theta_j$ is (H6)(ii) with $k = e_j$; $f$ is real at the real points of $T_{s_1} \times V$, i.e. on $\mathbb{R}^n \times U$, by (H6)(iii) and $U \subset \mathcal{B}_s(R)$. Thus $f|_{T_{s_1}\times V} \in \mathcal{O}(r_0,s_1)^{\mathbb{R}}$. *Norm.* Fix $I \in V$. The function $u = f(\cdot, I)$ is holomorphic, 1-periodic, and $|u| \leq \varepsilon$ on $T_\rho$ by (H6)(iv); Lemma 2.9 gives $\sum_k |\hat{f}_k(I)| e^{2\pi|k|_1 s_1} \leq \coth^n(\pi\delta)\varepsilon = \varepsilon'$ (the coefficients $\hat{f}_k(I)$ are those of Definition 5.2). Taking the supremum over $I \in V$: $|f|_{r_0,s_1} \leq \varepsilon'$. *Range.* Lemma 11.2(c) with (H7) gives $0 < \varepsilon' \leq \varepsilon_1$.

**Step S11.9: Theorem-1 setting.** $n \geq 3 \geq 2$; (a)–(e) are (S1)–(S4) of Definition 10.1 (with $m \to \mu$, $M \to B_c$); $0 < \mu \leq B_c$. The constants $A = 11B_c/\mu$, $\varepsilon_0 = \mu r_0^2/(2^{10}A^{2n}) = \varepsilon_1$, $R_0 = r_0/A$ of that definition are those of Definition 11.1, using Lemma 11.2(b).

**Step S11.10: (f): the exceptional branch.** For $I_0 \in B(0,R/2) \subset B(0,R)$, (H3) gives $\omega_{\min} \leq |\omega_h(I_0)| \leq \|\omega_h(I_0)\|$ (Lemma 2.1(a)). And $\frac{1}{8}\mu r_0 \leq \frac{1}{8}\mu \cdot \frac{4\omega_{\min}}{\mu} = \frac{1}{2}\omega_{\min} < \omega_{\min}$ by $r_0 \leq 4\omega_{\min}/\mu$ (Lemma 11.2(b)) and $\omega_{\min} > 0$.

**Step S11.11: (g): $\Omega_0$ and the time.** Let $I_0 \in B(0,R/2) \subset P$. Lemma 10.4 gives $\bar{B} = \{I \in \mathbb{R}^n : \|I - I_0\| \leq R_0\} \subset U$ and $\|\omega(I_0)\| \leq \Omega_0(I_0) = \max_{\bar{B}}\|\omega\|$; so $\Omega_0 \geq \omega_{\min}$ by Step S11.10. For $I \in \bar{B} \subset U \subset B(0,R) \subset \mathcal{B}_s(R)$, (H5) and Lemma 2.4 give $\|\omega_h(I)\| \leq \sqrt{n}M$; hence $\Omega_0 \leq \sqrt{n}M$. Then $T_0 = A^2 s_1/\Omega_0 \geq \frac{A^2 s}{2\sqrt{n}M} \geq \frac{A^2 s}{2nM} = 2 \cdot \frac{A^2 \pi s}{4\pi n M} = 2T$ ($s_0 = \pi s$, $\sqrt{n} \leq n$).

**Step S11.12: (h): the field.** For $(\theta, I) \in W \subset W_s$, $(\theta, I) \in D_s(R)$ (Lemma 2.2(a)), and Lemma 2.3(b) gives $H_{h,f}(\theta, I) = h(I) + f(\theta, I) \in \mathbb{R}$, which is $H^{\mathbb{R}}$. $W$ is open ($U = V \cap \mathbb{R}^n$ is open), and two functions that agree on an open set have the same `fderiv` there; so $X_{H_{h,f}}(z) = X_{H^{\mathbb{R}}}(z)$ for $z \in W$ (Definition 1.3; differentiability is Lemma 2.3(b)). □

11.4. **The a-priori confinement.**

**Proposition 11.4** (A-priori confinement for the frozen predicate)**.** *Let $\tau = T\exp(c\varepsilon^{-a})$ and let $h, f, \varepsilon$ be as in Proposition 11.3. Let $\theta_0 \in \mathbb{R}^n$, $I_0 \in B(0, R/2)$, $J \ni 0$ any interval and $y : \mathbb{R} \to \mathbb{R}^n \times \mathbb{R}^n$ with $y(0) = (\theta_0, I_0)$ and $\mathtt{IsSolutionOn}\, H_{h,f}\, B(0,R)\, J\, y$. Then for all $t \in J$ with $|t| \le \tau$: $|y(t)_2 - I_0| \le C\varepsilon^a$.*
*Source: Pos93 p. 198 (a priori estimate on the maximal solution) via Theorem 10.7; ours for the dictionary.*

*Proof.* **Step S11.13: The setting.** Proposition 11.3(a)–(e) say that Definition 10.1 applies with $\varepsilon' \in (0, \varepsilon_1]$ as the bound for $f$.

**Step S11.14: The orbit.** Put $F_* = X_{H_{h,f}} : \mathbb{R}^{2n} \to \mathbb{R}^{2n}$ (total); by Proposition 11.3(h), $F_* = X_{H^{\mathbb{R}}}$ on $W$. Definition 1.16 gives, for every $t \in J$, $y'(t) = F_*(y(t))$ (two-sided derivative); $y(0)_2 = I_0 \in B(0, R/2) \subset P$ (Proposition 11.3(a)). So the hypotheses of Theorem 10.7 hold for $z = y$, $F = F_*$, $J$ (no confinement of $y$ is assumed, as Remark 10.8 explains).

**Step S11.15: Branch (a).** Proposition 11.3(f) gives $\|\omega(I_0)\| > \frac{1}{8}\mu r_0$, so Theorem 10.7(a) applies (with $\mu, B_c, \varepsilon'$): for all $t \in J$ with $|t| \le T_0 \exp\big(\frac{\pi s_1}{3\sqrt{n}}(\varepsilon_1/\varepsilon')^a\big)$ one has $|I(t) - I_0| \le \frac{1}{2}R_0(\varepsilon'/\varepsilon_1)^a$, $I(t) = y(t)_2$.

**Step S11.16: Translating the constants.** By Lemma 11.2(c) the exponent equals $c\varepsilon^{-a}$ and the radius equals $\frac{1}{2}C\varepsilon^a \le C\varepsilon^a$; by Proposition 11.3(g), $T_0 \ge 2T \ge T$. So $|t| \le \tau = Te^{c\varepsilon^{-a}}$ implies $|t| \le T_0 e^{c\varepsilon^{-a}}$ ((E1): exp monotone), and the conclusion follows. □

11.5. **Existence and uniqueness.**

**Theorem 11.5** (Existence on $[-\tau, \tau]$ with confinement)**.** *Let $h, f, \varepsilon$ be as in Proposition 11.3, $\theta_0 \in \mathbb{R}^n$, $I_0 \in B(0, R/2)$, $\tau = T\exp(c\varepsilon^{-a})$. There is $z : \mathbb{R} \to \mathbb{R}^n \times \mathbb{R}^n$ with (a) $z(0) = (\theta_0, I_0)$, (b) $\mathtt{IsSolutionOn}\, H_{h,f}\, B(0,R)\, [-\tau, \tau]\, z$, (c) $|z(t)_2 - I_0| \le C\varepsilon^a$ for $|t| \le \tau$.*
*Source: BM11 p. 13 (two-case argument), Pos93 p. 198; ours.*

*Proof.* **Step S11.17: Positivity and size of the radius.** $\rho := C\varepsilon^a > 0$ and $\rho \le R/4$ (Lemma 11.2(d)); $\tau > 0$.

**Step S11.18: Closed ball in the action set.** For $|I - I_0| \le \rho$: $|I| \le |I_0| + \rho < R/2 + R/4 = \frac{3}{4}R < R$, so $\overline{B}(I_0, \rho) \subset B(0, R)$.

**Step S11.19: A-priori hypothesis.** For every open interval $J \ni 0$ and $y$ with $y(0) = (\theta_0, I_0)$ and $\mathtt{IsSolutionOn}\, H_{h,f}\, B(0,R)\, J\, y$: $|y(t)_2 - I_0| \le \rho$ for $t \in J$, $|t| \le \tau$ (Proposition 11.4).

**Step S11.20: Glue.** The standing assumptions of Part 4, §4.9 are those of the theorem: $R, s > 0$, (H1), (H6) with $\varepsilon \ge 0$, and $U = B(0, R)$. Theorem 4.34 (with $\rho$, $\tau$, Steps S11.17–S11.19) yields $z$ with $z(0) = (\theta_0, I_0)$, $\mathtt{IsSolutionOn}\, H_{h,f}\, B(0,R)\, [-\tau, \tau]\, z$ and $|z(t)_2 - I_0| \le \rho$ for $|t| \le \tau$. □

**Theorem 11.6** (Uniqueness on $[-L, L]$)**.** *In the situation of Theorem 11.5, let $z$ be any function satisfying (a), (b) there. For every $L \in [0, \tau]$ and every $y : \mathbb{R} \to \mathbb{R}^n \times \mathbb{R}^n$ with $y(0) = (\theta_0, I_0)$ and $\mathtt{IsSolutionOn}\, H_{h,f}\, B(0,R)\, [-L, L]\, y$: $y(t) = z(t)$ for all $t \in [-L, L]$. Here $L = 0$ is allowed (then the claim is $y(0) = z(0)$).*
*Source: Standard (Gronwall); Theorem 4.33.*

*Proof.* **Step S11.21: Restriction.** $[-L, L] \subset [-\tau, \tau]$, so $z$ satisfies $\mathtt{IsSolutionOn}\, H_{h,f}\, B(0,R)\, [-L, L]\, z$ (the predicate is monotone in the time set, Definition 1.16); $z(0) = y(0)$.

**Step S11.22: Uniqueness.** Theorem 4.33 (with $U = B(0, R)$; no confinement of $y$ is used) gives $y = z$ on $[-L, L]$. For $L = 0$ the interval is $\{0\}$ and the claim is the initial condition. □

11.6. **Proof of N-HEAD-C.**

**Theorem 11.7** (Proof of N-HEAD-C)**.** *Theorem 1.19 holds, with the constants $C, T, c, \varepsilon_0$ of Definition 11.1.*

*Frozen Lean declaration:* `KAM.Nekhoroshev.quasiConvex_nekhoroshev_complexBound`

*Source: Pos93 Theorem 1, pp. 190–191, proof pp. 203–204; the conversions are those of Part 2 and this part (*`sources.md`*, "Uniform conversion"); ours.*

*Proof.* **Step S11.23: Order of quantifiers.** Fix $n \geq 3$ and $R, s, M, m, M_{\mathbb{C}}, \omega_{\min} > 0$. Define $C, T, c, \varepsilon_0$ by Definition 11.1. They are functions of these seven data alone (Lemma 11.2(a)) and are chosen before any of $h, f, \varepsilon, \theta_0, I_0$, as the quantifier order of Theorem 1.19 requires.

**Step S11.24: The conditions on the constants.** $C, T, c, \varepsilon_0 > 0$ and $C\varepsilon_0^{1/(2n)} \leq R/4$: Lemma 11.2(a).

**Step S11.25: The universal statement.** Let $h, f, \varepsilon$ satisfy (H1)–(H7), $\theta_0 \in \mathbb{R}^n$, $I_0 \in B(0, R/2)$, $\tau = T\exp(c\varepsilon^{-1/(2n)})$ and $H = H_{h,f}$. Proposition 11.3 applies (its hypotheses are exactly (H1)–(H7); no other hypothesis is used, and nothing depends on the bound $A$ of Definition 1.11(iii)).

**Step S11.26: Existence, (a)–(c).** Theorem 11.5 gives $z$ with (a) $z(0) = (\theta_0, I_0)$, (b) `IsSolutionOn` $H\, B(0,R)\, [-\tau, \tau]\, z$, (c) $|z(t)_2 - I_0| \leq C\varepsilon^{1/(2n)}$ for $|t| \leq \tau$ (the max norm in $\mathbb{R}^n$ is the frozen norm).

**Step S11.27: Uniqueness, (d).** Theorem 11.6 gives (d) for this $z$, every $L \in [0, \tau]$ and every $y$ as in Theorem 1.19(d).

**Step S11.28: No other dependence.** $z$ may depend on $h, f, \varepsilon, \theta_0, I_0$ (it is produced by Theorem 4.34 from them), which the frozen statement allows ($\exists z$ lies inside all the universal quantifiers). The constants were not changed in Steps S11.26–S11.27: this is the uniformity in $h, f, \varepsilon, \theta_0, I_0$. □

*Remark* 11.8 (Where each hypothesis and each constant is used). (H1): holomorphy and reality of $h$ (Step S11.6), $h_{\mathbb{R}} \in C^3$ for the ODE layer (Lemma 2.3). (H2): $B_c$, the Hessian bound of (S1), via the factor $n$ of Lemma 2.5. (H3): quantitative quasi-convexity $\ell$ (Lemma 2.7) and emptiness of the exceptional branch. (H4): $\mu$. (H5): $\Omega_0 \leq \sqrt{n}M$, hence $T$; only the first-order part of $B(M)$ is used. (H6): the weighted norm of $f$ and the analytic set-up of the ODE layer. (H7): $\varepsilon' \leq \varepsilon_1$ and $C\varepsilon^a \leq R/4$. Each of $C, c, T, \varepsilon_0$ has the dependence of Lemma 2.10: $C, c, \varepsilon_0$ on $(n, R, s, m, M_{\mathbb{C}}, \omega_{\min})$, $T$ on $(n, s, m, M_{\mathbb{C}}, M)$. The radius $R$ enters only through $r_0 \leq R/8$ and the requirement $C\varepsilon_0^a \leq R/4$.

*Remark* 11.9 (The case $\varepsilon = 0$; consistency with N-ZERO). Theorem 1.19 requires $\varepsilon > 0$ in (H7): the exponent $\varepsilon^{-1/(2n)}$ in $\tau$ needs it, and Theorem 10.7 needs $\varepsilon' > 0$ for $K = (\varepsilon_1/\varepsilon')^a$. If `IsAnalyticPerturbation` $R\, s\, 0\, f$ then $f = 0$ on $D_s(R)$ (Lemma 2.3(e)) and N-ZERO (Theorem 3.7) supplies the global solution $\gamma_{\theta_0, I_0}$ with $\gamma(t)_2 = I_0$ for all $t \in \mathbb{R}$, so conclusions (a)–(c) hold at $\varepsilon = 0$ with any $C \geq 0$ and any $\tau$; uniqueness at $\varepsilon = 0$ is not asserted (Remark 3.8). N-ZERO is not used in the proof above.

*Remark* 11.10 (Interface summary). Results of other parts used: Part 2 (Lemmas 2.2, 2.3, 2.4, 2.5, 2.7, 2.9, 2.10), Part 4 (Theorems 4.34, 4.33; Theorems 4.32 and 4.30 enter through the glue), Part 10 (Theorem 10.7, Lemma 10.4), Part 3 (N-ZERO, remark only). There are no forward references. For the Lean provers: the content of this part that is not a plain citation is the arithmetic of Lemma 11.2(c), Proposition 11.3(g), and the independent max-tube verification of Proposition 11.3(a)–(e), which re-verifies the Part-9 setting on the max-norm tubes from Part-1 facts on the full frozen domains $B(0,R)_s$, $B(0,R)$; these proof steps are to be retained in graph extraction even though Part 1 now states the matching max-tube dictionary.

## 12. N-PARTIAL-CONVEX: route, normalisation, slow variables as parameters

This part proves the frozen theorem N-PARTIAL-CONVEX (Theorem 1.28, `partialConvex_nekhoroshev`) in its stopped form. It is written in three files: `p11a` (route decision, normalisation, constants), `p11b` (the extension of Parts 5–7 to functions that also depend on the slow variables as parameters, Normal Form Lemma with slow variables) and `p11c` (stability estimates in the extended phase space, with an $\omega$-free energy error; the block theorem; assembly of the frozen statement, constants, quantifier order). Conventions are the frozen ones of Parts 5–11: angles of period 1 (the frozen fast angles of period $2\pi$ are converted once, Lemma 12.4), weight $e^{2\pi|k|_1 s}$, max norms; the Euclidean norm $\|\cdot\|_2$ is used only where Part 8 uses it.

### 12.1. **Route decision.**

*Remark* 12.1 (Route (i): Nekhoroshev's §§5–10 adapted to the convex case. Route (ii): a Pöschel–type proof with the slow variables as parameters. Decision: (ii)). *What Nek77's proof consists of* (pinned file `sources/incoming-round3/nek77-eng.pdf`, printed pages = PDF pages). The main theorem is Theorem 4.4 (p. 30). Its proof (stated p. 28: "formulated in §4 and proved in §§5–9, based on the lemma about eliminating non-resonance harmonics and on certain technical lemmas in §10") has four layers: (a) *forbidden motions* (§5, pp. 30–33): Definition 5.1.D and Proposition 5.2.B, whose proof (pp. 32–33) is a consequence of *Lemma 10.3* (elimination of non-resonance harmonics, stated pp. 45–46 for $H = H_0(I) + H_1(I, p, \varphi, q)$ on a tube over a closed set $A$ of $(I, p, q)$ with a loss $b$ in the slow variables: "$B$ maps a set contained in $\Delta F$ onto $\Delta F_1 = \{I, p, q \in A - b,\ |\mathrm{Im}\,\varphi| < \rho_1\}$"); (b) *combinatorics of resonances*: zones and blocks (§6, pp. 33–35), discs of fast drift and Proposition 7.4 (§7, pp. 36–37; it calls Lemma 10.6, p. 47, and Lemma 10.5, p. 46–47), the non-overlapping condition and Proposition 8.6 (§8, pp. 37–40), traps and Proposition 9.2 (§9, pp. 41–43); (c) *assembly* §9.5 (pp. 43–44), where the parameters (9.7) must satisfy (9.1)–(9.5), (10.1)–(10.4), (5.2)–(5.4), (8.1)–(8.9); *Lemma 10.7* (pp. 47–49) asserts that this system is solvable for $M < M_0$ with $d, T$ given by (4.4)–(4.5); (d) the exponents $a, b$ of (4.4), (4.5) come out of (c).

*Why route (i) is not available as a complete blueprint from the pinned source.* The summary on p. 1 (and p. 3) of the pinned file says that the proofs of "certain lemmas used to prove the main theorem, . . . stated in §10" "will be given elsewhere" (". . . II", Proceedings of the Petrovskii Seminar, No. 5). Thus the proofs of Lemmas 10.3, 10.5 and 10.7 (including the numerical system with the formulas (10.1)–(10.4), (10.9), whose printed formulas are in addition images that the text layer of the scan loses) are *not* in the pinned source, and Proposition 5.2.B and §9.5 depend on them.

Source gap: Nek77 Lemma 10.3 (p. 45–46), Lemma 10.5, Lemma 10.7 (pp. 46–49): statements only; proofs announced for Part II. A blueprint of route (i) would have to supply an independent proof of Lemma 10.3 (a parametric normal form lemma with a loss in $(p, q)$), which is precisely the content of route (ii), and, in addition, the full block/disc/trap combinatorics of §§6–9 and the parameter system of Lemma 10.7.

*Route (ii).* The analytic input of Nek77 (Lemma 10.3) is a parametric resonant normal form: Hamiltonian $h(I) + f(\varphi, I, p, q)$, nonresonant harmonics eliminated on a tube over a set of slow variables, with a loss in the slow width. Pöschel's normal form (Parts 5–7), stability estimates (Part 8) and geometry (Part 9; Part 10 for the assembly) already provide, with explicit constants, everything except the slow variables. Because $h$ does not depend on the slow variables, they enter the homological equation only as parameters, and in the Poisson bracket through one extra term, the bracket of the slow pairs, whose Cauchy loss is controlled by the slow width $w$ (Lemma 12.9). We therefore blueprint:

(1) the Part 4–6 infrastructure for the extended class (Part 11b), with one new hypothesis on the slow width, (SL) of Definition 12.7, under which the slow bracket is dominated by the fast one and all numerical constants of Parts 6–7 survive unchanged (the constant $2/e$ in the bracket estimate is replaced by 1 and every place where Part 7 uses it only through $2/e \le 1$ is checked);

(2) the stability estimates of Part 8 in the extended phase space, with one new argument: an $\omega$-*free* energy error (Lemma 12.28, with explicit analytic margins), needed because the frozen domain $G$ may be unbounded and $\|\omega\|$ is then not bounded on $G'$ (Remark 12.2);
(3) the geometry of Part 9 at the single point $I_0 = I(0)$ (Part 10, Proposition 10.5).

*Exponents; what is claimed.* The frozen statement is proved (subject to audit of the new lemmas) with the frozen slow margin $d = \mu^{b_n}$. The proof gives, as a by-product, a time $\geq C_T \mu^{-1/2} e^{c_2 \mu^{-1/(2n)}}$ and a drift $\leq C' \mu^{1/(2n)}$ for that margin; since $a_n, b_n < \frac{1}{2n}$ (Lemma 12.5) the frozen conclusion follows by monotone comparison for small $\mu$ (Steps S12.70, S12.74). The frozen exponents are consequences, not reproduced, and *no claim is made for the exponents* $\frac{1}{2n}$ *beyond this derivation*; they are unverified until the audit of Lemma 12.28. With a slow margin $d = \mu^{1/(2n)}$ condition (SL) would fail, so no such stopped theorem is claimed. The slow width is $w = \mu^{b_n}/4$; (SL) reads $r_\Lambda \leq c\, w^2$ and holds for small $\mu$ because $r_\Lambda \leq R_0/K \sim \mu^{1/(2n)}$ and $2b_n < \frac{1}{2n}$.

*Remark* 12.2 (Unbounded $G$ and $\omega$-free time bound). The frozen hypotheses do not bound $G$; for $h = \frac{1}{2}|I|_2^2$ on $G = \{I_1 > 1\}$ all of them hold and $\|\omega\|$ is unbounded on $G$. Pöschel's resonant estimate (Part 8, Lemma 8.16, hypothesis (iii); Part 10, $T_\Lambda \propto 1/\Omega_\Lambda(I_0)$) has a time bound proportional to $1/\sup\|\omega\|$ on the ball, so it cannot be uniform in $I_0 \in G$. Lemma 12.28 removes $\|\omega(I_0)\|$ from the energy error by an exact identity ($\omega(I)\cdot\partial_\theta f_* = \frac{d}{dt} f_* - \{f_*, g\}$ along the normal-form orbit). Nek77's own argument (forbidden motions plus traps) never uses an upper bound for $\|\omega\|$; the new lemma is our replacement.

Source gap: Pos93 p. 198–199 and Part 10 give $T \propto 1/\Omega$; no uniform statement for unbounded $G$ is in the sources; Lemma 12.28 is ours and must be audited.

12.2. **External facts.**

(E1) *Numerics.* $e > 2.718$, $2/e < 0.736$, $1/e < 0.368$; $\coth(x) > 1$ for $x > 0$; for $0 < \beta < \gamma$ and $x \in (0,1)$: $x^\gamma < x^\beta$. Exists.
(E2) *Powers.* For $x > 0$, $x^\alpha x^\beta = x^{\alpha+\beta}$, $(x^\alpha)^{-1} = x^{-\alpha}$, $x \mapsto x^\alpha$ is monotone (increasing for $\alpha > 0$); $\exp(y) \geq 1 + y \geq y$; $y \mapsto \exp(\epsilon y)/y^N \to \infty$ as $y \to \infty$ for $\epsilon > 0$ (Mathlib `Real.rpow_add`, `Real.tendsto_exp_div_rpow_atTop`). Exists.
(E3) *Chain rule for a linear change of variables and for differentiable curves.* Exists ([K:`bp:p1:ext-mvt`] for the mean value inequality).

The conversion constant is $\rho/2 \geq r_0$ throughout. Besides these: Part 4 (Definition 4.2, Theorems 4.11, 4.13, Corollary 4.16, Lemma 4.19); Part 5 (Definition 5.1, Lemmas 5.3, 5.4, 5.5, 5.6, 5.8, 5.9); Part 6 (Lemma 6.4, Lemma 6.8); Part 7; Part 8; Part 9 (Lemma 9.15, Corollary 9.16); Part 10 (Definition 10.1, Lemma 10.2).

12.3. **From the frozen system to the period-one weighted system.**

**Definition 12.3** (Weighted Hamiltonian field)**.** For $c > 0$ and $\hat{H} : \mathbb{R}^n \times \mathbb{R}^n \times \mathbb{R}^k \times \mathbb{R}^k \to \mathbb{R}$ (points $y = (\theta, I; q, p)$: fast angles, fast actions, slow positions, slow momenta) differentiable at $y$,

$$X^c_{\hat{H}}(y) = \big(\partial_I \hat{H},\ -\partial_\theta \hat{H},\ c\,\partial_p \hat{H},\ -c\,\partial_q \hat{H}\big)(y),$$

and

$$\{u, v\}_c = \partial_\theta u \cdot \partial_I v - \partial_I u \cdot \partial_\theta v + c\big(\partial_q u \cdot \partial_p v - \partial_p u \cdot \partial_q v\big) = Du \cdot X^c_v,$$

the same formulas for holomorphic $u, v$ with complex partials ($\{u, v\}_c = -\{v, u\}_c$). $\Omega_c = \mathrm{diag}(J_n, cJ_k)$ with $J_d = \left(\begin{smallmatrix} 0 & \mathbf{1} \\ -\mathbf{1} & 0 \end{smallmatrix}\right)$ so that $X^c_u = \Omega_c \nabla u$ for $\nabla u = (\partial_\theta u, \partial_I u, \partial_q u, \partial_p u)$ in the pairing order $(\theta, I)$, $(q, p)$.

Source: Ours; $c = 1$ with $k = 0$ is Definition 6.2.

**Lemma 12.4** (Period conversion with slow variables)**.** *Let* $H : \mathrm{Ph}_{n+k} \to \mathbb{R}$, $\hat{H}(\theta, I; q, p) = \frac{1}{2\pi} H\big((2\pi\theta, q), (I, p)\big)$ *(so that for the frozen Hamiltonian* $H_\mu$, $\hat{H}_\mu = \frac{1}{2\pi}\,\mathrm{Re}\, h(I) + \frac{\mu}{2\pi}\,\mathrm{Re}\, P(\ldots)$*).*

*Let $\Theta : \mathrm{Ph}_{n+k} \to \mathbb{R}^n \times \mathbb{R}^n \times \mathbb{R}^k \times \mathbb{R}^k$, $\Theta((\varphi_{\mathrm{f}}, q), (I, p)) = (\varphi_{\mathrm{f}}/2\pi, I; q, p)$. A curve $z$ satisfies $z'(t) = X_H(z(t))$ at $t$ (with $H$ differentiable at $z(t)$) iff $y = \Theta \circ z$ satisfies $y'(t) = X^{2\pi}_{\hat{H}}(y(t))$ ($\hat{H}$ differentiable at $y(t)$). $\Theta$ is a linear isomorphism, $\Theta^{-1}$ also; the fast actions, slow variables and time are unchanged, the fast angles are divided by $2\pi$.*
Source: *Ours; Lemma 2.8 for $k = 0$ (there with the opposite normalisation). Nek77 §4.3: $\varphi$ of period $2\pi$.*

*Proof.* **Step S12.1: Chain rule.** $\hat{H} = \frac{1}{2\pi} H \circ \Theta^{-1}$ with $\Theta^{-1}(\theta, I; q, p) = ((2\pi\theta, q), (I, p))$ linear; so $\hat{H}$ is differentiable at $y = \Theta z$ iff $H$ is at $z$, and $\partial_\theta \hat{H} = \partial_{\varphi_{\mathrm{f}}} H$, $\partial_I \hat{H} = \frac{1}{2\pi} \partial_I H$, $\partial_q \hat{H} = \frac{1}{2\pi} \partial_q H$, $\partial_p \hat{H} = \frac{1}{2\pi} \partial_p H$ (all at corresponding points).

**Step S12.2: Equations.** The frozen field is $X_H = (\partial_{\mathrm{I}} H, -\partial_\varphi H)$ (Definition 1.3 for $\mathrm{Ph}_{n+k}$): $\dot{\varphi}_{\mathrm{f}} = \partial_I H$, $\dot{q} = \partial_p H$, $\dot{I} = -\partial_{\varphi_{\mathrm{f}}} H$, $\dot{p} = -\partial_q H$. Then $\dot{\theta} = \dot{\varphi}_{\mathrm{f}}/2\pi = \partial_I H/2\pi = \partial_I \hat{H}$; $\dot{I} = -\partial_{\varphi_{\mathrm{f}}} H = -\partial_\theta \hat{H}$; $\dot{q} = \partial_p H = 2\pi \partial_p \hat{H}$; $\dot{p} = -\partial_q H = -2\pi \partial_q \hat{H}$. These are the components of $X^{2\pi}_{\hat{H}}$. Reversing the computation gives the converse. □

**Lemma 12.5** (Exponents)**.** *For $n \geq 2$ put $a_n = \frac{2}{6n^2-3n+14}$, $b_n = \frac{3}{6n^2-3n+14}$ (Definition 1.27). Then $0 < a_n < b_n < \frac{1}{2n}$ and $2b_n < \frac{1}{2n}$; hence $\gamma_1 := \frac{1}{2n} - 2b_n > 0$, $\gamma_2 := \frac{1}{2n} - b_n > 0$, $\gamma_3 := \frac{1}{2n} - a_n > 0$.*

*Proof.* **Step S12.3: Denominator.** $q_n := 6n^2 - 3n + 14 \geq 32 > 0$. $a_n < b_n$ is $2 < 3$.

**Step S12.4: Comparison.** $2b_n < \frac{1}{2n} \iff 12n < 6n^2 - 3n + 14 \iff 6n^2 - 15n + 14 > 0$, true since the discriminant $225 - 336 < 0$ and the leading coefficient is positive. Then $a_n < b_n < 2b_n < \frac{1}{2n}$. □

12.4. **Constants and order of choice.**

**Definition 12.6** (Data and constants of the stopped proof)**.** Fix the data $n \geq 2$, $k \geq 1$, $G, G', D, \rho, g, m, M_{\mathbb{C}}, h$ of Theorem 1.28 with $\sigma \in \{1, -1\}$ from `HasDefiniteFastPart`. Put (all depend only on these data)

$$s = \frac{\rho}{4\pi}, \quad m_1 = \frac{m}{2\pi}, \quad M_1 = \frac{M_{\mathbb{C}}}{2\pi}, \quad g_1 = \frac{g}{2\pi}, \quad r_0 = \min\Big\{\frac{\rho}{2}, \frac{32g}{m}\Big\}, \quad A = \frac{40M_1}{m_1} = \frac{40M_{\mathbb{C}}}{m},$$

$$\varepsilon_0 = \frac{m_1 r_0^2}{2^{14} A^{2n}}, \qquad R_0 = \frac{r_0}{A}, \qquad \kappa_C = \coth(\pi s)^n, \qquad c_K = \Big(\frac{2\pi\varepsilon_0}{\kappa_C}\Big)^{1/(2n)}.$$

For $\mu > 0$ put $\varepsilon = \varepsilon(\mu) = \kappa_C \mu/(2\pi)$, $K = K(\mu) = (\varepsilon_0/\varepsilon)^{1/(2n)} = c_K \mu^{-1/(2n)}$ (when $\varepsilon \leq \varepsilon_0$), $d = \mu^{b_n}$, $w = d/4$, $K_a = K/\sqrt{n}$, $\kappa = 2\pi K_a s = 2\pi K s/\sqrt{n}$. The threshold $\mu_0$ is the minimum of the positive numbers $\mu_a, \dots, \mu_f$ of Step S12.69; it depends only on the data. The covering parameters are $p = \frac{9}{8}$, $q = 9$, $b_{\mathrm{c}} = \frac{1}{32}$ (Remark 12.35); On the branch $G \neq \emptyset$ (where $m \leq M_{\mathbb{C}}$, Lemma 12.30) one has $A \geq \frac{pM_1}{b_{\mathrm{c}} m_1} + \sqrt{2}$; the numerical definitions themselves are meaningful for every data.
Source: Ours; Part 10 (Definition 10.1); `docs/tracks/N/notes.md`, F-NADD-1 (there $L = (n+1)M_{\mathbb{C}}$; here the Euclidean operator norm of the complex Hessian is bounded by $M_{\mathbb{C}}$ itself, Lemma 12.30, because $|u|_\infty \leq \|u\|_2$).

12.5. **The extended class: slow variables as parameters.** Throughout §12.5–12.7: $n \geq 1$, $k \geq 1$, $I_c \in \mathbb{R}^n$ is a fixed base action, $S \subset \mathbb{R}^{2k}$ a fixed set of physical slow pairs $x = (q, p)$ (ordered $q_1..q_k, p_1..p_k$; later $S = \mathrm{core}(D, 3d/4)$). A *block* fixes once and for all a *starting radius* $r_B > 0$, a physical slow width $w > 0$ and

$$\lambda := \frac{w}{r_B}, \qquad c := \frac{2\pi}{\lambda^2} = \frac{2\pi r_B^2}{w^2}, \qquad \tilde{x} := x/\lambda, \qquad \tilde{S} := \lambda^{-1} S.$$

$\lambda$ and $c$ are *never recomputed* during an iteration: an iterated (smaller) radius $R \leq r_B$ has physical slow tube width $\lambda R$.

**Definition 12.7** (Extended class $\mathcal{O}(R,s)^e$, extended phase space, bracket)**.** $\hat D=\{I_c\}\times\tilde S\subset\mathbb{C}^n\times\mathbb{C}^{2k}$; extended actions $\hat I=(I,\tilde x)\in\mathbb{C}^{n+2k}$ (so $m=n+2k$ in Part 5); points $y=(\theta,I,\tilde x)\in\mathbb{C}^{2n+2k}$. For $0<R\le r_B$: $\mathcal{D}^e_{R,s}=T_s\times V_R\hat D=\{|\mathrm{Im}\,\theta|<s,\ |I-I_c|<R,\ \mathrm{dist}_\infty(\tilde x,\tilde S)<R\}$ (physically $\mathrm{dist}_\infty(x,S)<\lambda R$), and $\mathcal{O}(R,s)^e$, $|u|_{R,s}$ are the class and norm of Definition 5.2 for $D=\hat D$; $\mathcal{O}(R,s)^{e,\mathbb{R}}$ those with real values on $W^e_R:=\mathcal{D}^e_{R,s}\cap\mathbb{R}^{2n+2k}$. The bracket is $\{u,v\}_c$ of Definition 12.3 in the variables $(\theta,I,\tilde x)$, $\tilde x=(\tilde q,\tilde p)$; since $x=\lambda\tilde x$ and $c\lambda^2=2\pi$ it is the weight-$2\pi$ bracket of the physical slow pair. The conditions

$$(\mathrm{SL})_{\rho,\sigma}:\quad 16kc\sigma\le\rho,\qquad\qquad (\mathrm{SL}):\quad c\le\frac{r_B}{32ks}\ \Big(\iff 64\pi ksr_B\le w^2\Big)$$

are used; (SL) implies $(\mathrm{SL})_{\rho,\sigma}$ whenever $\sigma\le 2s\rho/r_B$.
Source: Ours (device for reusing Part 5); Nek77 Lemma 10.3 (p. 45–46) has the same tube with a slow loss $b$.

**Lemma 12.8** (Part 5 for the extended class)**.** *Lemmas 5.3, 5.4, 5.5, 5.6 ((a) pointwise in $\hat I$; (b) with $\max_{i\le n+2k}$), 5.8 and 5.9 hold for $\mathcal{O}(\cdot,s)^e$ with the same constants.* Proof: *Part 5 is proved for an arbitrary $D\subset\mathbb{C}^m$ with $m$ independent of $n$ (its introduction), and $\hat D\subset\mathbb{C}^{n+2k}$ is such a set; only the bracket uses $m=n$.* □

**Lemma 12.9** (Bracket estimate with slow pairs, B.4$^e$)**.** *Let $u\in\mathcal{O}(R_0,s_0)^e$, $v\in\mathcal{O}(R,s)^e$ have finite norms, $0<\rho<R$, $0<\sigma<s$, $R-\rho<R_0$, $s-\sigma<s_0$. Then $\{u,v\}_c\in\mathcal{O}(R-\rho,s-\sigma)^e$ and*

$$|\{u,v\}_c|_{R-\rho,s-\sigma}\le\Big[\frac1e\Big(\frac{1}{(R_0-R+\rho)\sigma}+\frac{1}{(s_0-s+\sigma)\rho}\Big)+\frac{2kc}{(R_0-R+\rho)\rho}\Big]|u|_{R_0,s_0}|v|_{R,s}.$$

*Under $(\mathrm{SL})_{\rho,\sigma}$ the bracket in [ ] is $\le\frac12\big(\frac{1}{(R_0-R+\rho)\sigma}+\frac{1}{(s_0-s+\sigma)\rho}\big)$. The pointwise values on $\mathcal{D}^e_{R-\rho,s-\sigma}$ are bounded by the weighted norm (Lemma 5.4(a)).*

*Proof.* **Step S12.5: Fast part.** Lemma 5.7 (its proof uses only Lemmas 5.5, 5.6).

**Step S12.6: Slow part.** Fix $\hat I\in V_{R-\rho}\hat D$, $t=s-\sigma$. Each of the $2k$ products $\partial_{\tilde q_j}u\,\partial_{\tilde p_j}v$, $\partial_{\tilde p_j}u\,\partial_{\tilde q_j}v$ has $|\cdot|_{\hat I,t}\le|\partial u|_{\hat I,t}|\partial v|_{\hat I,t}\le\frac{|u|_{R_0,s_0}}{R_0-R+\rho}\cdot\frac{|v|_{R,s}}{\rho}$ (Lemma 5.5 pointwise, Lemma 5.6(b)); multiply by $c$, sum, take $\sup_{\hat I}$ last.

**Step S12.7: Domination.** $\frac{2kc}{(R_0-R+\rho)\rho}=\frac{2kc\sigma}{\rho}\cdot\frac{1}{(R_0-R+\rho)\sigma}\le\frac18\cdot\frac{1}{(R_0-R+\rho)\sigma}$ and $\frac1e+\frac18<0.493<\frac12$. □

**Lemma 12.10** (Bracket with a decomposed $g$)**.** *In the situation of Lemma 7.5 for $\mathcal{O}(\cdot,\cdot)^e$ and $(\mathrm{SL})_{\rho,\sigma}$: $|\{g,\phi\}_c|_{R-\rho,s-\sigma}\le\frac{1}{\rho\sigma}|g|_*|\phi|_{R,s}$ ($|g|_*$ as in Definition 7.1, for the extended norms).*

*Proof.* Lemma 12.9 for each $g_j$: $\le\frac12\big(\frac{1}{(R_j-R+\rho)\sigma}+\frac{1}{(s_j-s+\sigma)\rho}\big)|g_j|,|\phi|,\ =\frac{1}{\rho\sigma}\cdot\frac12\big(\frac{\rho}{R_j-R+\rho}+\frac{\sigma}{s_j-s+\sigma}\big)|g_j|,|\phi|,$; sum over $j$. □

12.6. **The Lie transform for the extended class.**

**Definition 12.11** (Scaled norm, field)**.** $\nu_{\rho,\sigma}(a,b,\xi)=\max(|a|/\sigma,|b|/\rho,|\xi|/\rho)$ on $\mathbb{C}^n\times\mathbb{C}^n\times\mathbb{C}^{2k}$. $X^c_\phi=\Omega_c\nabla\phi$ has components $\partial_I\phi$, $-\partial_\theta\phi$, $c\,\partial_{\tilde p}\phi$, $-c\,\partial_{\tilde q}\phi$. The closed $\nu$-ball of radius $a$ about $y\in\mathcal{D}^e_{R,s}$ is the polydisc $\bar P(y;a\sigma,a\rho,a\rho)$; for $a\le1$, $\sigma<s$, $\rho<R$... (Definition 5.1(b) for $\hat D$): if $y\in\mathcal{D}^e_{R-\rho,s-\sigma}$ and $a\le1$ then $\bar B_\nu(y,a)\subset\mathcal{D}^e_{R-(1-a)\rho,s-(1-a)\sigma}$.

**Lemma 12.12** (Size of the field)**.** *Let $\phi\in\mathcal{O}(R,s)^e$ have finite norm, $0<\rho<R$, $0<\sigma<s$, $(\mathrm{SL})_{\rho,\sigma}$. On $\mathcal{D}^e_{R-\rho,s-\sigma}$: $|\partial_{I_i}\phi|\le|\phi|_{R,s}/\rho$, $|\partial_{\theta_i}\phi|\le|\phi|_{R,s}/(e\sigma)$, $|\partial_{\tilde x_j}\phi|\le|\phi|_{R,s}/\rho$, and $\nu_{\rho,\sigma}(X^c_\phi)\le|\phi|_{R,s}/(\rho\sigma)$; moreover $c|\phi|_{R,s}/\rho\le|\phi|_{R,s}/(16k\sigma)$.*

*Proof.* Lemma 12.8 (B.3) and B.1(a) as in Lemma 6.3; $c \le \rho/(16k\sigma)$ gives the slow bounds: $c|\phi|/\rho \le |\phi|/(16k\sigma)$, $\nu$-slow $\le c|\phi|/\rho^2 \le |\phi|/(16k\rho\sigma)$. □

**Lemma 12.13** (Flow package)**.** *Lemma 6.4 applies verbatim to $N = 2n + 2k$, the norm $\nu_{\rho,\sigma}$, the open set $\Omega = \mathcal{D}^e_{R',s'}$ and a holomorphic field $F$ with $\nu(F) \le \varepsilon \le \frac{1}{2}$ on $\Omega$; (d) (translations) with $\tau = (\ell, 0, 0)$, $\ell \in \mathbb{Z}^n$; (e) (reality) with the field real on $\Omega \cap \mathbb{R}^{2n+2k}$ (the proof is the conjugation symmetry and does not use $N = 2n$).*

*Proof.* Lemma 6.4 is stated for an arbitrary complex norm on $\mathbb{C}^N$ and an arbitrary open $\Omega$; the reality statement is proved by complex conjugation of the solution and uniqueness. □ □

**Lemma 12.14** (Symplectic flow; canonical maps)**.** *In Lemma 12.13 with $F = X^c_\phi$ ($\phi$ holomorphic on $\Omega$), $Y = DX^t(z)$ satisfies $Y^\top \Omega_c^{-1} Y = \Omega_c^{-1}$, $Y\Omega_c Y^\top = \Omega_c$, $\det Y = \pm 1$. For $\Phi = X^t$, $u, v, H$ holomorphic near $\Phi(z)$: (a) $\{u\circ\Phi, v\circ\Phi\}_c = \{u, v\}_c \circ \Phi$; (b) $D\Phi\, X^c_{H\circ\Phi} = X^c_H \circ \Phi$; (c) if $\dot y = X^c_{H\circ\Phi}(y)$ then $x = \Phi \circ y$ satisfies $\dot x = X^c_H(x)$.*

*Proof.* $Y' = DF\, Y$, $Y(0) = 1$ (Lemma 6.4(b)); $DF = \Omega_c S$, $S$ the symmetric Hessian ([K:bp:p1 :ext-schwarz]), $\Omega_c^\top = -\Omega_c$, so $DF^\top \Omega_c^{-1} = -S\Omega_c\Omega_c^{-1} = -S$ and $\Omega_c^{-1} DF = S$; $\frac{d}{dt}(Y^\top \Omega_c^{-1} Y) = Y^\top(-S + S)Y = 0$. Inverting: $Y\Omega_c Y^\top = \Omega_c$; determinants give $\det Y^2 = 1$. (a): $\nabla(u \circ \Phi) = Y^\top \nabla u \circ \Phi$ and $\{u, v\}_c = \nabla u^\top \Omega_c \nabla v$. (b), (c): chain rule and $Y\Omega_c Y^\top = \Omega_c$. □

**Lemma 12.15** (Iterated brackets)**.** *Let $\phi \in \mathcal{O}(R', s')^e$, $u \in \mathcal{O}(R, s)^e$ have finite norms, $0 < \rho < R$, $0 < \sigma < s$, $R + \rho \le R'$, $s + \sigma \le s'$, $(\mathrm{SL})_{\rho,\sigma}$, $\eta = 2|\phi|_{R',s'}/(\rho\sigma)$. Then $\mathrm{ad}^h_\phi u := \{\mathrm{ad}^{h-1}_\phi u, \phi\}_c \in \mathcal{O}(R - \rho, s - \sigma)^e$ and for $h \ge 1$*

$$\frac{1}{h!}|\mathrm{ad}^h_\phi u|_{R-\rho,s-\sigma} \le \Big(\frac{e^2\eta}{8}\Big)^h |u|_{R,s}.$$

*Proof.* **Step S12.8: Setup.** $\kappa = \rho/h$, $\lambda_h = \sigma/h$, $R_i = R - i\kappa$, $s_i = s - i\lambda_h$; $(\mathrm{SL})_{\kappa,\lambda_h}$ holds (same ratio). By induction $\mathrm{ad}^i_\phi u \in \mathcal{O}(R_i, s_i)^e$.

**Step S12.9: One step.** Apply Lemma 12.9 with $u := \phi$ (radius $R'$, width $s'$), $v := \mathrm{ad}^{i-1}_\phi u$ (radius $R_{i-1}$, width $s_{i-1}$) and losses $\kappa$, $\lambda_h$ (using $\{a, b\}_c = -\{b, a\}_c$). Since $R' - R_{i-1} + \kappa \ge (h + i)\kappa$ and $s' - s_{i-1} + \lambda_h \ge (h + i)\lambda_h$: $|\mathrm{ad}^i_\phi u|_{R_i,s_i} \le \frac{1}{2} \cdot \frac{2}{(h+i)\kappa\lambda_h}|\phi|_, |\mathrm{ad}^{i-1}_\phi u|_{R_{i-1},s_{i-1}}$.

**Step S12.10: Product.** $\prod_{i=1}^h (h + i) = (2h)!/h!$, $\kappa\lambda_h = \rho\sigma/h^2$: $|\mathrm{ad}^h_\phi u|_, \le (h^2/(\rho\sigma))^h \frac{h!}{(2h)!}|\phi|^h_, |u|_,$, so $\frac{1}{h!}|\mathrm{ad}^h_\phi u|_, \le \frac{(\eta h^2/2)^h}{(2h)!}|u|_, \le \big(\frac{\eta h^2}{2} \cdot \frac{e^2}{4h^2}\big)^h |u|_,$ using $(2h)! \ge (2h/e)^{2h}$ ((E1)). □

**Theorem 12.16** (Lemma B.5 for the extended class)**.** *Let $0 < \rho < R \le R' - \rho$, $0 < \sigma < s \le s' - \sigma$, $\phi \in \mathcal{O}(R', s')^e$ with finite norm, $\eta = 2|\phi|_{R',s'}/(\rho\sigma) \le \frac{1}{2}$, $(\mathrm{SL})_{\rho,\sigma}$; $\Omega = \mathcal{D}^e_{R'-\rho,s'-\sigma} \supset \mathcal{D}^e_{R,s}$, $\nu = \nu_{\rho,\sigma}$. (i) $\nu(X^c_\phi) \le \eta/2 \le \frac{1}{4}$ on $\Omega$; the flow $X^t$, $|t| \le 1$, is defined and holomorphic on $\Omega^{1/2} \supset \mathcal{D}^e_{R-\rho,s-\sigma}$ (this larger set is only the existence domain);* for *$z \in \mathcal{D}^e_{R-\rho,s-\sigma}$ one has $X^t(z) \in \mathcal{D}^e_{R-(1-\eta/2)\rho,s-(1-\eta/2)\sigma} \subset \mathcal{D}^e_{R,s}$ (no image claim is made for other points of $\Omega^{1/2}$). (ii) For $u \in \mathcal{O}(R, s)^e$ with finite norm, $u \circ X^1 = \sum_{h\ge0} \frac{1}{h!} \mathrm{ad}^h_\phi u$ on $\mathcal{D}^e_{R-\rho,s-\sigma}$, the series converges in $|\cdot|_{R-\rho,s-\sigma}$, and $|u \circ X^1|_{R-\rho,s-\sigma} \le (1-\eta)^{-1}|u|_{R,s}$. (iii) The same for $X^t$ with $\eta \to |t|\eta$. (iv) If $\phi, u$ are real on real points, so is $u \circ X^1$.*

*Proof.* **Step S12.11: Field and domain.** Lemma 12.12 (losses $\rho, \sigma$): $\nu(X^c_\phi) \le |\phi|_,/(\rho\sigma) = \eta/2 =: \varepsilon \le \frac{1}{4}$ on $\Omega$. For $z \in \mathcal{D}^e_{R-\rho,s-\sigma}$, $\bar B_\nu(z, \frac{1}{2}) \subset \mathcal{D}^e_{R-\rho/2,s-\sigma/2} \subset \mathcal{D}^e_{R,s} \subset \Omega$ (Definition 12.11), so $z \in \Omega^{1/2}$; Lemma 12.13(a): $X^t(z) \in \bar B_\nu(z, \varepsilon) \subset \mathcal{D}^e_{R-(1-\varepsilon)\rho,s-(1-\varepsilon)\sigma}$.

**Step S12.12: Series.** Lemma 12.15: $\frac{1}{h!}\mathrm{ad}^h_\phi u \in \mathcal{O}(R - \rho, s - \sigma)^e$ has norm $\le (e^2\eta/8)^h |u|_{R,s}$; $e^2/8 < 0.93$ so $\sum_h \le (1 - e^2\eta/8)^{-1} \le (1 - \eta)^{-1}$. By B.1(a) the partial sums converge uniformly

on $\mathcal{D}^e_{R-\rho,s-\sigma}$, the limit $S$ is holomorphic and periodic ((E2)), its Fourier coefficients are the sums of the coefficients (termwise integration), and the norm bound follows by the triangle inequality.

**Step S12.13: Remainder.** Let $z \in \mathcal{D}^e_{R-\rho,s-\sigma}$, $w = X^t(z) \in \mathcal{D}^e_{R-\hat\rho,s-\hat\sigma}$ with $\hat\rho = (1-\varepsilon)\rho$, $\hat\sigma = (1-\varepsilon)\sigma$ (ratio unchanged, so $(\mathrm{SL})_{\hat\rho,\hat\sigma}$), $R+\hat\rho \le R'$. Lemma 12.15 with $(\hat\rho,\hat\sigma)$, $\hat\eta = \eta/(1-\varepsilon)^2 \le \frac{16}{9}\eta \le \frac{8}{9}$: $|\mathrm{ad}^N_\phi u|_{R-\hat\rho,s-\hat\sigma} \le N!(e^2\hat\eta/8)^N|u|_{R,s}$ and $e^2\hat\eta/8 \le e^2/9 < 0.83$. The Taylor formula with integral remainder (Lemma 6.8, one real variable; derivatives $g^{(j)} = \mathrm{ad}^j_\phi u \circ X^t$ by Lemma 12.14 and $\frac{d}{dt}v(X^t z) = Dv \cdot X^c_\phi = \{v,\phi\}_c$) gives $|R_N(z)| \le (e^2\hat\eta/8)^N|u|_{R,s} \to 0$, so $u(X^1 z) = S(z)$. (iii) is (ii) for $t\phi$ ($X^t_\phi = X^1_{t\phi}$); (iv) by Lemma 12.13(e). □

**Proposition 12.17** (Lie transform with first-order remainder; integral remainders)**.** *In the setting of Theorem 12.16 let $h \in \mathcal{O}(R,s)^e$ (norm possibly infinite) with $w = \{h,\phi\}_c \in \mathcal{O}(R,s)^e$ of finite norm, $f_0 \in \mathcal{O}(R,s)^e$ with finite norm. On $\mathcal{D}^e_{R-\rho,s-\sigma}$: $(h+f_0)\circ X^1 = h + w + f_0 + \mathcal{R}$, $\mathcal{R} = \int_0^1 \{(1-t)w + f_0, \phi\}_c \circ X^t\,dt$, $|\mathcal{R}|_{R-\rho,s-\sigma} \le \frac{\eta}{1-\eta}(\frac12|w|_, + |f_0|_,)$. Moreover if $v_t \in \mathcal{O}(R,s)^e$ $(0 \le t \le 1)$ have $\sup_t |v_t|_{R,s} = V < \infty$ and $(t,z) \mapsto v_t(z), Dv_t(z)$ are continuous, then $G = \int_0^1 v_t \circ X^t\,dt \in \mathcal{O}(R-\rho,s-\sigma)^e$ with $|G|_{R-\rho,s-\sigma} \le (1-\eta)^{-1}V$.*

*Proof.* **Step S12.14: Taylor.** Lemma 6.8 (order 2 for $h$, order 1 for $f_0$) gives the identity pointwise (the trajectory stays in $\mathcal{D}^e_{R,s}$ by Theorem 12.16(i)). For the series form: $h \circ X^1 = h + \sum_{i\ge0}\frac{1}{(i+1)!}\mathrm{ad}^i_\phi w$ (remainder $\le \frac{1}{N!}\max|\mathrm{ad}^{N-1}_\phi w \circ X^t| \le (e^2\hat\eta/8)^{N-1}|w|_,/N \to 0$) and $f_0 \circ X^1 = \sum \frac{1}{j!}\mathrm{ad}^j_\phi f_0$; so $\mathcal{R} = \sum_{j\ge1}(\frac{1}{(j+1)!}\mathrm{ad}^j_\phi w + \frac{1}{j!}\mathrm{ad}^j_\phi f_0)$ and the bound is $\sum_{j\ge1}(e^2\eta/8)^j(\frac{|w|_,}{j+1} + |f_0|_,) \le \frac{\eta}{1-\eta}(\frac12|w|_, + |f_0|_,)$ $(e^2/8 < 1)$.

**Step S12.15: Integral remainder.** $g_t = v_t \circ X^t \in \mathcal{O}(R-\rho,s-\sigma)^e$, $|g_t|_, \le (1-\eta)^{-1}V$ by Theorem 12.16(iii); $(t,z) \mapsto g_t(z)$ and $D_z g_t = Dv_t(X^t z)D_z X^t(z)$ are continuous; (E1) gives holomorphy of $G$, Fubini and Tonelli give $|G|_{I,s-\sigma} \le \int_0^1 |g_t|_{I,s-\sigma}dt$. □

**Proposition 12.18** (Canonical map package)**.** *Let $\phi \in \mathcal{O}(R,s)^e$ have finite norm, $0<\rho<R/2$, $0<\sigma<s/2$, $(\mathrm{SL})_{\rho,\sigma}$, $\varepsilon := |\phi|_{R,s}/(\rho\sigma) \le \frac12$, $\Omega = \mathcal{D}^e_{R-\rho,s-\sigma}$, $\Phi = X^1_\phi$, $\Psi = X^{-1}_\phi$. Then: (a) $\Phi,\Psi$ are holomorphic on $\mathcal{D}^e_{R-\frac32\rho,s-\frac32\sigma}$ with values in $\Omega$, $\nu(\Phi-\mathrm{id}) \le \varepsilon$, and $|\Phi_\theta - \theta| \le |\phi|_,/\rho$, $|\Phi_I - I| \le |\phi|_,/(e\sigma)$, $|\Phi_{\tilde x} - \tilde x| \le c|\phi|_,/\rho \le \frac{1}{16k}|\phi|_,/\sigma$; (b) $\Phi(\theta+\ell,I,\tilde x) = \Phi(\theta,I,\tilde x) + (\ell,0,0)$, real points go to real points if $\phi$ is real; (c) $|D\Phi - 1|_\nu \le 2\varepsilon$ on $\mathcal{D}^e_{R-2\rho,s-2\sigma}$; (d) $D\Phi\,\Omega_c\,D\Phi^\top = \Omega_c$; (e) $\Psi\circ\Phi = \mathrm{id} = \Phi\circ\Psi$ on $\mathcal{D}^e_{R-2\rho,s-2\sigma}$, and $\Phi$ maps it into $\mathcal{D}^e_{R-\frac32\rho,s-\frac32\sigma}$.*

*Proof.* As Proposition 6.13, step by step: Lemma 12.12 gives $\nu(X^c_\phi) \le \varepsilon$ on the open set $\Omega$; $\bar B_\nu(z,c') \subset \Omega$ for $z \in \mathcal{D}^e_{R-(1+c')\rho,s-(1+c')\sigma}$ (Definition 12.11), so $\mathcal{D}_{R-\frac32\rho,\cdot} \subset \Omega^{1/2}$ and $\mathcal{D}_{R-2\rho,\cdot} \subset \Omega^1$; Lemma 12.13(a)–(e) gives (a), (b), (c), (e) (the componentwise displacements are the mean value inequality along the trajectory with the bounds of Lemma 12.12); (d) is Lemma 12.14. □

12.7. **Iterative Lemma and Normal Form Lemma with slow parameters.** Notation of Part 7: $(D,r,K,\alpha,L,\Lambda,h) \to (\{I_c\},R,K_a,\alpha,L,\Lambda,h)$, $K_a$ the analytic cut-off, $p>1$, $q = p/(p-1)$, $\bar\alpha = 2\pi\alpha$, $\kappa = 2\pi K_a s$; $h$ is holomorphic on $V_R(I_c)$, regarded as a function on $\mathcal{D}^e_{R,s}$ independent of $(\theta,\tilde x)$; $\omega = \partial_I h$.

**Proposition 12.19** (Homological equation with parameters)**.** *Let $f \in \mathcal{O}(R,s)^e$, $\delta = |f|_{R,s} < \infty$, $\Lambda \subset \mathbb{Z}^n$, $K_a \ge 0$, and assume (N-dir): $|k\cdot\omega(I)| \ge \alpha/q$ on $V_R(I_c)$ for $k \in Z_{K_a} \setminus \Lambda$. Put $\xi = P_\Lambda T_{K_a} f$, $\phi = \sum_{k\in Z_{K_a}\setminus\Lambda}\frac{\hat f_k(I,\tilde x)}{2\pi i k\cdot\omega(I)}\chi_k(\theta)$. Then $\xi,\phi \in \mathcal{O}(R,s)^e$, $\{\phi,h\}_c + \xi = T_{K_a}f$, $|\xi|_,, |T_{K_a}f|_, \le \delta$, $|\phi|_, \le q\delta/\bar\alpha$, $P_\Lambda\phi = 0$; if moreover $\Lambda = -\Lambda$, $f$ is real on real points* and the frequency $\omega(I)$ is real for every real $I \in V_R(I_c)$ *(true when $h$ is real on real points of the open set $V_R(I_c)$), then $\xi,\phi$ are*

*real on real points. Without real $\omega$ the last clause is false: $h = iI$, $f = a\cos 2\pi\theta$, $\Lambda = \{0\}$ gives $\phi = -ia\sin(2\pi\theta)/2\pi$.*

*Proof.* Identical to Proposition 7.4 (with Corollary 7.7): it is a statement about the Fourier coefficients in $\theta$ at fixed $\hat{I} = (I, \tilde{x})$ (Lemma 12.8); $\hat{\phi}_k$ is holomorphic in $\hat{I}$ since $k \cdot \omega$ is holomorphic and bounded below; $\{\phi, h\}_c = \partial_\theta \phi \cdot \omega$ because $h$ depends on neither $\theta$ nor $\tilde{x}$ (the slow bracket of $h$ vanishes); the coefficients of $\partial_\theta \phi \cdot \omega$ are $2\pi i(k \cdot \omega)\hat{\phi}_k = \hat{f}_k$ for $k \in Z_{K_a} \setminus \Lambda$; reality as Step S7.6, which uses exactly that $\omega(I)$ is real: $\overline{\hat{\phi}_k} = \hat{f}_{-k}/\overline{2\pi i k \cdot \omega} = \hat{\phi}_{-k}$. □ □

**Theorem 12.20** (Iterative Lemma$^e$)**.** *Let (N-dir) hold; $s, \rho, \sigma > 0$, $2\rho < R$, $2\sigma < s$, $(\mathrm{SL})_{\rho,\sigma}$; $H = h + g + f$ on $\mathcal{D}^e_{R,s}$, $g = \sum_{j\in J} g_j$ as in Definition 7.1 (extended norms), $f \in \mathcal{O}(R,s)^e$, $\delta = |f|_{R,s} \le \bar{\alpha}\rho\sigma/(4q)$. Let $\phi$ be that of Proposition 12.19 and $\Phi = X^{1,c}_\phi$. Then: (a) $\Phi$ is holomorphic on $\mathcal{D}^e_{R-\frac{3}{2}\rho,s-\frac{3}{2}\sigma}$ with values in $\mathcal{D}^e_{R-\rho,s-\sigma}$, $\theta$-translation equivariant, symplectic ($D\Phi\Omega_c D\Phi^\top = \Omega_c$), injective on $\mathcal{D}^e_{R-2\rho,s-2\sigma}$ (left inverse $X^{-1}_\phi$); (b) $|\Phi_\theta - \theta| \le q\delta/(\bar{\alpha}\rho)$, $|\Phi_I - I| \le q\delta/(e\bar{\alpha}\sigma)$, $|\Phi_{\tilde{x}} - \tilde{x}| \le \frac{e}{16k}\cdot\frac{q\delta}{e\bar{\alpha}\sigma}$; (c) $H \circ \Phi = h + g_+ + f_+$ on $\mathcal{D}^e_{R-2\rho,s-2\sigma}$, $g_+ = g + P_\Lambda T_{K_a} f$, $|g_+ - g|, \le \delta$; (d) $|f_+|_{R-2\rho,s-2\sigma} \le \big(1 - \frac{2q\delta}{\bar{\alpha}\rho\sigma}\big)^{-1}\big(\frac{q}{\bar{\alpha}\rho\sigma}(|g|_* + \delta) + e^{-2\pi K_a\sigma}\big)\delta$; (e) if $\Lambda = -\Lambda$ and $h, g, f$ are real on real points, so are $\phi, g_+, f_+$ and $\Phi$ preserves real points.*

*Proof.* **Step S12.16: Generating function.** Proposition 12.19: $|\phi|, \le q\delta/\bar{\alpha}$; $\eta := 2|\phi|,/(\rho\sigma) \le \frac{2q\delta}{\bar{\alpha}\rho\sigma} \le \frac{1}{2}$.

**Step S12.17: (a), (b).** Proposition 12.18 ($\rho < R/2$, $\sigma < s/2$, $\varepsilon \le \eta/2 \le \frac{1}{4}$).

**Step S12.18: Taylor identity.** $f_0 = g + T_{K_a} f$ (finite norm, $|g|, \le \sum |g_j|,$), $w = \{h, \phi\}_c = \xi - T_{K_a} f$. Proposition 12.17 with data $(R - \rho, s - \sigma; R, s)$ for ($u$-domain; $\phi$-domain) (admissible: $2\rho < R$, $2\sigma < s$): on $\mathcal{D}^e_{R-2\rho,s-2\sigma}$, $(h + f_0) \circ \Phi = h + g + \xi + \mathcal{R} = h + g_+ + \mathcal{R}$, $\mathcal{R} = \int_0^1 \{g + f_t, \phi\}_c \circ X^t \, dt$, $f_t = (1-t)\xi + tT_{K_a} f$. Then $H \circ \Phi = h + g_+ + f_+$, $f_+ = \mathcal{R} + (f - T_{K_a} f) \circ \Phi$.

**Step S12.19: Estimates.** $v_t = \{g + f_t, \phi\}_c$ is affine in $t$; Lemma 12.10 (decomposition $g_j$ plus the term $f_t$, $|f_t|, \le \delta$, weight 1): $|v_t|_{R-\rho,s-\sigma} \le \frac{1}{\rho\sigma}(|g|_* + \delta)\frac{q\delta}{\bar{\alpha}} = V$. Proposition 12.17: $|\mathcal{R}|, \le (1-\eta)^{-1} V$. $|f - T_{K_a} f|_{R,s-\sigma} \le e^{-2\pi K_a\sigma}\delta$ (Lemma 5.9), and Theorem 12.16(ii) gives $|(f - T_{K_a} f) \circ \Phi|, \le (1 - \eta)^{-1} e^{-2\pi K_a\sigma}\delta$; $(1-\eta)^{-1} \le (1 - 2q\delta/(\bar{\alpha}\rho\sigma))^{-1}$. This is (d); (e) as Step S7.15. □

**Theorem 12.21** (Normal Form Lemma$^e$)**.** *Let $K_a > 0$, $\Lambda = -\Lambda \subset \mathbb{Z}^n$, $h$ holomorphic on $V_R(I_c)$ and* real on the real points of $V_R(I_c)$ *(so $\omega$ is real there), $(\{I_c\}, R, K_a, \alpha, L, \Lambda, h)$ satisfy (N1), (N2), $K_a LR \le \alpha/p$ (so (N-dir), Lemma 7.3); $R = r_B$ (the starting radius of the block), $s > 0$, (SL) ($c \le R/(32ks)$); $f \in \mathcal{O}(R,s)^{e,\mathbb{R}}$ (real on real points) with $|f|_{R,s} \le \varepsilon$ for a number $\varepsilon > 0$, $H = h + f$, and*

$$\text{(i)}\quad \varepsilon \le \frac{\alpha R}{2^7 q K_a}, \qquad \text{(ii)}\quad \kappa = 2\pi K_a s \ge 6.$$

*Put $R_* = R/2$, $s_* = s/6$, $c_1 = 2qK_a/(\alpha R)$, $g_0 = P_\Lambda T_{K_a} f$. There are a holomorphic injective $\Psi : \mathcal{D}^e_{R_*,s_*} \to \mathcal{D}^e_{R,s}$ with $D\Psi\,\Omega_c\,D\Psi^\top = \Omega_c$, $\Psi(\theta + \ell, I, \tilde{x}) = \Psi(\theta, I, \tilde{x}) + (\ell, 0, 0)$, real at real points, and $g, f_* \in \mathcal{O}(R_*, s_*)^{e,\mathbb{R}}$ with $g$ a trigonometric polynomial in $\theta$ with frequencies in $\Lambda \cap Z_{K_a}$, such that $H \circ \Psi = h + g + f_*$ and (a) $|g - g_0|_{R_*,s_*} \le 16c_1\varepsilon^2$; (b) $|f_*|_{R_*,s_*} \le e^{-\kappa/6}\varepsilon$; (c) $|\Pi_I \Psi - I| \le qK_a\varepsilon/\alpha \le R/2^7$, $|\Pi_\theta \Psi - \theta| \le 10q\varepsilon/(2\pi\alpha R)$, $|\Pi_{\tilde{x}}\Psi - \tilde{x}| \le \frac{1}{8}qK_a\varepsilon/\alpha \le R/2^{10}$ (physical slow displacement $\le \lambda R/2^{10} = w/2^{10}$).*

*Source: Pos93 pp. 192–196 and Theorem 7.10; the displacement of the slow variables (c) is new.*

*Proof.* Write $\bar{\alpha} = 2\pi\alpha$; $\bar{\alpha}s = \alpha\kappa/K_a$, $K_a \ge 3/(\pi s)$. Indexing as in Theorem 7.10: $V_i = \mathcal{D}^e_{R_i,s_i}$, $f_0 = f$, $g_{-1} = 0$; step $i$ is Theorem 12.20 with losses $(\rho_i, \sigma_i)$ producing $\Phi_i : V_{i+1} \to V_i$, $g_i = g_{i-1} + P_\Lambda T_{K_a} f_i$, $R_{i+1} = R_i - 2\rho_i$, $s_{i+1} = s_i - 2\sigma_i$, $\varepsilon_i = |f_i|_{V_i}$, ($\varepsilon_0 \le \varepsilon$; we carry the upper bound $\varepsilon$).

**Step S12.20: (SL) at every step.** The losses are $(\rho_0,\sigma_0)=(\frac{R}{8},\frac{s}{6})$, $(\rho',\sigma')=(\frac{R}{4},\frac{s}{3})$, $(\rho,\sigma)=(\frac{R}{8N},\frac{s}{4N})$; $\sigma/\rho\in\{\frac{4s}{3R},\frac{4s}{3R},\frac{2s}{R}\}$, so $\sigma\le 2s\rho/R$ and (SL) gives $16kc\sigma\le\frac{R\sigma}{2s}\le\rho$.

**Step S12.21: Step 0: hypotheses.** $2\rho_0<R$, $2\sigma_0<s$, $\varepsilon\le\frac{\alpha R}{2^7qK_a}\le\frac{\alpha R}{2^7q}\frac{\pi s}{3}=\frac{\bar\alpha\rho_0\sigma_0}{16q}$ (as $\rho_0\sigma_0=Rs/48$), so $y_0:=\frac{q\varepsilon}{\bar\alpha\rho_0\sigma_0}\le\frac{1}{16}$, $u\le\frac14$.

**Step S12.22: Step 0: two cases.** Theorem 12.20(d) with $g=0$: $|f_1|_*\le(1-2y_0)^{-1}(y_0+e^{-\kappa/6})\varepsilon$, $(1-2y_0)^{-1}\le\frac87$. *Case A, $e^{-\kappa/6}\le y_0$:* $|f_1|_*\le\frac87\cdot 2y_0\varepsilon=\frac{16}{7}y_0\varepsilon$, so $\varepsilon_1\le\frac{16}{7}\cdot\frac{48q\varepsilon^2}{\bar\alpha Rs}\le\frac{2^7q\varepsilon^2}{\bar\alpha Rs}\le\frac{2^7q\varepsilon}{\bar\alpha Rs}\cdot\frac{\alpha R}{2^7qK_a}\varepsilon=\frac{\varepsilon}{\kappa}\le\frac{\varepsilon}{4}$ (6.1). *Case B, $e^{-\kappa/6}>y_0$:* repeat with losses $(\frac{R}{4},\frac{s}{3})$ (then $u'=y_0/4$): $|f_1|_{R/2,s/3}\le(1-\frac{y_0}{2})^{-1}(\frac{y_0}{4}+e^{-2\kappa/6})\varepsilon\le\frac{32}{31}(\frac14+\frac1e)e^{-\kappa/6}\varepsilon<0.64\,e^{-\kappa/6}\varepsilon$ (using $y_0<e^{-\kappa/6}$, $e^{-2\kappa/6}\le e^{-1}e^{-\kappa/6}$). Stop: $\Psi=\Phi_0'$ on $\mathcal{D}^e_{R/2,s/3}\supset\mathcal{D}^e_{R_*,s_*}$, $g=g_0$, $f_*=f_1$; (a), (b) hold; displacement in Step S12.26. From now on Case A.

**Step S12.23: Number of steps.** $N=\lfloor\kappa/(12\ln 2)\rfloor$; $\kappa\ge 12\ln 2\,N>8N$. If $N=0$: $\Psi=\Phi_0$, $g=g_0$, $f_*=f_1$, $|f_*|_*\le\varepsilon/4\le e^{-\kappa/6}\varepsilon$ since $\kappa<12\ln 2$ gives $e^{-\kappa/6}>\frac14$. For $N\ge1$, $i=1..N$: $\rho=\frac{R}{8N}$, $\sigma=\frac{s}{4N}$, $R_i=\frac34R-2(i-1)\rho$, $s_i=\frac23 s-2(i-1)\sigma$; $R_{N+1}=R_*$, $s_{N+1}=s_*$; $2\pi K_a\sigma=\frac{\kappa}{4N}\ge 3\ln 2$, $e^{-2\pi K_a\sigma}\le\frac18$; $\rho\sigma=Rs/(32N^2)$; from (i), $\kappa>8N$: $\varepsilon\le\frac{\bar\alpha Rs}{2^{10}qN}$, hence by (6.1) $\varepsilon_1\le\frac{2^7q}{\bar\alpha Rs}\big(\frac{\bar\alpha Rs}{2^{10}qN}\big)^2=\frac{\bar\alpha\rho\sigma}{2^8q}$ (6.2).

**Step S12.24: Induction.** Claim: step $i$ ($1\le i\le N$) is admissible and $\varepsilon_{i+1}\le\varepsilon_i/4$; so $\varepsilon_i\le 4^{1-i}\varepsilon_1$. Admissibility: $R_i\ge R/2>2\rho$, $s_i>2\sigma$, $\varepsilon_i\le\varepsilon_1\le\bar\alpha\rho\sigma/(2^8q)\le\bar\alpha\rho\sigma/(4q)$. $|g_{i-1}|_*$ (decomposition $g_0$ on $V_0$ and $g_j-g_{j-1}$ on $V_j$): the weights are $\le\frac12(\frac13+\frac13)\le\frac12$ for $j\ge1$ and $\le\frac12(\frac{R/8N}{R/4}+\frac{s/4N}{s/3})=\frac{5}{8N}\le\frac1N$ for $g_0$; so $|g_{i-1}|_*\le\frac12\cdot\frac43\varepsilon_1+\frac{\varepsilon}{N}\le\frac{\bar\alpha\rho\sigma}{2^8q}+\frac{\bar\alpha\rho\sigma}{2^5q}\le\frac{\bar\alpha\rho\sigma}{2^4q}$ (using $\varepsilon/N\le\bar\alpha Rs/(2^{10}qN^2)=2^5\bar\alpha\rho\sigma/(2^{10}q)$). Theorem 12.20(d): $\frac{q}{\bar\alpha\rho\sigma}(|g_{i-1}|_*+\varepsilon_i)\le\frac{1}{2^4}+\frac{1}{2^8}$, $e^{-2\pi K_a\sigma}\le\frac18$, $(1-2^{-7})^{-1}=\frac{128}{127}$, so $\varepsilon_{i+1}\le\frac{128}{127}(\frac{1}{16}+\frac{1}{256}+\frac18)\varepsilon_i<0.1930\,\varepsilon_i<\frac14\varepsilon_i$.

**Step S12.25: Conclusion.** $\Psi=\Phi_0\circ\cdots\circ\Phi_N$, $g=g_N$, $f_*=f_{N+1}$ on $\mathcal{D}^e_{R_*,s_*}$; $H_{i+1}=H_i\circ\Phi_i$ gives $H\circ\Psi=h+g+f_*$; Lemma 7.9 (subsets of $\mathbb{C}^{2n+2k}$, maps with $D\Phi_i\Omega_cD\Phi_i^\top=\Omega_c$) gives holomorphy, symplecticity, periodicity, injectivity. (b): $\varepsilon_{N+1}\le 4^{-N}\varepsilon_1\le 4^{-N-1}\varepsilon\le e^{-\kappa/6}\varepsilon$ ($N+1>\kappa/(12\ln 2)$). (a): $|g-g_0|_*\le\frac43\varepsilon_1\le\frac43\frac{2^7q\varepsilon^2}{\bar\alpha Rs}$ and $\bar\alpha s=\alpha\kappa/K_a$, $\kappa>8$: $\le 32\frac{qK_a\varepsilon^2}{\alpha R}=16c_1\varepsilon^2$.

**Step S12.26: Displacement.** Lemma 7.9 with the seminorms $|\Pi_I\cdot|$, $|\Pi_\theta\cdot|$, $|\Pi_{\tilde x}\cdot|$ and Theorem 12.20(b). Case A: $|\Pi_I\Psi-I|\le\frac{q\varepsilon}{e\bar\alpha\sigma_0}+\sum_{i=1}^N\frac{q\varepsilon_i}{e\bar\alpha\sigma}\le\frac{6q\varepsilon}{e\bar\alpha s}+\frac{q}{e\bar\alpha\sigma}\frac43\varepsilon_1$, with $\frac{1}{\bar\alpha s}\le\frac{K_a}{6\alpha}$, $\frac{1}{\bar\alpha\sigma}\le\frac{4K_a}{12\ln 2\,\alpha}$, $\varepsilon_1\le\varepsilon/4$: $\le\frac{qK_a\varepsilon}{e\alpha}(1+\frac{4}{12\ln 2}\cdot\frac43\cdot\frac14)\le 1.17\frac{qK_a\varepsilon}{e\alpha}\le\frac{qK_a\varepsilon}{\alpha}$. $|\Pi_\theta\Psi-\theta|\le\frac{q\varepsilon}{\bar\alpha\rho_0}+\frac{q}{\bar\alpha\rho}\frac43\varepsilon_1\le\frac{10q\varepsilon}{\bar\alpha R}$ (as in Part 7). Slow: each step's slow displacement is $\le\frac{e}{16k}$ times its $I$-displacement bound, so $|\Pi_{\tilde x}\Psi-\tilde x|\le\frac{e}{16}\cdot 1.17\frac{qK_a\varepsilon}{e\alpha}<\frac18\frac{qK_a\varepsilon}{\alpha}$; by (i) $\le R/2^{10}$. Case B: half of these.

**Step S12.27: Reality.** Theorem 12.20(e) at each step. □

12.8. **Real extended systems, conjugation, stopped transfer.** Setting of §12.5: $I_c$, $S$, $r_B$, $s$, $w$, $\lambda,c$, $\tilde S$ ($\lambda,c$ fixed for the block). For $0<\varrho\le r_B$ let $W^e_\varrho=\{(\theta,I,\tilde x)\in\mathbb{R}^{2n+2k}:|I-I_c|<\varrho,\ \mathrm{dist}_\infty(\tilde x,\tilde S)<\varrho\}=\mathcal{D}^e_{\varrho,s}\cap\mathbb{R}^{2n+2k}$ (open). For $G\in\mathcal{O}(r,s)^{e,\mathbb{R}}$ the real Hamiltonian $G^{\mathbb{R}}$ is $G|_{W^e_r}$ (and 0 off it). *A $G$-orbit in $W^e_\varrho$ on an interval $J$* is a $y:\mathbb{R}\to\mathbb{R}^{2n+2k}$ with $y(t)\in W^e_\varrho$ and $y'(t)=X^c_{G^{\mathbb{R}}}(y(t))$ for $t\in J$ ($y$ is defined on $\mathbb{R}$ and differentiable *two-sidedly* at every $t\in J$, also at endpoints, exactly as in Part 4 and the frozen `IsSolutionOn`; all helper statements below use this predicate, which the frozen input satisfies).

**Lemma 12.22** (Real extended systems)**.** *Let $G$ be holomorphic and 1-periodic in $\theta$ on $\mathcal{D}^e_{r,s}$, real on $W^e_r$ (finite norm is needed only in (c), (d)), $0<\varrho\le r$. (a) $G^{\mathbb{R}}\in C^\infty(W^e_r)$, $G^{\mathbb{R}}$ is 1-periodic in $\theta$, and $X^c_{G^{\mathbb{R}}}(y)=X^c_G(y)\in\mathbb{R}^{2n+2k}$ (complex partials at the real point $y$); $X^c_{G^{\mathbb{R}}}$ is $C^1$ and $\theta$-periodic on*

*$W^e_r$. (b) (Energy) If $y$ is a $G$-orbit in $W^e_\varrho$ on $J$ then $G^{\mathbb{R}} \circ y$ is constant on $J$. (c) For $u \in \mathcal{O}(\varrho,s)^e$ with finite norm and $y \in W^e_\varrho$: $|u(y)| \le |u|_{\varrho,s}$ and $\sum_j |\partial_{\theta_j} u(y)| \le |u|_{\varrho,s}/(es)$. (d) If $P_\Lambda g = g$ ($g \in \mathcal{O}(\varrho,s)^e$ finite norm, $\Lambda \subset \mathbb{Z}^n$) then $\Pi^\perp_\Lambda \partial_\theta g = 0$ on $W^e_\varrho$.*

*Proof.* **Step S12.28: (a).** Lemma 8.2(a),(b),(c) with $\mathcal{D}^e_{r,s} \cap \mathbb{R}^{2n+2k} = W^e_r$ and [K:`bp:p1:hamiltonian-real`]: the proof only uses holomorphy on an open set and reality at the real points, not the structure of the variables.

**Step S12.29: (b).** $G^{\mathbb{R}}$ is differentiable and $\frac{d}{dt} G^{\mathbb{R}}(y(t)) = \nabla G^{\mathbb{R}}(y) \cdot \Omega_c \nabla G^{\mathbb{R}}(y) = 0$ ($\Omega_c^\top = -\Omega_c$); a function with zero derivative on an interval is constant (mean value inequality, [K:`bp:p1:ext-mvt`]).

**Step S12.30: (c),(d).** Lemma 8.3 and Lemma 8.4: both are statements about Fourier coefficients in $\theta$ at fixed $\hat{I}$ and use Lemma 12.8 only. □

**Lemma 12.23** (Conjugation)**.** *Let $H \in \mathcal{O}(r,s)^{e,\mathbb{R}}$, $\tilde{H} \in \mathcal{O}(\tilde{r},\tilde{s})^{e,\mathbb{R}}$ ($\tilde{r} \le r$, $\tilde{s} \le s$ in the same extended structure), $\Psi : \mathcal{D}^e_{\tilde{r},\tilde{s}} \to \mathcal{D}^e_{r,s}$ holomorphic, real at real points, with $D\Psi\,\Omega_c\,D\Psi^\top = \Omega_c$ and $\tilde{H} = H \circ \Psi$. If $\tilde{y}$ is an $\tilde{H}$-orbit in $W^e_{\tilde{r}}$ on $J$ then $\Psi \circ \tilde{y}$ is an $H$-orbit in $W^e_r$ on $J$.*

*Proof.* **Step S12.31: Chain rule.** As Lemma 8.5: $Y = D\Psi(\tilde{y})$ is real, $\nabla \tilde{H}^{\mathbb{R}} = Y^\top \nabla H^{\mathbb{R}} \circ \Psi$, so $\frac{d}{dt}\Psi(\tilde{y}) = Y\Omega_c Y^\top \nabla H^{\mathbb{R}}(\Psi\tilde{y}) = X^c_{H^{\mathbb{R}}}(\Psi\tilde{y})$; $\Psi$ maps $W^e_{\tilde{r}}$ into $W^e_r$. □

**Lemma 12.24** (Real preimages of a near-identity map)**.** *Let $\Psi : \mathcal{D}^e_{\tilde{r},\tilde{s}} \to \mathbb{C}^{2n+2k}$ be holomorphic, real at real points, $G = \Psi - \mathrm{id}$, $|G_\theta| \le d_\theta$, $|G_I| \le d_I$, $|G_{\tilde{x}}| \le d_x$ on $\mathcal{D}^e_{\tilde{r},\tilde{s}}$. Let $0 < \varrho' \le \tilde{r}$, $0 < \sigma' \le 2\tilde{s}$ and $\kappa_* = \max(d_\theta/\sigma', d_I/\varrho', d_x/\varrho') \le \frac{1}{4}$. Then every $y = (\theta_y, I_c, \tilde{x}_y)$ with $\theta_y \in \mathbb{R}^n$, $\tilde{x}_y \in \tilde{S}$ equals $\Psi(x)$ for some $x \in W^e_{\tilde{r}}$ with $|\theta_x - \theta_y| \le d_\theta$, $|I_x - I_c| \le d_I$, $|\tilde{x}_x - \tilde{x}_y| \le d_x$.*

*Proof.* Proof of Lemma 8.6 with $\nu(a,b,\xi) = \max(|a|/\sigma', |b|/\varrho', |\xi|/\varrho')$: $X = \{x \in \mathbb{R}^{2n+2k} : \nu(x-y) \le \kappa_*\} \subset W^e_{\tilde{r}}$ (since $\tilde{x}_y \in \tilde{S}$ and $\kappa_*\varrho' \le \varrho'/4 < \tilde{r}$); $T(x) = y - G(x)$ maps $X$ to itself; the complex $\nu$-ball of radius $\frac{1}{2}$ about $x \in X$ lies in $\mathcal{D}^e_{\tilde{r},\tilde{s}}$ (imaginary parts $< \sigma'/2 \le \tilde{s}$, $\mathrm{dist}(\tilde{x}', \tilde{S}) < \frac{\varrho'}{4} + \frac{\varrho'}{2} < \tilde{r}$), so Lemma 5.8 gives $|DG|_\nu \le 2\kappa_* \le \frac{1}{2}$ and $T$ is a contraction on the convex set $X$ (mean value inequality); Banach's fixed point theorem ((E1)) gives $x$. □

**Lemma 12.25** (Continuation with independent periodic and transverse coordinates)**.** *Let $E = \mathbb{R}^n \times \mathbb{R}^N$ (max norm), $U \subset \mathbb{R}^N$ open, $W = \mathbb{R}^n \times U$, $F : W \to E$ of class $C^1$ with $F(\theta+\ell, v) = F(\theta, v)$ ($\ell \in \mathbb{Z}^n$), $x_0 \in W$, $(T_-, T_+)$ and $z = (z_\theta, z_v)$ the maximal interval and solution (Theorem 4.13). If $K \subset U$ is compact and $z_v(t) \in K$ for $t \in [t_*, T_+)$ for some $t_* \in [0, T_+)$, then $T_+ = +\infty$.*

*Proof.* **Step S12.32: Uniform data.** If $K = \emptyset$ the hypothesis is void ($z_v(t) \in K$ impossible); else choose $\delta \in (0,1]$ with $K_\delta = \{v : \mathrm{dist}_\infty(v, K) \le \delta\} \subset U$ (Step S4.6; $\delta = 1$ if $U = \mathbb{R}^N$); $C = [-1,2]^n \times K_\delta$ is compact in $W$; let $M, L$ be the bound and Lipschitz constant of $F$ on $C$ (Lemma 4.4). For $x = (\theta, v)$ with $v \in K$ and $\ell = (\lfloor \theta_i \rfloor)_i$: $x' = x - (\ell, 0) \in [0,1)^n \times K$, $\bar{B}(x', \delta) \subset C$ and $\bar{B}(x, \delta) = \bar{B}(x', \delta) + (\ell, 0) \subset W$; by periodicity $F$ is bounded by $M$ and $L$-Lipschitz on $\bar{B}(x, \delta)$. So $\Phi = \mathbb{R}^n \times K$ has uniform data $(\delta, M, L)$.

**Step S12.33: Conclusion.** $z(t) \in \Phi$ on $[t_*, T_+)$; Theorem 4.14 gives $T_+ = \infty$. □

**Lemma 12.26** (Stopped transfer through a symplectic map)**.** *Let $H, \tilde{H}, \Psi$ be as in Lemma 12.23, $|\cdot|_* \ge |\cdot|$ a norm on $\mathbb{R}^n$, and assume*

*(T1) $|\Pi_I \Psi(x) - I_x|_* \le d_*$ and $|\Pi_{\tilde{x}} \Psi(x) - \tilde{x}_x|_\infty \le d_x$ for all $x \in W^e_{\tilde{r}}$, with $d_x < \tilde{r}$;*

*(T2) every $y = (\theta_y, I_c, \tilde{x}_y)$ ($\theta_y \in \mathbb{R}^n$, $\tilde{x}_y \in \tilde{S}$) is $\Psi(x)$ for some $x \in W^e_{\tilde{r}}$ with $|I_x - I_c|_* \le d_*$;*

*(T3) (*conditional helper; a priori confinement for slowly displaced normal-form orbits*) there are* $\varrho > 0$ *with* $d_* + \varrho < \tilde{r}$ *and* $T \in (0, \infty]$ *such that every* $\tilde{H}$*-orbit* $\tilde{y}$ *in* $W^e_{\tilde{r}}$ *on an interval* $\tilde{J} \ni 0$ *with* $|\tilde{I}(0) - I_c|_* \le d_*$ *and* $\mathrm{dist}_\infty(\tilde{x}(t), \tilde{S}) \le d_x$ *for all* $t \in \tilde{J}$ *satisfies* $|\tilde{I}(t) - \tilde{I}(0)|_* \le \varrho$ *for* $t \in \tilde{J}$, $|t| \le T$.

*Let* $F$ *be any function with* $F = X^c_{H^{\mathbb{R}}}$ *on* $W^e_r$, $J \ni 0$ *an interval,* $y : \mathbb{R} \to \mathbb{R}^{2n+2k}$ *differentiable at every* $t \in J$ *(two-sided) with* $y'(t) = F(y(t))$, $y(0) = (\theta_0, I_c, \tilde{x}_0)$, $\tilde{x}_0 \in \tilde{S}$, $t_1 \in J$, $|t_1| \le T$, *with* $\tilde{x}(u) \in \tilde{S}$ *for all* $u$ *between* $0$ *and* $t_1$. *Then* $y(u) \in W^e_r$ *and* $|I(u) - I_c|_* \le \varrho + 2d_*$ *for all* $u$ *between* $0$ *and* $t_1$.

*Source: Ours; Lemma 8.8 with Corollary 8.9; (T3) is a helper hypothesis of this lemma only, never a premise of the frozen theorem.*

*Proof.* Take $t_1 > 0$ (for $t_1 < 0$ apply it to $u \mapsto y(-u)$, $-H$, $-\tilde{H}$; $t_1 = 0$ is trivial).

**Step S12.34: A normal-form orbit.** By (T2) choose $x$; $\tilde{y}$ the maximal $\tilde{H}$-orbit in $W^e_{\tilde{r}}$ through $x$ on $(T_-, T_+) \ni 0$ (Lemma 12.22(a), Theorem 4.13).

**Step S12.35: The good set.** $E = \{\tau \in [0, t_1] : \ [0, \tau] \subset (T_-, T_+),\ y([0, \tau]) \subset W^e_r,\ y = \Psi \circ \tilde{y} \text{ on } [0, \tau]\}$. $0 \in E$, and $E$ *is downward closed* (if $\tau \in E$ and $\tau' \le \tau$ then $\tau' \in E$ by definition). So $E$ is an interval containing $0$.

**Step S12.36: Bounds on every prefix.** Let $\tau \in E$ and $u \in [0, \tau]$. First the slow variables: $y(u) = \Psi(\tilde{y}(u))$ and (T1) give $|\tilde{x}_{\tilde{y}}(u) - \tilde{x}(u)| \le d_x$ with $\tilde{x}(u) \in \tilde{S}$, so $\mathrm{dist}(\tilde{x}_{\tilde{y}}(u), \tilde{S}) \le d_x$. Then (T3) applies to $\tilde{y}$ on $\tilde{J} = [0, \tau]$: $|\tilde{I}(u) - \tilde{I}(0)|_* \le \varrho$, hence $|\tilde{I}(u) - I_c| \le \varrho + d_* < \tilde{r}$.

**Step S12.37: $E$ is open in $[0, t_1]$.** Let $\tau \in E$, $\tau < t_1$. $\Psi \circ \tilde{y}$ is an $H$-orbit in $W^e_r$ on $(T_-, T_+)$ (Lemma 12.23), and $\tau < T_+$ because $[0, \tau] \subset (T_-, T_+)$. $y(\tau) = \Psi(\tilde{y}(\tau)) \in W^e_r$ (open); by continuity $y \in W^e_r$ near $\tau$, where $F = X^c_{H^{\mathbb{R}}}$ is $C^1$; $y$ and $\Psi \circ \tilde{y}$ solve the same equation with the same value at $\tau$, so they agree on a neighbourhood (Theorem 4.11). Hence $[\tau, \tau + \epsilon) \subset E$.

**Step S12.38: $E$ is closed.** Let $\tau^* = \sup E \le t_1$. By Step S12.36 on every prefix $[0, \tau]$, $\tau < \tau^*$, the transverse part $v = (\tilde{I}, \tilde{x}_{\tilde{y}})$ of $\tilde{y}$ lies on $[0, \tau^*)$ in $K = \{|I - \tilde{I}(0)|_* \le \varrho\} \times \{\tilde{x} : \mathrm{dist}_\infty(\tilde{x}, \tilde{x}([0, t_1])) \le d_x\}$, which is compact ($\tilde{x}([0, t_1])$ is a continuous image of $[0, t_1]$) and contained in the open transverse set $U = \{|I - I_c| < \tilde{r}\} \times \{\mathrm{dist}(\tilde{x}, \tilde{S}) < \tilde{r}\}$ (here $\tilde{x}([0, t_1]) \subset \tilde{S}$ and $d_x < \tilde{r}$). The field $X^c_{\tilde{H}^{\mathbb{R}}}$ is $C^1$ and $\theta$-periodic on $\mathbb{R}^n \times U \supset W^e_{\tilde{r}}$-part; $T_+ \ge \tau^*$ (all prefixes lie in $(T_-, T_+)$), and if $T_+ = \tau^*$ Lemma 12.25 would give $T_+ = \infty$: so $T_+ > \tau^*$. By continuity $y(\tau^*) = \Psi(\tilde{y}(\tau^*)) \in W^e_r$ and $y = \Psi \circ \tilde{y}$ on $[0, \tau^*]$: $\tau^* \in E$.

**Step S12.39: Conclusion.** $E = [0, t_1]$; $I(u) - I_c = (\Pi_I \Psi(\tilde{y}(u)) - \tilde{I}(u)) + (\tilde{I}(u) - \tilde{I}(0)) + (I_x - I_c)$, of $|\cdot|_*$-size $\le d_* + \varrho + d_*$ by (T1), Step S12.36, (T2). □

## 12.9. **Nonresonant stability (Proposition 1$^e$).**

**Proposition 12.27** (Nonresonant stability in the extended phase space)**.** *Let* $r, s, K_a, \alpha > 0$, $L \ge 0$, $p > 1$, $q = p/(p-1)$, $\kappa = 2\pi K_a s \ge 6$; $r = r_B$; $h$ *holomorphic on* $V_r(I_c)$, *real on real points;* $|k \cdot \omega(I_c)| \ge \alpha$ *for* $0 < |k|_1 \le K_a$ *((N1),* $\Lambda = \{0\}$*), (N2) with* $L$, $K_a L r \le \alpha/p$, *(SL);* $f \in \mathcal{O}(r, s)^{e,\mathbb{R}}$, $|f|_{r,s} \le \varepsilon$ *with a number* $\varepsilon > 0$, $\varepsilon \le \alpha r/(2^7 q K_a)$; $H = h + f$; $T_{\mathrm{nr}} = \frac{sr}{5\varepsilon} e^{\kappa/6}$. *Then for every* $F$ *with* $F = X^c_{H^{\mathbb{R}}}$ *on* $W^e_r$, $J \ni 0$, $y$ *with* $y' = F(y)$ *on* $J$, $y(0) = (\theta_0, I_c, \tilde{x}_0)$, $\tilde{x}_0 \in \tilde{S}$, *and* $t_1 \in J$ *with* $|t_1| \le T_{\mathrm{nr}}$ *and* $\tilde{x}(u) \in \tilde{S}$ *between* $0$ *and* $t_1$: $|I(u) - I_c| \le r/2$ *between* $0$ *and* $t_1$.

*Source: Pos93 p. 196 (Proposition 1), Proposition 8.11.*

*Proof.* **Step S12.40: Normal form.** Theorem 12.21 ($\Lambda = \{0\}$, $g_0 = \hat{f}_0$ is $\theta$-independent): $\Psi : \mathcal{D}^e_{r/2, s/6} \to \mathcal{D}^e_{r,s}$, $H \circ \Psi = h + g + f_*$, $\eta = |f_*|$, $\le e^{-\kappa/6}\varepsilon$, $d_I = qK_a\varepsilon/\alpha \le r/2^7$, $d_x = r/2^{10}$, $d_\theta = 10q\varepsilon/(2\pi\alpha r)$.

**Step S12.41: (T1), (T2).** $\tilde r = r/2$, $\tilde s = s/6$, $|\cdot|_* = |\cdot|$, $d_* = d_I$; $d_x < \tilde r$. (T2): Lemma 12.24 with $\varrho' = r/2$, $\sigma' = s/3$: $d_I/\varrho' \le \frac{1}{64}$, $d_x/\varrho' \le 2^{-9}$, $d_\theta/\sigma' \le \frac{30}{2^7\kappa} \le \frac{30}{768} < \frac14$ (from (i), $\kappa \ge 6$); the preimage has $|I_x - I_c| \le d_I = d_*$.

**Step S12.42: (T3).** $\varrho = \frac{15}{32}r$, $d_* + \varrho < \tilde r$. For an orbit $\tilde y$ in $W^e_{\tilde r}$ with $|\tilde I(0) - I_c| \le d_*$ violating the claim, Lemma 8.10 gives $t^*$ with $|\tilde I(t^*) - \tilde I(0)| = \varrho$ and $\tilde I(\tau)$ in the closed $\varrho$-ball before. There $\dot{\tilde I} = -\partial_\theta f_*(\tilde y)$ ($h, g$ are $\theta$-independent; the $I$-equation has no slow terms), $|\dot{\tilde I}| \le \eta/(e\tilde s) = 6\eta/(es)$ (Lemma 12.22(c)), so $\varrho \le \frac{6e^{-\kappa/6}\varepsilon}{es} \cdot \frac{sre^{\kappa/6}}{5\varepsilon} = \frac{6}{5e}r < 0.442r < \varrho$: contradiction.

**Step S12.43: Transfer.** Lemma 12.26: $|I(u) - I_c| \le \frac{15}{32}r + \frac{2}{2^7}r = \frac{31}{64}r \le \frac{r}{2}$. □

12.10. **The resonant case: an $\omega$-free energy error on explicit domains.**

**Lemma 12.28** (Energy confinement near a resonance, convex case, no bound on $\omega$)**.** *Let $\tilde r, \tilde s > 0$ (the analytic radius and width of the normal form), $I_c \in \mathbb{R}^n$, $\Lambda \subset \mathbb{Z}^n$, $\Pi = \Pi_\Lambda$ the Euclidean projection, $\Pi^\perp = 1 - \Pi$. Let $h$ be holomorphic on $V_{\tilde r}(I_c)$, real on real points, with on $U = \{|I - I_c| < \tilde r\}$: $m\|\xi\|_2^2 \le Q(I)[\xi,\xi]$, $\|Q(I)\|_{2\to 2} \le M$ $(0 < m \le M)$. Let $g, f_* \in \mathcal{O}(\tilde r, \tilde s)^{e,\mathbb{R}}$ with finite norms, $P_\Lambda g = g$, $\zeta_g = |g|_{\tilde r,\tilde s}$, $\eta = |f_*|_{\tilde r,\tilde s}$, and let $\varepsilon, \bar\eta > 0$ with*

$$\zeta_g \le \tfrac54\varepsilon, \qquad \eta \le \bar\eta \le \varepsilon/e, \qquad \varepsilon \le \frac{m\tilde r^2}{2^{12}}, \qquad 16kc\tilde s \le \tilde r.$$

*Let $\delta \in (0, m\tilde r/16]$ with $\|\Pi\omega(I_c)\|_2 \le \delta$, $\rho' = \delta/(4M)$ $(\le \tilde r/64)$, $\tilde H = h + g + f_*$, $T_* = \dfrac{\tilde s\, m\, \tilde r}{20\, M\, \bar\eta}$. Then for every $\tilde H$-orbit $\tilde y$ in $W^e_{\tilde r}$ on $\tilde J \ni 0$ with $\|\tilde I(0) - I_c\|_2 < \rho'$ **and** $\operatorname{dist}_\infty(\tilde x(t), \tilde S) < \tilde r/2$ for all $t \in \tilde J$ (*explicit slow margin*):*

$$\|\tilde I(t) - \tilde I(0)\|_2 \le \tilde r/4 \qquad \text{for } t \in \tilde J,\ |t| \le T_*.$$

*Source: Pos93 pp. 197–198 (Lemma 1), Lemma 8.16.* New*: the identity (2) replaces the factor $\|\omega_0\|$; the Cauchy estimates are applied only at points of the strictly smaller tube $\mathcal{D}^e_{\tilde r/2,\tilde s/2}$, which the orbit does not leave on the stretch considered (review N11-01).*

*Proof.* Suppose some $t_1 \in \tilde J$, $|t_1| \le T_*$ violates the claim. Lemma 8.10 gives $t^*$, $|t^*| \le T_*$, with $\|\tilde I(\tau) - \tilde I(0)\|_2 \le \Delta := \tilde r/4$ for $\tau$ between 0 and $t^*$ and equality at $t^*$. Put $\omega_0 = \omega(\tilde I(0))$, $\Delta I = \tilde I(t^*) - \tilde I(0)$.

**Step S12.44: The orbit stretch lies in the small tube.** For $\tau$ between 0 and $t^*$: $|\tilde I(\tau) - I_c| \le \|\tilde I(\tau) - \tilde I(0)\|_2 + \|\tilde I(0) - I_c\|_2 < \frac{\tilde r}{4} + \frac{\tilde r}{64} < \frac{\tilde r}{2}$; $\operatorname{dist}(\tilde x(\tau), \tilde S) < \frac{\tilde r}{2}$ by hypothesis; $\theta$ is real. So $\tilde y(\tau) \in \mathcal{D}^e_{\tilde r/2,\tilde s/2} \cap \mathbb{R}^{2n+2k}$, and the segment $[\tilde I(0), \tilde I(t^*)] \subset U$. This is the only domain on which the Cauchy-type bounds below are used.

**Step S12.45: Admissibility.** $\|\omega_0 - \omega(I_c)\|_2 \le M\rho' = \delta/4$ (mean value inequality on the convex ball $\{\|I - I_c\|_2 < \rho'\} \subset U$), so $\|\Pi\omega_0\|_2 \le \frac54\delta$; $\rho' \le \frac{m\tilde r}{64M} \le \frac{\tilde r}{64}$.

**Step S12.46: Energy inequality.** Lemma 8.14(a),(c),(d) (energy: Lemma 12.22(b); Taylor with Lagrange remainder along $I_{s'} = \tilde I(0) + s'\Delta I \in U$, $Q \ge m$): $\frac{m}{2}\Delta^2 \le A + |\langle \Pi\omega_0, \Delta I\rangle| + C$, $A = |(g+f_*)(\tilde y(t^*)) - (g+f_*)(\tilde y(0))| \le 2(\zeta_g + \eta)$, $C = |\langle \omega_0, \Pi^\perp \Delta I\rangle|$. Now $|\langle \Pi\omega_0, \Delta I\rangle| \le \frac54\delta\Delta \le \frac{5}{64}m\tilde r\Delta = \frac{5}{16}m\Delta^2$ ($\tilde r = 4\Delta$). Hence $\frac{3}{16}m\Delta^2 \le A + C$, and $\frac{3}{16}m\Delta^2 = \frac{3m\tilde r^2}{256} > 0.0117\, m\tilde r^2$.

**Step S12.47: The identity.** Along $\tilde y$: $\mathsf{F}(\tau) = f_*(\tilde y(\tau))$ has $\mathsf{F}' = Df_* \cdot X^c_{\tilde H^{\mathbb{R}}} = \{f_*, \tilde H\}_c = \{f_*, h\}_c + \{f_*, g\}_c$ ($\{f_*, f_*\}_c = 0$), and $\{f_*, h\}_c = \partial_\theta f_* \cdot \omega(\tilde I)$ since $h$ depends only on $I$:

$$\partial_\theta f_* \cdot \omega(\tilde I(\tau)) = \mathsf{F}'(\tau) - \{f_*, g\}_c(\tilde y(\tau)). \tag{2}$$

**Step S12.48: Estimate of $C$.** $\dot{\tilde I} = -\partial_\theta f_* - \partial_\theta g$, $\Pi^\perp \partial_\theta g = 0$ (Lemma 12.22(d)), so $\Pi^\perp \Delta I = -\int_0^{t^*} \Pi^\perp \partial_\theta f_*\, d\tau$ and, $\Pi$ being self-adjoint, $\langle \omega_0, \Pi^\perp \Delta I\rangle = -\int \langle \omega_0, \partial_\theta f_*\rangle + \int \langle \Pi\omega_0, \partial_\theta f_*\rangle$; with $\langle \omega_0, \partial_\theta f_*\rangle = \langle \omega(\tilde I), \partial_\theta f_*\rangle + \langle \omega_0 - \omega(\tilde I), \partial_\theta f_*\rangle$ and (2):

$$C \le |\mathsf{F}(t^*) - \mathsf{F}(0)| + |t^*|\Big(\sup|\{f_*, g\}_c| + M\Delta \sup\|\partial_\theta f_*\|_2 + \tfrac{5}{4}\delta \sup\|\partial_\theta f_*\|_2\Big),$$

suprema over $\tau$ between 0 and $t^*$. $|\mathsf{F}(t^*) - \mathsf{F}(0)| \le 2\eta$. *Angular derivative:* at real points $\|\partial_\theta f_*\|_2 \le \eta/(e\tilde s)$ (Lemma 12.22(c)). *Bracket:* Lemma 12.9 with $(u, v) = (f_*, g)$, $R_0 = R = \tilde r$, losses $(\rho, \sigma) = (\tilde r/2, \tilde s/2)$, whose $(\mathrm{SL})_{\rho,\sigma}$ is $16kc\tilde s \le \tilde r$, gives $|\{f_*, g\}_c|_{\tilde r/2, \tilde s/2} \le \frac{1}{\rho\sigma}\eta\zeta_g = \frac{4\eta\zeta_g}{\tilde r\tilde s}$; by Step S12.44 the points $\tilde y(\tau)$ lie in $\mathcal{D}^e_{\tilde r/2,\tilde s/2}$, so $\sup|\{f_*, g\}_c| \le \frac{4\eta\zeta_g}{\tilde r\tilde s} \le \frac{5\eta\varepsilon}{\tilde r\tilde s} \le \frac{5m\tilde r}{2^{12}\tilde s}\eta$. With $M\Delta = M\tilde r/4$, $\frac{5}{4}\delta \le \frac{5}{64}m\tilde r$, $\frac{1}{e} < 0.3679$ and $m \le M$, and the exact coefficient

$$\frac{5}{2^{12}} + \frac{5}{64}\cdot\frac{1}{e} + \frac{1}{4}\cdot\frac{1}{e} < 0.00123 + 0.02875 + 0.09198 < 0.1220,$$

$$C \le 2\eta + 0.1220\,\frac{M\tilde r\,\eta\,|t^*|}{\tilde s} \le 2\bar\eta + 0.0061\, m\tilde r^2 \qquad (|t^*| \le T_* = \tfrac{\tilde s m\tilde r}{20M\bar\eta},\ \eta \le \bar\eta).$$

**Step S12.49: Contradiction.** $A \le 2(\frac{5}{4} + \frac{1}{e})\varepsilon < 3.24\varepsilon \le 0.00080\, m\tilde r^2$, $2\bar\eta \le \frac{2\varepsilon}{e} < 0.00018\, m\tilde r^2$ ($\varepsilon \le m\tilde r^2/2^{12}$); so $A + C < (0.00080 + 0.00018 + 0.0061)m\tilde r^2 < 0.0071\, m\tilde r^2 < 0.0117\, m\tilde r^2$. Contradiction. □

**Proposition 12.29** (Resonant stability in the extended phase space, convex case)**.** *Let $r = r_B$, $s, K_a, \alpha, \delta > 0$, $p = \frac{9}{8}$, $q = 9$, $\kappa = 2\pi K_a s \ge 6$, $\Lambda \subset \mathbb{Z}^n$ a subgroup, $0 < m \le M$. Let $h$ be holomorphic on $V_r(I_c)$, real on real points, $\|\partial_I^2 h\|_{2\to 2} \le M$ on $V_r(I_c)$, $m$-convex on $U_r(I_c)$; $|k\cdot\omega(I_c)| \ge \alpha$ for $k \in Z_{K_a} \setminus \Lambda$, (N2) with $L = \sqrt{n}M$, $K_a L r \le \alpha/p$; $\|\Pi_\Lambda \omega(I_c)\|_2 \le mr/32$ (we put $\delta := mr/32$); (SL); $f \in \mathcal{O}(r,s)^{e,\mathbb{R}}$ with $|f|_{r,s} \le \varepsilon$, a number $\varepsilon > 0$ with $\varepsilon \le mr^2/2^{14}$; $H = h + f$; $T_\Lambda = \dfrac{s\,m\,r}{240\,M\,\varepsilon}e^{\kappa/6}$. Then for every $F$ with $F = X^c_{H^\mathbb{R}}$ on $W^e_r$, $J \ni 0$, $y$ with $y' = F(y)$ on $J$, $y(0) = (\theta_0, I_c, \tilde x_0)$, $\tilde x_0 \in \tilde S$, and $t_1 \in J$ with $|t_1| \le T_\Lambda$, $\tilde x(u) \in \tilde S$ between 0 and $t_1$: $\|I(u) - I_c\|_2 \le r/2$ (so $|I(u) - I_c| \le r/2$) between 0 and $t_1$.*
*Source: Pos93 pp. 199–200 (Proposition 2) in the convex case; time bound new.*

*Proof.* **Step S12.50: Reductions.** $\varepsilon \le \frac{mr^2}{2^{14}} \le \frac{\alpha r}{2^7 q K_a}$ since $\frac{\alpha r}{2^7 q K_a} \ge \frac{p\sqrt{n}Mr^2}{2^7 q} \ge \frac{Mr^2}{2^{10}}$ (inequality: $\alpha \ge p\sqrt{n}MK_a r$ from $K_a L r \le \alpha/p$). Theorem 12.21 applies ($\Lambda = -\Lambda$).

**Step S12.51: Normal form.** $\Psi, g, f_*$ on $\mathcal{D}^e_{r/2,s/6}$; $|g|, \le |g_0|, + 16c_1\varepsilon^2 \le \frac{5}{4}\varepsilon$ ($16c_1\varepsilon = \frac{32qK_a\varepsilon}{\alpha r} \le \frac{1}{4}$); $\eta \le \bar\eta := e^{-\kappa/6}\varepsilon \le \varepsilon/e$. $\tilde r = r/2$, $\tilde s = s/6$: $\varepsilon \le mr^2/2^{14} = m\tilde r^2/2^{12}$; $\delta = mr/32 = m\tilde r/16$, $\rho' = \delta/(4M) = \frac{mr}{128M}$; $16kc\tilde s \le \frac{r}{12} < \tilde r$ by (SL).

**Step S12.52: (T1), (T2) in $|\cdot|_* = \|\cdot\|_2$.** $\|\Pi_I \Psi - I\|_2 \le \sqrt{n}\frac{qK_a\varepsilon}{\alpha} \le \frac{q\varepsilon}{pMr} \le \frac{q}{p}\cdot\frac{mr}{2^{14}M} = \frac{mr}{2^{11}M} =: d_*$ ($\alpha \ge p\sqrt{n}MK_a r$; $\frac{q}{p} = 8$), so $d_* = \rho'/16 < \rho'$; $d_x = r/2^{10}$; (T2) is Lemma 12.24 as in Step S12.41.

**Step S12.53: (T3).** $\varrho = \tilde r/4$, $T = \tilde T_* = \frac{\tilde s m\tilde r}{20M\bar\eta} = \frac{smr}{240M\bar\eta} = \frac{smre^{\kappa/6}}{240M\varepsilon} = T_\Lambda$. An orbit with $\|\tilde I(0) - I_c\|_2 \le d_*$ and $\operatorname{dist}(\tilde x(\tilde t), \tilde S) \le d_x < \tilde r/2$ satisfies the hypotheses of Lemma 12.28 (as $d_* < \rho'$); then $\|\tilde I(t) - \tilde I(0)\|_2 \le \varrho$ for $|t| \le T_\Lambda$. $d_* + \varrho < \tilde r$.

**Step S12.54: Conclusion.** Lemma 12.26: $\|I(u) - I_c\|_2 \le \frac{\tilde r}{4} + 2d_* < \frac{r}{2}$. □

12.11. **The stopped block theorem.**

**Lemma 12.30** (Hessian and sizes (nonempty $G$))**.** *Let $h$ be as in Theorem 1.28, $\sigma$ as in* `HasDefiniteFastPart`*, and assume $G \neq \emptyset$. For $z \in G'_\rho$ the symmetric matrix $Q_{\mathbb{C}}(z) = (D^2h(z)[e_i, e_j])_{ij}$ has $\|Q_{\mathbb{C}}(z)\|_{2\to 2} \le M_{\mathbb{C}}$. For real $I \in U$ (the real slice of $G'_\rho$): $Q(I) = D^2h_{\mathbb{R}}(I) = Q_{\mathbb{C}}(I)$ (Lemma 2.3), $m\|\xi\|_2^2 \le \sigma Q(I)[\xi,\xi]$, $|Q(I)[\xi,\xi]| \le M_{\mathbb{C}}\|\xi\|_2^2$, $|\omega(I)|_\infty \ge g$. Consequently $m \le M_{\mathbb{C}}$ (take $I^* \in G \subset G' \subset U$ and any unit $\xi$). For $h_1 = h/2\pi$ the same with $(m_1, M_1, g_1) = (m, M_{\mathbb{C}}, g)/2\pi$, and $\sigma h_1$ is $m_1$-convex.*

*Proof.* **Step S12.55: Operator norm.** $|D^2h(z)[u,v]| \le M_{\mathbb{C}}|u|_\infty|v|_\infty \le M_{\mathbb{C}}\|u\|_2\|v\|_2$ and $\|Q_{\mathbb{C}}u\|_2 = \sup_{\|v\|_2=1}|D^2h(z)[u,\bar v]|$. $G \subset G'$: $I \in \bar B(I,\rho) \subset G'$ and a real $I \in G'$ lies in $G'_\rho$. The rest is the definition of `HasDefiniteFastPart`, and $|\xi|_\infty \le \|\xi\|_2$. The inequality $m \le \sigma Q[\xi,\xi] \le M_{\mathbb{C}}$ at a real point of $G$ for a unit vector gives $m \le M_{\mathbb{C}}$; this needs a point of $G$. □

**Definition 12.31** (Sets and the extended system for the block theorem)**.** Data and constants of Definition 12.6, $G \neq \emptyset$, $\mu > 0$ with $\varepsilon = \varepsilon(\mu) \le \varepsilon_0$ and $d = \mu^{b_n} \le \rho$. $S = \mathrm{core}(D, 3d/4) \subset \mathbb{R}^{2k}$, $w = d/4$. The *perturbation pair* is $(\breve h, \breve f) = (\sigma h_1, \sigma f_1)$, $f_1(\theta, I, x) = \frac{\mu}{2\pi}P\big((2\pi\theta, q), (I, p)\big)$, $x = (q,p)$. For $I_c \in G$ and a block starting radius $r_B \le r_0$: $\lambda = w/r_B$, $c = 2\pi/\lambda^2$, $W^e_\varrho$ as in §12.8, and the physical real set $\mathcal{W}(I_c, r_B) = \{(\theta, I; x) : |I - I_c| < r_B,\ \mathrm{dist}_\infty(x, S) < w\}$.

**Lemma 12.32** (Analyticity, norm and field on the block domain)**.** *In Definition 12.31, $I_c \in G$, $0 < r_B \le r_0$. (a) $\breve h$ is holomorphic on $V_{r_B}(I_c) \subset G'_\rho$, real on real points, $\|\partial_I^2\breve h\|_{2\to 2} \le M_1$, $m_1$-convex on $U_{r_B}(I_c)$, $|\omega|_\infty \ge g_1$. (b) $\breve f \in \mathcal{O}(r_B, s)^{e,\mathbb{R}}$ and $|\breve f|_{r_B,s} \le \varepsilon(\mu) = \kappa_C\mu/(2\pi)$. (c) On the open set $\mathcal{W}(I_c, r_B)$ the frozen $\hat H_\mu = \frac{1}{2\pi}H_\mu \circ \Theta^{-1}$ equals $\mathrm{Re}(h_1 + f_1)$ and is real analytic, so $\sigma X^{2\pi}_{\hat H_\mu} = X^{2\pi}_{(\breve h + \breve f)^{\mathbb{R}}}$ there (sign $\sigma$); in $\tilde x = x/\lambda$ this is $X^c_{(\breve h+\breve f)^{\mathbb{R}}}$ on $W^e_{r_B}$.*

*Proof.* **Step S12.56: Complex points.** Let $(\theta, I, x)$ be complex with $|\mathrm{Im}\,\theta| < 2s = \frac{\rho}{2\pi}$, $|I - I_c| < r_B \le \frac{\rho}{2}$, $\mathrm{dist}_\infty(x, S) < w$. Then $z = ((2\pi\theta, q), (I, p)) \in \mathcal{F}(G', D, \rho)$: $|\mathrm{Im}\,2\pi\theta_i| < \rho$; $\mathrm{Re}\,I$ is within $\rho/2$ of $I_c \in G$ so lies in $G'$ (collar); $|\mathrm{Im}\,I_i| < \rho/2$; with $x_0 \in S$, $|x - x_0| < w$, $\mathrm{Re}\,x \in \bar B(x_0, d/4) \subset D$ ($\bar B(x_0, 3d/4) \subset D$); $|\mathrm{Im}\,x_j| < w < \rho$. So $f_1$ is holomorphic (P1), $|f_1| \le \frac{\mu}{2\pi}$ (P2), 1-periodic in $\theta$ (P4), real at real points (P3). The same points give $V_{r_B}(I_c) \subset G'_\rho$; (a) follows from Lemma 12.30.

**Step S12.57: Norm.** B.1(b) (Lemma 12.8) with $\sigma = s$: $|f_1|_{r_B,s} \le \coth(\pi s)^n \frac{\mu}{2\pi}$ (the bound holds on $\mathcal{D}^e_{r_B,2s}$).

**Step S12.58: Field.** $\mathcal{W}(I_c, r_B) \subset \{\mathrm{cx}\, z \in \mathcal{F}\}$ (open); on its real points $H_\mu = \mathrm{Re}(h + \mu P \circ \mathrm{cx})$, so $\hat H_\mu$ is the real part of a holomorphic function and Lemma 12.22(a) applies; rescaling $x = \lambda\tilde x$ turns weight $2\pi$ into $c$ (Definition 12.7). □

**Lemma 12.33** (Block data from the covering lemma)**.** *In Definition 12.31 assume $\kappa = 2\pi K_a s \ge 6$ ($K_a = K/\sqrt{n}$, $K = (\varepsilon_0/\varepsilon)^{1/(2n)} \ge 1$). Let $I_0 \in G$, $p = \frac{9}{8}$, $q = 9$, $b_{\mathrm{c}} = \frac{1}{32}$. Apply Lemma 9.15 with $(P, \omega) = (\{I_0\}, \omega_{h_1})$, parameters $(p, b_{\mathrm{c}}, r_0, M_1, m_1, K, A)$ ($A = 40M_1/m_1 \ge \frac{pM_1}{b_{\mathrm{c}}m_1} + \sqrt{2} = 36\frac{M_1}{m_1} + \sqrt{2}$). Then there is $\Lambda \in \mathbb{M}_K$ with $I_0 \in D_\Lambda$, $\Lambda \neq \mathbb{Z}^n$, and with $r = r_\Lambda$, $\alpha = \alpha_\Lambda = pM_1Kr$: (1) $r_\Theta := r_0/(AK)^n \le r \le R_0/K$; (2) $|k \cdot \omega(I_0)| \ge \alpha$ for $k \in Z_{K_a} \setminus \Lambda$ (for $\Lambda = \Theta$: $0 < |k|_1 \le K_a$, with $\alpha_\Theta$); (3) $\|\Pi_\Lambda\omega(I_0)\|_2 < \frac{m_1 r}{32}$ (for $\Lambda = \Theta$ the projection is 0, so this is the trivial $0 < m_1 r/32$, not the nontrivial-lattice closeness clause); (4) (N2) holds on $V_r(I_0)$ with $L = \sqrt{n}M_1$ and $K_aLr = \alpha/p$; (5) $\varepsilon \le m_1 r^2/2^{14}$ and $\varepsilon \le \alpha r/(2^7 q K_a)$.*

*Proof.* **Step S12.59: Covering.** Lemma 9.15(a),(d) (valid for $K \ge 1$): $\Lambda \in \mathbb{M}_K$, $I_0 \in D_\Lambda$, $r_\Theta \le r_\Lambda \le r_0/(AK) = R_0/K$ for $\Lambda \neq \mathbb{Z}^n$, $r_{\mathbb{Z}^n} = r_0$. If $\Lambda = \mathbb{Z}^n$ then $\Pi_\Lambda = 1$ and (c) gives $\|\omega(I_0)\|_2 < \delta_{\mathbb{Z}^n} = b_{\mathrm{c}}m_1 r_0 = \frac{m_1 r_0}{32} \le g_1$ (as $r_0 \le 32g/m$), contradicting $\|\omega(I_0)\|_2 \ge |\omega(I_0)|_\infty \ge g_1$ (Lemma 12.30): so $\Lambda \neq \mathbb{Z}^n$.

**Step S12.60: (2)–(4).** Corollary 9.16(b),(c) with $K_g = K$, $K_a = K/\sqrt{n} \le K_g$: (2), (3) ($\delta_\Lambda = b_c m_1 r$). Lemma 7.2-type: for $v \in \mathbb{C}^n$, $|\partial_I^2 h_1 v|_\infty \le \|\cdot\|_2 \le M_1\|v\|_2 \le \sqrt{n}M_1|v|_\infty$. $\alpha = pM_1Kr = p\sqrt{n}M_1K_ar$, i.e. $K_aLr = \alpha/p$. *Sign transport:* the lemma is applied to $\omega_{\sigma h_1} = \sigma\omega_{h_1}$; all quantities used (absolute divisors $|k\cdot\omega|$, lengths $\|\Pi\omega\|_2$, $\|\omega\|_2 \ge g_1$, Hessian operator norms) are unchanged by $\omega \mapsto \sigma\omega$, and $\omega_{\sigma h_1}$ is real on the real slice, as required by Proposition 12.19.

**Step S12.61: (5).** $m_1 r_\Theta^2/2^{14} = \varepsilon_0 K^{-2n} = \varepsilon$ ($K^{2n} = \varepsilon_0/\varepsilon$) and $r \ge r_\Theta$; $\frac{\alpha r}{2^7 q K_a} = \frac{p\sqrt{n}M_1 r^2}{2^7 q} \ge \frac{M_1 r^2}{2^{10}} \ge \frac{m_1 r^2}{2^{14}} \ge \varepsilon$. □

**Theorem 12.34** (Stopped block stability (Pos93 Theorem 1 with slow parameters))**.** *In Definition 12.31 assume (B1) $\kappa = 2\pi K_a s \ge 6$ and (B2) $K \ge 256\,k\,\rho\,R_0/d^2$. Put $r_\Theta = r_0/(AK)^n$, $T_{\mathrm{P}} = \dfrac{s\,m_1\,r_\Theta}{240\,M_1\,\varepsilon}e^{\kappa/6}$. Let $I_0 \in G$, $F : \mathbb{R}^{2n+2k} \to \mathbb{R}^{2n+2k}$ any function with $F = X^{2\pi}_{(\check h+\check f)^{\mathbb{R}}}$ on $\mathcal{W}(I_0, r_0)$, $J \ni 0$ an interval, $y = (\theta, I; x)$ differentiable on $J$ with $y' = F(y)$ there, $I(0) = I_0$, $x(0) \in S$, and $t_1 \in J$ with $|t_1| \le T_{\mathrm{P}}$ and $x(u) \in \mathrm{core}(D, d)$ for all $u$ between $0$ and $t_1$. Then $|I(u) - I_0|_\infty \le \frac{R_0}{2K}$ for all $u$ between $0$ and $t_1$.*

*Source: Pos93 Theorem 1 (pp. 190–191, proof pp. 203–204); the covering is Part 9 with $b_{\mathrm{c}} = \frac{1}{32}$ in place of Pos93's $\frac{1}{8}$ (Remark 12.35).*

*Proof.* We write $h, f$ for $\check h, \check f$.

**Step S12.62: Block.** Lemma 12.33: $\Lambda$, $r = r_\Lambda \in [r_\Theta, R_0/K]$, $\alpha$. Take $r_B = r$, $\lambda = w/r$, $c = 2\pi r^2/w^2$.

**Step S12.63: (SL).** $64\pi k s r \le 64\pi k s \frac{R_0}{K} = 16 k\rho\frac{R_0}{K} \le \frac{d^2}{16} = w^2$ by (B2) ($s = \rho/4\pi$). This is the only place where $d$, hence $b_n$, enters.

**Step S12.64: Extended system.** By Lemma 12.32: $h$ on $V_r(I_0)$, $f \in \mathcal{O}(r,s)^{e,\mathbb{R}}$ with $|f|_{r,s} \le \varepsilon$; $\tilde y = (\theta, I, x/\lambda)$ satisfies $\tilde y' = F^\sim(\tilde y)$ with $F^\sim = X^c_{(h+f)^{\mathbb{R}}}$ on $W_r^e \subset$ image of $\mathcal{W}(I_0, r_0)$; $\tilde x(u) \in \tilde S$ since $\mathrm{core}(D, d) \subset S$.

**Step S12.65: Case $\Lambda = \Theta$.** Proposition 12.27 with $(r, K_a, \alpha, L) \to (r_\Theta\text{-block: } r, K_a, \alpha_\Theta, \sqrt{n}M_1)$, $p = \frac{9}{8}$: hypotheses by Lemma 12.33(2),(4),(5), (B1), (SL); $T_{\mathrm{nr}} = \frac{sr}{5\varepsilon}e^{\kappa/6} \ge T_{\mathrm{P}}$ ($r = r_\Theta$ for $\Theta$; $\frac{m_1}{M_1} \le 1$). Conclusion $|I(u) - I_0| \le r/2 \le R_0/(2K)$.

**Step S12.66: Case $\Lambda \ne \Theta, \mathbb{Z}^n$.** Proposition 12.29 with $(r, K_a, \alpha, m, M) = (r_\Lambda, K_a, \alpha_\Lambda, m_1, M_1)$: hypotheses by Lemma 12.33(2)–(5), (B1), (SL), $m_1$-convexity (Lemma 12.32(a)); $T_\Lambda = \frac{s m_1 r_\Lambda}{240 M_1 \varepsilon}e^{\kappa/6} \ge T_{\mathrm{P}}$ since $r_\Lambda \ge r_\Theta$. Conclusion $\|I(u) - I_0\|_2 \le r_\Lambda/2 \le R_0/(2K)$. □

## 12.12. Assembly: proof of the frozen theorem.

*Proof of Theorem 1.28.* **Step S12.67: Empty $G$.** If $G = \emptyset$ then $\mathrm{core}(G, \cdot) = \emptyset$ and (Z2) is impossible: any $\mu_0 > 0$ (say 1) works. From now on $G \ne \emptyset$; Lemma 12.30 (in particular $m \le M_{\mathbb{C}}$) holds.

**Step S12.68: Data and constants.** Definition 12.6 ($A = 40M_{\mathbb{C}}/m \ge 11$, $\varepsilon_0 = m_1 r_0^2/(2^{14}A^{2n})$, $r_0 = \min\{\rho/2, 32g/m\}$, $R_0 = r_0/A$, $c_K = (2\pi\varepsilon_0/\kappa_C)^{1/(2n)}$).

**Step S12.69: The threshold.** With $\gamma_1, \gamma_2$ of Lemma 12.5 let $\mu_0 = \min\{\mu_a, \dots, \mu_f\}$: $\mu_a = 2\pi\varepsilon_0/\kappa_C$; $\mu_b = \rho^{1/b_n}$; $\mu_c = (\pi s c_K/(3\sqrt{n}))^{2n}$; $\mu_d = (c_K/(256k\rho R_0))^{1/\gamma_1}$; $\mu_e = (c_K/R_0)^{1/\gamma_2}$; $\mu_f$ from Step S12.70. For $0 < \mu < \mu_0$: $\varepsilon \le \varepsilon_0$; $d = \mu^{b_n} \le \rho$; $\kappa = 2\pi c_K s \mu^{-1/(2n)}/\sqrt{n} \ge 6$ (B1); $K = c_K\mu^{-1/(2n)} \ge 256k\rho R_0\mu^{-2b_n}$ iff $\mu^{\gamma_1} \le c_K/(256k\rho R_0)$ (B2); $\frac{R_0}{2K} \le \frac{1}{2}\mu^{b_n}$ iff $\mu^{\gamma_2} \le c_K/R_0$.

**Step S12.70: Time.** $T_{\mathrm{P}} = C_T \mu^{-1/2} \exp(c_2 \mu^{-1/(2n)})$ with $C_T = \frac{s\, m\, r_0}{240\, M_{\mathbb{C}} (A c_K)^n} \cdot \frac{2\pi}{\kappa_C} > 0$ and $c_2 = \frac{\pi s c_K}{3\sqrt{n}} > 0$ ($r_\Theta = r_0 (A c_K)^{-n} \mu^{1/2}$, $\varepsilon = \kappa_C \mu / 2\pi$, $m_1/M_1 = m/M_{\mathbb{C}}$). As $a_n < \frac{1}{2n}$, $c_2 \mu^{-1/(2n)} - \mu^{-a_n} \geq \frac{c_2}{2} \mu^{-1/(2n)}$ for small $\mu$, so $\mu T_{\mathrm{P}} e^{-\mu^{-a_n}} \geq C_T \mu^{1/2} e^{\frac{c_2}{2} \mu^{-1/(2n)}} \to \infty$ ((E2)); this defines $\mu_f$.

**Step S12.71: Given $\mu, P, C, z, t$.** If $t = 0$ the claim is $0 \leq \frac{1}{2} \mu^{b_n}$. Let $t > 0$; $t \leq \mu^{-1} e^{\mu^{-a_n}} \leq T_{\mathrm{P}}$, $[0, t] \subset [0, C]$.

**Step S12.72: The orbit.** By (Z1), $z$ is differentiable on $[0, C]$ with $z' = X_{H_\mu}(z)$; $I_0 := \mathrm{fast}(z(0)_2) \in G$ by (Z2). $y = \Theta \circ z$ satisfies $y' = F_0(y)$ with $F_0 = X^{2\pi}_{\hat{H}_\mu}$, an everywhere defined function (Lemma 12.4); by Lemma 12.32(c), $\sigma F_0 = X^{2\pi}_{(\check{h}+\check{f})^{\mathbb{R}}}$ on $\mathcal{W}(I_0, r_0)$. Slow variables of $y$ lie in $\mathrm{core}(D, d)$ on $[0, t]$.

**Step S12.73: Sign.** $\sigma = +1$: Theorem 12.34 with $F = F_0$, $J = [0, C]$, $t_1 = t$. $\sigma = -1$: $\bar{y}(u) = y(-u)$, $\bar{J} = [-C, 0]$, $\bar{y}' = -F_0(\bar{y}) = X^{2\pi}_{(\check{h}+\check{f})^{\mathbb{R}}}(\bar{y})$ on $\mathcal{W}$; Theorem 12.34 with $F = -F_0$, $t_1 = -t$.

**Step S12.74: Drift.** $|I(t) - I_0|_\infty \leq \frac{R_0}{2K} \leq \frac{1}{2} \mu^{b_n}$.

**Step S12.75: Quantifier order.** $\mu_0$ depends only on $(n, k, \rho, g, m, M_{\mathbb{C}})$ and is chosen before $\mu, P, C, z, t$. □

12.13. **Deviations, errata, doubts, status.**

*Remark* 12.35 (Change of the covering parameter). Pos93's $b = \frac{1}{8}$ is replaced by $b_{\mathrm{c}} = \frac{1}{32}$ (so $\delta_\Lambda = \frac{1}{32} m r_\Lambda$, $A = 40M/m$, $\varepsilon_0$ with $2^{14}$) because the energy argument now needs the explicit margins of Lemma 12.28: orbit exit radius $\tilde{r}/4$ with $\delta \leq m\tilde{r}/16$. Part 9 (Lemma 9.15) is stated for arbitrary $b > 0$; Part 10's block hypotheses are *not* cited, their items are re-derived in Lemma 12.33.

*Remark* 12.36 (What is proved; exponents; deviations). (1) *Proved (subject to audit):* the frozen statement, via the stopped block theorem, with time $T_{\mathrm{P}} = C_T \mu^{-1/2} e^{c_2 \mu^{-1/(2n)}}$ and drift $\leq \frac{R_0}{2c_K} \mu^{1/(2n)}$, *under the slow margin $d = \mu^{b_n}$ of the frozen statement*; these are used only through the monotone comparisons Steps S12.70, S12.74, which give the frozen exponents $a_n, b_n$. No statement with a smaller slow margin is claimed (with $d = \mu^{1/(2n)}$, (SL) fails). The larger exponents are a by-product, not a claim of this blueprint, and are unverified until audited. (2) The extended bracket constant is 1 instead of $\frac{2}{e}$ (checked at Step S12.22, S12.24); the slow bracket is dominated by (SL). (3) $\omega$-free resonant time: Lemma 12.28 with Cauchy estimates only on the strictly smaller tube $\mathcal{D}^e_{\tilde{r}/2, \tilde{s}/2}$ (the orbit is confined to radius $\tilde{r}/4 + \rho'$ and slow distance $< \tilde{r}/2$), and covering parameter $b_{\mathrm{c}} = \frac{1}{32}$. (4) Convex Lemma 1 instead of Lemma 2; initial erosion used only as $\mathrm{fast}(z(0)_2) \in G$; Euclidean Hessian bound $M_{\mathbb{C}}$; $2\pi$ converted once; imaginary widths inside the open tube; empty $G$ treated separately.

*Remark* 12.37 (Suspected defects in the sources; doubts). Nek77 p. 1 (and p. 3): proofs of the §10 lemmas are announced for Part II and are not in the pinned file (acquisition gap, not an error). Pos93 pp. 198–200: time bound $\propto 1/\Omega$, not uniform for unbounded $G$ (replaced here). No inconsistency of the frozen statement was found; $k \geq 1$ is not used.

*Remark* 12.38 (Status and remaining obligations). The blueprint now states and proves, in full and not as lists of changes, the extended Fourier class (by Part 5 for arbitrary base sets), bracket, field, symplectic, adjoint, composition, transform, canonical-map, homological, iterative and normal-form results, the continuation lemma with independent dimensions, the stopped transfer, both stability propositions and the block theorem. *Unaudited new mathematics:* Lemma 12.28 (repaired after review N11-01; constants $0.00123 + 0.02875 + 0.09198 < 0.1220$, $0.0061$, $0.00080$, $0.00018$ against $0.0117$) and Lemma 12.26. *Lean obligations:* generalise Parts 4–6 to a base set in $\mathbb{C}^{n+2k}$ with the

matrix $\Omega_c$ (Lemmas 12.8–12.21), then Lemmas 12.25–12.34 and the assembly; Part 6's Lemmas on flows, Taylor, and Part 7's telescoping and divisor lemmas are used as stated. No claim is made that the audit of Lemma 12.28 is complete.

## 13. N-AVERAGING: THE MEAN ERROR AND THE MOST-DATA EXISTENCE THEOREM (D28)

This part proves the two frozen statements of Part 0, Theorems 1.33 and 1.34, for the fixed analytic system $H_\varepsilon = \mathrm{Re}(h(I)+\varepsilon f(\theta,I))$. The route is the one of Neishtadt 1976 (Nei76, pp. 1295–1298) and Kasuga, specialised to our setting, with the repair of E-NAVG-NEI-17.

Mechanism. Because $f$ is a function of the phase point alone, the slow equation is $\dot I = -\varepsilon\partial_\theta f =: \varepsilon a$ and $a$ has no Fourier mode $k=0$ (it is $-2\pi i\sum_k k f_k \mathrm{e}_k$). So the averaged action orbit is constant, $J(t)=I_0$ (Remark 13.2), and there is nothing to solve in the averaged system: no Gronwall step is needed. Fast oscillations are removed by a corrector $w$ with $\partial_\theta w\cdot\omega \approx a$, built from Fourier modes with the *regularised* small divisor $\psi_\kappa(k\cdot\omega)$, $\psi_\kappa(y)=y/(y^2+\kappa^2)$; it is smooth on the resonant surfaces. Then

$$I(t)-I_0 = \varepsilon\big(w(t)-w(0)\big) - \varepsilon^2\int_0^t Q\,ds + \varepsilon\int_0^t r\,ds,$$

with $Q$, $r$ explicit (Lemma 13.12). Each term is bounded *pointwise along the trajectory* by an envelope function of the action alone; *integrating over initial data*, the flow is volume preserving (Liouville, Lemma 13.3), so the integral over initial data of an envelope evaluated along the trajectory at a fixed time is at most the integral of the envelope over the action ball (Lemma 13.14). The envelope integrals are controlled by the rank hypothesis: Lemma 13.9 turns it into a uniform volume bound for resonant slabs $\{|k\cdot\omega|\le u\}$ of size $u/|k|_2$, with no global chart. Taking $\kappa=\sqrt{\varepsilon}$ gives the stopped mean estimate $\int_K \Psi \le A_1\sqrt{\varepsilon}$ (Proposition 13.15); Markov's inequality and the first-exit argument finish both theorems (§13.8, §13.9).

Summary of deviations from the sources. (i) No Gronwall and no cohort-by-cohort continuation: $a_0=0$ and the flow is exactly volume preserving, so Nei76's Jacobian estimates (15)–(16) and the continuation steps of p. 1298 are replaced by one stopping time (Remark 13.19). (ii) The reciprocal regularisation $\psi_\kappa$ is a rational function, not Nei76's piecewise cubic (which is only $C^1$). (iii) No Fourier truncation. (iv) Conversions: period one, max norms, Euclidean norm only in the slab lemma; all constants are recomputed here.

### 13.1. Setting, constants and external facts.

Standing data. $n\ge 1$ ($n\ge 2$ is not used), $R,s>0$, $h,f$, $K$ as in Theorem 1.33 with `IsAnalyticIntegrable` $R\,s\,h$, `IsAnalyticPerturbation` $R\,s\,1\,f$, the rank alternative, $K$ compact, $K\subset\mathbb{R}^n\times B(0,R/2)$. For $\varepsilon>0$: $T=1/\varepsilon$, $H=H_\varepsilon$, $W_0=\mathbb{R}^n\times B(0,R)$ (open), $F=X_H$ the Hamiltonian vector field, and

$$\rho = R/8, \qquad G=\overline{B}(0,3R/4)\subset B(0,R), \qquad \overline{B}(I_0,\rho)\subset G \ \text{ for } |I_0|<R/2.$$

Real points: $f(\theta,I)\in\mathbb{R}$ for real $\theta$ and $I\in B(0,R+s)$, and $\omega=\omega_h$, $M(I)=D\omega(I)$ (a real $n\times n$ matrix, $M_{ij}=\partial_j\omega_i$, continuous on $B(0,R+s)$, Lemma 2.3(b),(c): $h_{\mathbb{R}}$ is $C^\infty$ and $\omega_i=\partial_i h_{\mathbb{R}}$, $M_{ij}=\partial_i\partial_j h_{\mathbb{R}}$).

Constants from $h$. $\Omega_0=\max_{|I|\le R}|\omega(I)|$, $M_2=\max_{|I|\le R}\|M(I)\|$ with $\|M\|=\max_i\sum_j|M_{ij}|$ (finite: $\overline{B}(0,R)\subset B(0,R+s)$ is compact, $\omega\in C^2$).

**Lemma 13.1** (Real-variable derivatives of $f$)**.** *For real $\theta\in\mathbb{R}^n$, $I\in B(0,R)$ and all $\ell,j$: $|\partial_{\theta_\ell}f(\theta,I)|\le 2/s$, $|\partial_{I_j}f(\theta,I)|\le 2/s$; the real partial derivatives of* $\mathrm{Re}\,f$ *equal these complex partials. Hence $|a|_\infty\le 2/s$ for $a:=-\partial_\theta f$ and $|\partial_I f|_\infty\le 2/s$.*

*Proof.* **Step S13.1: Polydiscs.** For $(\theta,I)$ real with $I\in B(0,R)$ the closed polydisc of radius $s/2$ about $(\theta,I)$ lies in $D_s(R)=T_s\times B(0,R)_s$ (Definition 1.7: $|\mathrm{Im}\,\theta_i'|\le s/2<s$ and $|I'-I|\le s/2<s$ with $I\in B(0,R)$ real).

**Step S13.2: Cauchy.** $|f| \le 1$ there, so [K:`bp:p1:cauchy-open`] gives $|\partial f| \le 1/(s/2)$.

**Step S13.3: Real partials.** Lemma 2.3 (and [K:`bp:p1:real-restriction`]): at real points $f$ is real, the real derivative is the restriction of the complex one, so $X_H = (\omega + \varepsilon \partial_I f,\ -\varepsilon \partial_\theta f)$ on $W_0$. □

External facts used. Besides those of Parts 4, 5 and track K's `bp:p1:ext-`... list, the proofs use:

(X1) *Change of variables.* If $U \subset \mathbb{R}^d$ is open, $\Gamma : U \to \mathbb{R}^d$ is injective and $C^1$, $D \subset U$ is Borel and $g \ge 0$ is Borel, then $\Gamma(D)$ is Borel and $\int_{\Gamma(D)} g = \int_D g \circ \Gamma \, |\det D\Gamma|$ (`MeasureTheory.lintegral_image_eq_lintegral_abs_det_fderiv_mul`). Exists.

(X2) *Termwise differentiation.* If $\sum_k \sup|Dg_k| < \infty$ locally uniformly and $\sum g_k$ converges, then $\sum g_k$ is $C^1$ with derivative $\sum Dg_k$ (`hasFDerivAt_tsum`). Exists.

(X3) *Differentiable dependence on initial data, short time (infrastructure lemma, to be proved separately).* If $F$ is $C^1$ on an open $W \subset \mathbb{R}^N$ and $x_0 \in W$, there are $a, r > 0$ and a continuous $y : (-a, a) \times B(x_0, r) \to W$ with $y(0, x) = x$, $\partial_t y = F(y)$, such that $D_x y$ exists, is continuous on $(-a, a) \times B(x_0, r)$ and satisfies $D_x y(t, x) = \mathbf{1} + \int_0^t DF(y(\tau, x)) D_x y(\tau, x)\, d\tau$. **Not available in Mathlib** as stated: `Analysis/ODE` supplies local flows, Picard–Lindelöf, Gronwall, uniqueness and Lipschitz dependence on the initial datum, but not the initial-data derivative and its variational equation. It is neither a premise of the averaging theorems nor to be assumed; it is a standard result with the following proof obligation. *Proof sketch.* Choose $\delta > 0$ with $\overline{B}(x_0, 2\delta) \subset W$ (compact ball $\mathcal{B}$), put $L = \sup_{\mathcal{B}} |F|$, $\Lambda = \sup_{\mathcal{B}} \|DF\|$ (finite), take $r = \delta/2$, and $a > 0$ with $aL < \delta/2$ and $q := a\Lambda < 1$. Then Picard iterates $y_0 = x$, $y_{m+1}(t, x) = x + \int_0^t F(y_m(\tau, x))\, d\tau$ stay in $\overline{B}(x, \delta/2) \subset \mathcal{B}$ for $|t| \le a$, $x \in B(x_0, r)$, and converge uniformly (contraction, factor $q$). They are $C^1$ in $x$, with $Y_m = D_x y_m$ satisfying $Y_0 = \mathbf{1}$, $Y_{m+1} = \mathbf{1} + \int_0^t DF(y_m) Y_m$, hence $\sup\|Y_m\| \le 1/(1-q)$. The linear map $Z \mapsto \mathbf{1} + \int_0^t DF(y) Z$ is a contraction (factor $q$) on bounded continuous matrix families, so has a continuous fixed point $Y$; by uniform continuity of $DF$ on $\mathcal{B}$, $d_m := \sup\|DF(y_m) - DF(y)\| \to 0$ and $\sup\|Y_{m+1} - Y\| \le q \sup\|Y_m - Y\| + a\, d_m/(1-q) \to 0$. Uniform convergence of $y_m$ and of $D_x y_m$ on the open ball gives $D_x y = Y$, continuous, with the stated integral equation. Alternatively for the analytic field of Part 6 one may use Lemma 6.4(b) after explicit rescaling (not needed here).

(X4) *Markov, Tonelli, monotone convergence* for nonnegative Borel functions; a continuous function on a compact set is bounded, attains its maximum, and *(compact-time maximum)*: if $U$ is a topological (here metrisable) space, $T > 0$ and $g : [0, T] \times U \to \mathbb{R}$ is jointly continuous, then $x \mapsto \max_{0 \le u \le T} g(u, x)$ is continuous on $U$ (uniform continuity of $g$ on $[0, T] \times$ a compact neighbourhood of any $x$). Variable time: if $g$ is jointly continuous on $\{(u, x) : (u, x) \in \mathcal{D},\ u \ge 0\}$ and $(t, x) \mapsto (vt, x)$ maps into it for $v \in [0, 1]$, then $(t, x) \mapsto \max_{0 \le u \le t} g(u, x) = \max_{0 \le v \le 1} g(vt, x)$ is continuous on $\{(t, x) \in \mathcal{D} : t \ge 0\}$ by the fixed-time statement applied to $(v; (t, x)) \mapsto g(vt, x)$. (The unqualified statement with no compact time domain is false, e.g. $g(u, x) = -u$ on $\mathbb{R}$.)

(X5) *Fubini on boxes, mean value theorem, Heine–Borel* ([K:`bp:p1:ext-measure`], [K:`bp:p1:ext-mvt`]).

*Remark* 13.2 (The averaged orbit is constant). In the standard form of AKN06 pp. 208–209 (period $2\pi$, $\varphi = 2\pi\theta$) the slow field is $\varepsilon a$, $a = -\partial_\theta f$, and $\bar{a}(I) = \int_{\mathbb{T}^n} a\, d\theta = 0$ since $\int \partial_{\theta_\ell} f\, d\theta = 0$ ([K:`bp:p1:averages`]). The averaged system is $\dot{J} = 0$, $J(t) = I_0$, which is at distance $\ge R/4$ from $\partial B(0, R)$ for $|I_0| < R/2$: the continuability condition of AKN Remark 6.4, pp. 217–218 (and Nei76 4°) holds with room. The conversion $\varphi = 2\pi\theta$, $\Omega = 2\pi\omega$ preserves time and both rank conditions; here we keep period one and never use $\varphi$.

13.2. **The flow of $X_H$ and Liouville's theorem.** Fix $\varepsilon > 0$ (the results of this subsection hold for any $\varepsilon \ge 0$). By Lemma 4.28, $H$ is $C^\infty$ on $W_s \supset W_0$, $X_H$ is $C^1$ on $W_0$ (Lemma 4.22), 1-periodic in $\theta$, and by Proposition 4.25, `IsSolutionOn` $H\, B(0, R)\, J\, z \iff$ `IsSolIn` $X_H\, W_0\, J\, z$. For

$x \in W_0$ let $(T_-(x), T_+(x))$ and $\gamma_x$ be the maximal interval and solution (Theorem 4.13) and put $\Phi_t(x) = \gamma_x(t) = (\theta(t,x), I(t,x))$, $\mathcal{D} = \{(t,x) : x \in W_0,\ T_-(x) < t < T_+(x)\}$, $\Omega_t = \{x : (t,x) \in \mathcal{D}\}$.

**Lemma 13.3** (Flow, regularity, volume preservation). *(a) $\mathcal{D}$ is open in $\mathbb{R} \times W_0$, $\Phi : \mathcal{D} \to W_0$ is continuous, $\Omega_t$ is open, $\Phi_t : \Omega_t \to \Omega_{-t}$ is a bijection with inverse $\Phi_{-t}$, hence a homeomorphism.*

*(b) $\Phi_t$ is $C^1$ on $\Omega_t$; $Y(t,x) = D\Phi_t(x)$ is continuous on $\mathcal{D}$ and $\partial_t Y = DF(\Phi_t x)\, Y$, $Y(0,x) = \mathbf{1}$.*

*(c) $\det D\Phi_t(x) = 1$ on $\Omega_t$.*

*(d) (Translation) $\Omega_t + (k,0) = \Omega_t$ and $\Phi_t(x + (k,0)) = \Phi_t(x) + (k,0)$ for $k \in \mathbb{Z}^n$.*

*Source: Standard (flow of a $C^1$ field; Liouville for Hamiltonian flows). Used by Nei76 p. 1296 (15)–(16) in the form of a Jacobian bound.*

*Proof.* **Step S13.4: Short time.** By (X3) each $x_0 \in W_0$ has $a, r > 0$ and a $C^1$-in-$x$ family $y$ on $(-a,a) \times B(x_0, r)$; by uniqueness (Theorem 4.11) $y(t,x) = \Phi_t(x)$.

**Step S13.5: Chaining.** Let $t_1 > 0$, $(t_1, x_1) \in \mathcal{D}$ (the case $t_1 < 0$ is time reversal, Lemma 4.3(b), $t_1 = 0$ is trivial). The orbit arc $\gamma_{x_1}([0,t_1])$ is compact; cover it by finitely many balls $B(\gamma(u_i), r_i)$ with data $a_i, r_i$ as above, choose a partition $0 = u_0 < \cdots < u_N = t_1$ with $\gamma([u_{i-1}, u_i])$ inside the ball $i$ and $u_i - u_{i-1} < a_i$. Define inductively $\Gamma_i = \Phi_{u_i - u_{i-1}} \circ \Gamma_{i-1}$: by continuity of the short-time map, $\Gamma_i$ is defined and $C^1$ on a neighbourhood of $x_1$ and takes values near $\gamma(u_i)$ (chain rule). By the flow property (uniqueness) $\Gamma_N = \Phi_{t_1}$ there. Hence $\Omega_{t_1} \ni x_1$ has a neighbourhood in $\Omega_{t_1}$; the same for $t$ near $t_1$ gives openness of $\mathcal{D}$ and continuity of $\Phi$ and of $Y = D\Gamma_N = \prod DF$-propagators; $Y$ satisfies the variational equation because each factor does. This proves (b) and the openness in (a).

**Step S13.6: Inverse.** Let $t > 0$ and $x \in \Omega_t$, so $\gamma_x$ is defined on $[0,t]$ (indeed on a neighbourhood). Put $y(u) = \gamma_x(t+u)$ for $u \in [-t, 0]$. Then $y'(u) = F(y(u))$ ($y$ is a solution of $F$, not of $-F$), $y(0) = \Phi_t(x)$, $y(-t) = x$, and $y$ is a solution on the closed interval $[-t,0]$ in $W_0$. By the maximality statement of Theorem 4.13 (every solution on an interval containing 0 is contained in the maximal one, applied to the closed interval $[-t,0] \ni 0$): $[-t,0] \subset (T_-(\Phi_t x), T_+(\Phi_t x))$, so $T_-(\Phi_t x) < -t$, $\Phi_t x \in \Omega_{-t}$, and $y = \gamma_{\Phi_t x}$ on $[-t,0]$. In particular $\Phi_{-t}(\Phi_t x) = y(-t) = x$. The same argument with $-t$ in place of $t$ (a solution on $[0,t]$ through $\Phi_{-t}z$, $z \in \Omega_{-t}$) gives $\Phi_{-t}z \in \Omega_t$ and $\Phi_t(\Phi_{-t}z) = z$. Hence $\Phi_t : \Omega_t \to \Omega_{-t}$ is a bijection with inverse $\Phi_{-t}$, and both are continuous by the openness already proved: (a). The case $t < 0$ is the same with signs exchanged, $t = 0$ is trivial.

**Step S13.7: Symplectic form.** $H$ is $C^2$, so $F = J\nabla H$ with $J = \begin{pmatrix} 0 & \mathbf{1} \\ -\mathbf{1} & 0 \end{pmatrix}$ and $DF = JS$, $S = \nabla^2 H$ symmetric ([K:bp:p1:ext-schwarz]). Then $DF^\top J = S^\top J^\top J = S$ and $J\,DF = J^2 S = -S$, so $DF^\top J + J\,DF = 0$ and $\frac{d}{dt}(Y^\top J Y) = Y^\top (DF^\top J + J\,DF) Y = 0$ and $Y^\top J Y = J$ (the computation of Lemma 6.5, Step S6.10, for real matrices). Taking determinants $\det Y = \pm 1$; $t \mapsto \det Y(t,x)$ is continuous on the interval $(T_-, T_+) \ni 0$ with value 1 at $t = 0$, so $\det Y \equiv 1$: (c).

**Step S13.8: Translation.** Lemma 4.24 (with Lemma 4.28) gives that $\gamma_x + (k,0)$ is a solution through $x + (k,0)$, so by Theorem 4.13 (d). □

**Lemma 13.4** (First exit from the tube; no blow-up inside). *Let $x = (\theta_0, I_0) \in W_0$ with $|I_0| < R/2$, $\varepsilon > 0$. Put $V = \{(\theta, I) : |I - I_0| < \rho\}$ and $\tau(x) = \sup\{t \in [0, T_+(x)) :\ |I(u,x) - I_0| < \rho\ \forall u \in [0,t]\} \in (0, \infty]$. Then: (i) if $\tau(x) < \infty$ then $\tau(x) < T_+(x)$ and $|I(\tau, x) - I_0| = \rho$; (ii) if $\tau(x) = \infty$ then $T_+(x) = \infty$; (iii) $|I(u,x)| \le 3R/4$ for $0 \le u \le \tau(x)$ ($u < \infty$, $u < T_+$). Moreover the set $S = \{(t,x) : x \in W_0,\ |I_0| < R/2,\ 0 \le t < T_+(x),\ |I(u,x) - I_0| < \rho\ \forall u \in [0,t]\} = \{(t,x) : 0 \le t < \tau(x)\}$ is open in $[0,\infty) \times \{x \in W_0 : |I_0| < R/2\}$.*

*Proof.* **Step S13.9: First exit.** Lemma 4.19 with $J = [0, T_+)$ and the open set $V$ (the solution is continuous on $J$): either $z(t) \in V$ for all $t \in [0, T_+)$ (then $\tau = T_+$), and the solution stays in

$\mathbb{R}^n \times \overline{B}(I_0,\rho)$ with $\overline{B}(I_0,\rho) \subset G \subset B(0,R)$ compact; Corollary 4.16 (with $U = B(0,R)$, $K = \overline{B}(I_0,\rho)$) gives $T_+ = \infty$: this is (ii). Or a first exit time $t^* \in (0,T_+)$ with $z(t^*) \in \partial V$, i.e. $|I(t^*) - I_0| = \rho$ (closure minus $V$ of a set of the form $\mathbb{R}^n\times$ open ball): $\tau = t^*$, which is (i).

**Step S13.10: (iii).** $|I| \le |I_0| + \rho < R/2 + R/8$.

**Step S13.11: Openness of $S$.** On the open set $\mathcal{D} \cap \{t \ge 0\}$ the function $m(t,x) = \max_{0\le u\le t}|I(u,x) - I_0|$ is continuous ((X4)); $S = \{m < \rho\}$. □

13.3. **Analytic data: Fourier coefficients and bounds.** For $k \in \mathbb{Z}^n$ and $I' \in B(0,R)_s$ let $f_k(I') = \int_{[0,1]^n} f(x,I')e^{-2\pi ik\cdot x}dx$ (Fourier coefficient in $\theta$) and $\beta_k = e^{-\pi s|k|_1}$, $\mathrm{e}_k(\theta) = e^{2\pi ik\cdot\theta}$, $q_k(I) = k\cdot\omega(I)$ for real $I$.

**Lemma 13.5** (Coefficients)**.** *(a) $f_k$ is holomorphic on $B(0,R)_s$ and $|f_k| \le \beta_k$ there. (b) For real $I \in B(0,R)$: $|\partial_{I_j} f_k(I)| \le \frac{2}{s}\beta_k$. (c) For real $\theta$ and $I \in B(0,R)$: $f(\theta,I) = \sum_k f_k(I)\mathrm{e}_k(\theta)$ and $a(\theta,I) = -\partial_\theta f = -2\pi i\sum_{k\ne0} f_k(I)\,k\,\mathrm{e}_k(\theta)$ (absolutely convergent, $a \in \mathbb{R}^n$). In particular the Fourier coefficient of $a$ at $k = 0$ is 0. (d) $\sigma := \sum_{k\ne0}(1+|k|_1)^3\beta_k < \infty$.*

*Proof.* **Step S13.12: (a).** Holomorphy: [N:`bpN:p4:coefficients`](a) with base set $D = \iota(B(0,R))$, action width $r = s$ and angle width $s$, so that $V_rD = B(0,R)_s$ and $\mathcal{D}_{r,s} = T_s\times B(0,R)_s = D_s(R)$; the frozen hypothesis gives $f \in \mathcal{O}(s,s)$ for this $D$ (periodic and holomorphic on $D_s(R)$). The domain of $f$ is not enlarged and no relation between $R$ and $s$ is used. For fixed $I' \in B(0,R)_s$, $f(\cdot,I')$ is holomorphic, 1-periodic and bounded by 1 on $T_s$, so [K:`bp:p1:decay`] with $\rho' = s/2$ gives $|f_k(I')| \le e^{-2\pi|k|_1 s/2}$.

**Step S13.13: (b).** The closed polydisc of radius $s/2$ about the real $I \in B(0,R)$ lies in $B(0,R)_s$, where $|f_k| \le \beta_k$: [K:`bp:p1:cauchy-open`].

**Step S13.14: (c).** $f(\cdot,I)$ is continuous, periodic, and $\sum|f_k| \le \sum\beta_k < \infty$ ([K:`bp:p1:exp-sums`](b)), so it equals its Fourier series ([K:`bp:p1:fourier-basic`](b)). The function $\partial_{\theta_\ell}f(\cdot,I)$ is continuous, periodic with coefficients $2\pi ik_\ell f_k$ ([K:`bp:p1:fourier-basic`](a)), which are summable; so it equals its series.

**Step S13.15: (d).** $(1+x)^3e^{-\pi sx/2} \le c(s)$ and [K:`bp:p1:exp-sums`](b): $\sigma \le c(s)\sum_k e^{-\pi s|k|_1/2} < \infty$. □

**Definition 13.6** (Corrector and its pieces)**.** For $0 < \kappa \le 1$ let $\psi_\kappa(y) = \dfrac{y}{y^2+\kappa^2}$ ($y \in \mathbb{R}$). Since $\kappa > 0$ the denominator never vanishes, so $\psi_\kappa$ is smooth on $\mathbb{R}$ (also at $y = 0$, where $\psi_\kappa = 0$; resonant $q_k = 0$ is allowed). *Global bounds (all $y \in \mathbb{R}$):* $|\psi_\kappa(y)| \le \frac{1}{2\kappa}$, $|\psi_\kappa'(y)| \le \kappa^{-2}$ and $0 \le 1 - y\psi_\kappa(y) = \dfrac{\kappa^2}{y^2+\kappa^2} \le 1$. *Bounds away from 0 (only for $y \ne 0$):* $|\psi_\kappa(y)| \le \frac{1}{|y|}$, $|\psi_\kappa'(y)| \le y^{-2}$, $\frac{\kappa^2}{y^2+\kappa^2} \le \kappa^2y^{-2}$ (elementary: $\psi_\kappa' = (\kappa^2-y^2)/(y^2+\kappa^2)^2$, $|\psi_\kappa'| \le \max(\kappa^2,y^2)/(y^2+\kappa^2)^2 \le (y^2+\kappa^2)^{-1}$, and $2\kappa|y| \le y^2+\kappa^2$). With total division in Lean these negative-power bounds are false at $y = 0$ and are never used there: the shell estimates use them only on $Z_j$, $j \ge 1$, where $|q_k| > \kappa > 0$, and the global bounds on $Z_0$ and in Lemma 13.7. For $(\theta,I) \in \mathbb{R}^n \times B(0,R)$ define

$$u_k(I) = f_k(I)\,\psi_\kappa(q_k(I))\,k \in \mathbb{C}^n, \qquad w(\theta,I) = -\sum_{k\ne0} u_k(I)\,\mathrm{e}_k(\theta) \in \mathbb{C}^n,$$

$$r(\theta,I) = a(\theta,I) - \partial_\theta w(\theta,I)\,\omega(I).$$

Here $\partial_\theta w\,\omega = \sum_\ell \omega_\ell \partial_{\theta_\ell} w$. Envelopes (functions of $I \in B(0,R)$, values in $[0,\infty]$):

$$\widehat{W}(I) = \sum_{k\neq 0} |k|_\infty \beta_k\, |\psi_\kappa(q_k(I))|, \qquad \widehat{W}_1(I) = \sum_{k\neq 0} 2\pi |k|_\infty |k|_1 \beta_k\, |\psi_\kappa(q_k(I))|,$$

$$\widehat{B}(I) = \sum_{k\neq 0} |k|_\infty \beta_k \Big[\tfrac{2n}{s}|\psi_\kappa(q_k(I))| + M_2 |k|_1 |\psi'_\kappa(q_k(I))|\Big], \qquad \widehat{R}(I) = \sum_{k\neq 0} 2\pi |k|_\infty \beta_k \,\frac{\kappa^2}{q_k(I)^2+\kappa^2},$$

and $\widehat{\Phi}_{\varepsilon,\kappa} = \varepsilon \tfrac{4}{s}\widehat{B} + \varepsilon \tfrac{2}{s}\widehat{W}_1 + \widehat{R}$ (the integrand of the time integral multiplied by $\varepsilon$ in Proposition 13.13) (all Borel; the pointwise sums of nonnegative continuous functions).

**Lemma 13.7** (Regularity of $w$ and the divisor identity)**.** *On $\mathbb{R}^n \times B(0,R)$ the function $w$ is $C^1$, 1-periodic in $\theta$, with termwise derivatives, and for $I \in G$ (the bounds with $M_2$ need only $|I| \le R$):*

*(a) $|w(\theta,I)|_\infty \le \widehat{W}(I)$;*

*(b) $r(\theta,I) = -2\pi i \sum_{k\neq 0} f_k(I)\, \dfrac{\kappa^2}{q_k^2+\kappa^2}\, k\, \mathrm{e}_k(\theta)$, hence $|r|_\infty \le \widehat{R}(I)$;*

*(c) $|\partial_\theta w\, v|_\infty \le \widehat{W}_1(I)|v|_\infty$ for $v \in \mathbb{R}^n$ (more precisely $|\partial_\theta w\, \partial_I f|_\infty \le \tfrac{2}{s}\widehat{W}_1$);*

*(d) for $v \in \mathbb{R}^n$: $|D_I w(\theta,I)v|_\infty \le \widehat{B}(I)|v|_\infty$, and for every $k \neq 0$, $|D_I u_k(I)v|_\infty \le b_k(I)|v|_\infty$ with $b_k$ the $k$-th summand of $\widehat{B}$.*

*Proof.* **Step S13.16: Smoothness.** Each $u_k$ is $C^1$ on $B(0,R)$: $f_k$ is holomorphic hence real-analytic, $q_k$ is $C^2$, $\psi_\kappa$ is smooth. $\partial_{I_j} u_k = (\partial_j f_k\, \psi_\kappa(q_k) + f_k\, \psi'_\kappa(q_k)\, \partial_j q_k)\, k$ and $\partial_j q_k = (M^\top k)_j$, so $\sum_j |\partial_j q_k| \le \|M\||k|_1 \le M_2|k|_1$ on $\overline{B}(0,R)$. On every $\overline{B}(0,R')$, $R' < R$ the terms $u_k \mathrm{e}_k$ and their $\theta$- and $I$-derivatives are bounded by $c(\kappa,R')\,(1+|k|_1)^3\beta_k$, summable by Lemma 13.5(d) (here $|\psi_\kappa| \le 1/(2\kappa)$, $|\psi'_\kappa| \le \kappa^{-2}$ are used). By (X2) $w$ is $C^1$ with termwise derivatives, $\partial_{\theta_\ell} w = -2\pi i \sum k_\ell u_k \mathrm{e}_k$.

**Step S13.17: (a),(c).** $|u_k|_\infty \le |k|_\infty \beta_k |\psi_\kappa(q_k)|$; sum. For (c): $\partial_\theta w\, v = -2\pi i \sum_k (k\cdot v) u_k \mathrm{e}_k$, $|k\cdot v| \le |k|_1 |v|_\infty$.

**Step S13.18: (b).** $\partial_\theta w\,\omega = -2\pi i \sum_k q_k u_k \mathrm{e}_k = -2\pi i \sum_k f_k q_k \psi_\kappa(q_k)\, k\, \mathrm{e}_k$ and $a = -2\pi i \sum f_k k \mathrm{e}_k$ (Lemma 13.5(c)); subtract and use $1 - q\psi_\kappa(q) = \kappa^2/(q^2+\kappa^2)$ and $|f_k| \le \beta_k$.

**Step S13.19: (d).** $D_I u_k v = \big(\sum_j v_j \partial_j f_k\, \psi_\kappa(q_k) + f_k \psi'_\kappa(q_k) \sum_j v_j \partial_j q_k\big) k$; by Lemma 13.5(b) and the row-sum bound its norm is $\le |k|_\infty \beta_k [\tfrac{2n}{s}|\psi_\kappa| + M_2|k|_1|\psi'_\kappa|]|v|_\infty = b_k|v|_\infty$; sum over $k$. □

13.4. **From the rank hypothesis to slab volumes.** Let $\mathbb{S} = \{e \in \mathbb{R}^n : |e|_2 = 1\}$ (Euclidean unit sphere); $|\cdot|_2$ is used only here and in $|k|_2 \ge 1$, $|k|_1 \le \sqrt{n}|k|_2$, $|k|_2 \le \sqrt{n}|k|_\infty$ (Lemma 2.1).

**Lemma 13.8** (Uniform transversality)**.** *There is $\lambda > 0$ with $|M(I)^\top e|_2^2 + (e\cdot\omega(I))^2 \ge \lambda^2$ for all $I \in G$, $e \in \mathbb{S}$.*

*Source: Ours; this is the coordinate-free replacement of Nei76 p. 1296 3° (chart $\omega_1 \neq 0$), notes.md "Rank geometry".*

*Proof.* **Step S13.20: Adjoint injective.** In branch (A) of Definition 1.30 $M(I)$ is onto, so $M(I)^\top$ is injective by the same argument ($A = M$, $A^\top e = 0 \Rightarrow e = 0$): $M^\top e \neq 0$ for $e \neq 0$. In branch (B) the map $(v,c) \mapsto Mv + c\omega$ is onto, so is the map $A : \mathbb{R}^{n+1} \to \mathbb{R}^n$, $A(v,c) = Mv + c\omega$, with adjoint $A^\top e = (M^\top e,\ \omega\cdot e)$. $A^\top$ is injective by a direct argument covering both branches (square $M$ in (A), rectangular $A$ in (B)): if $A$ is onto and $A^\top e = 0$, choose $z$ with $Az = e$; then $|e|_2^2 = e\cdot Az = (A^\top e)\cdot z = 0$, so $e = 0$. Thus $(M^\top e, \omega\cdot e) \neq 0$ for $e \neq 0$ (no further rank assumption). In both branches $N(I,e) := |M(I)^\top e|_2^2 + (e\cdot\omega(I))^2 > 0$ on $G \times \mathbb{S}$ (the branch is the same at all $I$, but this is not even needed).

**Step S13.21: Compactness.** $N$ is continuous on the compact $G\times\mathbb{S}$ ($M, \omega$ continuous on $B(0,R+s) \supset G$), positive, so has a positive minimum $\lambda^2$. □

**Lemma 13.9** (Slab volumes, uniform in the direction)**.** *There is $C_1 \geq 1$ (depending on $n, R, \lambda, \omega$ only) such that for all $e \in \mathbb{S}$ and $v > 0$: $\mathrm{vol}\{I \in G : |e \cdot \omega(I)| \leq v\} \leq C_1 v$. Consequently for $k \in \mathbb{Z}^n \setminus \{0\}$ and $u > 0$,*

$$V_k(u) := \mathrm{vol}\{I \in G : |q_k(I)| \leq u\} \leq C_1\, u/|k|_2.$$

*Source: Ours (rank-to-volume, notes.md and F-NAVG-4); replaces the integral estimates of Nei76 Lemma 1, p. 1296.*

*Proof.* Write $\varphi_e(I) = e \cdot \omega(I)$; $\nabla \varphi_e = M^\top e$ (a $C^1$ function on $B(0, R+s)$).

**Step S13.22: Cover of the resonant set.** Let $K_0 = \{(I,e) \in G \times \mathbb{S} : |\varphi_e(I)| \leq \lambda/2\}$, compact. For $(I,e) \in K_0$ Lemma 13.8 gives $|M^\top e|_2^2 \geq \frac{3}{4}\lambda^2$, so some coordinate $j$ has $|(M^\top e)_j| \geq \frac{\sqrt{3}\lambda}{2\sqrt{n}} > c_0 := \frac{\lambda}{2\sqrt{n}}$. By continuity of $(I', e') \mapsto (M(I')^\top e')_j$ there are an open cube $Q \subset B(0,R)$ centred at $I$ (side $\ell_Q \leq 1$) and an open $U \ni e$ in $\mathbb{S}$ such that on $Q \times U$ this coordinate has constant sign and modulus $\geq c_0$. By compactness finitely many $(Q_m, U_m, j_m)$, $m \leq p$, cover $K_0$.

**Step S13.23: One box.** Fix $e \in \mathbb{S}$, $0 < v \leq \lambda/2$, and $m$ with $e \in U_m$. Fubini along coordinate $j_m$ ((X5)): for fixed other coordinates, the section $\{t : I + te_{j_m} \in Q_m,\ |\varphi_e| \leq v\}$ lies in a segment on which $\partial_{j_m}\varphi_e$ has constant sign and modulus $\geq c_0$; by the mean value theorem two points of the section are at distance $\leq 2v/c_0$. So $\mathrm{vol}(Q_m \cap \{|\varphi_e| \leq v\}) \leq \frac{2v}{c_0}\ell_{Q_m}^{n-1}$.

**Step S13.24: Sum.** If $I \in G$, $|\varphi_e(I)| \leq v \leq \lambda/2$, then $(I,e) \in K_0$ lies in some $Q_m \times U_m$. Hence $\mathrm{vol} \leq \sum_m \frac{2}{c_0}\ell_{Q_m}^{n-1} v \leq \frac{2p}{c_0} v$. For $v > \lambda/2$: $\mathrm{vol} \leq \mathrm{vol}(G) \leq \frac{2\,\mathrm{vol}\,G}{\lambda} v$. Put $C_1 = \max(1, 2p/c_0, 2\,\mathrm{vol}(G)/\lambda)$.

**Step S13.25: Integer vectors.** $q_k = |k|_2\, \varphi_{k/|k|_2}$, so $V_k(u) = \mathrm{vol}\{|\varphi_e| \leq u/|k|_2\} \leq C_1 u/|k|_2$. □

13.5. **Envelope integrals.** Let $\ell_\kappa = 1 + \ln(1/\kappa)$ for $0 < \kappa \leq 1$ (so $\ell_\kappa \leq 1/\kappa$).

**Lemma 13.10** (Dyadic shell estimates)**.** *For $k \neq 0$, $0 < \kappa \leq 1$ and $\Omega_0, C_1$ as above:*

$$\textit{(i)}\ \int_G |\psi_\kappa(q_k)|\, dI \leq C_1\big(\tfrac{5}{2} + 3\ln^+ \tfrac{\Omega_0|k|_1}{\kappa}\big) \leq 3C_1(1+\Omega_0)(1+|k|_1)\,\ell_\kappa,$$

$$\textit{(ii)}\ \int_G |\psi_\kappa'(q_k)|\, dI \leq \frac{5C_1}{\kappa|k|_2}, \qquad \textit{(iii)}\ \int_G \frac{\kappa^2}{q_k^2+\kappa^2}\, dI \leq \frac{5C_1\kappa}{|k|_2}.$$

*Proof.* **Step S13.26: Shells.** Let $Z_0 = \{I \in G : |q_k| \leq \kappa\}$, $Z_j = \{I \in G : 2^{j-1}\kappa < |q_k| \leq 2^j\kappa\}$ ($j \geq 1$). Lemma 13.9: $\mathrm{vol}\, Z_0 \leq C_1\kappa/|k|_2$, $\mathrm{vol}\, Z_j \leq V_k(2^j\kappa) \leq C_1 2^j\kappa/|k|_2$. Since $|q_k| \leq |k|_1\Omega_0$ on $G$, $Z_j = \emptyset$ unless $2^{j-1}\kappa < |k|_1\Omega_0$, i.e. $j \leq J := 1 + \log_2^+(\Omega_0|k|_1/\kappa)$.

**Step S13.27: (i).** On $Z_0$: $|\psi| \leq \frac{1}{2\kappa}$, contribution $\leq \frac{C_1}{2|k|_2}$; on $Z_j$: $|\psi| \leq \frac{1}{2^{j-1}\kappa}$, contribution $\leq \frac{2C_1}{|k|_2}$. Sum $\leq C_1(\frac{1}{2} + 2J)$, with $|k|_2 \geq 1$ and $2\log_2^+ \leq 3\ln^+$ ($2/\ln 2 < 3$). The second inequality: $\frac{5}{2} + 3\ln^+(xy) \leq 3(1 + \Omega_0|k|_1 + \ln\frac{1}{\kappa})$ for $x = \Omega_0|k|_1$, $y = 1/\kappa \geq 1$, using $\ln^+ x \leq x$.

**Step S13.28: (ii).** $|\psi'| \leq \kappa^{-2}$ on $Z_0$ (contribution $\leq \frac{C_1}{\kappa|k|_2}$) and $\leq (2^{j-1}\kappa)^{-2}$ on $Z_j$ (contribution $\leq \frac{4C_1}{2^j\kappa|k|_2}$); $\sum_{j\geq 1} 2^{-j} = 1$.

**Step S13.29: (iii).** $\frac{\kappa^2}{q^2+\kappa^2} \leq 1$ on $Z_0$ and $\leq 4\cdot 4^{-j}$ on $Z_j$; the contributions are $\leq \frac{C_1\kappa}{|k|_2}$ and $\frac{4C_1\kappa}{2^j|k|_2}$. □

**Proposition 13.11** (Envelope integrals)**.** *With $\sigma$ of Lemma 13.5(d) put $c_W = 3C_1(1+\Omega_0)\sigma$, $c_B = \sigma C_1\big(6n(1+\Omega_0)/s + 5\sqrt{n}M_2\big)$, $c_R = 10\pi C_1\sigma$. For $0 < \kappa \leq 1$:*

$$\int_G \widehat{W} \leq c_W\ell_\kappa, \quad \int_G \widehat{W}_1 \leq 2\pi c_W\ell_\kappa, \quad \int_G \widehat{B} \leq \frac{c_B}{\kappa}, \quad \int_G \widehat{R} \leq c_R\kappa,$$

*hence, for* $\kappa=\sqrt{\varepsilon}$, $0<\varepsilon\le 1$, *with* $c_\Phi=\frac{4}{s}c_B+\frac{8\pi}{s}c_W+c_R$:

$$2\varepsilon\int_G \widehat{W}\le 4c_W\sqrt{\varepsilon},\qquad \int_G \widehat{\Phi}_{\varepsilon,\kappa}\le c_\Phi\sqrt{\varepsilon}.$$

*Proof.* **Step S13.30: Termwise.** Tonelli ((X4)) and Lemma 13.10, using $|k|_\infty(1+|k|_1)\le(1+|k|_1)^2$, $|k|_1|k|_\infty(1+|k|_1)\le(1+|k|_1)^3$ and $|k|_1/|k|_2\le\sqrt{n}$: $\int\widehat{W}\le\sum_k\beta_k(1+|k|_1)^2\,3C_1(1+\Omega_0)\ell_\kappa\le c_W\ell_\kappa$; $\int\widehat{W}_1\le 2\pi c_W\ell_\kappa$ likewise; $\int\widehat{B}\le\sum|k|_\infty\beta_k\big[\frac{2n}{s}3C_1(1+\Omega_0)(1+|k|_1)\ell_\kappa+M_2|k|_1\frac{5C_1}{\kappa|k|_2}\big]\le c_B/\kappa$ (as $\ell_\kappa\le 1/\kappa$); $\int\widehat{R}\le 2\pi\sum|k|_\infty\beta_k\,5C_1\kappa\le c_R\kappa$.

**Step S13.31: The choice $\kappa=\sqrt{\varepsilon}$.** $\varepsilon/\kappa=\sqrt{\varepsilon}$. And $\varepsilon\ell_\kappa=\varepsilon+\frac{\varepsilon}{2}\ln\frac{1}{\varepsilon}\le\sqrt{\varepsilon}+\frac12\sqrt{\varepsilon}\cdot\sqrt{\varepsilon}\ln\frac{1}{\varepsilon}\le 2\sqrt{\varepsilon}$ since $\sqrt{\varepsilon}\ln\frac{1}{\varepsilon}=2\sqrt{\varepsilon}\ln\frac{1}{\sqrt{\varepsilon}}\le 2/e<1$. Insert: $\int\widehat{\Phi}=\frac{4\varepsilon}{s}\int\widehat{B}+\frac{2\varepsilon}{s}\int\widehat{W}_1+\int\widehat{R}\le\frac{4}{s}c_B\sqrt{\varepsilon}+\frac{2}{s}2\pi c_W 2\sqrt{\varepsilon}+c_R\sqrt{\varepsilon}$. □

13.6. **The corrector identity and the pointwise stopped estimate.**

**Lemma 13.12** (Corrector identity)**.** *Let* $\varepsilon>0$, $0<\kappa\le 1$, $x=(\theta_0,I_0)\in W_0$, $\gamma_x=(\theta,I)$ *and* $0\le t<T_+(x)$. *Put* $\Lambda(u)=w(\theta(u),I(u))$. *Then* $\Lambda$ *is* $C^1$ *and*

$$I(t)-I_0=\varepsilon\big(\Lambda(t)-\Lambda(0)\big)-\varepsilon^2\int_0^t Q(u)\,du+\varepsilon\int_0^t r(\theta(u),I(u))\,du,$$

$$Q=\partial_\theta w\,\partial_I f+D_I w\,a\quad\text{evaluated at }(\theta(u),I(u)).$$

*Proof.* **Step S13.32: The equations.** By Lemma 13.1 and Part 4: $\dot\theta=\omega(I)+\varepsilon\partial_I f$, $\dot I=-\varepsilon\partial_\theta f=\varepsilon a$, at $(\theta(u),I(u))$.

**Step S13.33: Chain rule.** $w$ is $C^1$ (Lemma 13.7), so $\dot\Lambda=\partial_\theta w\,\dot\theta+D_I w\,\dot I=\partial_\theta w\,\omega+\varepsilon\,\partial_\theta w\,\partial_I f+\varepsilon D_I w\,a$.

**Step S13.34: Divisor identity.** $\partial_\theta w\,\omega=a-r$ by the definition of $r$. Hence $\varepsilon a=\varepsilon\dot\Lambda-\varepsilon^2 Q+\varepsilon r$; integrate $\dot I=\varepsilon a$ over $[0,t]$. □

**Proposition 13.13** (Pointwise estimate up to a stopping time)**.** *Let* $\varepsilon\in(0,1]$, $\kappa\in(0,1]$, $x=(\theta_0,I_0)\in W_0$ *with* $|I_0|<R/2$, *and* $0\le t<T_+(x)$ *with* $|I(u,x)|\le 3R/4$ *for all* $u\in[0,t]$. *Then*

$$\max_{0\le u\le t}|I(u,x)-I_0|_\infty\ \le\ 2\varepsilon\,\widehat{W}(I_0)+\varepsilon\int_0^t\widehat{\Phi}_{\varepsilon,\kappa}\big(I(s,x)\big)\,ds.$$

*Proof.* Write $\widehat{\Phi}=\widehat{\Phi}_{\varepsilon,\kappa}$ and use the notation of Lemma 13.12; all points $I(u)$ lie in $G$.

**Step S13.35: Sup of $\Lambda$.** For each $k$ the function $u\mapsto u_k(I(u))$ is $C^1$ with derivative $D_I u_k\dot I$, so $|u_k(I(u))|_\infty\le|u_k(I_0)|_\infty+\int_0^u b_k(I(s))\,\varepsilon\frac{2}{s}\,ds$ (Lemma 13.7(d), $|\dot I|_\infty\le\varepsilon\frac{2}{s}$ by Lemma 13.1). Therefore, with $|\Lambda(u)|_\infty\le\sum_k|u_k(I(u))|_\infty$, $\max_{u\le t}|\Lambda(u)|_\infty\le\widehat{W}(I_0)+\varepsilon\frac{2}{s}\int_0^t\widehat{B}(I(s))\,ds$.

**Step S13.36: The other terms.** Lemma 13.7(c),(d) and $|a|_\infty,|\partial_I f|_\infty\le\frac{2}{s}$: $|Q|\le\frac{2}{s}(\widehat{W}_1+\widehat{B})$; $|r|\le\widehat{R}$.

**Step S13.37: Sum.** By Lemma 13.12 for each $u\le t$ (all integrals monotone in $u$): $|I(u)-I_0|\le\varepsilon(\max|\Lambda|+|\Lambda(0)|)+\varepsilon^2\int_0^t|Q|+\varepsilon\int_0^t|r|\le 2\varepsilon\widehat{W}(I_0)+\varepsilon\int_0^t\big[\varepsilon\frac{2}{s}\widehat{B}+\varepsilon\frac{2}{s}(\widehat{W}_1+\widehat{B})+\widehat{R}\big]ds$, and the bracket is $\widehat{\Phi}$. □

13.7. **Transport and the stopped mean estimate.** Let $N_K = \#\{\ell \in \mathbb{Z}^n : K \cap ((\ell + [0,1)^n) \times \mathbb{R}^n) \neq \emptyset\}$ (finite: $K$ is bounded), $K_\ell = K \cap ((\ell + [0,1)^n) \times \mathbb{R}^n)$ (Borel).

**Lemma 13.14** (Transport of envelope integrals)**.** *Let $g : \mathbb{R}^n \to [0,\infty]$ be Borel, $\ell \in \mathbb{Z}^n$, $t \geq 0$, and $D \subset \Omega_t \cap \big((\ell + [0,1)^n) \times B(0,R)\big)$ Borel with $I(t,x) \in G$ for $x \in D$. Then $\int_D g(I(t,x))\,dx \leq \int_G g(I)\,dI$.*
*Source: Ours; replaces Nei76 p. 1296 (15)–(16) (Jacobian bounded) by exact Liouville and periodic cell decomposition.*

*Proof.* **Step S13.38: Change of variables.** By Lemma 13.3(a)–(c), $\Phi_t : \Omega_t \to W_0$ is injective, $C^1$ with $\det = 1$; (X1): $\int_D g(I(t,x))dx = \int_{\Phi_t(D)} g(y_2)\,dy$ ($y = (y_1, y_2)$), $\Phi_t(D)$ Borel, $\Phi_t(D) \subset \mathbb{R}^n \times G$.

**Step S13.39: Cells.** For $m \in \mathbb{Z}^n$ let $P_m = \big(\Phi_t(D) \cap ((m + [0,1)^n) \times \mathbb{R}^n)\big) - (m,0) \subset [0,1)^n \times G$. The $P_m$ are pairwise disjoint: if $y \in P_m \cap P_{m'}$, $y + (m,0) = \Phi_t(x)$, $y + (m',0) = \Phi_t(x')$ with $x, x' \in D$, then by Lemma 13.3(d) $\Phi_t(x - (m,0)) = y = \Phi_t(x' - (m',0))$, so $x - x' = (m - m', 0)$ by injectivity; the angle parts of $x, x'$ lie in one half-open unit cube, so $m = m'$.

**Step S13.40: Sum.** $g(y_2)$ does not depend on $y_1$ and Lebesgue measure is translation invariant, so $\int_{\Phi_t(D)} g(y_2)dy = \sum_m \int_{P_m} g(y_2)dy = \int_{\cup P_m} g(y_2)dy \leq \int_{[0,1)^n \times G} g(y_2) = \int_G g$. □

For $x \in K$ put, with $T = 1/\varepsilon$,

$$\Psi(x) = 2\varepsilon\,\widehat{W}(I_0) + \varepsilon \int_0^T \mathbf{1}_S(s,x)\,\widehat{\Phi}_{\varepsilon,\kappa}\big(I(s,x)\big)\,ds \qquad (\kappa = \sqrt{\varepsilon}),$$

with $S$ of Lemma 13.4 ($\mathbf{1}_S(s,x) = 1$ iff $s < \tau(x)$; the integrand is evaluated only there, where $I(s,x) \in G$).

**Proposition 13.15** (Stopped mean estimate)**.** *Let $0 < \varepsilon \leq 1$. $\Psi : K \to [0,\infty]$ is Borel and $\int_K \Psi\,dx \leq A_1\sqrt{\varepsilon}$ with $A_1 = \max\big(1, N_K(c_\Phi + 4c_W)\big)$, a constant depending only on $n, R, s, h, K$ (not on $f$ beyond $|f| \leq 1$, nor on $\varepsilon$). Moreover for every $x \in K$ and $t_1 = \min(T, \tau(x))$ (so $t_1 < T_+(x)$, $I(u,x) \in G$ for $u \leq t_1$),*

$$e_{t_1}(x) := \max_{0 \leq u \leq t_1} |I(u,x) - I_0| \leq \Psi(x).$$

*Proof.* **Step S13.41: Measurability.** $K \subset W_0$ and $S$ is open (Lemma 13.4); $(s,x) \mapsto I(s,x)$ is continuous on $\mathcal{D}$ (Lemma 13.3(a)); $\widehat{\Phi}$, $\widehat{W}$ Borel. So $(s,x) \mapsto \mathbf{1}_S\widehat{\Phi}(I(s,x))$ is Borel on $[0,T] \times K$ and $\Psi$ is Borel by Tonelli.

**Step S13.42: The pointwise bound.** For $x \in K$, $t_1 < T_+(x)$ by Lemma 13.4(i),(ii) (if $t_1 = T < \tau$, then $T < \tau \leq T_+$; if $t_1 = \tau < \infty$, $\tau < T_+$), and $|I(u) - I_0| \leq \rho$ on $[0,t_1]$, so $|I(u)| \leq 3R/4$. Proposition 13.13 with $t = t_1$ and $\int_0^{t_1} = \int_0^T \mathbf{1}_S$ (the set $\{s < \tau\} \cap [0,T] = [0,t_1)$ up to a null set) gives the bound.

**Step S13.43: Integral.** For each $\ell$, by Tonelli and Lemma 13.14 with $D = D_s = \{x \in K_\ell : (s,x) \in S\}$ (Borel; $I(s,x) \in G$ by Lemma 13.4(iii)) and $g = \widehat{\Phi}_{\varepsilon,\kappa}$:

$$\int_{K_\ell} \varepsilon \int_0^T \mathbf{1}_S\,\widehat{\Phi}\big(I(s,x)\big)\,ds\,dx = \varepsilon \int_0^T \int_{D_s} \widehat{\Phi}\big(I(s,x)\big)\,dx\,ds \leq \varepsilon T \int_G \widehat{\Phi} = \int_G \widehat{\Phi} \leq c_\Phi\sqrt{\varepsilon}.$$

Also $\int_{K_\ell} 2\varepsilon\widehat{W}(I_0)\,dx \leq 2\varepsilon \int_{B(0,R/2)} \widehat{W} \leq 4c_W\sqrt{\varepsilon}$ ($K_\ell$ lies in a unit cell times $B(0,R/2)$; Proposition 13.11). Sum over the $N_K$ cells. □

13.8. **Proof of Theorem 1.34 (N-AVERAGING-MOST).**

Constants. After the fixed data: $\rho = R/8$, $\varepsilon_0 = 1$, $\eta_0 = R/8$ ($\le R/4$), $A = A_1$ of Proposition 13.15 (they depend on $n, R, s, h, K$ only). Let $0 < \varepsilon \le 1$, $0 < \eta \le \eta_0$, $T = 1/\varepsilon$, $\kappa = \sqrt{\varepsilon}$.

**Lemma 13.16** (The good set is open in $K$). $\Gamma_\eta = \{x \in K : T < T_+(x)$ *and* $\max_{0\le t\le T}|I(t,x) - I_0| < \eta\}$ *is relatively open in* $K$*, hence Borel;* $E := K \setminus \Gamma_\eta$ *is measurable.*

*Proof.* $\Omega_T$ is open and $x \mapsto \max_{[0,T]}|I(t,x) - I_0|$ is continuous on $\Omega_T$ (Lemma 13.3(a), (X4)); $\Gamma_\eta = K \cap \Omega_T \cap \{\max < \eta\}$. □

**Lemma 13.17** (Bad data have large $\Psi$). *For* $\eta \le \rho$ *and* $x \in K \setminus \Gamma_\eta$*:* $\Psi(x) \ge \eta$*. For* $x \in K \setminus \Gamma_\rho$ *the same with* $\rho$*.*

*Proof.* **Step S13.44: Case $\tau(x) \le T$.** Then $t_1 = \tau < T_+$ and $|I(\tau) - I_0| = \rho \ge \eta$ (Lemma 13.4(i)); so $e_{t_1}(x) \ge \rho \ge \eta$.

**Step S13.45: Case $\tau(x) > T$.** Then $T < T_+(x)$ ($T < \tau \le T_+$) and $|I(u) - I_0| < \rho$ on $[0,T]$; as $x \notin \Gamma_\eta$, $\max_{[0,T]}|I - I_0| \ge \eta$, so $e_{t_1}(x) = e_T(x) \ge \eta$ with $t_1 = T$.

**Step S13.46: Conclusion.** $\Psi(x) \ge e_{t_1}(x) \ge \eta$ by Proposition 13.15. (For $\Gamma_\rho$ the same, case 2 giving $\max < \rho$, a contradiction, so only case 1 occurs.) □

*Proof of Theorem 1.34.* **Step S13.47: The exceptional set.** $E = K\setminus\Gamma_\eta$ is measurable and $E \subset K$ (Lemma 13.16). By Lemma 13.17, Markov and Proposition 13.15: $\eta \operatorname{vol} E \le \int_E \Psi \le \int_K \Psi \le A\sqrt{\varepsilon}$, so $\operatorname{vol} E \le A\sqrt{\varepsilon}/\eta$, i.e. $\le \operatorname{ofReal}(A\sqrt{\varepsilon}/\eta)$.

**Step S13.48: The solution.** For $x \in K \setminus E = \Gamma_\eta$ let $z = \gamma_x$ (the maximal solution, extended arbitrarily, e.g. by $x$, outside $(T_-, T_+)$). Then $z(0) = x$ and $[0,T] \subset (T_-(x), T_+(x))$ with $T_- < 0$, so $z$ is a solution of $X_H$ in $W_0$ with two-sided derivative at every $t \in [0,T]$, hence `IsSolutionOn` $H\, B(0,R)\, [0,T]\, z$ by Proposition 4.25. And $|z(t)_2 - x_2| < \eta$ on $[0,T]$ by definition of $\Gamma_\eta$. □

13.9. **Proof of Theorem 1.33 (N-AVERAGING-MEAN).** Constants: $\varepsilon_0 = 1$, $A = 13A_1$. Fix $0 < \varepsilon \le 1$ and $\Phi$ as in the theorem.

*Proof.* **Step S13.49: The family is the flow.** For $x \in K$, $\Phi x$ is a solution on $[0,T]$ with $\Phi x\, 0 = x$, so by Proposition 4.25 and Theorem 4.13: $[0,T] \subset (T_-(x), T_+(x))$ and $\Phi x = \gamma_x$ on $[0,T]$. In particular $K \subset \Omega_T$ and for $x \in K$, $t \in [0,T]$: $(\Phi x\, t)_2 = I(t,x)$.

**Step S13.50: The error is continuous on $K$.** By the previous step $e_\varepsilon^\Phi(x) = \sup_{[0,T]}|I(t,x) - I_0| = \max_{[0,T]}|I(t,x) - I_0| =: m(x)$ for $x \in K$ (a continuous function on the compact interval attains its maximum; the supremum is the real supremum, Remark 1.32), and $m$ is continuous on $\Omega_T \supset K$ ((X4)). $K$ is compact, so $m$ is bounded and Borel on $K$ and $\operatorname{vol} K < \infty$: $e_\varepsilon^\Phi$ agrees on $K$ with a continuous function, so it is integrable on $K$ (it is a.e.-strongly measurable and bounded on a finite-measure set).

**Step S13.51: Bound by $\Psi$ on good data.** Let $\Gamma_\rho$ be as in Lemma 13.16 with $\eta = \rho$. For $x \in \Gamma_\rho$: $\tau(x) > T$, $t_1 = T$ and $m(x) = e_T(x) \le \Psi(x)$ (Proposition 13.15).

**Step S13.52: Bound on bad data.** For $x \in K \setminus \Gamma_\rho$: $m(x) \le \max(|I(t)| + |I_0|) < R + R/2$ since the solution is in $B(0,R)$ (it is the supplied solution with actions in $B(0,R)$); and $\rho \operatorname{vol}(K \setminus \Gamma_\rho) \le \int \Psi \le A_1\sqrt{\varepsilon}$ (Lemma 13.17, Markov).

**Step S13.53: The mean.** $\int_K m \le \int_{\Gamma_\rho} \Psi + \frac{3R}{2}\operatorname{vol}(K \setminus \Gamma_\rho) \le A_1\sqrt{\varepsilon}\big(1 + \frac{3R/2}{R/8}\big) = 13A_1\sqrt{\varepsilon} = A\sqrt{\varepsilon}$. As $m \ge 0$, $\int_K^- \operatorname{ofReal}(m) = \operatorname{ofReal}\int_K m \le \operatorname{ofReal}(A\sqrt{\varepsilon})$ (the Lean `lintegral` of a nonnegative integrable function).

**Step S13.54: The Markov clause.** Let $\kappa > 0$, $\lambda = A\sqrt{\varepsilon}/\kappa > 0$, $E = \{x \in K : m(x) > \lambda\}$: measurable (continuous $m$ on measurable $K$), $E \subset K$, $\lambda \operatorname{vol} E \le \int_K m \le A\sqrt{\varepsilon}$, so $\operatorname{vol} E \le \kappa$. For $x \in K \setminus E$ and $t \in [0,T]$: $|(\Phi x\, t)_2 - x_2| \le m(x) \le \lambda$. □

13.10. **Remarks: errata, deviations, doubts, forward references.**

*Remark* 13.18 (Constants and order of choice). $\Omega_0, M_2$ (from $h$ on $\overline{B}(0,R)$), $\lambda$ (rank, Lemma 13.8), $C_1$ (Lemma 13.9), $\sigma$ (from $s$), $c_W, c_B, c_R, c_\Phi$, $N_K$, then $A_1$; $\varepsilon_0 = 1$, $\eta_0 = R/8$. Nothing depends on $\varepsilon, \eta, \kappa, \Phi$, and $f$ enters only through $|f| \le 1$ and holomorphy (so the constants are in fact uniform over the class of $f$); $\varepsilon_0 = 1$ is allowed since no estimate used smallness of $\varepsilon$. All exceptional measures are absolute volumes of subsets of $K$, exactly as frozen: $\operatorname{vol} E \le A\sqrt{\varepsilon}/\eta$ and $\operatorname{vol} E \le \kappa$; the factor $N_K$ is the number of period cells met by $K$ and $A_1$ contains the Jacobian of nothing (Liouville gives 1; the passage $\varphi = 2\pi\theta$ of the sources is never used).

*Remark* 13.19 (Errata and deviations). **E-NAVG-NEI-17 (confirmed, repaired).** Nei76 p. 1298 (17): $\sup_t z(t) = \int_{U'} h$, with $z(t) = \int_{U'} |I(t) - \bar{I}(t)|$ (p. 1297) and $h = \sup_t |I(t) - \bar{I}(t)|$ (p. 1295 (4)). Only $\sup_t \int \le \int \sup_t$ holds (two initial points with errors $t$ and $1-t$ on $[0,1]$: $\sup_t \int = 1$, $\int \sup = 2$); the Gronwall bound for $z$ at fixed time therefore does not bound the mean of the maximum, which is what is claimed. Repair: here the *maximum* is bounded pointwise first (Proposition 13.13) by $2\varepsilon\widehat{W}(I_0) + \varepsilon \int_0^t \widehat{\Phi}(I(s))ds$, and the integration over initial data is done afterwards at *fixed* $s$ (Tonelli, Lemma 13.14). For a general system with $\bar{a} \neq 0$ one would apply Gronwall to $M_A(t) = \int_A \sup_{u \le t} |I(u) - J(u)|$ as in notes.md; since $\bar{a} = 0$ here the Gronwall term vanishes ($L_0 = 0$) and is not needed. **Regularisation:** Nei76's $\theta_\varkappa$ is $C^1$ but not $C^2$ at $\pm\varkappa$ (the printed cubic: second derivatives $-6/\varkappa^3$ inside and $2/\varkappa^3$ outside); we use the rational $\psi_\kappa$, smooth. **Jacobian:** Nei76 (15)–(16) bound the Jacobian of the shift by $e^c$; exact for Hamiltonian flows is 1 (Lemma 13.3(c)). **Continuation steps:** Nei76 p. 1298 extends cohorts by time steps $\delta = 0.25c^{-1}\varepsilon^{-1}$; here the first exit time and the dichotomy of Lemma 13.4 replace them. **Chart:** Nei76 3° uses $\omega_1 \neq 0$; AKN p. 243 the intrinsic projective rank; Lemma 13.8 proves the quantitative form from the intrinsic one (D-NAVG-2). No error in AKN06 pp. 242–244 was found.

*Remark* 13.20 (Doubts about the frozen statements). None. $n \ge 2$ is not used; `IsAnalyticIntegrable` (iii), (v) and continuity on closures are not used beyond $C^3$ regularity of $h_{\mathbb{R}}$ on $B(0, R+s)$ and holomorphy on $B(0,R)_s$; the hypothesis `IsAnalyticPerturbation` gives holomorphy and $|f| \le 1$ on $D_s(R)$, periodicity and reality. The statement of the mean theorem with a family $\Phi$ defined only through its values on $K$ is consistent: values of $\Phi$ off $K$ never enter ($e_\varepsilon^\Phi = m$ on $K$).

*Remark* 13.21 (Forward and external references). Used: Part 4 (maximal solutions, continuation 4.16, first exit 4.19, uniqueness, reduction 4.25, 4.28, 4.24), Part 2 (2.3, 2.4), Part 5 (5.3), track K ([K:`bp:p1:decay`], [K:`bp:p1:fourier-basic`], [K:`bp:p1:cauchy-open`], [K:`bp:p1:exp-sums`], [K:`bp:p1:ext-measure`]). Parts 6 and 7 (Lie transform, homological equation) are not needed: the first-order change of variables here is the explicit corrector $w$, and the homological equation is solved coefficientwise with a regularised divisor. Not in Mathlib and written out here: (X3) (differentiable dependence), Lemmas 13.3, 13.9, 13.14. No forward references to unwritten parts.